# MCAS BEAUFORT
# REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M)

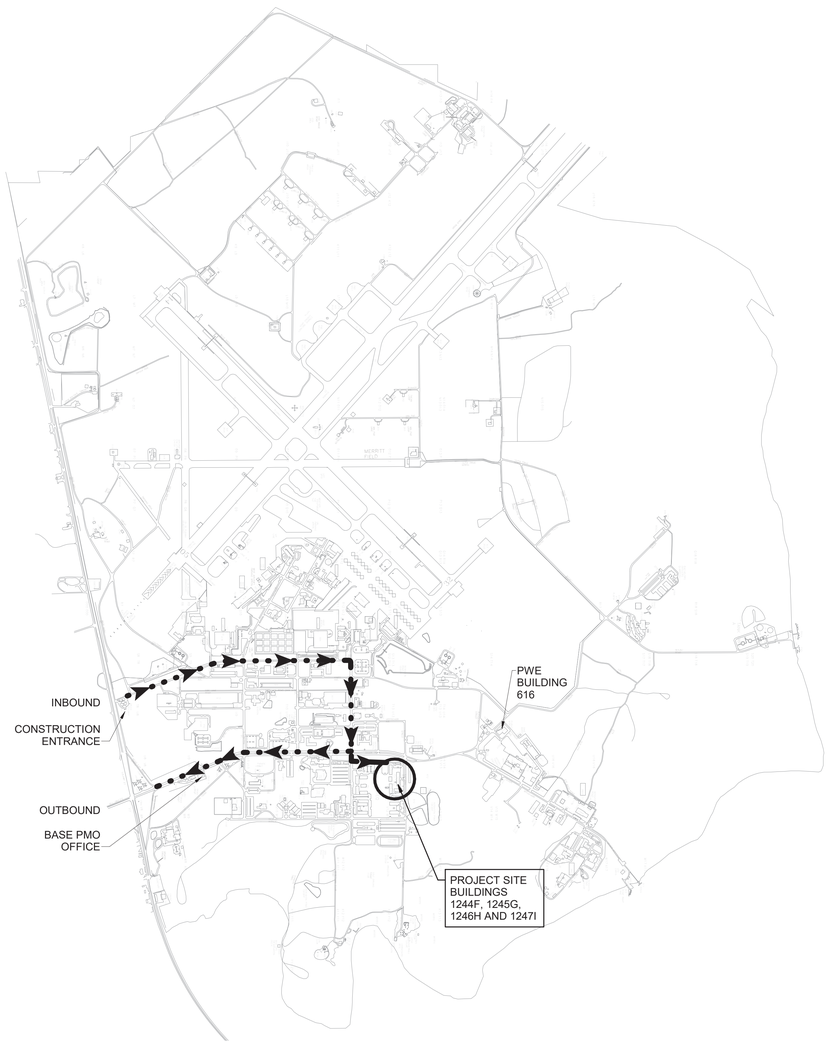
MARINE CORPS AIR STATION
BEAUFORT, SOUTH CAROLINA
08 SEPT 2023

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |

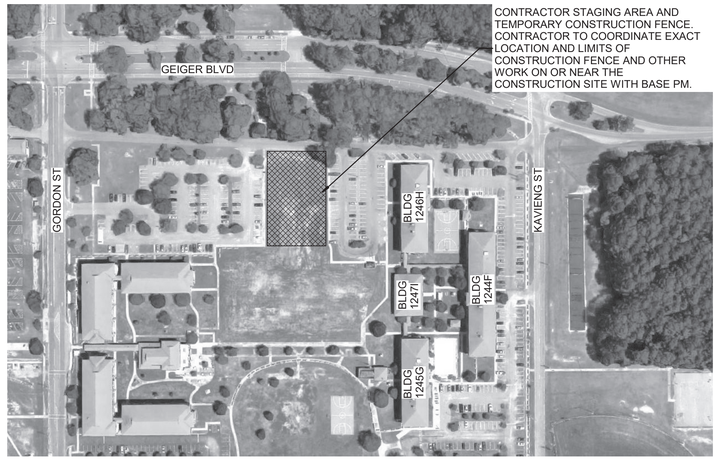
## GENERAL NOTES

THE DRAWINGS IN THIS SET ARE A STAND ALONE PACKAGE. THE ADDITIONAL PACKAGE DESIGNATIONS BELOW ARE FOR INFORMATION ONLY:

A. REPAIR PHASE 4 BEQ BUILDING 1244F (BE1812M) AND 1247I (BE1815M)
B. REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M)
C. REPAIR PHASE 4 BEQ BUILDING 1246H (BE1814M)



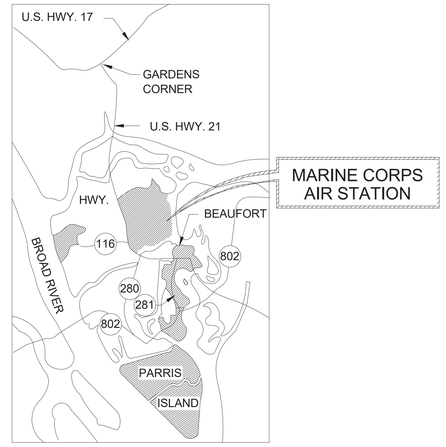

BASE LOCATION MAP
SCALE: NOT TO SCALE




LOCATION MAP
SCALE: NOT TO SCALE

PLAN NORTH

YOUR SIGNATURE ON THIS COVER SHEET APPROVES ALL SHEETS ASSOCIATED WITH THIS SET OF DRAWINGS.




VICINITY MAP
NOT TO SCALE

STATE LOCATION MAP
NOT TO SCALE

WISCONSIN
DEAN C. VON TERSCH
A-7903
CHARLESTON SC
ARCHITECT
04-OCT-2023

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE<br>VAIGNEUR.CHRISTOPHER.L.1264998332<br>Digitally signed by VAIGNEUR.CHRISTOPHER.L.1264998332<br>Date: 2023.09.26 08:54:45 -04'00' | P.W. DWG. NO.<br>28455 | CEMS | G-001 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE<br>LANOIE.RONALD.L.1011923352<br>Digitally signed by LANOIE.RONALD.L.1011923352<br>Date: 2023.09.26 13:07:36 -04'00' | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY<br>NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE<br>CRAFT.RICHARD.S.1058599340<br>Digitally signed by CRAFT.RICHARD.S.1058599340<br>Date: 2023.09.27 07:32:16 -04'00' | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M)<br>COVER SHEET | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE<br>RAMIREZ.PHILIP.JOHN.1395586064<br>Digitally signed by RAMIREZ.PHILIP.JOHN.1395586064<br>Date: 2023.09.26 11:12:01 -04'00' | SUPERVISOR<br>D. VON TERSCH | | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE<br>WEBB.OWEN.H.1229842990<br>Digitally signed by WEBB.OWEN.H.1229842990<br>Date: 2023.09.27 07:37:25 -04'00' | APPROVED DATE<br>GATES.JEREMY.BRYAN.1172057701<br>Digitally signed by GATES.JEREMY.BRYAN.1172057701<br>Date: 2023.09.27 15:40:45 -04'00' | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| PUBLIC WORKS OFFICER | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 | |
| | SCALE: As indicated | MACC. | T.O. # SHEET 1 OF 53 |

# INDEX OF DRAWINGS - BLDG 1245G

| # OF TOTAL | P.W.# | SHEET NO. | SHEET TITLE |
|---|---|---|---|
| GENERAL | | | |
| 1 | 28455 | G-001 | COVER SHEET |
| 2 | 28456 | G-002B | INDEX OF DRAWINGS - BLDG 1245G |
| CIVIL | | | |
| 3 | 28457 | C-001B | LEGEND AND GENERAL NOTES |
| 4 | 28458 | C-002B | GENERAL NOTES |
| 5 | 28459 | VF101B | EXISTING TOPOGRAPHIC SURVEY |
| 6 | 28460 | CD101B | DEMOLITION / EROSION CONTROL PLAN |
| 7 | 28461 | CS101B | SITE PLAN |
| 8 | 28462 | C-501B | EROSION CONTROL DETAILS |
| 9 | 28463 | C-502B | DETAILS |
| 10 | 28464 | C-503B | DETAILS |
| STRUCTURAL | | | |
| 11 | 28465 | S-001 | GENERAL NOTES AND DESIGN LOADS |
| 12 | 28466 | S-002 | SPECIAL INSPECTIONS AND NOTES |
| 13 | 28467 | S-101B | FIRST, SECOND AND THIRD FLOOR PLANS - BLDG 1245G |
| 14 | 28468 | S-102B | ROOF AND CEILING FRAMING PLANS - BLDG 1245G |
| 15 | 28469 | S-301 | DETAILS AND SECTIONS |
| 16 | 28470 | S-302 | SECTIONS |
| 17 | 28471 | S-401 | ENLARGED STAIR PLAN AND SECTIONS |
| 18 | 28472 | S-402 | ENLARGED STAIR PLAN AND SECTIONS |
| 19 | 28473 | S-501 | TYPICAL DETAILS |
| ARCHITECTURAL | | | |
| 20 | 28474 | A-001 | ABBREVIATIONS, SYMBOLS AND LEGENDS |
| 21 | 28475 | A-002 | GENERAL NOTES |
| 22 | 28476 | AD101B | DEMOLITION FIRST, SECOND AND THIRD FLOOR PLANS - BLDG 1245G |
| 23 | 28477 | AD102B | DEMOLITION ATTIC AND ROOF PLANS - BLDG 1245G |
| 24 | 28478 | AD103B | DEMOLITION FIRST, SECOND AND THIRD FLOOR REFLECTED CEILING PLANS - BLDG 1245G |
| 25 | 28479 | AD201B | DEMOLITION EXTERIOR ELEVATIONS - BLDG 1245G |
| 26 | 28480 | AD401 | TYPICAL DEMOLITION ENLARGED SLEEPING ROOM PLANS |
| 27 | 28481 | AD901 | DEMOLITION PHOTOS AND DETAIL |
| 28 | 28482 | A-101B | FIRST, SECOND AND THIRD FLOOR PLANS - BLDG 1245G |
| 29 | 28483 | A-102B | ATTIC AND ROOF PLANS - BLDG 1245G |
| 30 | 28484 | A-103B | FIRST, SECOND AND THIRD FLOOR REFLECTED CEILING PLANS - BLDG 1245G |
| 31 | 28485 | A-201B | EXTERIOR ELEVATIONS - BLDG 1245G |
| 32 | 28486 | A-301 | SECTIONS |
| 33 | 28487 | A-401 | ENLARGED SLEEPING ROOM PLANS |
| 34 | 28488 | A-501 | DETAILS |
| 35 | 28489 | A-502 | DETAILS |
| 36 | 28490 | A-601 | FINISH SCHEDULE AND CABINET DETAILS |
| INTERIORS | | | |
| 37 | 28491 | I-001 | INTERIORS GENERAL NOTES AND ABBREVIATIONS |
| FIRE PROTECTION | | | |
| 38 | 28492 | F-101B | FIRST, SECOND AND THIRD FLOOR SUPPRESSION PLAN -BLDG 1245G |
| MECHANICAL | | | |
| 39 | 28493 | M-001 | LEGEND, ABBREVIATIONS, AND GENERAL NOTES |
| 40 | 28494 | M-101B | FIRST, SECOND, AND THIRD FLOOR PLANS - BLDG 1245G |
| 41 | 28495 | M-401 | ENLARGED PLANS |
| 42 | 28496 | MP402 | TYPICAL SLEEPING ROOM PLAN, ABBREVIATIONS, LEGEND AND SCHEDULE |
| 43 | 28497 | M-501 | SECTIONS AND ENLARGED PLANS |
| 44 | 28498 | M-601B | SCHEDULES - BLDG 1245G |
| 45 | 28499 | M-701 | CONTROLS LEGEND AND ABBREVIATIONS |
| 46 | 28500 | M-702 | CONTROLS |
| 47 | 28501 | M-703 | CONTROLS |
| ELECTRICAL | | | |
| 48 | 28502 | E-001 | GENERAL NOTES, LEGENDS AND MOUNTING HEIGHTS |
| 49 | 28503 | E-101B | FIRST, SECOND AND THIRD FLOOR PLANS - BLDG 1245G |
| 50 | 28504 | E-102B | ATTIC AND ROOF PLANS - BLDG 1245G |
| 51 | 28505 | E-401 | ENLARGED PLAN |
| 52 | 28506 | E-501 | LIGHTING SCHEDULES AND DETAILS |
| 53 | 28507 | E-502 | DETAILS |
| TOTAL: 53 | | | |

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

WISCONSIN
DEAN C. VON TERSCH
E-17903
CHARLESTON SC
ARCHITECT
04-OCT-2023

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28456 | CEMS | G-002B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY<br>NAVAL FACILITIES ENGINEERING COMMAND<br>**MARINE CORPS AIR STATION**<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G<br>(BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | INDEX OF DRAWINGS - BLDG 1245G | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F / CODE IDENT. NO. / N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: / MACC. / T.O. # / SHEET 2 OF 53 | |

# SURVEY LEGEND

EXISTING

- BUILDING OUTLINE
- CONCRETE CURB AND GUTTER
- EDGE OF PAVEMENT
- FENCE
- CONTOUR
- UNDERGROUND ELECTRICAL (REC DRAWING)
- UNDERGROUND ELECTRICAL
- OVERHEAD ELECTRICAL
- UNDERGROUND COMMUNICATIONS
- UNDERGROUND COMMUNICATIONS (REC DRAWING)
- WATER LINE (REC DRAWING)
- WATER LINE
- SANITARY SEWER (REC DRAWING)
- SANITARY SEWER
- STORM DRAIN (REC DRAWING)
- STORM DRAIN
- HIGH PRESSURE STEAM (REC DRAWING)
- GAS LINE
- UNKNOWN UNDERGROUND UTILITY
- BENCHMARK / CONTROL POINT
- BOLLARD
- DOWNSPOUT
- SIGN
- ELECTRIC TRANSFORMER
- ELECTRICAL MANHOLE
- ELECTRICAL DEVICE
- LIGHT POLE
- POWER POLE
- GUY WIRE
- COMMUNICATIONS MANHOLE
- TELEPHONE PEDESTAL
- TELEVISION PEDESTAL
- FIRE HYDRANT
- WATER VALVE
- DRAINAGE MANHOLE
- DROP INLET
- SANITARY SEWER CLEANOUT
- SANITARY SEWER MANHOLE
- TREE
- WHEELSTOP
- SPOT ELEVATION
- ASPHALT PAVEMENT
- CONCRETE SIDEWALK

# CIVIL LEGEND

EXISTING

- BUILDING OUTLINE
- CONCRETE CURB AND GUTTER
- EDGE OF PAVEMENT
- FENCE
- CONTOUR
- UNDERGROUND ELECTRICAL (REC DRAWING)
- UNDERGROUND ELECTRICAL
- OVERHEAD ELECTRICAL
- UNDERGROUND COMMUNICATIONS
- UNDERGROUND COMMUNICATIONS (REC DRAWING)
- WATER LINE (REC DRAWING)
- WATER LINE
- SANITARY SEWER (REC DRAWING)
- SANITARY SEWER
- STORM DRAIN (REC DRAWING)
- STORM DRAIN
- HIGH PRESSURE STEAM (REC DRAWING)
- GAS LINE
- UNKNOWN UNDERGROUND UTILITY
- BENCHMARK / CONTROL POINT
- BOLLARD
- SIGN
- ELECTRIC TRANSFORMER
- ELECTRICAL MANHOLE
- ELECTRICAL DEVICE
- LIGHT POLE
- POWER POLE
- GUY WIRE
- COMMUNICATIONS MANHOLE
- TELEPHONE PEDESTAL
- TELEVISION PEDESTAL
- FIRE HYDRANT
- WATER VALVE
- DRAINAGE MANHOLE
- DROP INLET
- SANITARY SEWER CLEANOUT
- SANITARY SEWER MANHOLE
- TREE
- STORM DRAIN CLEANOUT
- DOWNSPOUT
- LIMITS OF CONSTRUCTION
- LIMITS OF DISTURBANCE
- SILT FENCE
- INLET PROTECTION
- WHEELSTOP
- SPOT ELEVATION
- TREE PROTECTION
- ASPHALT PAVEMENT
- CONCRETE SIDEWALK
- TO BE REMOVED
- BUILDING COORDINATE POINT (SHOWN ON CS101B AND TABLE BELOW)

NEW

- BUILDING OUTLINE
- CONCRETE CURB AND GUTTER
- EDGE OF PAVEMENT / SIDEWALK
- CONTOUR
- UNDERGROUND ELECTRICAL (REC DRAWING)
- UNDERGROUND ELECTRICAL
- OVERHEAD ELECTRICAL
- UNDERGROUND COMMUNICATIONS
- UNDERGROUND COMMUNICATIONS (REC DRAWING)
- WATER LINE (REC DRAWING)
- WATER LINE
- SANITARY SEWER (REC DRAWING)
- SANITARY SEWER
- STORM DRAIN (REC DRAWING)
- STORM DRAIN
- HIGH PRESSURE STEAM (REC DRAWING)
- GAS LINE
- UNKNOWN UNDERGROUND UTILITY
- BENCHMARK / CONTROL POINT
- BOLLARD
- SIGN
- ELECTRIC TRANSFORMER
- ELECTRICAL MANHOLE
- ELECTRICAL DEVICE
- LIGHT POLE
- POWER POLE
- GUY WIRE
- COMMUNICATIONS MANHOLE
- TELEPHONE PEDESTAL
- TELEVISION PEDESTAL
- FIRE HYDRANT
- WATER VALVE
- DRAINAGE MANHOLE
- DROP INLET
- SANITARY SEWER CLEANOUT
- SANITARY SEWER MANHOLE
- TREE
- STORM DRAIN CLEANOUT
- DOWNSPOUT
- LIMITS OF CONSTRUCTION
- LIMITS OF DISTURBANCE
- SILT FENCE
- INLET PROTECTION
- WHEELSTOP
- SPOT ELEVATION
- TREE PROTECTION
- ASPHALT PAVEMENT
- CONCRETE SIDEWALK
- TO BE REMOVED
- BUILDING COORDINATE POINT (SHOWN ON CS101B AND TABLE BELOW)

# ABBREVIATIONS

STORM DRAIN = SD
MANHOLE = MH
TYP. = TYPICAL
REINFORCED CONCRETE PIPE = RCP
DUCTILE IRON PIPE = DIP
HIGH DENSITY POLYETHYLENE = HDPE
TRANSF. = TRANSFORMER
INVERT ELEVATION = IE
ELEV. = ELEVATION
FRAME (RIM ELEVATION) = FR

# INSTALLATION ACCESS

1. CONTRACTOR AND SUB-CONTRACTOR PERSONNEL ARE REQUIRED TO FOLLOW THE PROCEDURES FOR BASE ACCESS AND DBIDS PROCESS TO ACCESS THE BASE.

# DEMOLITION NOTES

1. UNLESS NOTED OTHERWISE, DEMOLITION WORK ENTAILS COMPLETE REMOVAL OF THE ITEMS OR STRUCTURES IN THEIR ENTIRETY. REPAIR, REMOVAL AND REPLACEMENT OF PAVEMENT AND SITE ITEMS SHOWN IS THE MINIMUM AMOUNT REQUIRED TO COMPLETE THE PROJECT. ALL DEMOLISHED MATERIAL SHALL BE REMOVED FROM THE INSTALLATION.
2. SAW CUT AND REMOVE ALL EXISTING PAVEMENT AS NOTED ON PLANS. PROVIDE NEAT AND CLEAN JOINTS AT PAVEMENT TO REMAIN. REMOVE AND REPLACE THE FULL SECTION.
3. ALL DEPRESSIONS RESULTING FROM DEMOLITION ACTIVITIES SHALL BE FILLED & COMPACTED WITH SUITABLE SOILS TO EXISTING GRADE AND SLOPED FOR POSITIVE DRAINAGE. OFFSITE BORROW NECESSARY FOR THE CONSTRUCTION OF THIS PROJECT SHALL BE PROVIDED BY THE CONTRACTOR.
4. AFTER GRADING ACTIVITY HAS BEEN COMPLETED, THE CONTRACTOR SHALL PLACE NECESSARY TOPSOIL AND SOD AS SPECIFIED ON THE SHEETS IN ALL DISTURBED LOCATIONS NOT TO BE PAVED.

# LAND DISTURBANCE NOTES

1. THE PROJECT AREA IS APPROXIMATELY 7.8 ACRES AS SHOWN BY THE LIMITS OF CONSTRUCTION. DISTURBED AREA IS APPROXIMATELY 0.6 ACRES AS SHOWN BY SILT FENCE UNLESS OTHERWISE SHOWN BY THE LIMITS OF DISTURBANCE.
2. ALL AREAS DISTURBED BY CONSTRUCTION OPERATIONS AND NOT PAVED SHALL BE SODDED / GRASSED. PROVIDE PERMANENT SODDING AND TEMPORARY SEEDING PER COVERAGE REQUIREMENTS, SEE SHEET C-002B.
3. THE PROPERTY IS LOCATED WITHIN BEAUFORT COUNTY, SOUTH CAROLINA LOCATED ON BEAUFORT USGS QUAD SHEET AND OUTFALLS TO BRICKYARD CREEK.
4. THIS DEVELOPED PROPERTY LIES IN ZONE X PER FIRM COMMUNITY NO. 450025, PANEL 007D, DATED EFFECTIVE DATE SEPTEMBER 29, 1986.
5. THE DISTURBED ACREAGE IS 0.6 ACRES, OF WHICH NO NEW IMPERVIOUS AREA OR BUILT UPON AREA (BUA) WILL BE CONSTRUCTED IN PROJECT AREA; THEREFORE, DOES NOT REQUIRE SCDHEC PERMITTING FOR SEDIMENT & EROSION CONTROL AND STORMWATER, ONLY NOTIFICATION LETTERS. A COASTAL AREA MANAGEMENT ACT (CAMA) MAJOR DEVELOPMENT PERMIT IS NOT REQUIRED.
6. SOUTH CAROLINA DEPARTMENT OF HEALTH AND ENVIRONMENTAL CONTROL (SCDHEC) PERMITTING SUBMISSION AND ACQUISITION SHALL BE HANDLED BY MCAS BEAUFORT, NATURAL RESOURCES AND ENVIRONMENTAL AFFAIRS OFFICE WITH ASSISTANCE FROM ENGINEER OF RECORD (EOR).

# SCDHEC NOTES

1. STORMWATER, SEDIMENT AND EROSION CONTROL MEASURES SHALL BE IN PLACE FOR THE DURATION OF SITE WORK IN ACCORDANCE WITH THE DRAWINGS. MEASURES SHALL BE ESTABLISHED PRIOR TO SITE DISTURBANCE (TOTALING 0.6 ACRES) AND MAINTAINED THRU STABILIZATION CLOSEOUT.
2. I HAVE PLACED MY SIGNATURE AND SEAL ON THE DESIGN DOCUMENTS SUBMITTED SIGNIFYING THAT I ACCEPT RESPONSIBILITY FOR THE DESIGN OF THE SYSTEM. FURTHER, I CERTIFY TO THE BEST OF MY KNOWLEDGE AND BELIEF THAT THE DESIGN IS CONSISTENT WITH THE REQUIREMENTS OF TITLE 48, CHAPTER 14 OF THE CODE OF LAWS OF SC, 1976 AS AMENDED, PURSUANT TO REGULATION 72-300 ET SEQ. (IF APPLICABLE), AND IN ACCORDANCE WITH THE TERMS AND CONDITIONS OF SCR100000.

# CONSTRUCTION SEQUENCE NOTES

1. PLACE SCDHEC LAND DISTURBANCE NOTIFICATION, APPROVED PLANS AND THE ON-SITE SWPPP (OS-SWPPP) AT A LOCATION ON THE SITE.
2. BEFORE LAND DISTURBING ACTIVITIES, THE LOCAL EQC MUST BE NOTIFIED, A DIG PERMIT SHALL BE SUBMITTED TO MCAS WITH ADEQUATE NOTICE, AND A SWPPP PRECONSTRUCTION MEETING PERFORMED.
3. INSTALL TEMPORARY CONSTRUCTION CHAIN LINK FENCING UPON APPROVAL BY CONTRACTING OFFICER AND MCAS BEAUFORT FEAD.
4. INSTALL SILT FENCE, INLET PROTECTION AND TEMPORARY GRAVEL CONSTRUCTION ENTRANCE WHERE INDICATED ON THE PLANS OR AS NECESSARY. DISTURBANCE SHOULD BE MINIMAL AS NECESSARY TO INSTALL MEASURES.
5. ESTABLISH PROPER CONSTRUCTION LIMITS TO LIMIT NON-CONSTRUCTION PERSONNEL FROM WORK AREA.
6. REMOVE ALL CONCRETE SIDEWALKS AND ANY SITE STRUCTURES / FEATURES SPECIFIED TO BE REMOVED ON THE PLANS.
7. TRENCHING AND INSTALLATION OF NEW STORM DRAINAGE LINES AND STRUCTURES MAY BEGIN. SEQUENCE SHALL BE DICTATED BY WHETHER PRODUCTS / SUBMITTALS HAVE BEEN APPROVED. SEE MECHANICAL PLANS FOR MORE DETAILS AND IMPROVEMENTS WITHIN 5 FEET OF BUILDINGS.
8. FILL AND COMPACT ALL TRENCHES.
9. INSTALL ALL NEW CONCRETE SIDEWALK.
10. COMPLETE FINISHED GRADING OPERATIONS AND PAVEMENT REPAIRS.
11. INITIATE PERMANENT VEGETATIVE STABILIZATION MEASURES.
12. REMOVE ACCUMULATED SEDIMENT AND MAINTAIN SEDIMENT DEVICES UNTIL SITE HAS BEEN STABILIZED.
13. REMOVE SEDIMENT DEVICES ONCE SITE HAS BEEN STABILIZED AND GIVEN APPROVAL BY GOVERNMENT.
14. SUBMIT NOTICE OF TERMINATION OF SCDHEC AND MEET ALL OF THE LAND DISTURBANCE PERMIT REQUIREMENTS.

# GENERAL NOTES

1. SEE SHEET C-501B THROUGH C-503B FOR CIVIL DETAILS.
2. ALL DISTURBED AREAS TO BE GRADED FOR POSITIVE DRAINAGE TO STORM INLETS.
3. CONTRACTOR WILL BE RESPONSIBLE FOR ADDITIONAL REMOVAL AND REPLACEMENT OF CONCRETE PAVEMENT AS NECESSARY TO CONSTRUCT THE PROJECT. PAVEMENT REMOVAL AND REPLACEMENT SHOWN IS THE MINIMUM AMOUNT REQUIRED TO CONSTRUCT THE PROJECT.
4. THE LOCATION, SIZES, AND DEPTHS OF EXISTING UTILITIES, INCLUDING SERVICE LATERALS, AND DRAINAGE STRUCTURES SHOWN ON THE PLANS ARE APPROXIMATE ONLY. THE CONTRACTOR SHALL VERIFY THE EXACT LOCATION, SIZES, AND DEPTHS OF UNDERGROUND UTILITIES AT LEAST 48 HOURS PRIOR TO CONSTRUCTION WHETHER SHOWN ON PLANS OR NOT, AND TO PROTECT THE SAME DURING CONSTRUCTION.
5. THE CONTRACTOR IS HEREBY CAUTIONED THAT ALL SITE FEATURES SHOWN HEREON ARE BASED ON FIELD OBSERVATIONS BY THE SURVEYOR AND BY INFORMATION PROVIDED BY OTHERS. THE INFORMATION IS NOT TO BE RELIED ON AS BEING EXACT OR COMPLETE AND WILL BE RESPONSIBLE FOR FIELD VERIFICATION PRIOR TO CONSTRUCTION.
6. CONTRACTOR TO COORDINATE LOCATION/MARKING OF EXISTING SITE UTILITIES IN AREAS WHERE WORK IS CONDUCTED, AND ENSURE THEY ARE TURNED OFF PROPERLY BEFORE BEGINNING CONSTRUCTION.
7. THE CONTRACTOR WILL BE HELD RESPONSIBLE FOR THE REPAIR OF ANY DAMAGE DONE TO ANY NEW AND EXISTING SITE ITEMS IN THE EXECUTION OF HIS WORK UNDER THIS SECTION.
8. THE CONTRACTOR SHALL FAMILIARIZE HIMSELF WITH THE EXISTING CONDITIONS AND BE PREPARED TO ADEQUATELY CARE FOR AND SAFEGUARD HIMSELF AND THE GOVERNMENT FROM DAMAGE. CONTRACTOR TO VERIFY EXISTING UTILITY (STORM DRAIN, WATER LINE, ETC. ..) LINE SIZES AND NOTIFY THE ENGINEER OF RECORD OF ANY DISCREPANCIES BEFORE ORDERING MATERIALS. ADVANCED PERMISSION FROM THE INSTALLATION FOR PHOTOGRAPHY TO DOCUMENT EXISTING CONDITIONS PRIOR TO START OF ANY WORK IS REQUIRED.
9. PRIOR TO STORM DRAIN LINE INSTALLATION, THE CONTRACTOR MUST VERIFY ELEVATIONS OF ALL THE EXISTING UTILITIES AT ALL UTILITY CROSSINGS. IF CONFLICT EXISTS, CONTRACTOR SHALL RELOCATE UTILITIES AS NECESSARY AND NOTIFY THE CONTRACTING OFFICER (COR). CONTRACTOR TO TAKE EVERY MEASURE TO LOCALIZE OUTAGES TO BUILDINGS. COORDINATE ALL NECESSARY OUTAGES WITH COR.

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |

## BEQ 1245G COORDINATE TABLE

| POINT # | LOCATION / BLDG CORNER | NORTHING | EASTING |
|---|---|---|---|
| 1 | NORTHWEST | 227539.303 | 2087242.460 |
| 2 | NORTHEAST | 227539.671 | 2087290.609 |
| 3 | SOUTHEAST | 227362.276 | 2087291.964 |
| 4 | SOUTHWEST | 227361.908 | 2087243.816 |

SOUTH CAROLINA 33329 REGISTERED LANDSCAPE ARCHITECT
10/4/2023

IF SHEET IS LESS THAN (34"x22") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28457 | CEMS www.cems-se.com \| Summerville, SC \| Wilmington, NC \| 843.875.3027 | C-001B |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY J. BARNES | DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND MARINE CORPS AIR STATION BEAUFORT, S.C. | |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY D. FARLEY | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR C. MOORING | LEGEND AND GENERAL NOTES | |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F \| CODE IDENT. NO. \| N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: AS NOTED \| MACC. \| T.O. # | SHEET 3 OF 53 |

## SITE INVESTIGATION NOTES

1. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING AN AS-BUILT SURVEY (PERFORMED BY A SURVEYOR LICENSED IN THE STATE OF SOUTH CAROLINA) OF ALL ABOVE AND BELOW GROUND SITE FEATURES, INCLUDING UTILITIES, WITHIN THE PROJECT SITE AND AREA AFFECTED BY THIS PROJECT SCOPE. SURVEY SHALL CAPTURE ALL UTILITIES AND STRUCTURES. SURVEY SHALL PROPERLY VERIFY THE LOCATION, ELEVATION / DEPTH, RIM, INVERT(S) AND ITEM SIZE(S).

2. THE CONTRACTOR SHALL HIRE A UTILITY LOCATE COMPANY TO LOCATE AND MARK ALL SUBSURFACE UTILITIES OR OTHER NOTEWORTHY ITEMS FOR AWARENESS AND TO ENSURE PROTECTION IN THE VICINITY OF EARTHWORK, DEMOLITION, REPAIR AND CONSTRUCTION.

3. ALL EXISTING UTILITIES ARE SHOWN IN AN APPROXIMATE LOCATIONS PER GIS DATA, LIDAR DATA, AND INSTALLATION RECORD DRAWINGS. THE CONTRACTOR SHALL VERIFY LOCATIONS, SIZES, AND DEPTHS OF ALL EXISTING ABOVE & BELOW GROUND UTILITIES AT LEAST 48 HOURS PRIOR TO THE START OF EARTHWORK, DEMOLITION, REPAIR AND CONSTRUCTION. THE CONTRACTING OFFICER SHALL BE NOTIFIED OF ANY DISCREPANCIES AFFECTING EXCAVATION, DEMOLITION, REPAIR OR CONSTRUCTION.

4. THE CONTRACTOR SHALL CALL PUPS, 811 OR (888) 721-7877, BEFORE ANY EXCAVATION AND COORDINATE WITH MARINE CORPS AIR STATION (MCAS) ON LOCATING REQUIREMENTS. SOUTH CAROLINA STATE LAW REQUIRES ANY ACTIVITY THAT RESULTS IN THE MOVEMENT OR REMOVAL OF EARTH, ROCK OR OTHER MATERIALS IN OR ON THE GROUND REQUIRES THE EXCAVATOR TO CONTACT THE UTILITY COMPANY OR ONE CALL CENTER WITH ADEQUATE INFORMATION REGARDING THE DIG. EACH EXCAVATOR IS REQUIRED TO WAIT 72-HOURS OR 3 BUSINESS DAYS BEFORE STARTING THE DIGGING WORK.

5. CONTRACTOR SHALL BE RESPONSIBLE FOR COORDINATING WITH UTILITY PROVIDERS (AS NECESSARY) WITHIN THE PROJECT AREA. SOME UTILITY CONTACT INFORMATION IS AS FOLLOWS:
   COMCAST: 4400 BELLE OAKS DR. CHARLESTON, SC 29405. 843-266-3010
   DOMINION ENERGY: PO BOX 1168, BEAUFORT, SC 29901. 843-525-7731
   CENTURYLINK: PO DRAWER 1659, BEAUFORT, SC 29902. 843-525-7979
   HARGRAY COMMUNICATIONS GROUP, INC: PO BOX 346 BLUFFTON, SC 29910. 843-815-1676
   BJWSA: 6 SNAKE RD OKATIE, SC 29909. 843-987-9229.

6. KNOWN UNDERGROUND UTILITY LOCATIONS ARE SHOWN ON THE PLAN. THE CONTRACTOR IS HEREBY NOTIFIED THAT NOT ALL UTILITY LOCATIONS ARE SHOWN AND/OR KNOWN, BUT COULD POSSIBLY BE ENCOUNTERED IN THE AREA OF WORK. ALL UTILITIES ENCOUNTERED SHALL BE PROTECTED DURING EXCAVATION, DEMOLITION, REPAIR AND CONSTRUCTION UNLESS OTHERWISE STATED TO BE REMOVED.

## ENVIRONMENTAL NOTES

1. IF ANY SUSPECTED ARTIFACTS ARE DISCOVERED DURING GROUND DISTURBING ACTIVITIES, ALL WORK SHOULD STOP AND NREAO (AND PROJECT MANAGER / CONTRACTING OFFICER) SHOULD BE NOTIFIED IMMEDIATELY.

2. ALLIGATORS HAVE BEEN KNOWN TO INHABIT VARIOUS STORM WATER OUTFALLS. WORKERS SHOULD BE AWARE OF THEIR POTENTIAL PRESENCE. IF ENCOUNTERED, CONTACT NREAO PERSONNEL IMMEDIATELY FOR ALLIGATOR REMOVAL/RELOCATION.

3. AREA IN-SITU MATERIAL IS KNOWN TO BE CONTAMINATED WITH PER- AND POLYFLUOROALKYL SUBSTANCES (PFAS). EXCAVATED EARTHWORK SHALL BE BACKFILLED IN PLACE. ALTERNATIVELY, CONTAMINATED MATERIAL SHALL BE CONTAINED, ENCLOSED, AND FOLLOW REQUISITE STATE AND FEDERAL REQUIREMENTS FOR REMOVAL AND DISPOSAL.

4. BENTONITE CLAY AND/OR OTHER MATERIALS USED FOR DIRECTIONAL DRILLING MUST BE REMOVED FROM THE INSTALLATION. FOLLOW REQUISITE STATE AND FEDERAL REQUIREMENTS FOR REMOVAL AND DISPOSAL.

5. THE CONTRACTOR SHALL BE RESPONSIBLE TO ACQUIRE ALL NECESSARY PERMITS PER STATE, FEDERAL REGULATIONS, AND OSHA STANDARDS INCLUDING HAZARDOUS MATERIAL PERMITTING. A COPY OF THE LEAD AND ASBESTOS REPORT MUST BE SUBMITTED TO SCDHEC AT LEAST (10) WORKING DAYS PRIOR TO DEMOLITION WHEN APPLYING FOR A DEMOLITION PERMIT. ALL PERMITS/ RECEIPTS/ INSPECTION DOCUMENTS SHALL BE COORDINATED WITH MCAS PROGRAM MANAGER, CHRIS DOSHER AT CHRISTOPHER.DOSHER@USMC.MIL. PROPER LABELING, TRANSPORTING, AND DISPOSAL OF HAZARDOUS CONTAINING MATERIALS ARE THE RESPONSIBILITY OF THE CONTRACTOR.

6. UNLESS EXPLICITLY STATED, VAULTS SHALL BE CONSIDERED AS POSSIBLE SOURCES OF ASBESTOS CONTAINING MATERIAL. WET METHODS MUST BE EMPLOYED IF VAULT IS NOT ALREADY MOISTENED OR UNDERWATER. DUST SUPPRESSION VIA AMENDED WATER SHALL BE USED AT THE POINT OF DUST GENERATION. HEPA VACUUMS AT THE POINT OF WORK SHALL BE USED TO COLLECT ANY REMAINING AIRBORNE FIBERS. SHOULD THERE BE A CONCERN FOR SIGNIFICANT DUST GENERATION, THE GENERAL SETUP OF VAULTS AND MANHOLES MUST ENABLE THE USE OF NEGATIVE PRESSURE CONTAINMENT. EXPOSURE MONITORING OF AIRBORNE FIBERS MUST BE CONDUCTED. THE MONITORING WILL BE USED TO CREATE A "NEGATIVE EXPOSURE ASSESSMENT" AND WILL DICTATE WHAT THE APPROPRIATE LEVEL OF RESPIRATORY PROTECTION IS. RUN INITIAL AIRBORNE ASBESTOS "EXCURSION" SAMPLES TO DEVELOP A NEGATIVE EXPOSURE ASSESSMENT. UNDER NO CIRCUMSTANCES SHALL PERSONNEL USE COMPRESSED AIR OR OTHER GASES IN THE VAULTS FOR THE PURPOSE OF "PUSHING" A DEVICE THROUGH THE CONDUIT.

7. ALL DEMOLITION DEBRIS SHALL BE HAULED OFFSITE AND DISPOSED OF OFFSITE AT A FACILITY LICENSED AND PERMITTED TO ACCEPT LIKE MATERIALS, WITH A STATE APPROVED EROSION CONTROL PLAN IN PLACE. ALL MATERIALS SHALL BE TRANSPORTED OFF GOVERNMENT PROPERTY FOR FURTHER PROCESSING BY THE CONTRACTOR OR OTHER PARTIES IN ACCORDANCE WITH DOD REGULATIONS.

8. HAZARDOUS COMPONENT ABATEMENT FOR MATERIALS SUCH AS ASBESTOS, PETROLEUM DERIVATIVES, LEAD PAINT, PCB LIQUIDS, AND OTHER HAZARDOUS MATERIALS SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR WITH A HAZARDOUS WASTE MANAGEMENT PLAN (HWMP) IN PLACE. THE CONTRACTOR SHALL ACQUIRE A COMPETENT/ CURRENT SCDHEC ASBESTOS SUPERVISOR DURING ANY POSSIBLE DISTURBANCE OF ACM TO ASSURE COMPLIANCE FOR BOTH STATE AND FEDERAL REGULATIONS. IF ASBESTOS IS PRESENT OR PRESUMED, THIS PROJECT IS SUBJECT TO SOUTH CAROLINA REGULATION 61-86.1, STANDARDS OF PERFORMANCE FOR ASBESTOS.

9. THE CONTRACTOR SHALL ABATE, REMOVE, AND MANAGE THE HAZARDOUS MATERIALS IN ACCORDANCE WITH FEDERAL AND LOCAL ENVIRONMENTAL PROTECTION LAWS AND REGULATIONS. THE CONTRACTOR SHALL DETERMINE IF PERSONAL PROTECTIVE EQUIPMENT IS RECOMMENDED OR REQUIRED BASED ON EVALUATIONS OF THE COMPLETED REPORTS. IF PROJECT IS CLASSIFIED BY SCDHEC ASBESTOS STANDARDS AS NESHAP SIZE PROJECT A SCDHEC LICENSED PROJECT.

10. CONTRACTOR WILL BE REQUIRED BY MCAS TO PROVIDE ASBESTOS PLAN FOR REVIEW AND ACCEPTANCE FROM NREAO WITH SAFETY BEFORE APPLYING FOR PERMIT. ESTABLISH DESIGNATED LIMITS FOR THE ASBESTOS REGULATED WORK AREA WITH THE USE OF RED BARRIER TAPE, CRITICAL BARRIERS AND SIGNS, AND MAINTAIN ALL OTHER REQUIREMENTS FOR ASBESTOS CONTROL AREA EXCEPT LOCAL EXHAUST.

## STANDARD NOTES:

1. IF NECESSARY, SLOPES, WHICH EXCEED EIGHT (8) VERTICAL FEET SHOULD BE STABILIZED WITH SYNTHETIC OR VEGETATIVE MATS, IN ADDITION TO HYDROSEEDING. IT MAY BE NECESSARY TO INSTALL TEMPORARY SLOPE DRAINS DURING CONSTRUCTION. TEMPORARY BERMS MAY BE NEEDED UNTIL THE SLOPE IS BROUGHT TO GRADE.

2. STABILIZATION MEASURES SHALL BE INITIATED AS SOON AS PRACTICABLE IN PORTIONS OF THE SITE WHERE CONSTRUCTION ACTIVITIES HAVE TEMPORARILY OR PERMANENTLY CEASED, BUT IN NO CASE MORE FOURTEEN (14) DAYS AFTER WORK HAS CEASED, EXCEPT AS STATED BELOW.
   - WHERE STABILIZATION BY THE 14TH DAY IS PRECLUDED BY SNOW COVER OR FROZEN GROUND CONDITIONS STABILIZATION MEASURES MUST BE INITIATED AS SOON AS PRACTICABLE.
   - WHERE CONSTRUCTION ACTIVITY ON A PORTION OF THE SITE IS TEMPORARILY CEASED, AND EARTH-DISTURBING ACTIVITIES WILL BE RESUMED WITHIN 14 DAYS, TEMPORARY STABILIZATION MEASURES DO NOT HAVE TO BE INITIATED ON THAT PORTION OF THE SITE.

3. ALL SEDIMENT AND EROSION CONTROL DEVICES SHALL BE INSPECTED ONCE EVERY CALENDAR WEEK. IF PERIODIC INSPECTION OR OTHER INFORMATION INDICATES THAT A BMP HAS BEEN INAPPROPRIATELY, OR INCORRECTLY, THE PERMITTEE MUST ADDRESS THE NECESSARY REPLACEMENT OR MODIFICATION REQUIRED TO CORRECT THE BMP WITHIN 48 HOURS OF IDENTIFICATION.

4. PROVIDE SILT FENCE AND/OR OTHER CONTROL DEVICES, AS MAY BE REQUIRED, TO CONTROL SOIL EROSION DURING UTILITY CONSTRUCTION. ALL DISTURBED AREAS SHALL BE CLEANED, GRADED AND STABILIZED WITH GRASSING IMMEDIATELY AFTER THE UTILITY INSTALLATION. FILL, COVER, AND TEMPORARY SEEDING AT THE END OF EACH DAY ARE RECOMMENDED. IF WATER IS ENCOUNTERED WHILE TRENCHING, THE WATER SHOULD BE FILTERED TO REMOVE ANY SEDIMENTS BEFORE BEING PUMPED BACK INTO THE WATERS OF THE STATE.

5. ALL EROSION CONTROL DEVICES SHALL BE PROPERLY MAINTAINED DURING ALL PHASES OF CONSTRUCTION UNTIL THE COMPLETION OF ALL CONSTRUCTION ACTIVITIES AND ALL DISTURBED AREAS HAVE BEEN STABILIZED. ADDITIONAL CONTROL DEVICES MAY BE REQUIRED DURING CONSTRUCTION IN ORDER TO CONTROL EROSION AND/OR OFFSITE SEDIMENTATION. ALL TEMPORARY CONTROL DEVICES SHALL BE REMOVED ONCE CONSTRUCTION IS COMPLETE AND THE SITE IS STABILIZED.

6. THE CONTRACTOR MUST TAKE NECESSARY ACTION TO MINIMIZE THE TRACKING OF MUD ONTO PAVED ROADWAY(S) FROM CONSTRUCTION AREAS AND THE GENERATION OF DUST. THE CONTRACTOR SHALL DAILY REMOVE MUD/SOIL FROM PAVEMENT, AS MAY BE REQUIRED.

7. RESIDENTIAL SUBDIVISIONS REQUIRE EROSION CONTROL FEATURES FOR INFRASTRUCTURE AS WELL AS FOR INDIVIDUAL LOT CONSTRUCTION. INDIVIDUAL PROPERTY OWNERS SHALL FOLLOW THESE PLANS DURING CONSTRUCTION OR OBTAIN APPROVAL OF AN INDIVIDUAL PLAN IN ACCORDANCE WITH S.C. REG. 72-300 ET SEQ. AND SCR100000.

8. TEMPORARY DIVISION BERMS AND/OR DITCHES WILL BE PROVIDED AS NEEDED DURING CONSTRUCTION TO PROTECT WORK AREAS FROM UPSLOPE RUNOFF AND/OR DIVERT SEDIMENT-LADEN WATER TO APPROPRIATE TRAPS OR STABLE OUTLETS.

9. ALL WATERS OF THE STATE (WOS), INCLUDING WETLANDS, ARE TO BE FLAGGED OR OTHERWISE CLEARLY MARKED IN THE FIELD. A DOUBLE ROW OF SILT FENCE IS TO BE INSTALLED IN ALL AREAS WHERE A 50-FOOT BUFFER CAN'T BE MAINTAINED BETWEEN THE DISTURBED AREA AND ALL WOS. A 10-FOOT BUFFER SHOULD BE MAINTAINED BETWEEN THE LAST ROW OF SILT FENCE AND ALL WOS.

10. LITTER, CONSTRUCTION DEBRIS, OILS, FUELS AND BUILDING PRODUCTS WITH SIGNIFICANT POTENTIAL FOR IMPACT (SUCH AS STOCKPILES OF FRESHLY TREATED LUMBER) AND CONSTRUCTION CHEMICALS THAT COULD BE EXPOSED TO STORM WATER MUST BE PREVENTED FROM BECOMING A POLLUTANT SOURCE IN STORM WATER DISCHARGES.

11. A COPY OF THE SWPPP, INSPECTIONS RECORDS, AND RAINFALL DATA MUST BE RETAINED AT THE CONSTRUCTION SITE OR A NEARBY LOCATION EASILY ACCESSIBLE DURING NORMAL BUSINESS HOURS, FROM THE DATE OF COMMENCEMENT OF CONSTRUCTION ACTIVITIES TO THE DATE THAT FINAL STABILIZATION IS REACHED.

12. INITIATE STABILIZATION MEASURES ON ANY EXPOSED STEEP SLOPE (3H:1V OR GREATER) WHERE LAND-DISTURBING ACTIVITIES HAVE PERMANENTLY OR TEMPORARILY CEASED, AND WILL NOT RESUME FOR A PERIOD OF 7 CALENDAR DAYS.

13. MINIMIZE SOIL COMPACTION AND, UNLESS INFEASIBLE, PRESERVE TOPSOIL.

14. MINIMIZE THE DISCHARGE OF POLLUTANTS FROM EQUIPMENT AND VEHICLE WASHING, WHEEL WASH WATER, AND OTHER WASH WATERS. WASH WATERS MUST BE TREATED IN A SEDIMENT BASIN OR ALTERNATIVE CONTROL THAT PROVIDES EQUIVALENT OR BETTER TREATMENT PRIOR TO DISCHARGE;

15. MINIMIZE THE DISCHARGE OF POLLUTANTS FROM DEWATERING OF TRENCHES AND EXCAVATED AREAS. THESE DISCHARGES ARE TO BE ROUTED THROUGH APPROPRIATE BMPS (SEDIMENT BASIN, FILTER BAG, ETC.).

16. THE FOLLOWING DISCHARGES FROM SITES ARE PROHIBITED:
   - WASTEWATER FROM WASHOUT OF CONCRETE, UNLESS MANAGED BY AN APPROPRIATE CONTROL;
   - WASTEWATER FROM WASHOUT AND CLEANOUT OF STUCCO, PAINT, FORM RELEASE OILS, CURING
   - COMPOUNDS AND OTHER CONSTRUCTION MATERIALS;
   - FUELS, OILS, OR OTHER POLLUTANTS USED IN VEHICLE AND EQUIPMENT OPERATION AND
   - MAINTENANCE; AND
   - SOAPS OR SOLVENTS USED IN VEHICLE AND EQUIPMENT WASHING.

17. AFTER CONSTRUCTION ACTIVITIES BEGIN, INSPECTIONS MUST BE CONDUCTED AT A MINIMUM OF AT LEAST ONCE EVERY CALENDAR WEEK AND MUST BE CONDUCTED UNTIL FINAL STABILIZATION IS REACHED ON ALL AREAS OF THE CONSTRUCTION SITE.

18. IF EXISTING BMPS NEED TO BE MODIFIED OR IF ADDITIONAL BMPS ARE NECESSARY TO COMPLY WITH THE REQUIREMENTS OF THIS PERMIT AND/OR SC'S WATER QUALITY STANDARDS, IMPLEMENTATION MUST BE COMPLETED BEFORE THE NEXT STORM EVENT WHENEVER PRACTICABLE. IF IMPLEMENTATION BEFORE THE NEXT STORM EVENT IS IMPRACTICABLE, THE SITUATION MUST BE DOCUMENTED IN THE SWPPP AND ALTERNATIVE BMPS MUST BE IMPLEMENTED AS SOON AS REASONABLY POSSIBLE.

19. A PRE-CONSTRUCTION CONFERENCE MUST BE HELD FOR EACH CONSTRUCTION SITE WITH AN APPROVED ON-SITE SWPPP PRIOR TO THE IMPLEMENTATION OF CONSTRUCTION ACTIVITIES. FOR NON-LINEAR PROJECTS THAT DISTURB 10 ACRES OR MORE THIS CONFERENCE MUST BE HELD ON-SITE UNLESS THE DEPARTMENT HAS APPROVED OTHERWISE.

## SEDIMENT AND EROSION CONTROL NOTES:

1. GENERAL
   A. ALL AREAS WHICH ARE DISTURBED BY THE CONTRACTOR SHALL BE FINAL STABILIZED WITH SOD.

2. GRASS AND COASTAL SEEDING SPECIFICATIONS

   TEMPORARY SEEDING (EXCEPT FOR FLIGHTLINE LOCATIONS):

   "COOL SEASON": PLANTED BETWEEN 1 SEPTEMBER AND 3 MARCH

   | SPECIES | RATE |
   |---|---|
   | RYEGRASS W/ BERMUDA | 90-135 LBS./AC. |

   CONTRACTOR SHALL ENSURE EFFECTIVE WATERING AND PERIODIC INSPECTION TO ENSURE APPROPRIATE GROWTH OF GRASS.

3. EROSION CONTROL PLAN
   A. INSTALL PERMANENT VEGETATIVE COVER AND THE LONG-TERM EROSION PROTECTION MEASURES OR STRUCTURES AS SOON AS PRACTICAL IN THE DEVELOPMENT PROCESS.
   B. PROVIDE FOR HANDLING THE INCREASED RUNOFF CAUSED BY CHANGED SOIL AND SURFACE CONDITIONS. USE EFFECTIVE MEANS TO CONSERVE EXISTING ON-SITE SOIL INCLUDING THE USE OF DIVERSION DITCHES, GRASSED WATERWAYS AND STORM SEWERS.
   C. PLACE SILT FENCE BARRIERS AT LOCATIONS SHOWN ON PLAN. SILT BARRIERS SHALL BE MAINTAINED IN PLACE AND IN GOOD CONDITION UNTIL GROUND COVER IS ESTABLISHED.
   D. ALL DISTURBED AREAS NOT PAVED SHALL BE GRASSED. USE TEMPORARY PLANT COVER, MULCHING, AND/OR STRUCTURES TO CONTROL RUNOFF AND PROTECT AREAS SUBJECT TO EROSION DURING CONSTRUCTION.
   E. CONTRACTOR SHALL PROVIDE DUMPSTER OF SUFFICIENT SIZE AND NUMBER TO CONTAIN THE SOLID WASTE GENERATED BY THE PROJECT.
   F. CONTRACTOR SHALL FREQUENTLY REMOVE GARBAGE TO MAINTAIN A CLEAN CONSTRUCTION SITE AND MINIMIZE THE EXPOSURE OF WASTE TO STORM WATER.

**REVISIONS**

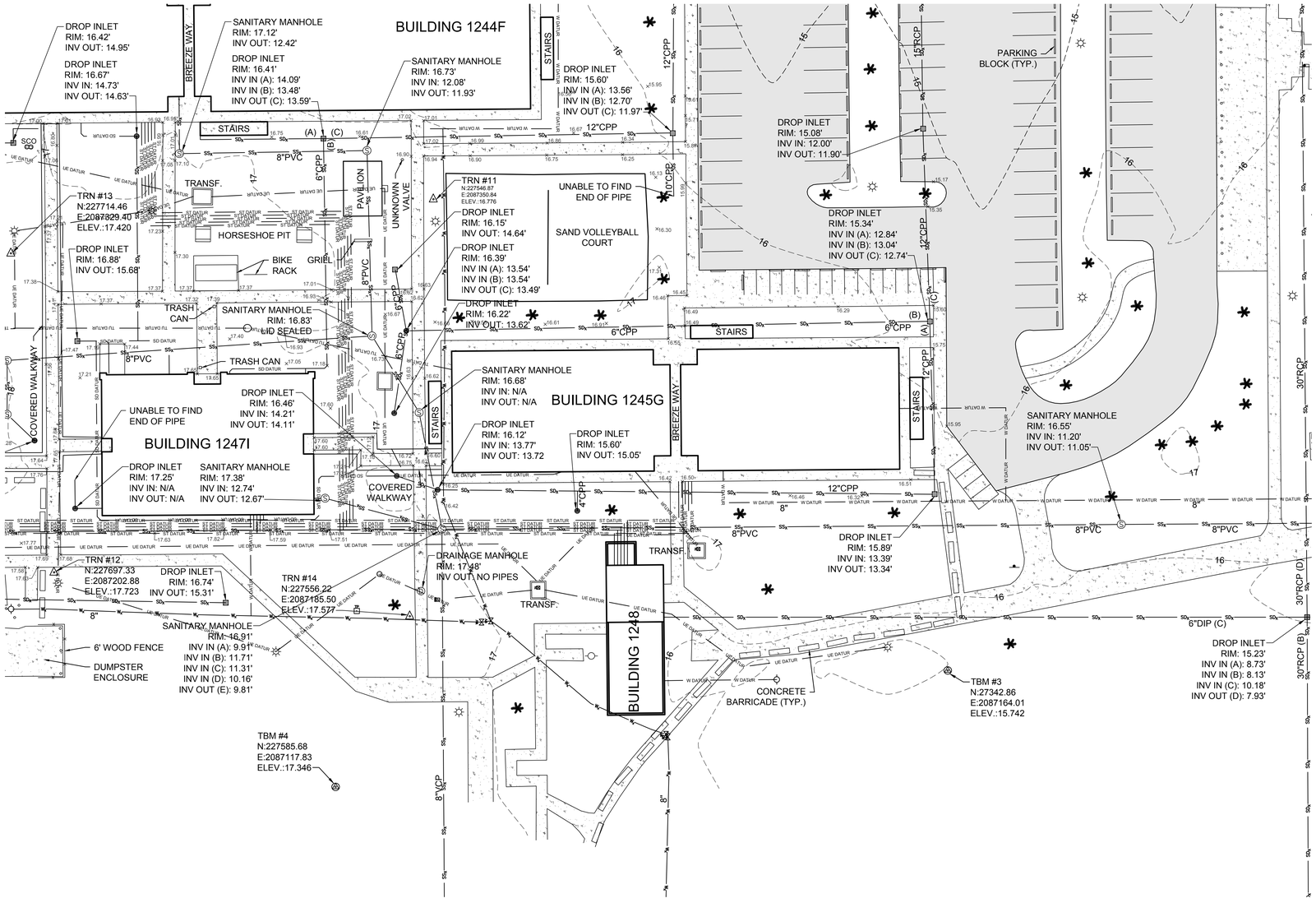
| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

NORTH CAROLINA REGISTERED LANDSCAPE ARCHITECT No. 3329 — 10/4/2023

IF SHEET IS LESS THAN (34"x22") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28458 | CEMS<br>www.cems-se.com \| Summerville, SC \| Wilmington, NC \| 843.875.3837 | C-002B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>J. BARNES | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. FARLEY | REPAIR PHASE 4 BEQ BUILDING 1245G<br>(BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>C. MOORING | GENERAL NOTES | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE: F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: AS NOTED; MACC.; T.O. #; SHEET 4 OF 53 | |




## 1 EXISTING TOPOGRAPHIC SURVEY

SCALE: 1" = 20'-0"

NOTE:
EXISTING TOPOGRAPHIC SURVEY SHOWS EXISTING CONDITIONS OF EACH BEQ 1244 1245 1246 AND 1247 PRIOR TO ANY CONTRACT WORK. FOLLOWING SHEETS SHOW BEQ 1244 AND 1247 AS COMPLETED. CONTRACTOR SHALL ACQUIRE ALL THREE SETS OF PLANS ASSOCIATED WITH BEQ 1244, 1245, 1246 AND 1247 TO COMPLETE ALL REQUIRED WORK SHOWN.

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES:

1. ALL DISTANCES ARE HORIZONTAL GROUND IN INTERNATIONAL FEET UNLESS OTHERWISE SHOWN.
2. HORIZONTAL DATUM: NAD 1983/2011(SC) VERTICAL DATUM: NAVD 1988
3. SURVEY COMPLETED BY McKIM & CREED IS BASED ON PHYSICAL EVIDENCE FOUND IN NOVEMBER 2022.
4. SURVEYOR HAS MADE NO INVESTIGATION OR INDEPENDENT SEARCH FOR EASEMENTS OF RECORD, ENCUMBRANCES, RESTRICTIVE COVENANTS, OWNERSHIP, TITLE EVIDENCE, OR ANY OTHER FACTS THAT AN ACCURATE AND CURRENT TITLE SEARCH MAY DISCLOSE.
5. BROKEN LINES INDICATE PROPERTY LINES NOT SURVEYED.
6. THE OFF-SITE RIGHT-OF-WAY/PROPERTY LINE SHOWN HEREON IS FOR ILLUSTRATIVE PURPOSES ONLY. THE UNDERSIGNED CERTIFIES ONLY TO THE RIGHT-OF-WAY/PROPERTY LINES SURVEYED, AND DOES NOT CERTIFY TO THE RIGHT-OF-WAY WIDTH/PROPERTY LINE OF ANY ADJACENT PROPERTIES.
7. THE SUBJECT PROPERTY IS LOCATED WITHIN 2000' OF A HORIZONTAL NGS MONUMENT.
8. HORIZONTAL COORDINATES BASED UPON SC-RTN GPS OBSERVATIONS USING TRIMBLE "R12i" UNITS. THE NETWORK POSITIONAL ACCURACY OF THE SC-RTN DERIVED POSITIONAL COORDINATES MEETS OR EXCEEDS THE CLASS A HORIZONTAL POSITIONS. COORDINATES BASED UPON GEOID 18 AND HAVE A 95% CONFIDENCE RATING. VERTICAL DATUM HAS BEEN ADJUSTED TO NAVD '88.
9. UNDERGROUND UTILITIES ARE SHOWN ONLY WHERE ABOVEGROUND SERVICES WERE VISIBLE.
10. SUBSURFACE AND ENVIRONMENTAL CONDITIONS WERE NOT EXAMINED OR CONSIDERED AS A PART OF THIS SURVEY. NO STATEMENT IS MADE CONCERNING THE EXISTENCE OF UNDERGROUND OR OVERHEAD CONTAINERS OR FACILITIES THAT MAY AFFECT THE USE OR DEVELOPMENT OF THIS TRACT.

SOUTH CAROLINA PROFESSIONAL LAND SURVEYOR No. 28600 DAVID E. LUKAC JR. 10/04/2023

McKIM & CREED INC. No. 453 CERTIFICATE OF AUTHORIZATION

I HEREBY STATE THAT TO THE BEST OF MY PROFESSIONAL KNOWLEDGE, INFORMATION, AND BELIEF, THE SURVEY SHOWN HEREON WAS MADE IN ACCORDANCE WITH THE REQUIREMENTS OF THE STANDARDS OF PRACTICE MANUAL FOR SURVEYING IN SOUTH CAROLINA, AND MEETS OR EXCEEDS THE REQUIREMENTS FOR A CLASS A SURVEY AS SPECIFIED THEREIN; ALSO THERE ARE NO VISIBLE ENCROACHMENTS OR PROJECTIONS OTHER THAN SHOWN.

DAVID E. LUKAC JR., PLS. #28600

NORTH

20' 0 20' 40'

SCALE: 1" = 20'

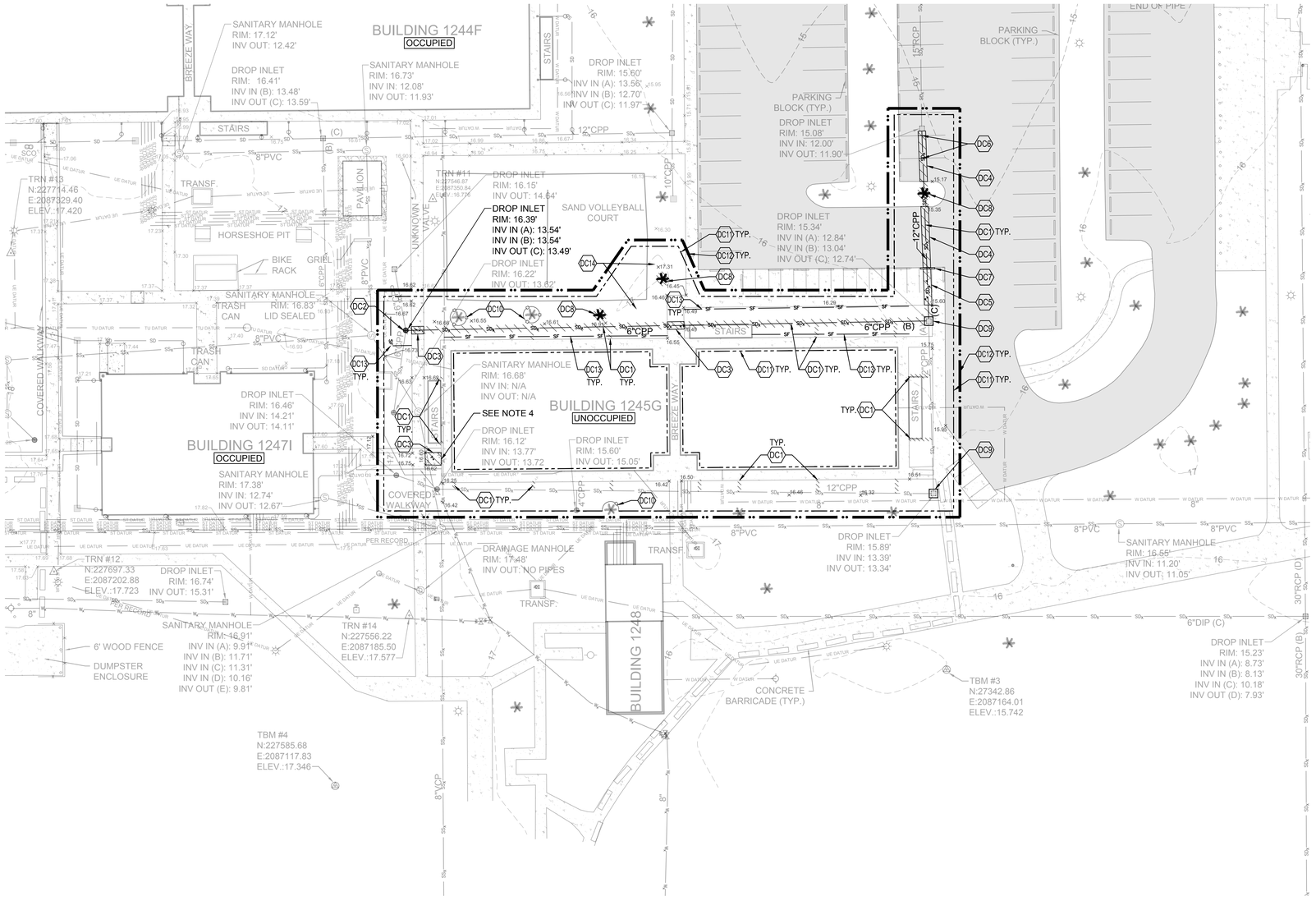
**GRAPHIC SCALE**

IF SHEET IS LESS THAN (34"x22") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

| | | |
|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28459 | CEMS www.cems-se.com \| Summerville, SC \| Wilmington, NC \| 843.875.3637 — VF101B |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY - | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND — MARINE CORPS AIR STATION — BEAUFORT, S.C. |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY - | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR - | EXISTING TOPOGRAPHIC SURVEY |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F \| CODE IDENT. NO. \| N A V F A C DRAWING NO. |
| | | HQMC PROJ. NO.: \| CONSTR. CONTR. NO.: N40085-18-D-0616 |
| PUBLIC WORKS OFFICER | SCALE: AS NOTED | MACC. \| T.O. # \| SHEET 5 OF 53 |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |




## 1 DEMOLITION / EROSION CONTROL PLAN

SCALE: 1" = 20'-0"

## GENERAL NOTES:

1. SEE SHEET C-001B FOR PROJECT NOTES AND LEGEND.
2. BUILDING(S) HAVE MULTIPLE ALUMINUM DOWNSPOUTS THAT CONNECT TO EXISTING UNDERGROUND STORM SYSTEM. ALL DOWNSPOUTS WILL BE REMOVED AND REPLACED AS PART OF THIS RENOVATION WORK (SOW). SEE ARCHITECTURAL FOR MORE INFORMATION.
3. CONTRACTOR SHALL VERIFY ALL EXTERIOR SITE FEATURES AND UTILITIES PRIOR TO DEMOLITION WORK.
4. CONTRACTOR TO BE AWARE THESE LOCATIONS ARE UNDER A COVERED WALKWAY AND PROPOSED WORK MAY REQUIRE SMALLER EQUIPMENT AND WORK WITH HAND TOOLS.
5. CONTRACTOR TO BE AWARE THIS AREA CONTAINS SEVERAL CONCRETE PROTECTIVE BARRIER(S). BARRIER(S) NOT INDICATED TO BE TEMPORARILY MOVED FOR CONSTRUCTION EFFORTS MUST BE COORDINATED WITH MCAS BEAUFORT PHYSICAL SECURITY (MR. MILLER) PRIOR TO MOVING. BARRIER(S) TO BE REPLACED IN PREVIOUS LOCATION AFTER INSTALLATION OF NEW WORK.
6. CONTRACTOR IS REQUIRED TO PROVIDE AND MAINTAIN TEMPORARY CHAIN LINK FENCING FOR THE DURATION OF CONSTRUCTION. FENCING SHALL NOT BE PERMANENTLY MOUNTED IN THE GROUND. THE FENCING AND POSTS SHALL BE A PORTABLE SELF STANDING SYSTEM. SEE DETAIL 6/C-501A.
7. BUILDING 1245G SHALL BE UN-OCCUPIED FOR REPAIRS / CONSTRUCTION UNDER THIS CONTRACT.
8. BUILDING 1247I "DUTY HUT" MUST BE ACCESSIBLE FOR BASE PERSONNEL AT ALL TIMES FOR LAUNDRY USAGE DURING CONSTRUCTION.
9. BUILDING 1244F, 1246H, AND 1247I SHALL BE OCCUPIED DURING REPAIRS / CONSTRUCTION UNDER THIS CONTRACT. ACCESS TO THESE BUILDINGS SHALL NOT BE IMPEDED AT ANY TIME DURING CONSTRUCTION.

## SHEET KEYNOTES

DC1 REMOVE STORM DRAIN LINE

DC2 REMOVE ENTIRE STORM DRAIN INLET STRUCTURE

DC3 REMOVE CONCRETE SIDEWALK

DC4 SAWCUT AND REMOVE ASPHALT PAVEMENT

DC5 REMOVE CONCRETE PAVEMENT

DC6 REMOVE PAVEMENT MARKINGS

DC7 REMOVE CONCRETE WHEELSTOP

DC8 REMOVE TREE

DC9 INLET PROTECTION (SEE DETAIL 4/C-501B)

DC10 TREE PROTECTION (SEE DETAIL 5/C-501B)

DC11 LIMITS OF DISTURBANCE

DC12 LIMITS OF CONSTRUCTION AND MIN. LIMITS OF TEMPORARY CONSTRUCTION CHAIN LINK FENCING. SEE NOTE 6. (SEE DETAIL 6/C-501)

DC13 SILT FENCE (SEE DETAIL 1/C-501B)

DC14 CONTRACTOR SHALL TEMPORARILY MOVE CONCRETE PROTECTIVE BARRIER(S), AS NECESSARY. MOVE SHALL BE COORDINATED WITH MCAS BEAUFORT PHYSICAL SECURITY MR. MILLER). BARRIER(S) TO BE PLACED IN PREVIOUS LOCATION AFTER INSTALLATION OF NEW WORK

NORTH

20' 0 20' 40'

SCALE: 1" = 20'

**GRAPHIC SCALE**

SOUTH CAROLINA 24827 REGISTERED PROFESSIONAL ENGINEER CHRISTOPHER A. MOORE No. 35329

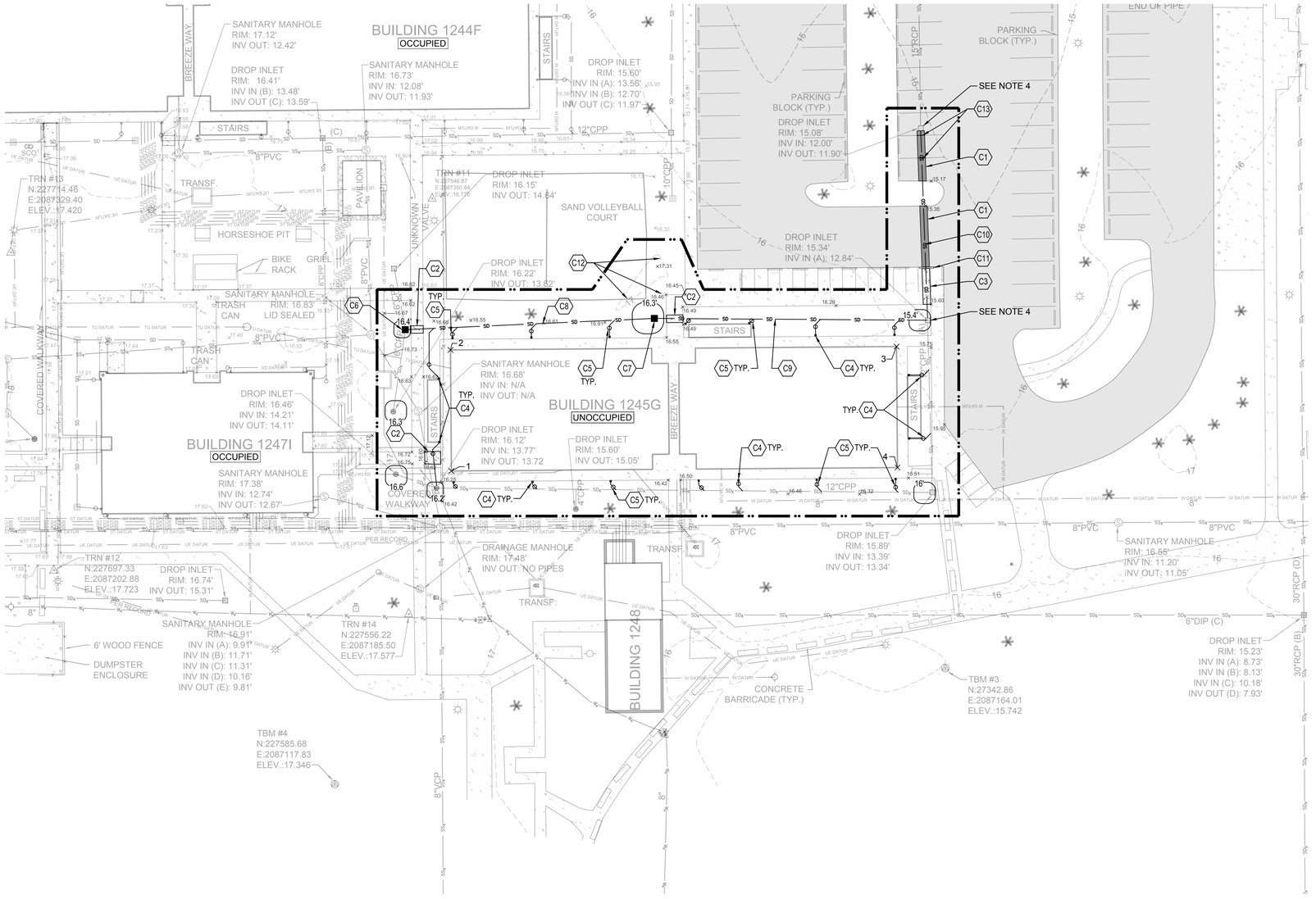
10/4/2023

IF SHEET IS LESS THAN (34"x22") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28460 | CEMS www.cems-se.com \| Summerville, SC \| Wilmington, NC \| 843.875.3837 | CD101B |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY J. BARNES | DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND MARINE CORPS AIR STATION BEAUFORT, S.C. | |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY D. FARLEY | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR C. MOORING | DEMOLITION / EROSION CONTROL PLAN | |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F \| CODE IDENT. NO. \| N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: \| CONSTR. CONTR. NO.: N40085-18-D-0616 | |
| PUBLIC WORKS OFFICER | SCALE: AS NOTED | MACC. \| T.O. # \| SHEET 6 OF 53 | |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |




## 1 SITE PLAN

SCALE: 1" = 20'-0"

## GENERAL NOTES:

1. SEE SHEET C-001B FOR PROJECT NOTES AND LEGEND.
2. CONCRETE SIDEWALKS AND/OR PAVEMENTS DAMAGED DUE TO CONSTRUCTION, BUT NOT DIRECTLY IDENTIFIED ON PLANS, SHALL BE REMOVED & RESTORED TO ORIGINAL CONDITIONS AT NO ADDITIONAL COST TO PROJECT (CONTRACTOR EXPENSE).
3. A MINIMUM 18" SEPARATION SHALL BE MAINTAINED WHEN WATER LINE AND SEWER LINE CROSS IN A PERPENDICULAR ALIGNMENT. A MINIMUM 10' HORIZONTAL SEPARATION SHALL BE MAINTAINED WHEN WATER LINE AND SEWER LINE ROUTE IN A PARALLEL ALIGNMENT. IF REQUIRED CLEARANCES CANNOT BE MAINTAINED, FERROUS PIPE MATERIAL (DUCTILE IRON OR STEEL ENCASEMENT) SHALL BE REQUIRED.
4. CONTRACTOR TO CORE EXISTING INLETS FOR NEW CONNECTIONS AND SHALL BE WATERTIGHT SEALS WITH BOOTS OR GROUT FILL.
5. SEE SHEET C-502B FOR STORM DRAIN INLET TYPE AND STORM PIPE BEDDING DETAIL.
6. ALL CORRUGATED HIGH-DENSITY POLYETHYLENE (HDPE) PIPE SHOWN ON THE DRAWINGS SHALL BE DUAL WALL PIPE WITH SMOOTH INTERIOR WALL AND CORRUGATED EXTERIOR.
7. ALL FITTINGS SHALL BE OF LIKE MATERIAL (PIPING) AND / OR PER DETAILS ON SHEET C-502B UNLESS SPECIFIED OTHERWISE ON DRAWINGS.
8. BUILDING 1245G SHALL BE UN-OCCUPIED FOR REPAIRS / CONSTRUCTION UNDER THIS CONTRACT.
9. BUILDING 1247I "DUTY HUT" MUST BE ACCESSIBLE FOR BASE PERSONNEL AT ALL TIMES FOR LAUNDRY USAGE.
10. BUILDING 1244F, 1246H, AND 1247I SHALL BE OCCUPIED DURING REPAIRS / CONSTRUCTION UNDER THIS CONTRACT. ACCESS TO THESE BUILDINGS SHALL NOT BE IMPEDED AT ANY TIME DURING CONSTRUCTION.

## SHEET KEYNOTES

C1 INSTALL ASPHALT PAVEMENT TO MATCH EXISTING (SEE DETAIL 5/C-503B)

C2 INSTALL CONCRETE SIDEWALK TO MATCH EXISTING (SEE DETAIL 2/C-503B AND 3/C-503B)

C3 INSTALL CONCRETE PAVEMENT TO MATCH EXISTING (SEE DETAIL 4/C-503B)

C4 INSTALL AND CONNECT MIN. 6" HDPE STORM PIPE FROM ROOF DOWNSPOUT TO STORM DRAINAGE NETWORK (SEE DETAIL 2/C-502B)

C5 INSTALL STORM DRAIN CLEANOUTS BETWEEN DOWNSPOUT AND STORM PIPING (SEE DETAIL 4/C-502B)

C6 STORM DRAIN BASIN
FR: 16.30'
I.E.: 13.47'
I.E.: (2) EX. 6" CPP (13.54')

C7 STORM DRAIN BASIN
FR: 16.00'
I.E.: 13.12'

C8 99 LF OF 12" HDPE AT MIN. 0.35% SLOPE

C9 109 LF OF 15" HDPE AT MIN. 0.35% SLOPE

C10 75 LF OF 15" RCP CLASS IV AT MIN. 1.00% SLOPE

C11 INSTALL CONCRETE WHEELSTOP (SEE DETAIL 1/C-503B)

C12 CONTRACTOR SHALL REMOVE RAISED BERM (ELEVATION 17') AND GRADE AREA TO DRAIN TO NEW INLET - KEYNOTE C7

C13 INSTALL PARKING STRIPES TO MATCH EXISTING (SEE DETAIL 7/C-503B)

ALL DISTURBED AREAS TO BE STABILIZED WITH SOD AND GRADED FOR POSITIVE DRAINAGE TO STORM INLETS. CONTRACTOR TO REMOVE ALL GRASS / VEGETATIVE OVERGROWTH ALONG SIDEWALKS / WALKWAYS TO REDUCE PONDING (TRAPPED WATER) AND PROVIDE POSITIVE SLOPED DRAINAGE AWAY FROM BUILDINGS. GRASSING SHALL BE SLIGHTLY LOWER THAN SURROUNDING SIDEWALKS / WALKWAYS. CONTRACTOR RESPONSIBLE FOR POSITIVE SITE FLOW TO STORM INLETS.

NORTH

20' 0 20' 40'

SCALE: 1" = 20'

**GRAPHIC SCALE**

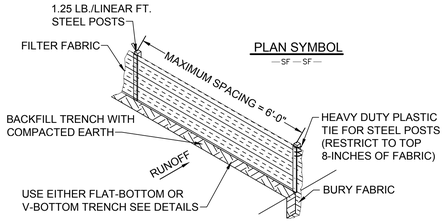
IF SHEET IS LESS THAN (34"x22") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

NORTH CAROLINA REGISTERED LANDSCAPE ARCHITECT No. 3339 CHRISTOPHER A. ACKERMAN

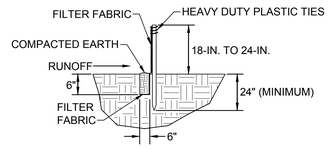
10/4/2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28461 | CEMS www.cems-se.com \| Summerville, SC \| Wilmington, NC \| 843.875.3837 | CS101B |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY J. BARNES | DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND MARINE CORPS AIR STATION BEAUFORT, S.C. | |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY D. FARLEY | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) SITE PLAN | |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR C. MOORING | | |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F \| CODE IDENT. NO. \| N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: \| CONSTR. CONTR. NO.: N40085-18-D-0616 | |
| | PUBLIC WORKS OFFICER | SCALE: AS NOTED \| MACC. \| T.O. # \| SHEET 7 OF 53 | |



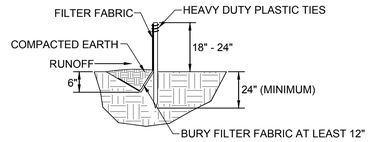

## 1 SILT FENCE INSTALLATION

SCALE: NOT TO SCALE




## 2 FLAT BOTTOM TRENCH DETAIL

SCALE: NOT TO SCALE




## 3 V-SHAPED TRENCH DETAIL

SCALE: NOT TO SCALE

### SILT FENCE - GENERAL NOTES

1. DO NOT PLACE SILT FENCE ACROSS CHANNELS OR IN OTHER AREAS SUBJECT TO CONCENTRATED FLOWS. SILT FENCE SHOULD NOT BE USED AS A VELOCITY CONTROL BMP. CONCENTRATED FLOWS ARE ANY FLOWS GREATER THAN 0.5 CFS.
2. MAXIMUM SHEET OR OVERLAND FLOW PATH LENGTH TO THE SILT FENCE SHALL BE 100-FEET.
3. MAXIMUM SLOPE STEEPNESS (NORMAL [PERPENDICULAR] TO THE FENCE LINE) SHALL BE 2:1.
4. SILT FENCE JOINTS, WHEN NECESSARY, SHALL BE COMPLETED BY ONE OF THE FOLLOWING OPTIONS: WRAP EACH FABRIC TOGETHER AT A SUPPORT POST WITH BOTH ENDS FASTENED TO THE POST, WITH A 1-FOOT MINIMUM OVERLAP;
   - OVERLAP SILT FENCE BY INSTALLING 3-FEET PASSED THE SUPPORT POST TO WHICH THE NEW SILT FENCE ROLL IS ATTACHED. ATTACH OLD ROLL TO NEW ROLL WITH HEAVY-DUTY PLASTIC TIES; OR,
   - OVERLAP ENTIRE WIDTH OF EACH SILT FENCE ROLL FROM ONE SUPPORT POST TO THE NEXT SUPPORT POST.
5. ATTACH FILTER FABRIC TO THE STEEL POSTS USING HEAVY-DUTY PLASTIC TIES THAT ARE EVENLY SPACED WITHIN THE TOP 8-INCHES OF THE FABRIC.
6. INSTALL THE SILT FENCE PERPENDICULAR TO THE DIRECTION OF THE STORMWATER FLOW AND PLACE THE SILT FENCE THE PROPER DISTANCE FROM THE TOE OF STEEP SLOPES TO PROVIDE SEDIMENT STORAGE AND ACCESS FOR MAINTENANCE AND CLEANOUT.
7. INSTALL SILT FENCE CHECKS (TIE-BACKS) EVERY 50-100 FEET, DEPENDENT ON SLOPE, ALONG SILT FENCE THAT IS INSTALLED WITH SLOPE AND WHERE CONCENTRATED FLOWS ARE EXPECTED OR ARE DOCUMENTED ALONG THE PROPOSED/INSTALLED SILT FENCE.

### SILT FENCE - POST REQUIREMENTS

1. SILT FENCE POSTS MUST BE 48-INCH LONG STEEL POSTS THAT MEET, AT A MINIMUM, THE FOLLOWING PHYSICAL CHARACTERISTICS.
   - COMPOSED OF A HIGH STRENGTH STEEL WITH A MINIMUM YIELD STRENGTH OF 50,000 PSI.
   - INCLUDE A STANDARD "T" SECTION WITH A NOMINAL FACE WIDTH OF 1.38-INCHES AND A NOMINAL "T" LENGTH OF 1.48-INCHES.
   - WEIGH 1.25 POUNDS PER FOOT (± 8%)
2. POSTS SHALL BE EQUIPPED WITH PROJECTIONS TO AID IN FASTENING OF FILTER FABRIC.
3. STEEL POSTS MAY NEED TO HAVE A METAL SOIL STABILIZATION PLATE WELDED NEAR THE BOTTOM WHEN INSTALLED ALONG STEEP SLOPES OR INSTALLED IN LOOSE SOILS. THE PLATE SHOULD HAVE A MINIMUM CROSS SECTION OF 17-SQUARE INCHES AND BE COMPOSED OF 15 GAUGE STEEL, AT A MINIMUM. THE METAL SOIL STABILIZATION PLATE SHOULD BE COMPLETELY BURIED.
4. INSTALL POSTS TO A MINIMUM OF 24-INCHES. A MINIMUM HEIGHT OF 1-TO 2- INCHES ABOVE THE FABRIC SHALL BE MAINTAINED, AND A MAXIMUM HEIGHT OF 3 FEET SHALL BE MAINTAINED ABOVE THE GROUND.
5. POST SPACING SHALL BE AT A MAXIMUM OF 6-FEET ON CENTER.

### SILT FENCE - INSPECTION & MAINTENANCE

1. THE KEY TO FUNCTIONAL SILT FENCE IS WEEKLY INSPECTIONS, ROUTINE MAINTENANCE, AND REGULAR SEDIMENT REMOVAL.
2. REGULAR INSPECTIONS OF SILT FENCE SHALL BE CONDUCTED ONCE EVERY CALENDAR WEEK AND, AS RECOMMENDED, WITHIN 24-HOURS AFTER EACH RAINFALL EVEN THAT PRODUCES 1/2-INCH OR MORE OF PRECIPITATION.
3. ATTENTION TO SEDIMENT ACCUMULATIONS ALONG THE SILT FENCE IS EXTREMELY IMPORTANT. ACCUMULATED SEDIMENT SHOULD BE CONTINUALLY MONITORED AND REMOVED WHEN NECESSARY.
4. REMOVE ACCUMULATED SEDIMENT WHEN IT REACHES 1/3 THE HEIGHT OF THE SILT FENCE.
5. REMOVED SEDIMENT SHALL BE PLACED IN STOCKPILE STORAGE AREAS OR SPREAD THINLY ACROSS DISTURBED AREA. STABILIZE THE REMOVED SEDIMENT AFTER IT IS RELOCATED.
6. CHECK FOR AREAS WHERE STORMWATER RUNOFF HAS ERODED A CHANNEL BENEATH THE SILT FENCE, OR WHERE THE FENCE HAS SAGGED OR COLLAPSED DUE TO RUNOFF OVERTOPPING THE SILT FENCE. INSTALL CHECKS/TIE-BACKS AND/OR REINSTALL SILT FENCE, AS NECESSARY.
7. CHECK FOR TEARS WITHIN THE SILT FENCE, AREAS WHERE SILT FENCE HAS BEGUN TO DECOMPOSE, AND FOR ANY OTHER CIRCUMSTANCE THAT MAY RENDER THE SILT FENCE INEFFECTIVE. REMOVED DAMAGED SILT FENCE AND REINSTALL NEW SILT FENCE IMMEDIATELY.
8. SILT FENCE SHOULD BE REMOVED WITHIN 30 DAYS AFTER FINAL STABILIZATION IS ACHIEVED AND ONCE IT IS REMOVED, THE RESULTING DISTURBED AREA SHALL BE PERMANENTLY STABILIZED.

### SILT FENCE - FABRIC REQUIREMENTS

1. SILT FENCE MUST BE COMPOSED OF WOVEN GEOTEXTILE FILTER FABRIC THAT CONSISTS OF THE FOLLOWING REQUIREMENTS:
   - COMPOSED OF FIBERS CONSISTING OF LONG CHAIN SYNTHETIC POLYMERS OF AT LEAST 85% BY WEIGHT OF POLYOLEFINS, POLYESTERS, OR POLYAMIDES THAT ARE FORMED INTO A NETWORK SUCH THAT THE FILAMENTS OR YARNS RETAIN DIMENSIONAL STABILITY RELATIVE TO EACH OTHER;
   - FREE OF ANY TREATMENT OR COATING WHICH MIGHT ADVERSELY ALTER ITS PHYSICAL PROPERTIES AFTER INSTALLATION;
   - FREE OF ANY DEFECTS OR FLAWS THAT SIGNIFICANTLY AFFECT ITS PHYSICAL AND/OR FILTERING PROPERTIES; AND,
   - HAVE A MINIMUM WIDTH OF 36-INCHES.
2. USE ONLY FABRIC APPEARING ON SC DOT'S QUALIFIED PRODUCTS LISTING (QPL), APPROVAL SHEET #34, MEETING THE REQUIREMENTS OF THE MOST CURRENT EDITION OF THE SC DOT STANDARD SPECIFICATIONS FOR HIGHWAY CONSTRUCTION.
3. 12-INCHES OF THE FABRIC SHOULD BE PLACED WITHIN EXCAVATED TRENCH AND TOED IN WHEN THE TRENCH IS BACKFILLED.
4. FILTER FABRIC SHALL BE PURCHASED IN CONTINUOUS ROLLS AND CUT TO THE LENGTH OF THE BARRIER TO AVOID JOINTS.
5. FILTER FABRIC SHALL BE INSTALLED AT A MINIMUM OF 24-INCHES ABOVE THE GROUND.

### SOUTH CAROLINA DEPARTMENT OF TRANSPORTATION: LIST OF QUALIFIED PROVIDERS OF SILT FENCING / LOW FLOW DROP INLET PROTECTION GEOTEXTILE FABRICS

BELTON INDUSTRIES, INC.
PO BOX 127 (2) STYLE 941
BELTON, SC 29627 (3) STYLE 945
(864) 338-5711
ATTN: MR. DAVID GALLOWAY

THRACE-LINQ
2550 WEST FIFTH NORTH ST.
SUMMERVILLE, SC 29483-9699
(803) 875-8280
(800) 445-4675

CONSTRUCTION FABRICS & SUPPLY
1603 JAMES P. ROGERS DRIVE
VALDOSTA, GEORGIA 31601
(229) 244-0004
ATTN: ANGIE ANDERSON OR
JEFF COPPAGE

HANES GEOCOMPONENTS
10800 S. COMMERCE BLVD. #9
SUITE L, CHARLOTTE, NC 28273
(704)588-7626
FAX: (704)588-7972
ATTN: KEITH GREGORY
WWW.HANESGEO.COM

PROPEX, INC.
P O BOX 22788
6025 LEE HIGHWAY, SUITE 425
CHATTANOOGA, TN 37422
(800)621-1273
FAX: (423)899-7619
WWW.GEOTEXTILE.COM

COASTAL SITE SUPPLY, INC.
P O BOX 60988
NORTH CHARLESTON, SC 29419
877-744-9747
FAX: 843-744-9750
ATTN: BILLY DENSON

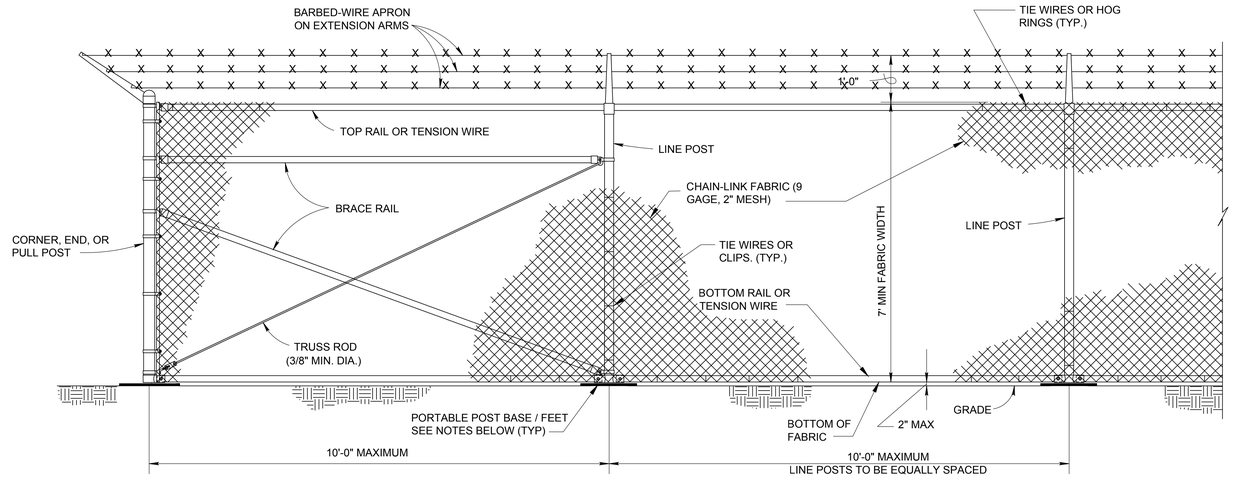
PROTECH ENVIRONMENTAL SUPPLY
1500 CONTINENTAL BLVD., SUITE C/F
CHARLOTTE, NC 28273
(704)676-9788
FAX: (704)676-9791
ATTN: MITCH PERKINS

WILLACOOCHEE INDUSTRIAL FABRICS
P. O. BOX 599
HIGHWAY 82 WEST
WILLACOOCHEE, GA 31650
(912) 534-5757
FAX: (912) 534-5533

ACF ENVIRONMENTAL
2831 CARDWELL ROAD
RICHMOND, VA 23234
(800)448-3636
FAX: (804)743-7779
WWW.ACFENVIRONMENTAL.COM

DDD EROSION CONTROL, INC.
P O BOX 694
1383 INDUSTRIAL DRIVE
ASHBURN, GA 31714
(229) 567-0751
FAX: (229) 567-0752
EMAIL: DDDEROSION@WINDSTREAM.NET
WWW.3DEROSION.COM
ATTN: DON DAVIS

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |

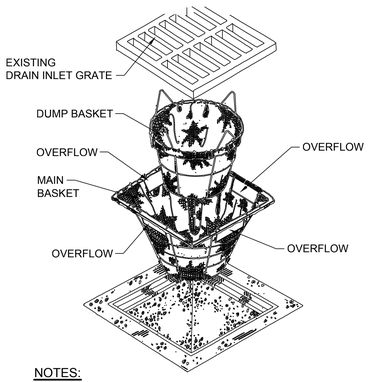



NOTES:

1. THE TEMPORARY CHAIN LINK FENCING SHALL NOT BE PERMANENTLY MOUNTED IN THE GROUND. THE FENCING AND POSTS SHALL BE A PORTABLE SELF STANDING SYSTEM. SYSTEM SHALL BE MOUNTED ON PEG STANDS / FEET TO ALLOW MOVEMENT OF FENCING TO ACCOMMODATE BOTH GOVERNMENT AND CONTRACTOR EQUIPMENT TRAFFIC. CONTRACTOR SHALL BE ABLE TO TEMPORARILY MOVE THE FENCING TO ACCOMMODATE LARGE EQUIPMENT MOVEMENT WITH APPROVAL FROM THE CONTRACTING OFFICER AND MCAS BEAUFORT FEAD. INSTALLATION AND MOVEMENT OF FENCING SHALL BE DONE BY THE CONTRACTOR.
2. CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING APPROVAL OF A "CONSTRUCTION SITE PLAN" PER SPECIFICATION 01 50 00 (1.3) TO SHOW LOCATION, MATERIAL, AND INFORMATION PERTAINING TO TEMPORARY CONSRUCTION CHAIN LINK FENCING. THE TEMPORARY FENCE BOUNDARY AND CHARACTERISTICS SHALL BE APPROVED BY THE CONTRACTING OFFICER AND MCAS BEAUFORT FEAD PRIOR TO INSTALLATION.
3. PROVIDE TEMPORARY FENCING COMPONENTS, CONNECTIONS, AND CHARACTERISTICS AS SHOWN IN THIS DETAIL.
4. PROVIDE VISUAL SCREENING TO MINIMUM 6' HEIGHT IN ACCORDANCE WITH FOD REQUIREMENTS.

## 6 TEMPORARY FENCING DETAIL

SCALE: NOT TO SCALE

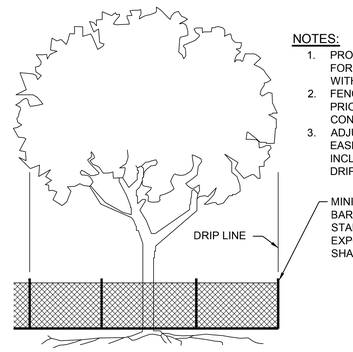



NOTES:

1. INLET PROTECTION SHALL BE INSTALLED PRIOR TO COMMENCING ANY DEMOLITION AND CONSTRUCTION WORKS, OR AT THE TIME ANY NEW DRAIN STRUCTURE IS SET.
2. INSTALLATION: REMOVE GRATE, DROP MAIN SEDIMENT BASKET IN OPENING, PLACE DUMP BASKET WITHIN MAIN BASKET. REPLACE GRATE.
3. INSPECT WEEKLY AND AFTER EACH RAIN. IF THERE IS AN ACCUMULATION OF SEDIMENT; SLIDE THE GRATE OFF THE INLET, LIFT THE DUMP BASKET OUT OF THE MAIN BASKET, EMPTY CONTENTS IN A SUITABLE LOCATION (TO BE REMOVED FROM SITE), REPLACE DUMP BASKET, REPLACE GRATE.
4. AFTER APPROXIMATELY 10 DUMPS, OR AS NEEDED, EMPTY THE MAIN BASKET OF ALL ACCUMULATED SEDIMENT.
5. VERIFY WATER LEVEL TO ENSURE DRAIN INLET FUNCTIONS PROPERLY, AND IS NOT BLOCKED DURING STORM EVENTS. REMOVE / DO NOT INSTALL AND NOTIFY COR / ENGINEER OF RECORD IF AREA PONDS, FLOODS, OR COULD POSE A SAFETY HAZARD TO VEHICLE TRAFFIC.

## 4 INLET SEDIMENT PROTECTION DEVICE

BASIS OF DESIGN: SEDCATCH - SQUARE GRATE (SEDCATCH ENVIRONMENTAL) OR PROVIDE APPROVED EQUIVALENT




NOTES:

1. PROVIDE PROTECTION FENCE FOR ALL TREES TO REMAIN WITHIN CONSTRUCTION AREA.
2. FENCES SHALL BE IN PLACE PRIOR TO BEGINNING CONSTRUCTION.
3. ADJUST FENCE AS NEEDED FOR EASE OF CONSTRUCTION TO INCLUDE NO LESS THAN 75% OF DRIP LINE AREA.

MINIMUM 4' HEIGHT ORANGE PLASTIC BARRIER FENCING. SET WITH #4 REBAR STAKES. PLACE REBAR SAFETY CAPS ON ALL EXPOSED REBAR ENDS. FENCE BARRIERS SHALL BE SQUARE (4 SIDED)

## 5 TREE PROTECTION DETAIL

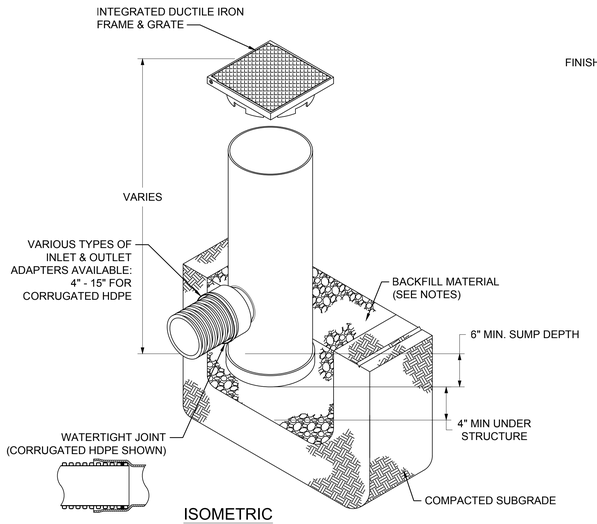
SCALE: NOT TO SCALE

IF SHEET IS LESS THAN (34"x22") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

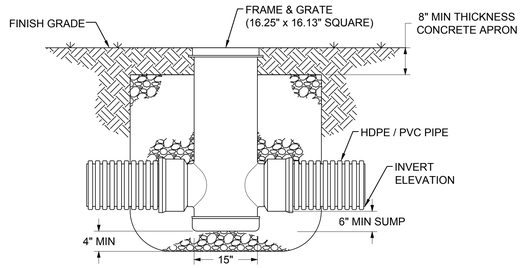
ISSUE DATE: 08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28462 | CEMS www.cems-se.com \| Summerville, SC \| Wilmington, NC \| 843.875.3837 | C-501B |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY J. BARNES | DEPARTMENT OF THE NAVY / NAVAL FACILITIES ENGINEERING COMMAND: MARINE CORPS AIR STATION, BEAUFORT, S.C. | |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY D. FARLEY | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR C. MOORING | EROSION CONTROL DETAILS | |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | PUBLIC WORKS OFFICER | HQMC PROJ. NO.:; CONSTR. CONTR. NO.: N40085-18-D-0616 | |
| | | SCALE: AS NOTED; MACC.; T.O. #; SHEET 8 OF 53 | |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |




ISOMETRIC

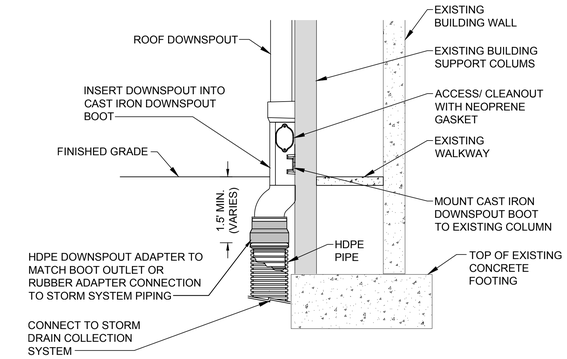



ELEVATION

NOTES:
1. DRAIN BASIN SHALL BE MANUFACTURED FROM PVC PIPE STOCK, UTILIZING A THERMO-MOLDING PROCESS. MATERIAL SHALL CONFORM TO ASTM D1784 CELL CLASS 12454.
2. SOLID COVER SHALL BE DUCTILE IRON PER ASTM A536, GRADE 70-50-05. COVERS SHALL MEET H-20 LOAD RATING.
3. FRAMES SHALL BE DUCTILE IRON PER ASTM A536, GRADE 70-50-05.
4. DRAIN BASIN TO BE CUSTOM MANUFACTURED ACCORDING TO PLAN DETAILS.
5. DRAINAGE CONNECTION STUB JOINT TIGHTNESS SHALL CONFORM TO ASTM D3212 FOR CORRUGATED HDPE (ADS N-12/HANCOR DUAL WALL), N-12 HP, & PVC SEWER.
6. THE BACKFILL MATERIAL SHALL BE CRUSHED STONE OR OTHER GRANULAR MATERIAL MEETING THE REQUIREMENTS OF CLASS I, CLASS II, OR CLASS III MATERIAL AS DEFINED IN ASTM D2321. BEDDING & BACKFILL FOR SURFACE DRAINAGE INLETS SHALL BE PLACED & COMPACTED UNIFORMLY IN ACCORDANCE WITH ASTM D2321.

## 1 DRAIN BASIN DETAIL

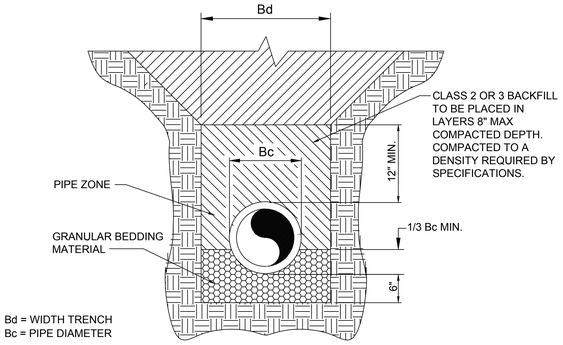
SCALE: NOT TO SCALE




NOTES:
1. DOWNSPOUT SIZE SHALL BE FIELD VERIFIED PRIOR TO ORDERING DOWNSPOUT BOOTS AND STORM PIPE ADAPTERS.
2. DOWNSPOUT BOOT SHALL BE OF CAST IRON MATERIAL AND PRE-COATED WITH A RUST INHIBITIVE PRIMER.
3. DOWNSPOUT LAYOUT AND BOOT TYPE SHALL BE COORDINATED WITH ARCHITECTURAL PLANS.
4. UNDERGROUND ADAPTER CONNECTION AND PIPE JOINTS SHALL BE WATERTIGHT (WT).
5. CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING THE NECESSARY FITTINGS TO INSTALL ROOF DRAIN PIPING TO STORM DRAIN COLLECTION SYSTEM. SEE PLANS FOR COLLECTION SYSTEM LAYOUT AND LOCATIONS.
6. CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING THE NECESSARY FASTENERS TO INSTALL CAST IRON DOWNSPOUT BOOT TO BUILDING / POST / COLUMN PER MANUFACTURERS RECOMMENDATIONS.
7. UNDERGROUND STORM PIPING INSTALLED PAST ADAPTERS SHALL BE CORRUGATED HDPE (DUAL WALL, SMOOTH INTERIOR).

## 2 DOWNSPOUT CONNECTION

SCALE: NOT TO SCALE




Bd = WIDTH TRENCH
Bc = PIPE DIAMETER

Bd = Bc +24" FOR PIPE DIA. < 24"
Bd = Bc +36" FOR PIPE DIA > 24"

CROSS SECTION

NOTES:
1. ALL STORM PIPES SHALL BE INSTALLED IN ACCORDANCE WITH ASTM D2321 (LATEST EDITION), WITH EXCEPTION THAT THE INITIAL BACKFILL MAY EXTEND TO THE CROWN OF THE PIPE.
2. Bd SHALL NOT BE GREATER THAN Bc PLUS 2' FOR PIPES UP TO AND INCLUDING 30" AND 3' FOR PIPES GREATER THAN 30" TO PERMIT SATISFACTORY JOINING AND TAMPING OF BEDDING MATERIAL UNDER AND AROUND PIPES.
3. BEDDING MATERIAL:
   3.1. 6" DEPTH - COMPACT MATERIAL PRIOR TO PIPE PLACEMENT
   3.2. 1/3Bc MIN. - LOOSE MATERIAL PLACEMENT AROUND PIPE

## 3 STORM PIPE BEDDING

SCALE: NOT TO SCALE




PLAN




"INLINE" STORM DRAIN CLEANOUT ELEVATION

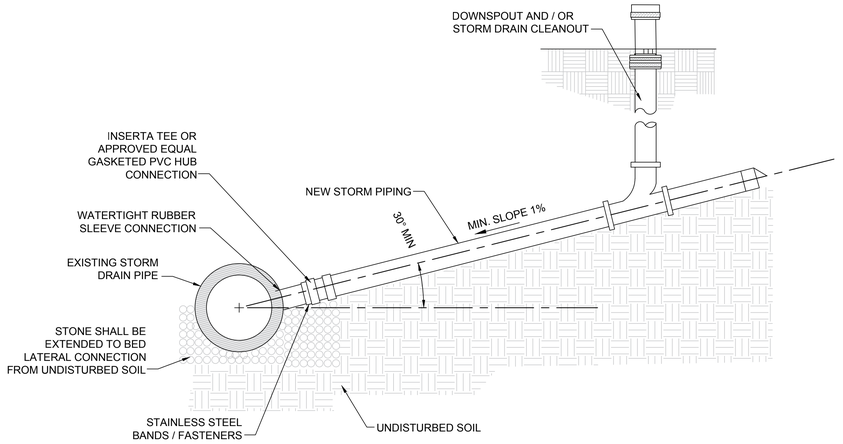



"DEAD-END" STORM DRAIN CLEANOUT ELEVATION

NOTES:
1. PIPE SIZES SHALL CORRESPOND WITH PLANS AND SPECIFICATIONS.
2. PIPE JOINTS AND FITTINGS SHALL BE WATERTIGHT (WT) CONNECTIONS.
3. CONTRACTOR SHALL USE CAST IRON BOX W/ COVER AND THREADED CLEANOUT PLUG (END CAP) IF LOCATED IN PAVEMENT OR CONCRETE. KEEP 1" MIN. CLEARANCE OF PLUG.
4. CONCRETE COLLAR SHALL DESIGNED TO HAVE #3 BARS ALL AROUND W/ 6" OVERLAP. COLLAR SHALL BE MIN. 8" THICK. TOP OF COLLAR SHALL BE A MIN. 2" ABOVE FINISH GRADE.

## 4 STORM DRAIN CLEANOUT

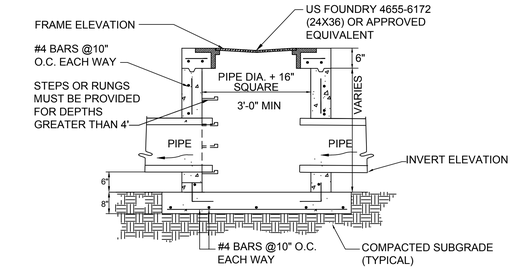
SCALE: NOT TO SCALE




NOTES:
1. CONNECTION SHALL BE COMPRESSION FIT INTO THE CORED WALL OF EXISTING STORM DRAIN PIPING. CONNECTION SHALL PROVIDE A WATERTIGHT SEAL TO EXISTING PIPING.
2. CONTRACTOR SHALL BE RESPONSIBLE FOR ANY DAMAGE TO EXISTING STORM DRAIN PIPING.

## 5 HDPE CONNECTION TO EXISTING STORM DRAIN PIPING

SCALE: NOT TO SCALE




## 6 DRAIN INLET DETAIL

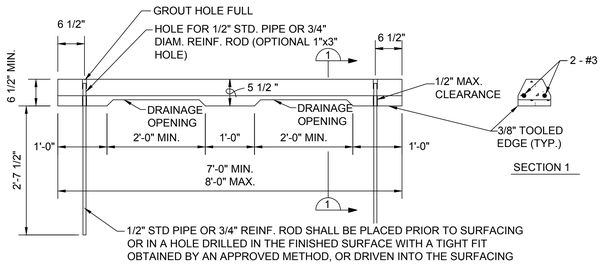
SCALE: NOT TO SCALE

IF SHEET IS LESS THAN (34"x22") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

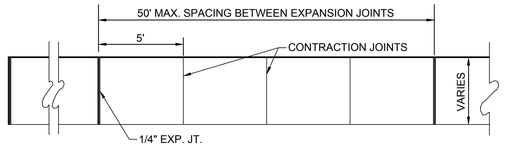
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28463 | CEMS www.cems-se.com \| Summerville, SC \| Wilmington, NC \| 843.875.3637 | C-502B |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY J. BARNES | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY D. FARLEY | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR C. MOORING | DETAILS | |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F \| CODE IDENT. NO. | N A V F A C DRAWING NO. |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: AS NOTED \| MACC. \| T.O. # | SHEET 9 OF 53 |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |




## 1 PRECAST CONCRETE WHEEL STOP DETAIL

SCALE: NOT TO SCALE

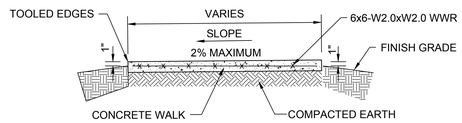



NOTES:

1. ALL SIDEWALKS SHALL BE 4" THICK.
2. CONTRACTION JOINTS SHALL BE 1" DEEP AND EDGED WITH 1/8" RADIUS.
3. EXPANSION MATERIAL 1/2" THICK SHALL BE REQUIRED WHERE SIDEWALK ABUTS CONCRETE STRUCTURE.
4. CONTRACTION JOINTS SHALL BE SPACED AT 5'-0".
5. SIDEWALKS TO RECEIVE LIGHT BROOM FINISH.
6. ALL JOINTS AND EDGES TO BE TOOLED.
7. SIDEWALKS TO BE BRUSHED AND TOOLED.

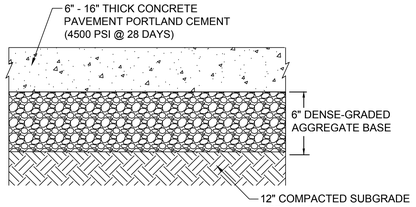
## 2 SIDEWALK PLAN

SCALE: NOT TO SCALE




## 3 TYPICAL SIDEWALK SECTION

SCALE: NOT TO SCALE

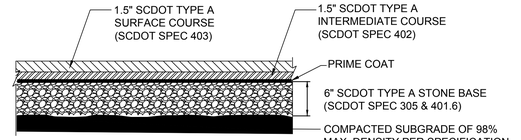



NOTES

JOINTS IN CONCRETE PAVEMENTS SHOULD BE WATERTIGHT TO PREVENT PONDING OF WATER WITHIN THE SUB-BASE STONE. THE SUBGRADE SURFACE SHOULD BE SLOPED TO PROVIDE POSITIVE DRAINAGE PRIOR TO PLACEMENT OF THE STONE BASE MATERIALS.

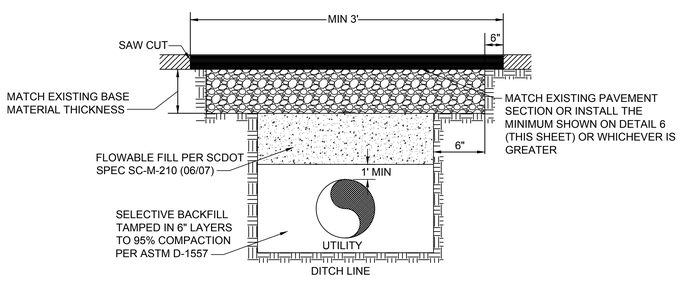
## 4 CONCRETE PAVEMENT DETAIL

SCALE: NOT TO SCALE




## 5 ASPHALT PAVEMENT SECTION

SCALE: NOT TO SCALE




## 6 PAVEMENT REPLACEMENT DETAIL

SCALE: NOT TO SCALE

ALL PAVEMENT MARKING COMPOUND SHALL BE THERMOPLASTIC PAVEMENT MARKING MATERIAL, EITHER HEATED IN PLACE OR EXTRUDED, IN ACCORDANCE WITH SCDOT SPECIFICATIONS AND LISTED ON THEIR APPROVED QUALITY PRODUCTS LIST.

| SCDOT DETAIL | DWG NO. |
|---|---|
| PAVEMENT MARKING DETAILS | 625-000 |
| | |
| | |
| | |
| | |

1. SCDOT STANDARD DRAWINGS AND SPECIFICATIONS CAN BE LOCATED HERE: HTTP://WWW.DOT.STATE.SC.US/DOING/SD_BOOK.ASPX

## 7 SCDOT PAVEMENT MARKING NOTES

SCALE: NOT TO SCALE

IF SHEET IS LESS THAN (34"x22") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

SOUTH CAROLINA REGISTERED PROFESSIONAL ENGINEER No. 33329

10/4/2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28464 | CEMS<br>www.cems-se.com \| Summerville, SC \| Wilmington, NC \| 843.875.3637 | C-503B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>J. BARNES | DEPARTMENT OF THE NAVY<br>NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. FARLEY | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M)<br>DETAILS | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>C. MOORING | | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE<br>PUBLIC WORKS OFFICER | SIZE F \| CODE IDENT. NO. \| N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | | SCALE: AS NOTED \| MACC. \| T.O. # | SHEET 10 OF 53 |

# DESIGN REFERENCES:

CODES AND STANDARDS:

| | |
|---|---|
| ACI 318-19 | BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE |
| AISC 325-17 | STEEL CONSTRUCTION MANUAL, 15TH EDITION |
| AISI S100-16 | SPECIFICATION FOR THE DESIGN OF COLD-FORMED STEEL STRUCTURAL MEMBERS, 2016 EDITION |
| AISI 214-12 | STANDARD FOR THE DESIGN OF COLD-FORMED STEEL FRAMING - TRUSS DESIGN, 2012 EDITION |
| ASCE/SEI 7-16 | MINIMUM DESIGN LOADS AND ASSOCIATED CRITERIA FOR BUILDINGS AND OTHER STRUCTURES |
| IBC 2021 | INTERNATIONAL BUILDING CODE, 2021 EDITION |
| IEBC 2021 | INTERNATIONAL EXISTING BUILDING CODE, 2021 EDITION |
| UFC 1-200-01 | DoD BUILDING CODE, 1 SEP. 2022 |
| UFC 3-301-01 | STRUCTURAL ENGINEERING, CHANGE 1, 4 FEB. 2022 |
| UFC 4-010-01 | DoD MINIMUM ANTITERRORISM STANDARDS FOR BUILDINGS, CHANGE 2, 30 JULY 2022 |

EXISTING DRAWINGS:
"BEQ-PHASE IV" MCAS BEAUFORT, SOUTH CAROLINA (2001) LOCKWOOD GREENE CONSULTING-DESIGN-CONSTRUCTION

# GENERAL NOTES:

1. THE STRUCTURE IS DESIGNED AS A STABLE UNIT AFTER ALL COMPONENT PARTS ARE IN PLACE. THEREFORE, THE CONTRACTOR/ERECTOR SHALL PROVIDE ALL NECESSARY SHORING AND BRACING AS REQUIRED TO ENSURE STABILITY DURING ERECTION.
2. WHEN A DETAIL IS SHOWN ON THE STRUCTURAL DRAWINGS FOR ONE CONDITION, IT SHALL APPLY TO ALL SIMILAR OR LIKE CONDITIONS, UNLESS NOTED OR SHOWN OTHERWISE ON THE DRAWINGS.
3. CONTRACTOR SHALL FULLY COORDINATE ALL DISCIPLINES' DRAWINGS (INCLUDING DIMENSIONS AND ELEVATIONS) AND SHALL FULLY RESOLVE ANY CONFLICTS PRIOR TO COMMENCING WORK.
4. FABRICATION DRAWINGS SHALL NOT BE REPRODUCTION OF STRUCTURAL CONTRACT DRAWINGS.
5. MISCELLANEOUS STRUCTURAL ITEMS, CONDITIONS, AND/OR PIECES OF STEEL MAY BE SHOWN ON OTHER DISCIPLINES' DRAWINGS.
6. CONTRACTOR SHALL CONSULT GOVERNMENT FOR DIRECTION SHOULD UNCHARTED OR INCORRECTLY CHARTED PIPING OR OTHER UTILITIES BE ENCOUNTERED DURING SITE CLEARING, GRADING AND EXCAVATION.
7. DESIGN CRITERIA:
   - A: FLOOR LIVE LOAD:
     - SLEEPING ROOMS 40 PSF
     - BALCONY AND DECK 100 PSF
     - CORRIDOR AND LOBBY 100 PSF
     - STAIRWAYS 100 PSF
     - MECHANICAL ROOM 125 PSF
   - B: ROOF LIVE LOAD: 20 PSF
   - C: ATTIC LIVE LOAD
     - ORIGINAL 90 PSF
     - NEW 50 PSF
   - D: GROUND SNOW LOAD, Pg: 5 PSF
   - E: RAIN INTENSITY: 4.3 IN/HR
   - F: WIND LOADS
     - a. ULTIMATE WIND SPEED, Vult: 139 MPH
     - b NOMINAL WIND SPEED, Vasd: 108 MPH
     - c. RISK CATEGORY: II
     - d. WIND EXPOSURE: C
     - e. INTERNAL PRESSURE COEFFICIENT: ±0.18
     - f. PRESSURE COEFFICIENT ZONE WIDTH, a: 4.65 FT
     - g. COMPONENTS AND CLADDING WIND PRESSURE:

| | | EFFECTIVE AREA (SF) | | | | | |
|---|---|---|---|---|---|---|---|
| | | 10 | | 20 | | 100 | |
| | ROOF | + | - | + | - | + | - |
| WIND ZONES (SEE DIAGRAMS BELOW) | 1 | 37 | -88 | 32 | -87 | 21 | -84 |
| | 2r | 37 | -113 | 32 | -108 | 21 | -97 |
| | 2e | 37 | -113 | 32 | -108 | 21 | -97 |
| | 3 | 37 | -138 | 32 | -123 | 21 | -87 |
| | WALL | + | - | + | - | + | - |
| | 4 | 50 | -54 | 48 | -52 | 43 | -47 |
| | 5 | 50 | -65 | 48 | -62 | 43 | -52 |

   - G: SEISMIC LOADS
     - a. RISK CATEGORY: II
     - b. SEISMIC IMPORTANCE FACTOR, Ie: 1.00
     - c. MAPPED SPECTRAL RESPONSE, $S_S$ = 0.632, $S_1$ = 0.200
     - d. SITE CLASS: E
     - e. DESIGN SPECTRAL RESPONSE, $S_{DS}$ = 0.627, $S_{D1}$ = 0.440
     - f. SEISMIC DESIGN CATEGORY: D
     - g. BASIC SEISMIC FORCE RESISTING SYSTEM: N/A
     - h. RESPONSE MODIFICATION COEFFICIENT, N/A
     - i. SEISMIC RESPONSE COEFFICIENT, Cs: N/A
     - j. DESIGN BASE SHEAR: N/A
     - k. ANALYSIS PROCEDURE: N/A

# CONCRETE NOTES:

1. FOUNDATIONS AND SIDEWALK/STAIR LANDING CONCRETE:
   - A. EXPOSURE CLASSES: F1,S0,W0,C1
   - B. CONCRETE COMPRESSIVE STRENGTH AT 28 DAYS: f'c = 4,500 PSI
   - C. AIR CONTENT: (SEE SPECS)
   - D. MAXIMUM WATER/CEMENTITIOUS RATIO (w/cm): 0.45
2. REINFORCING STEEL SHALL BE ASTM A615, GRADE 60.
3. WELDED WIRE REINFORCEMENT SHALL BE ASTM A1064 AND PLACED AT MID DEPTH OF SLAB, UNLESS OTHERWISE NOTED.
4. PROPORTION AND DESIGN CONCRETE MIXES TO RESULT IN CONCRETE SLUMP AT POINT OF PLACEMENT NOT LESS THAN 3" AND NOT MORE THAN 5".
5. ADDITION OF WATER TO READY-MIX IN THE FIELD SHALL NOT BE ALLOWED.
6. DEPOSIT CONCRETE IN A CONTINUOUS OPERATION UNTIL THE PLACING OF CONCRETE IS COMPLETE.
7. CONCRETE MUST BE CURED FOR A MINIMUM 21 DAYS PRIOR TO ANY ADHESIVE ANCHOR INSTALLATION.
8. SEE SECTION 20.6 OF ACI 318 FOR MINIMUM REINFORCEMENT PROTECTION REQUIREMENTS.
9. SEE SPECIFICATION SECTION 03 30 00 FOR ADDITIONAL INFORMATION.
10. WHERE SPECIFIED, GROUT SHALL BE NONSHRINK, ASTM C1107 WITH MINIMUM 4,500 PSI COMPRESSIVE STRENGTH AT 28 DAYS. PROVIDE GROUT FLOWABLILITY AS REQUIRED TO ENSURE FULL CONSOLIDATION.
11. TYPICAL CLASS B, NON-TOP BAR, REINFORCEMENT SPLICES SHALL BE AS FOLLOWS UNLESS OTHERWISE NOTED:

| | | LAP DISTANCE BY BAR SIZE | | | | |
|---|---|---|---|---|---|---|
| | | #4 | #5 | #6 | #7 | #8 |
| f'c (PSI) | 4,500 | 18" | 23" | 27" | 40" | 45" |

# STEEL DECK NOTES:

1. STEEL DECK SHALL BE MINIMUM 3 SPAN CONDITION WITH THE FOLLOWING MINIMUM PROPERTIES.
   ROOF:
   TYPE 1.5B-36
   $I_d$+ = 0.155 in⁴/ft. $I_d$- = 0.178 in⁴ f/t. 22GA.
   $S_s$+ = 0.169 in³/ft. $S_s$- = 0.179 in³/ft. Fy = 50 ksi
   $\Phi M_n$+ = 634 lb-ft/ft $\Phi M_n$- = 671 lb-ft/ft
2. SEE SPECIFICATION SECTION 05 30 00 FOR ADDITIONAL INFORMATION.

# STRUCTURAL STEEL:

1. ANCHOR RODS SHALL BE ASTM F1554 Gr. 36.
2. ALL WELDED CONSTRUCTION SHALL BE IN ACCORDANCE WITH AWS D1.1 AND AISC.
3. WELDING SHALL BE PERFORMED ONLY BY OPERATORS QUALIFIED BY THE AWS STANDARD QUALIFICATION PROCEDURE TO PERFORM THE PARTICULAR TYPE OF WORK REQUIRED.
4. MEMBERS EXPOSED TO WEATHER IN FINISHED STRUCTURE, INCLUDING LINTELS AND MECHANICAL EQUIPMENT SUPPORT FRAMES, SHALL BE PAINTED AFTER FABRICATION.
5. GROUT UNDER BEARING PLATES SHALL BE OF NON-SHRINK, NON-METALLIC COMPOSITION.
6. NO SPLICES IN STRUCTURAL FRAMING MEMBERS ARE PERMITTED EXCEPT AS INDICATED OR APPROVED.
7. SEE SPECIFICATION SECTION 05 12 00 FOR ADDITIONAL INFORMATION.
8. STRUCTURAL STEEL SHALL BE AS FOLLOWS:
   - A. W-SHAPES ASTM A992
   - B. PLATES, ANGLES, CHANNELS, MISC. STEEL ASTM A36
   - C. HSS ASTM A1085
   - D. WELDING ELECTRODE E70XX

# METAL TRUSS NOTES:

1. METAL TRUSSES, ASSOCIATED BRACING, AND ADDITIONAL FRAMING MEMBERS SHALL BE DESIGNED, FABRICATED AND INSTALLED AS INDICATED ON THE DRAWINGS AND IN ACCORDANCE WITH THE AMERICAN IRON AND STEEL INSTITUTE, NORTH AMERICAN SPECIFICATION FOR THE DESIGN OF COLD-FORMED STEEL STRUCTURAL MEMBERS 2016 AND AMERICAN IRON AND STEEL INSTITUTE STANDARD FOR COLD-FORMED STEEL FRAMING - TRUSS DESIGN 2012.
2. SUBMIT SHOP DRAWINGS AND CALCULATIONS SIGNED AND SEALED BY A REGISTERED ENGINEER.
3. PROVIDE HORIZONTAL BRIDGING (PERMANENT AND TEMPORARY) AS RECOMMENDED BY THE TRUSS MANUFACTURER.
4. THE TRUSS FRAMING SHOWN IS FOR INTENT OF DESIGN PURPOSES ONLY. FINAL FRAMING LAYOUT MAY VARY BASED UPON TRUSS MANUFACTURER'S DESIGN.
5. ROOF TRUSSES TO BE SPACED AT 48" O.C. MAXIMUM UNLESS SHOWN OTHERWISE.
6. TRUSS TOP CHORDS SHALL BE 16 GA. MINIMUM.
7. UNIFORM LOADS INDICATED BELOW TO BE USED FOR DESIGN OF HIP AND/OR GIRDERS TRUSSES AS REQUIRED.
8. MAXIMUM LIVE LOAD DEFLECTION SHALL BE SPAN/240 FOR ROOF TRUSSES.
9. ROOF TRUSSES SHALL BE DESIGNED FOR APPLICABLE WIND LOADS AS INDICATED IN GENERAL NOTES, IN COMBINATION WITH DEAD LOADS SHOWN BELOW.
10. TRUSS TO TRUSS CONNECTIONS AND TRUSS TO STRUCTURE CONNECTIONS SHALL BE DESIGNED BY TRUSS MANUFACTURER.
11. TRUSS FRAMING SHALL BE GALVANIZED IN ACCORDANCE WITH A653, G60.
12. SEE SPECIFICATION SECTION 05 40 00 FOR ADDITIONAL INFORMATION.
13. ROOF TRUSS DESIGN LOADS TO BE AS FOLLOWS:

| | |
|---|---|
| TOP CHORD LIVE LOAD | 20 PSF |
| TOP CHORD DEAD LOAD | 10 PSF |
| BOTTOM CHORD DEAD LOAD | 5 PSF |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |

# STRUCTURAL ABBREVIATIONS:

| A | |
|---|---|
| A.F.F. | ABOVE FINISHED FLOOR |
| A.B. | ANCHOR BOLT |
| & | AND |
| APPROX. | APPROXIMATE |
| ARCH. | ARCHITECTURAL |
| @ | AT |

| B | |
|---|---|
| BM. | BEAM |
| BLKG. | BLOCKING |
| BOT. | BOTTOM |
| B.O. | BOTTOM OF |
| BRDG. | BRIDGING |
| BLDG. | BUILDING |

| C | |
|---|---|
| CLG. | CEILING |
| CL. | CENTERLINE |
| CLR. | CLEAR |
| COL. | COLUMN |
| CONC. | CONCRETE |
| CMU | CONCRETE MASONRY UNIT |
| CONN. | CONNECTION |
| CONT. | CONTINUOUS |
| C.J. | CONTROL JOINT |
| COORD. | COORDINATE |

| D | |
|---|---|
| ° | DEGREE |
| DEMO. | DEMOLITION |
| DET. | DETAIL |
| Ø | DIAMETER |
| DIM. | DIMENSION |

| E | |
|---|---|
| EA. | EACH |
| E.F. | EACH FACE |
| E.W. | EACH WAY |
| ELEC. | ELECTRICAL OR ELECTRIC |
| EL. | ELEVATION |
| E.O.R. | ENGINEER OF RECORD |
| EQ. | EQUAL |
| EX. | EXISTING |
| E.J. (EXP. JT.) | EXPANSION JOINT |
| EXP. | EXPANSION/EXPOSED |
| EXT. | EXTERIOR |

| F | |
|---|---|
| F.S. | FAR SIDE |
| FT. | FEET |
| FRP | FIBER REINFORCED POLYMER |
| FIN. | FINISH |
| F.F. | FINISHED FLOOR |
| FLR. | FLOOR |
| FTG. | FOOTING |

| G | |
|---|---|
| GA. | GAGE OR GAUGE |
| GALV. | GALVANIZED |
| Gr. | GRADE |

| H | |
|---|---|
| HORIZ. | HORIZONTAL |

| I | |
|---|---|
| IN. | INCH |
| I.D. | INSIDE DIAMETER |
| INT. | INTERIOR |

| J | |
|---|---|
| JT. | JOINT |
| JST. | JOIST |

| K | |
|---|---|
| KSI | KIPS/SQUARE INCH |

| L | |
|---|---|
| LLH | LONG LEG HORIZONTAL |
| LLV | LONG LEG VERTICAL |
| LONG. | LONGITUDINAL |

| M | |
|---|---|
| MATL. | MATERIAL |
| MAX. | MAXIMUM |
| MECH. | MECHANICAL |
| MTL | METAL |
| MPH | MILE PER HOUR |
| MIN. | MINIMUM |
| MISC. | MISCELLANEOUS |

| N | |
|---|---|
| N.S. | NEAR SIDE |
| N.I.C. | NOT IN CONTRACT |
| NTS | NOT TO SCALE |
| NO. (#) | NUMBER |

| O | |
|---|---|
| O.C. | ON CENTER |
| O.H. | OPPOSITE HAND |
| O.D. | OUTSIDE DIAMETER |

| P | |
|---|---|
| PL | PLATE |
| LBS. | POUNDS |
| PCF | POUNDS/CUBIC FOOT |
| PSF | POUNDS/SQUARE FOOT |
| PSI | POUNDS/SQUARE INCH |
| PEMB | PRE-ENGINEERED METAL BUILDING |
| PT | PRESSURE TREATED |
| PROJ. | PROJECTION |

| R | |
|---|---|
| REF. | REFERENCE |
| RC. | REINFORCED CONCRETE |
| REINF. | REINFORCEMENT |
| REQ'D | REQUIRED |
| REV. | REVISIONS / REVISED |

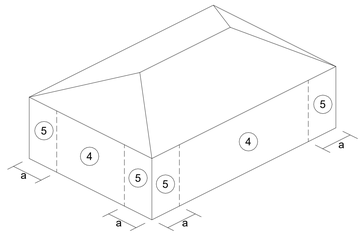
| S | |
|---|---|
| SECT. | SECTION |
| SIM. | SIMILAR |
| S.O.G. | SLAB ON GRADE |
| SL. | SLOPE |
| SPEC. | SPECIFICATION |
| SQ. | SQUARE |
| S.S. | STAINLESS STEEL |
| STD. | STANDARD |
| STL. | STEEL |
| STRUC. | STRUCTURAL |

| T | |
|---|---|
| T&B | TOP & BOTTOM |
| T.O. | TOP OF |
| TRANS. | TRANSVERSE |
| TYP. | TYPICAL |

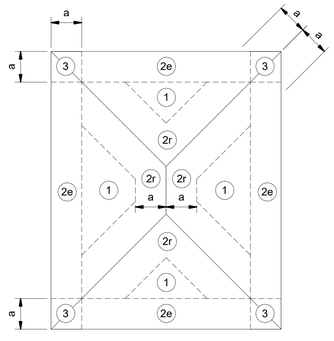
| U | |
|---|---|
| U.N.O. | UNLESS NOTED OTHERWISE |

| V | |
|---|---|
| V.I.F. | VERIFY IN FIELD |
| VERT. | VERTICAL |

| W | |
|---|---|
| WWR | WELDED WIRE REINFORCEMENT |
| W/ | WITH |
| W/O | WITHOUT |
| WP | WORKING POINT |




HIP ROOF WALL WIND ZONES




HIP ROOF WIND ZONES (7° < Θ < 45°)

1 COMPONENTS & CLADDING WIND ZONES
SCALE: NOT TO SCALE

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

04-OCT-2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28465 | CEMS www.cems.com 1765 Sam Houston Parkway, 29707 (843) 871-3007 | S-001 |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY C. STOCKMAN | DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND MARINE CORPS AIR STATION BEAUFORT, S.C. | |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY J. LI | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR B. MINOR | GENERAL NOTES AND DESIGN LOADS | |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: 12" = 1'-0" MACC. T.O. # | SHEET 11 OF 53 |

## STEEL - WELDING

*REFERENCE: 2021 IBC SECT. 1705.2.1; AISC 360-16 TABLE N5.4-1 - N5.4-3*

| STEEL INSPECTION PRIOR TO WELDING | | |
|---|---|---|
| **TASK** | **INSPECTION TYPE** | **NOTES** |
| VERIFY THAT THE WELDING PROCEDURE(S) SPECIFICATIONS (WPS) ARE AVAILABLE. | PERFORM | |
| VERIFY MANUFACTURER CERTIFICATIONS FOR WELDING CONSUMABLES ARE AVAILABLE. | PERFORM | |
| VERIFY MATERIAL IDENTIFICATION | OBSERVE | TYPE AND GRADE. |
| WELDER IDENTIFICATION SYSTEM | OBSERVE | THE FABRICATOR OR ERECTOR, AS APPLICABLE, SHALL MAINTAIN A SYSTEM BY WHICH A WELDER WHO HAS WELDED A JOINT OR MEMBER CAN BE IDENTIFIED. STAMPS, IF USED, SHALL BE THE LOW-STRESS TYPE. |
| FIT-UP OF GROOVE WELDS (INCLUDING JOINT GEOMETRY) | OBSERVE | - JOINT PREPARATION<br>- DIMENSIONS (ALIGNMENT, ROOT OPENING, ROOT FACE, BEVEL)<br>- CLEANLINESS (CONDITION OF STEEL SURFACES)<br>- TACKING (TACK WELD QUALITY AND LOCATION)<br>- BACKING TYPE AND FIT (IF APPLICABLE) |
| CONFIGURATION AND FINISH OF ACCESS HOLES | OBSERVE | |
| FIT-UP OF FILLET WELDS | OBSERVE | - DIMENSIONS (ALIGNMENT, GAPS AT ROOT)<br>- CLEANLINESS (CONDITION OF STEEL SURFACES)<br>- TACKING (TACK WELD QUALITY AND LOCATION) |
| CHECK WELDING EQUIPMENT | OBSERVE | |

| STEEL INSPECTION DURING WELDING | | |
|---|---|---|
| **TASK** | **INSPECTION TYPE** | **NOTES** |
| USE OF QUALIFIED WELDERS | PERFORM | WELDING BY WELDERS, WELDING OPERATORS, AND TACK WELDERS WHO ARE QUALIFIED IN CONFORMANCE WITH REQUIREMENTS. |
| CONTROL AND HANDLING OF WELDING CONSUMABLES | OBSERVE | - PACKAGING<br>- ELECTRODE ATMOSPHERIC EXPOSURE CONTROL |
| NO WELDING OVER CRACKED TACK WELDS | OBSERVE | |
| ENVIRONMENTAL CONDITIONS | OBSERVE | - WIND SPEED WITHIN LIMITS<br>- PRECIPITATION AND TEMPERATURE |
| WELDING PROCEDURES SPECIFICATION FOLLOWED | OBSERVE | - SETTINGS ON WELDING EQUIPMENT<br>- TRAVEL SPEED<br>- SELECTED WELDING MATERIALS<br>- SHIELDING GAS TYPE/FLOW RATE<br>- PREHEAT APPLIED<br>- INTERPASS TEMPERATURE MAINTAINED (MIN./MAX.)<br>- PROPER POSITION (F, V, H, OH)<br>- INTERMIX OF FILLER METALS AVOIDED |
| WELDING TECHNIQUES | OBSERVE | - INTERPASS AND FINAL CLEANING<br>- EACH PASS WITHIN PROFILE LIMITATIONS<br>- EACH PASS MEETS QUALITY REQUIREMENTS |

| STEEL INSPECTION AFTER WELDING | | |
|---|---|---|
| **TASK** | **INSPECTION TYPE** | **NOTES** |
| WELDS CLEANED | OBSERVE | |
| SIZE, LENGTH, AND LOCATION OF ALL WELDS | PERFORM | SIZE, LENGTH, AND LOCATION OF ALL WELDS CONFORM TO THE REQUIREMENTS OF THE DETAIL DRAWINGS. |
| WELDS MEET VISUAL ACCEPTANCE CRITERIA | PERFORM AND DOCUMENT | - CRACK PROHIBITION<br>- WELD/BASE-METAL FUSION<br>- CRATER CROSS SECTION<br>- WELD PROFILES<br>- WELD SIZE<br>- UNDERCUT<br>- POROSITY |
| ARC STRIKES | PERFORM | |
| K-AREA | PERFORM | WHEN WELDING OF DOUBLER PLATES, CONTINUITY PLATES, OR STIFFENERS HAS BEEN PERFORMED IN THE K-AREA, VISUALLY INSPECT THE WEB K-AREA FOR CRACKS. |
| BACKING REMOVED, WELD TABS REMOVED AND FINISHED, AND FILLET WELDS ADDED WHERE REQUIRED | PERFORM | |
| REPAIR ACTIVITIES | PERFORM AND DOCUMENT | |
| DOCUMENT ACCEPTANCE OR REJECTION OF WELDED JOINT OR MEMBER | PERFORM | |

## STEEL - OTHER INSPECTIONS

*REFERENCE: 2021 IBC SECT. 1705.2.1; AISC 341-16 TABLES J8.1 & J10.1*

| **TASK** | **INSPECTION TYPE** | **NOTES** |
|---|---|---|
| ANCHOR RODS AND OTHER EMBEDMENTS SUPPORTING STRUCTURAL STEEL | PERFORM | VERIFY THE DIAMETER, GRADE, TYPE, AND LENGTH OF THE ANCHOR ROD OR EMBEDDED ITEM, AND THE EXTENT OR DEPTH OF EMBEDMENT PRIOR TO PLACEMENT OF CONCRETE. |
| FABRICATED STEEL OR ERECTED STEEL FRAME | OBSERVE | VERIFY COMPLIANCE WITH THE DETAILS SHOWN ON THE CONSTRUCTION DOCUMENTS SUCH AS BRACES, STIFFENERS, MEMBER LOCATIONS, AND PROPER APPLICATION OF JOINT DETAILS AT EACH CONNECTION. |

## CONCRETE

*REFERENCE: 2021 IBC TABLE 1705.3; ACI 318-19 SECT. 26.13.3*

| **TASK** | **INSPECTION TYPE** | **NOTES** |
|---|---|---|
| ADHESIVE ANCHORS | PERFORM | - INSPECT AS REQUIRED PER APPROVED ICC-ES REPORT<br>- INSPECT PROOF LOADING AS REQUIRED BY THE CONTRACT DOCUMENTS<br>-TEST PER "ANCHOR PROOF LOAD" NOTES AS REQUIRED BY THE CONSTRUCTION DRAWINGS |

## COLD-FORMED METAL DECK

*REFERENCE: 2021 IBC SECT. 1705.2.2; SDI QA/QC-2017, APP. 1, TABLE 1.1-TABLE 1.8*

| PRIOR TO DECK PLACEMENT | | |
|---|---|---|
| **TASK** | **INSPECTION TYPE** | **NOTES** |
| VERIFY COMPLIANCE OF MATERIALS (DECK AND ALL DECK ACCESSORIES) WITH CONSTRUCTION DOCUMENTS, INCLUDING PROFILES, MATERIAL PROPERTIES, AND BASE METAL THICKNESS | PERFORM | |
| DOCUMENT ACCEPTANCE OR REJECTION OF DECK AND DECK ACCESSORIES | DOCUMENT | |

| DURING/AFTER DECK PLACEMENT | | |
|---|---|---|
| VERIFY COMPLIANCE OF DECK AND ALL DECK ACCESSORIES INSTALLATION WITH CONSTRUCTION DOCUMENTS | PERFORM | |
| VERIFY DECK MATERIALS ARE REPRESENTED BY THE MILL CERTIFICATIONS THAT COMPLY WITH THE CONSTRUCTION DOCUMENTS | PERFORM | |
| DOCUMENT ACCEPTANCE OR REJECTION OF INSTALLATION OF DECK AND DECK ACCESSORIES | DOCUMENT | |

| PRIOR TO DECK MECHANICAL FASTENING | | |
|---|---|---|
| MANUFACTURER INSTALLATION INSTRUCTIONS AVAILABLE FOR MECHANICAL FASTENERS | OBSERVE | |
| PROPER TOOLS AVAILABLE FOR FASTENER INSTALLATION | OBSERVE | |
| PROPER STORAGE FOR MECHANICAL FASTENERS | OBSERVE | |

| DURING DECK MECHANICAL FASTENING | | |
|---|---|---|
| FASTENERS ARE POSITIONED AS REQUIRED | OBSERVE | |
| FASTENERS ARE INSTALLED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS | OBSERVE | |

| AFTER DECK MECHANICAL FASTENING | | |
|---|---|---|
| CHECK SPACING, TYPE, AND INSTALLATION OF SUPPORT FASTENERS | PERFORM | |
| CHECK SPACING, TYPE, AND INSTALLATION OF SIDELAP FASTENERS | PERFORM | |
| CHECK SPACING, TYPE, AND INSTALLATION OF PERIMETER FASTENERS | PERFORM | |
| VERIFY REPAIR ACTIVITIES | PERFORM | |
| DOCUMENT ACCEPTANCE OR REJECTION OF MECHANICAL FASTENERS | DOCUMENT | |

## SPECIAL INSPECTION NOTES:

1. PERFORM ALL SPECIAL INSPECTIONS AND TESTING AS NOTED IN SPECIFICATION 01 45 35 AND AS SUMMARIZED BELOW.
2. KEY TERMS WITHIN THE SPECIAL INSPECTIONS AND TESTING SUMMARIES ARE DEFINED BELOW:
   A. PERFORM: PERFORM THE INDICATED TASK FOR EACH WELD, FASTENER, BOLTED CONNECTION OR MEMBER WITH REQUIRED VERIFICATION.
   B. OBSERVE: OBSERVE THE INDICATED TASK ON A RANDOM SAMPLING BASIS DAILY TO ENSURE THAT APPLICABLE REQUIREMENTS ARE MET. OPERATIONS NEED NOT BE DELAYED PENDING THESE INSPECTIONS AT THE CONTRACTOR'S RISK.
   C. DOCUMENT: DOCUMENT IN A REPORT THAT THE WORK HAS BEEN PERFORMED AS REQUIRED. THIS IS IN ADDITION TO ALL OTHER REQUIRED REPORTS.
   D. CONTINUOUS: CONSTANT MONITORING OF IDENTIFIED TASKS BY A SPECIAL INSPECTOR OVER THE DURATION OF PERFORMANCE OF SAID TASKS.

## ANCHOR PROOF LOAD NOTES:

1. ADHESIVE ANCHORS SHALL BE PROOF TESTED AS REQUIRED HEREIN. TEN PERCENT OF EACH TYPE AND SIZE OF ADHESIVE ANCHOR ASSEMBLY SHALL BE PROOF TESTED IN TENSION BY INDEPENDENT TESTING LABORATORY.
   A. TENSION TESTING SHALL BE PERFORMED IN ACCORDANCE WITH ASTM E488.
   B. THE INDEPENDENT TESTING LABORATORY SHALL SUBMIT AN ANCHORAGE TESTING PLAN TO THE ENGINEER OF RECORD TO ENSURE THE TESTING REQUIREMENTS ARE FULFILLED.
2. THE ADHESIVE ANCHORS SHALL BE TENSION PROOF LOADED TO 1.5 KIPS. PROOF LOADING SHALL BE PERFORMED AFTER A MINIMUM CURING PERIOD SPECIFIED BY THE MANUFACTURER.
3. ANCHORS SHALL HAVE NO VISIBLE INDICATIONS OF DISPLACEMENT OR DAMAGE DURING OR AFTER PROOF LOAD APPLICATION. CONCRETE CRACKING IN THE VICINITY OF THE ANCHOR AFTER LOADING SHALL BE CONSIDERED A FAILURE.
4. IF MORE THAN 10 PERCENT OF THE TESTED ADHESIVE ANCHORS FAIL TO ACHIEVE THE SPECIFIED PROOF LOAD WITHIN THE LIMITS DEFINED ON THESE DRAWINGS, 100 PERCENT OF THE ANCHORS OF THE SAME DIAMETER AND THE TYPE AS THE FAILED ANCHOR SHALL BE PROOF TESTED, UNLESS OTHERWISE DIRECTED BY THE ENGINEER OF RECORD.

**REVISIONS**

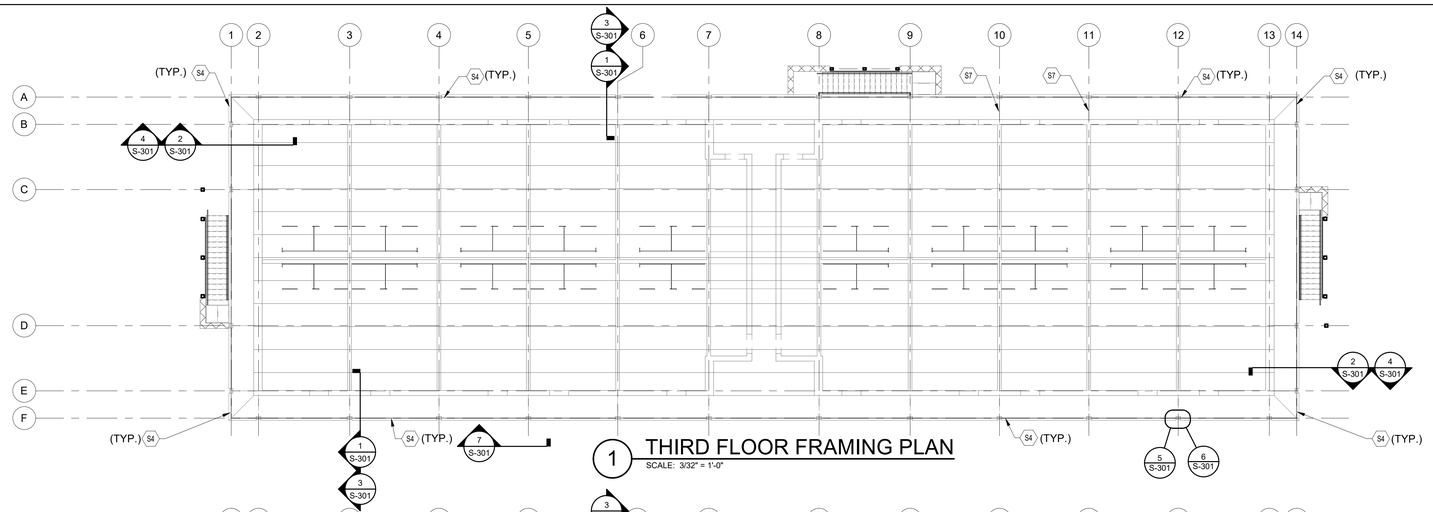
| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

04-OCT-2023

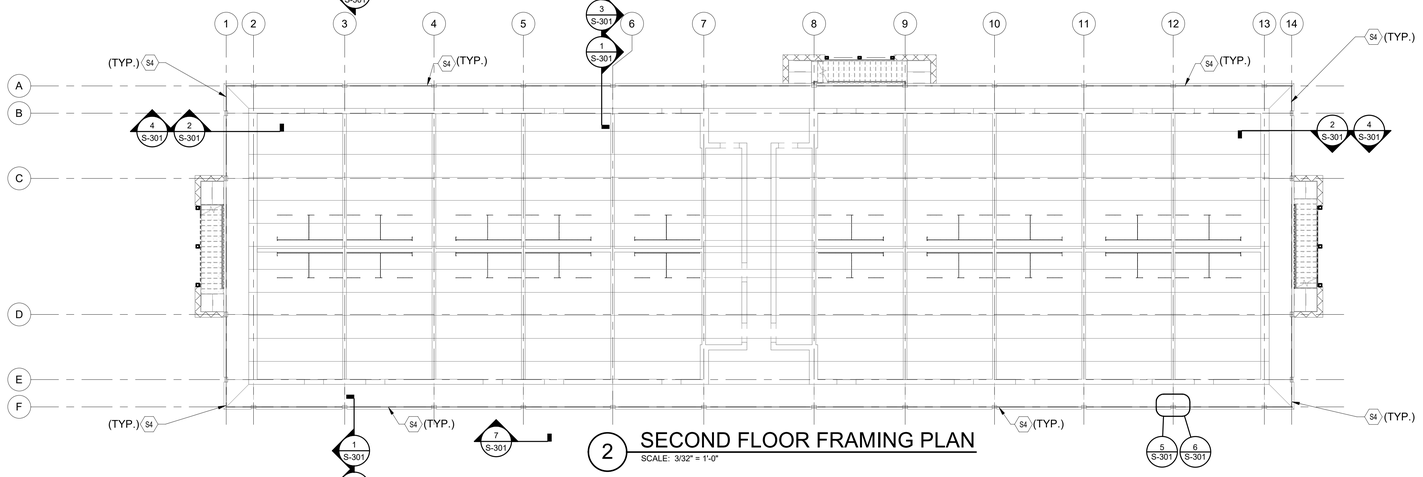
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

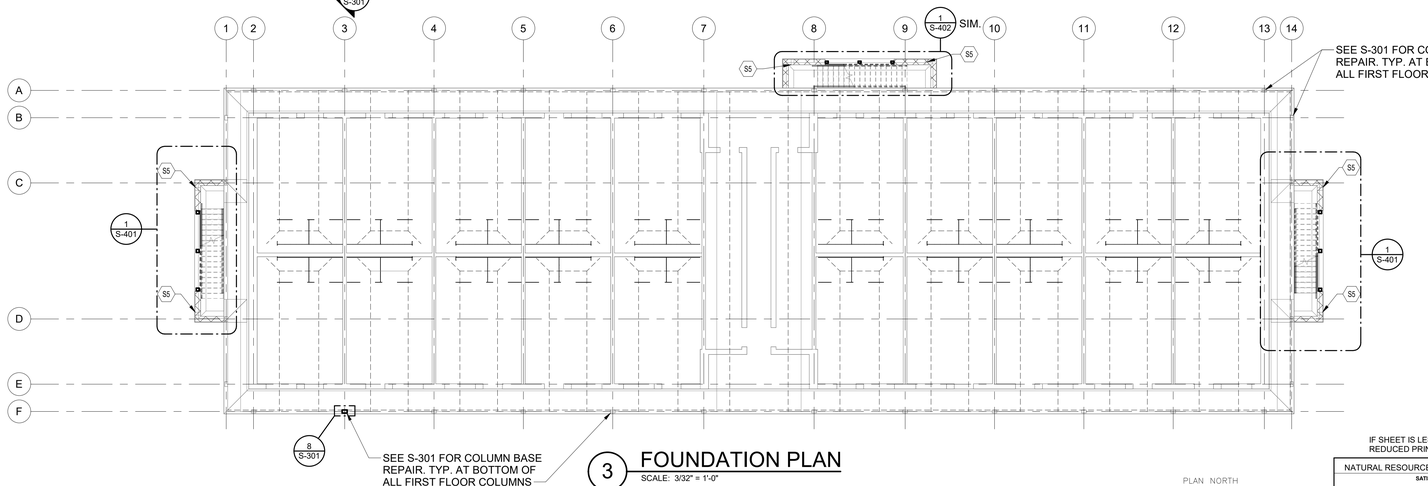
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28466 | CEMS | S-002 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>C. STOCKMAN | DEPARTMENT OF THE NAVY<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | NAVAL FACILITIES ENGINEERING COMMAND |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>J. LI | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>B. MINOR | SPECIAL INSPECTIONS AND NOTES | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F / CODE IDENT. NO. / N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: 3" = 1'-0" / MACC. / T.O. # | SHEET 12 OF 53 |




1 THIRD FLOOR FRAMING PLAN
SCALE: 3/32" = 1'-0"




2 SECOND FLOOR FRAMING PLAN
SCALE: 3/32" = 1'-0"




3 FOUNDATION PLAN
SCALE: 3/32" = 1'-0"

PLAN NORTH

24"0 4' 8' 16'

SCALE: 3/32"=1'-0"

GRAPHIC SCALE

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. ALL EXISTING CONDITIONS PER EXISTING DRAWINGS AND NON-DESTRUCTIVE OBSERVATION.
2. COORDINATE DEMOLITION WITH ARCHITECTURAL DRAWINGS.
3. DO NOT CUT ANY EXISTING STRUCTURAL MEMBERS UNLESS DIRECTLY AUTHORIZED BY ENGINEER OF RECORD.
4. CLEAN, REMOVE CORROSION FROM, AND PREPARE ALL EXPOSED TO VIEW EXISTING TO REMAIN STEEL FOR NEW PAINT, INCLUDING BUT NOT LIMITED TO COLUMNS, BEAMS, CONNECTIONS AND OTHER EXPOSED STEEL FRAMING ELEMENTS.

## SHEET KEYNOTES

S4 REMOVE EXISTING C STEEL PLATE AT EDGE OF WALKWAY ENTIRE PERIMETER AT SECOND AND THIRD FLOORS. SEE DETAILS ON SHEET S-301.

S5 4"+/- x 4"+/- OPENINGS IN EXISTING CMU WALLS FOR DRAINAGE. INSTALL THROUGH EXISTING UNGROUTED CELLS (SEE S-401 AND S-402).

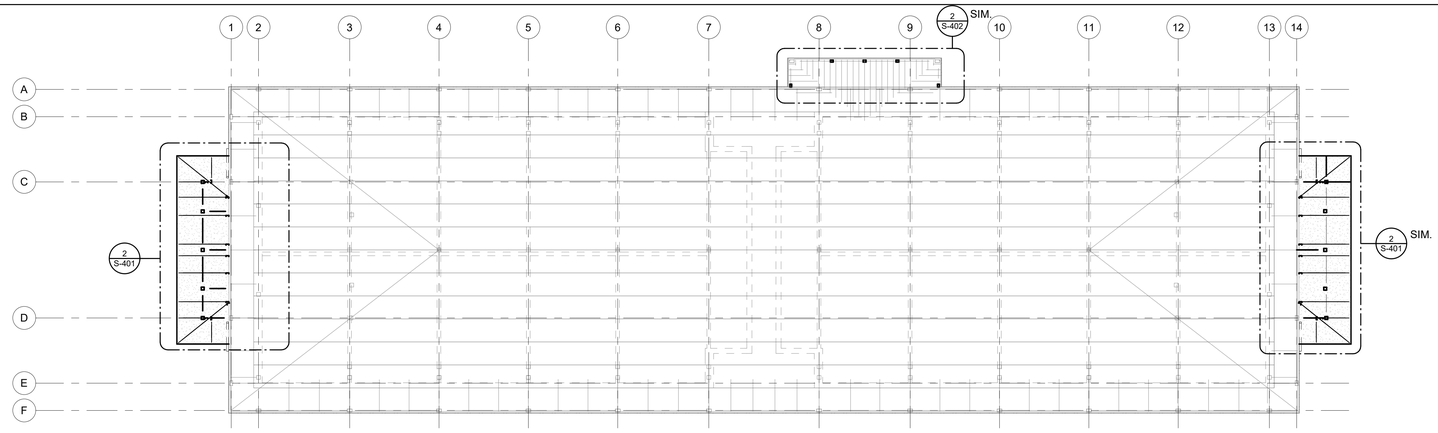
S7 REMOVE CAULKING AT EXISTING WT BEAM IN WALKWAY AREA.

STATE OF SOUTH CAROLINA No. 29581 04-OCT-2023

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

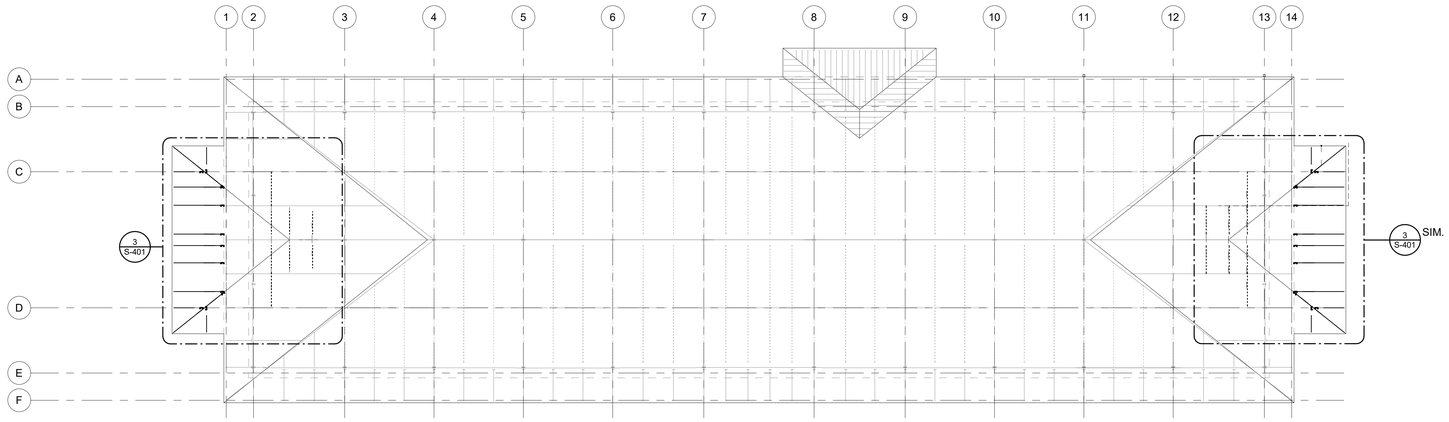
ISSUE DATE:08 SEPT 2023

| | | |
|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28467 | CEMS<br>S-101B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>C. STOCKMAN | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>**MARINE CORPS AIR STATION**<br>BEAUFORT, S.C. |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>J. LI | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>B. MINOR | FIRST, SECOND AND THIRD FLOOR PLANS - BLDG 1245G |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. |
| | | HQMC PROJ. NO.:; CONSTR. CONTR. NO.: N40085-18-D-0616 |
| PUBLIC WORKS OFFICER | | SCALE: 3/32" = 1'-0"; MACC.; T.O. #; SHEET 13 OF 53 |




## 1 CEILING FRAMING PLAN

SCALE: 3/32" = 1'-0"




## 2 ROOF FRAMING PLAN

SCALE: 3/32" = 1'-0"

PLAN NORTH

24"0 4' 8' 16'

**SCALE: 3/32"=1'-0"**

**GRAPHIC SCALE**

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |

# GENERAL NOTES

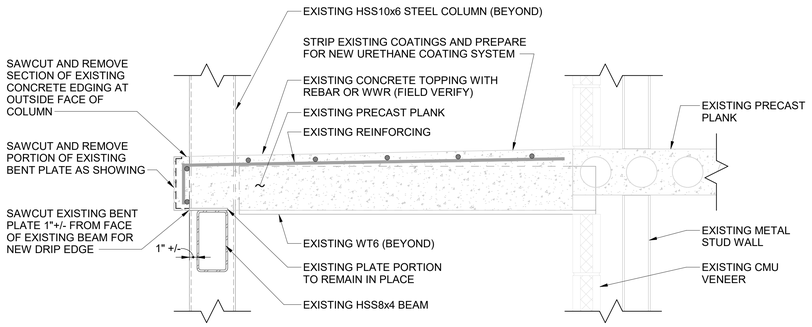
1. ALL EXISTING CONDITIONS PER ORIGINAL DOCUMENTS AND NON-DESTRUCTIVE OBSERVATION.
2. COORDINATE DEMOLITION WITH ARCHITECTURAL DRAWINGS.
3. DO NOT CUT ANY EXISTING STRUCTURAL MEMBERS UNLESS DIRECTLY AUTHORIZED BY ENGINEER OF RECORD.
4. CLEAN, REMOVE CORROSION FROM, AND PREPARE ALL EXPOSED TO VIEW EXISTING TO REMAIN STEEL FOR NEW PAINT, INCLUDING BUT NOT LIMITED TO COLUMNS, BEAMS, CONNECTIONS AND OTHER EXPOSED STEEL FRAMING ELEMENTS.

04-OCT-2023

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

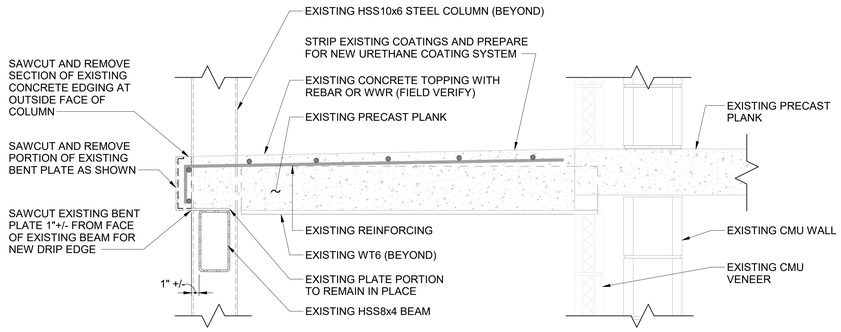
ISSUE DATE:08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28468 | CEMS | S-102B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>C. STOCKMAN | DEPARTMENT OF THE NAVY<br>**MARINE CORPS AIR STATION**<br>BEAUFORT, S.C. | NAVAL FACILITIES ENGINEERING COMMAND |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>J. LI | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M)<br>ROOF AND CEILING FRAMING PLANS - BLDG 1245G | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>B. MINOR | | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE<br>PUBLIC WORKS OFFICER | SIZE F / CODE IDENT. NO. / N A V F A C DRAWING NO.<br>HQMC PROJ. NO.: / CONSTR. CONTR. NO.: N40085-18-D-0616<br>SCALE: 3/32" = 1'-0" / MACC. / T.O. # / SHEET 14 OF 53 | |



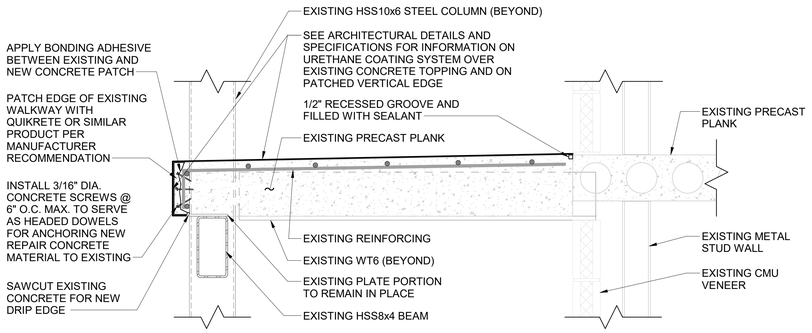

## TYPICAL WALKWAY SECTION AT LONGITUDINAL WALLS - DEMO

1 / S-101 — 1 — SCALE: 1 1/2" = 1'-0"



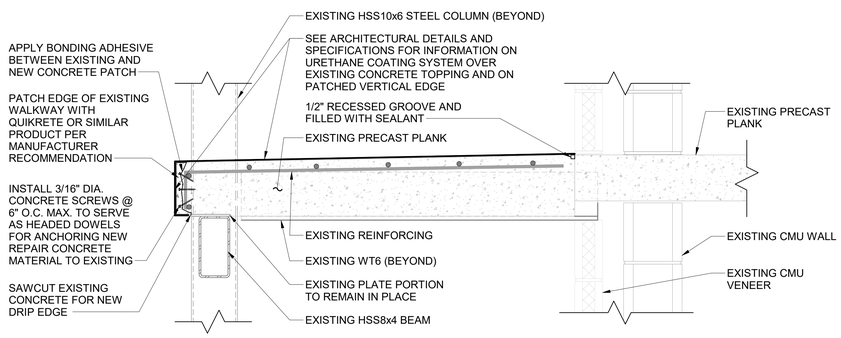

## TYPICAL WALKWAY SECTION AT END WALLS - DEMO

1 / S-101 — 2 — SCALE: 1 1/2" = 1'-0"



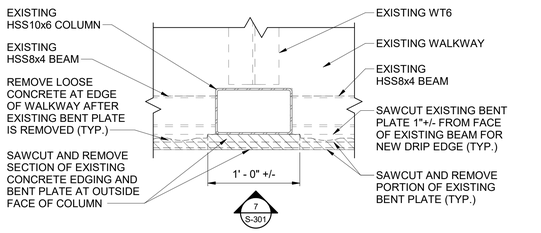

## TYPICAL WALKWAY SECTION AT LONGITUDINAL WALLS - NEW WORK

1 / S-101 — 3 — SCALE: 1 1/2" = 1'-0"



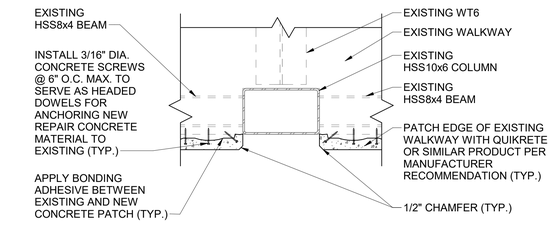

## TYPICAL WALKWAY SECTION AT END WALLS - NEW WORK

1 / S-101 — 4 — SCALE: 1 1/2" = 1'-0"



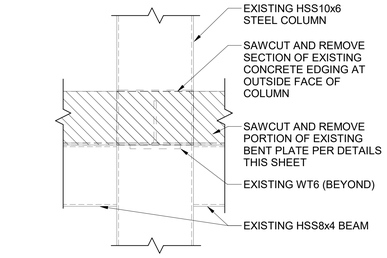

## TYPICAL COLUMN PLAN VIEW AT SECOND AND THIRD FLOORS - DEMO

1 / S-101 — 5 — SCALE: 1 1/2" = 1'-0"



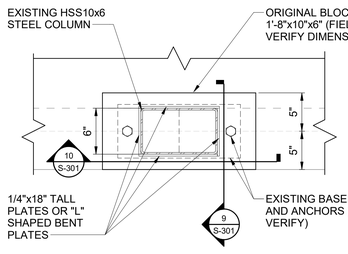

## TYPICAL COLUMN PLAN VIEW AT SECOND AND THIRD FLOORS - NEW WORK

1 / S-101 — 6 — SCALE: 1 1/2" = 1'-0"




## TYPICAL COLUMN ELEVATION VIEW AT SECOND AND THIRD FLOORS - DEMO

2 / S-101 — 7 — SCALE: 1 1/2" = 1'-0"

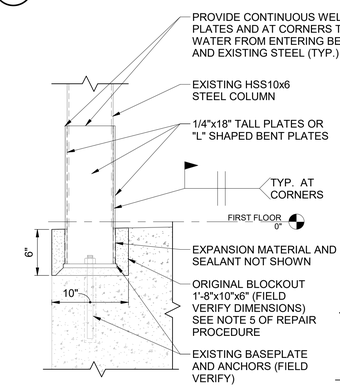



## TYPICAL COLUMN PLAN VIEW AT FIRST FLOOR

3 / S-101 — 8 — SCALE: 1 1/2" = 1'-0"

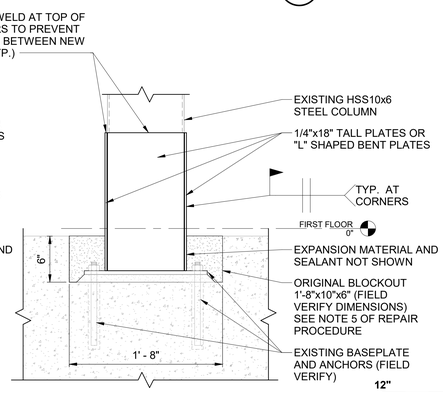
REPAIR PROCEDURE:

1. REMOVE EXISTING GROUT AND CONCRETE WITHIN ORIGINAL 20"x10"x6" DEEP BLOCKOUT , EXPOSING BOTTOM OF COLUMN AND BASEPLATE. THEN CLEAN ALL EXISTING STEEL TO PREPARE FOR NEW WELDING.
2. WELD NEW 1/4" THICK PLATES OR 1/4" THICK "L" SHAPED BENT PLATES, 18" IN HEIGHT TO ALL 4 SIDE OF COLUMN FACES AND BASEPLATE.
3. CLEAN AND PREPARE ALL STEEL TO RECEIVE NEW PRIMER, PAINT, FINISH, ETC.
4. WRAP NEW STEEL WITH AN EXPANSION JOINT MATERIAL.
5. REFILL EXISTING BLOCKOUT WITH NEW NON-SHRINK GROUT UP TO NEW EXPANSION JOINT MATERIAL.
6. REMOVE TOP 1/2" OF EXPANSION JOINT MATERIAL, INSTALL SEALANT BOND BREAKER MATERIAL, AND PLACE SEALANT IN NEW WORKING JOINT BETWEEN STEEL COLUMN AND NEW GROUT FILL.




## SECTION

8 / S-301 — 9 — SCALE: 1 1/2" = 1'-0"




## SECTION

8 / S-301 — 10 — SCALE: 1 1/2" = 1'-0"

12" 0 6" 1'

**SCALE: 1 1/2"=1'-0"**

**GRAPHIC SCALE**

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## LEGEND

EXISTING ITEM TO BE REMOVED

NEW CONCRETE PATCH

## GENERAL NOTES

1. ALL EXISTING CONDITIONS PER ORIGINAL DOCUMENTS AND NON-DESTRUCTIVE OBSERVATION.
2. COORDINATE DEMOLITION WITH ARCHITECTURAL DRAWINGS.
3. DO NOT CUT ANY EXISTING STRUCTURAL MEMBERS UNLESS DIRECTLY AUTHORIZED BY ENGINEER OF RECORD.
4. CLEAN AND PREPARE ALL EXPOSED TO VIEW EXISTING TO REMAIN STEEL FOR NEW PAINT. INCLUDING BUT NOT LIMITED TO COLUMNS, BEAMS, CONNECTIONS AND OTHER EXPOSED STEEL FRAMING ELEMENTS.

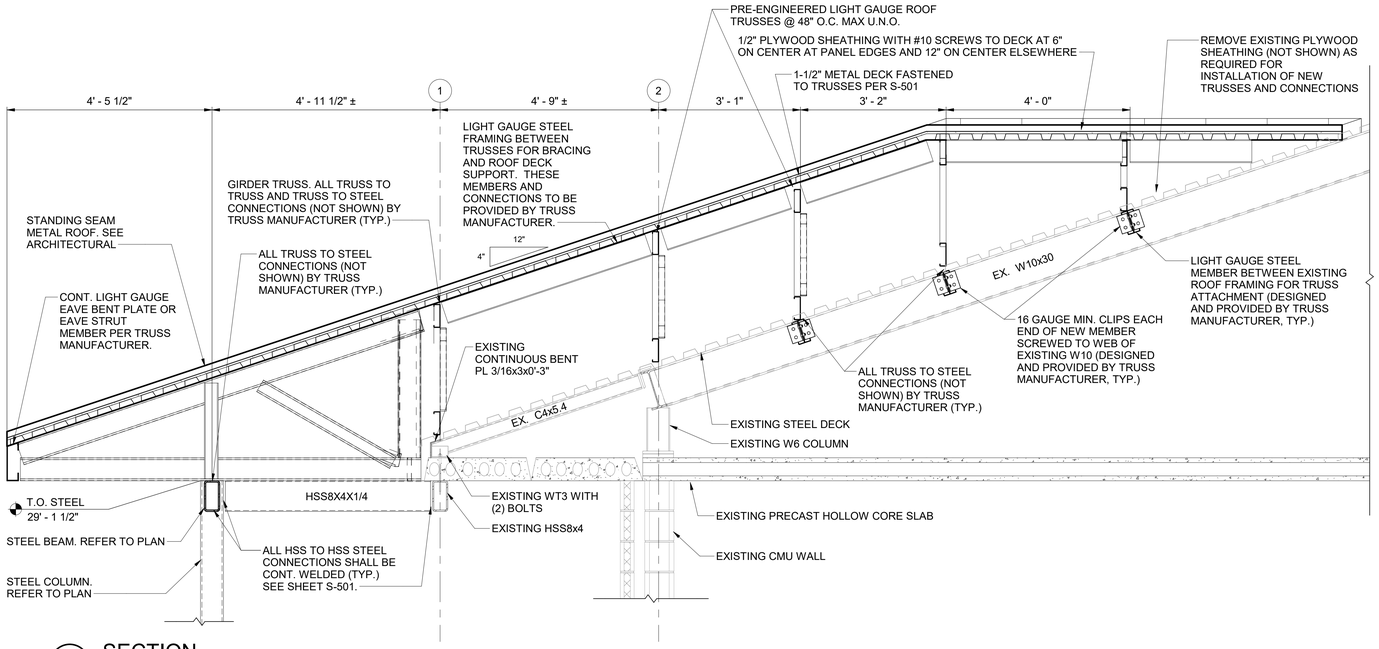
04-OCT-2023

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

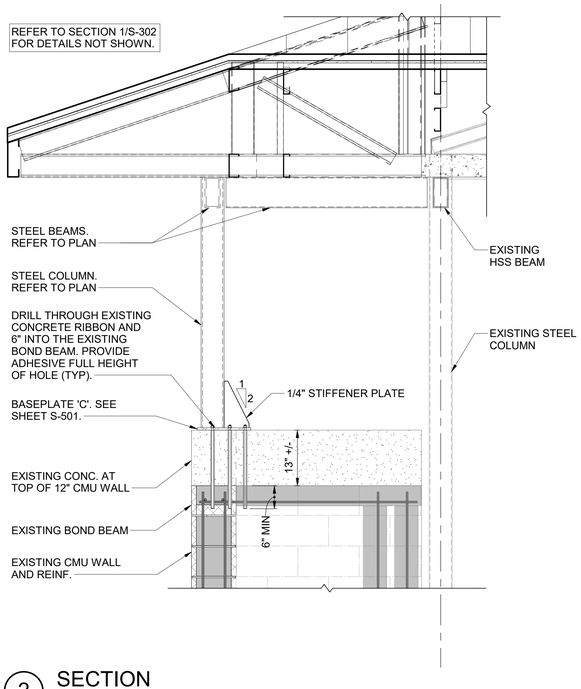
ISSUE DATE:08 SEPT 2023

| | | |
|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28469 | S-301 |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY J. LI | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND — MARINE CORPS AIR STATION BEAUFORT, S.C. |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY J. LI | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR B. MINOR | DETAILS AND SECTIONS |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. |
| | PUBLIC WORKS OFFICER | HQMC PROJ. NO.:; CONSTR. CONTR. NO.:N40085-18-D-0616; SCALE: As indicated; MACC.; T.O. #; SHEET 15 OF 53 |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |




2 / S-401 — 1 **SECTION**
SCALE: 3/4" = 1'-0"




2 / S-401 — 2 **SECTION**
SCALE: 3/4" = 1'-0"

12" 0 1' 2'

**SCALE: 3/4"=1'-0"**

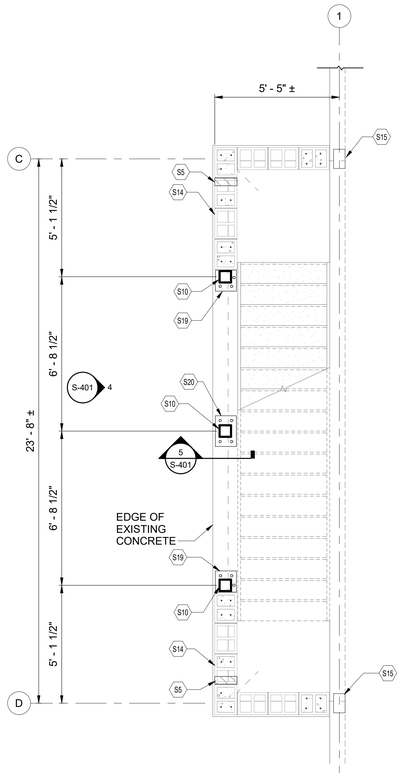
**GRAPHIC SCALE**

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

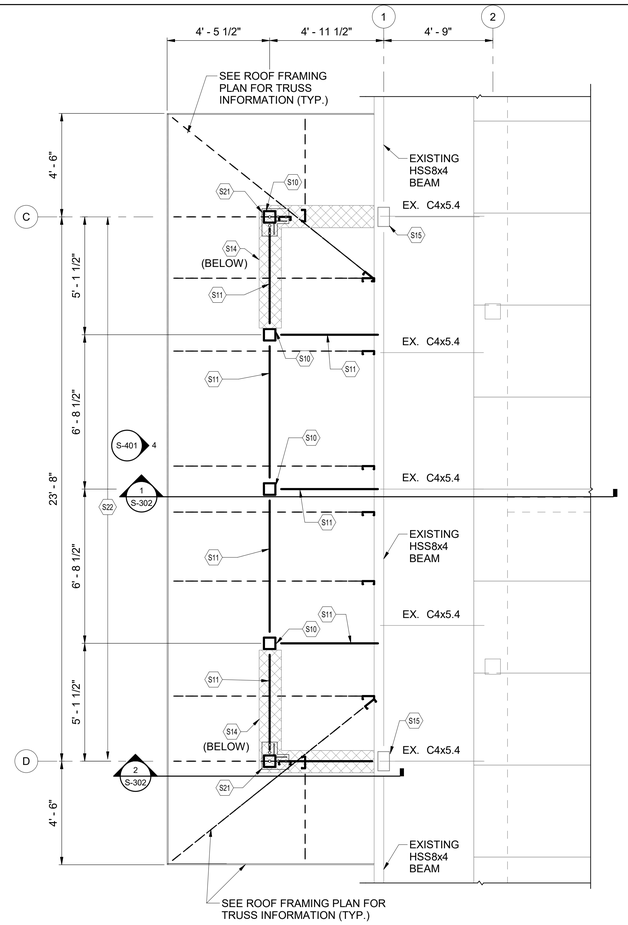
ISSUE DATE:08 SEPT 2023

04-OCT-2023

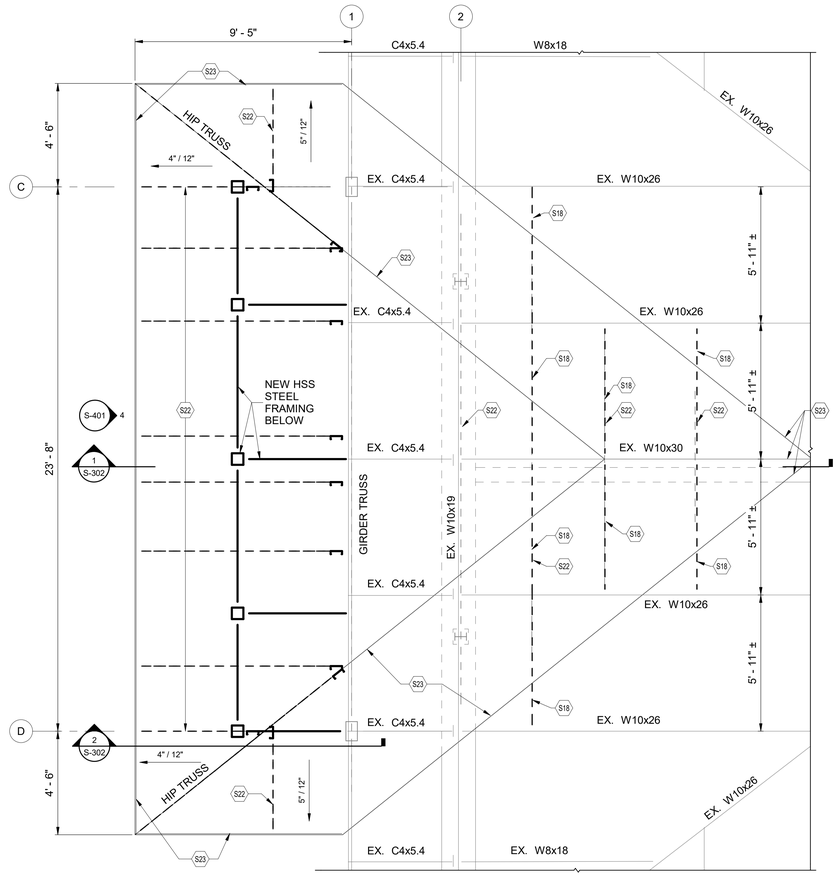
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28470 | CEMS | S-302 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>E. SHIPMAN | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>E. SHIPMAN | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>B. MINOR | SECTIONS | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: 3/4" = 1'-0" | MACC. T.O. # SHEET 16 OF 53 |



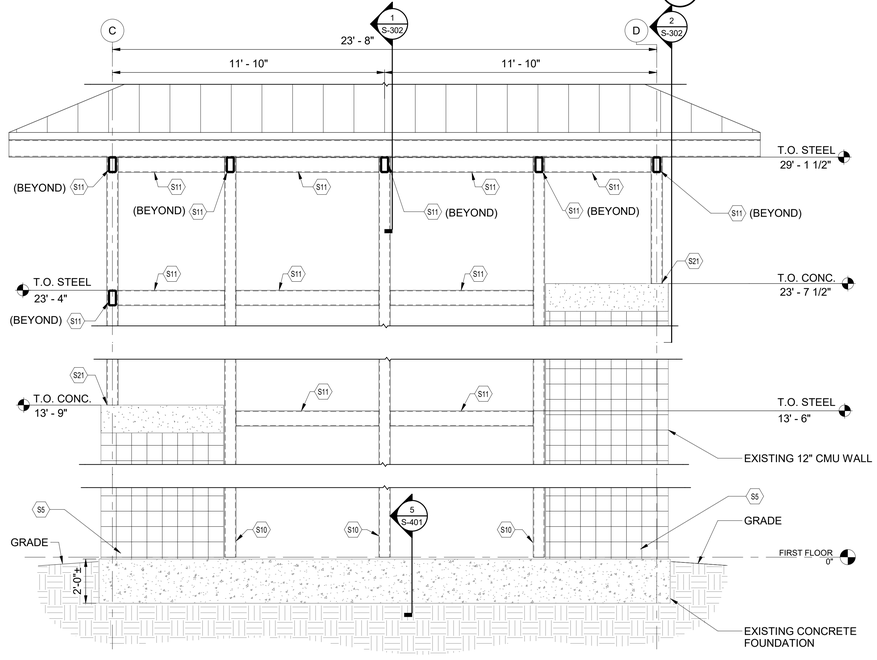

3 S-101 — 1 FOUNDATION PLAN
SCALE: 3/8" = 1'-0"



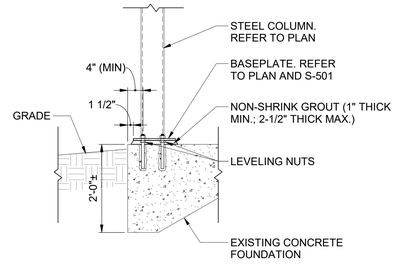

1 S-102 — 2 CEILING FRAMING PLAN
SCALE: 3/8" = 1'-0"




2 S-102 — 3 ROOF FRAMING PLAN
SCALE: 3/8" = 1'-0"




1 S-401 — 4 ELEVATION
SCALE: 3/8" = 1'-0"




1 S-401 — 5 SECTION
SCALE: 3/4" = 1'-0"

PLAN NORTH

12" 0 1' 2'
SCALE: 3/4"=1'-0"

12" 0 1' 2' 4'
SCALE: 3/8"=1'-0"

GRAPHIC SCALES

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. ALL EXISTING CONDITIONS PER ORIGINAL DOCUMENTS AND NON-DESTRUCTIVE OBSERVATION.
2. COORDINATE DEMOLITION WITH ARCHITECTURAL DRAWINGS.
3. DO NOT CUT ANY EXISTING STRUCTURAL MEMBERS UNLESS DIRECTLY AUTHORIZED BY ENGINEER OF RECORD.
4. ALL HSS TO HSS STEEL CONNECTIONS SHALL BE CONT. WELDED U.N.O. (TYP.)
5. SEE SHEET S-101 FOR FOUNDATION PLAN.
6. SEE SHEET S-102 FOR CEILING AND ROOF FRAMING PLAN.
7. PROVIDE INTERMEDIATE LIGHT GAUGE MEMBERS SUCH THAT THE NEW DECK SPANS 48" MAX.

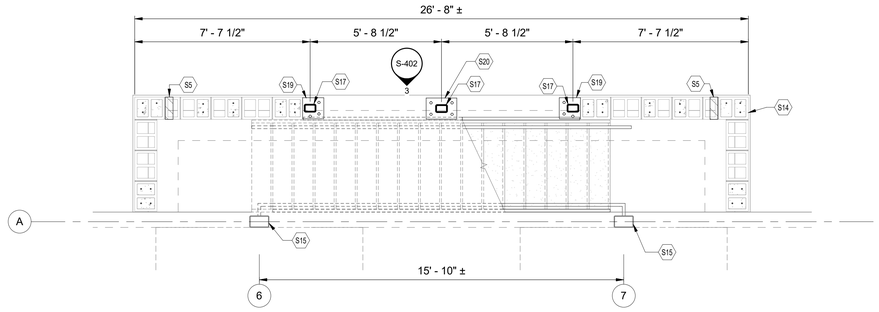
## SHEET KEYNOTES

- S5 4"+/- x 4"+/- OPENINGS IN EXISTING CMU WALLS FOR DRAINAGE. INSTALL THROUGH EXISTING UNGROUTED CELLS (SEE S-401 AND S-402).
- S10 STEEL COLUMN HSS6x6x3/8.
- S11 STEEL BEAM HSS8x4x1/4.
- S14 EXISTING 12" CMU WALL.
- S15 EXISTING STEEL COLUMN HSS10x6x1/4.
- S18 LIGHT GAUGE STEEL MEMBER BELOW.
- S19 BASEPLATE TYPE 'A1' OR 'A2'. REFER TO TYPICAL DETAIL ON S-501.
- S20 BASEPLATE TYPE 'B'. REFER TO TYPICAL DETAIL ON S-501.
- S21 ANCHOR NEW COLUMN TO TOP OF EXISTING CONCRETE PER BASEPLATE TYPE 'C' PER TYPICAL DETAIL ON S-501.
- S22 PRE-ENGINEERED LIGHT GAUGE STEEL TRUSSES @ 48" O.C. U.N.O. TRUSS TO SUPPORTING STRUCTURE CONNECTIONS SHALL BE DESIGNED BY TRUSS MANUFACTURER.
- S23 LIGHT GAUGE STEEL EAVE, HIP, VALLEY, AND RIDGE MEMBERS AND CONNECTIONS BY TRUSS MANUFACTURER.

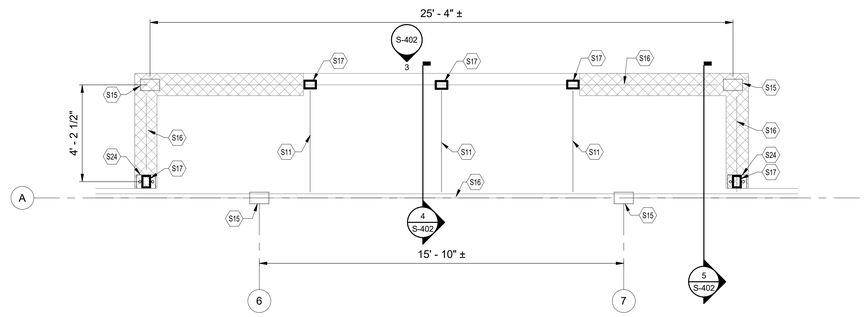
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

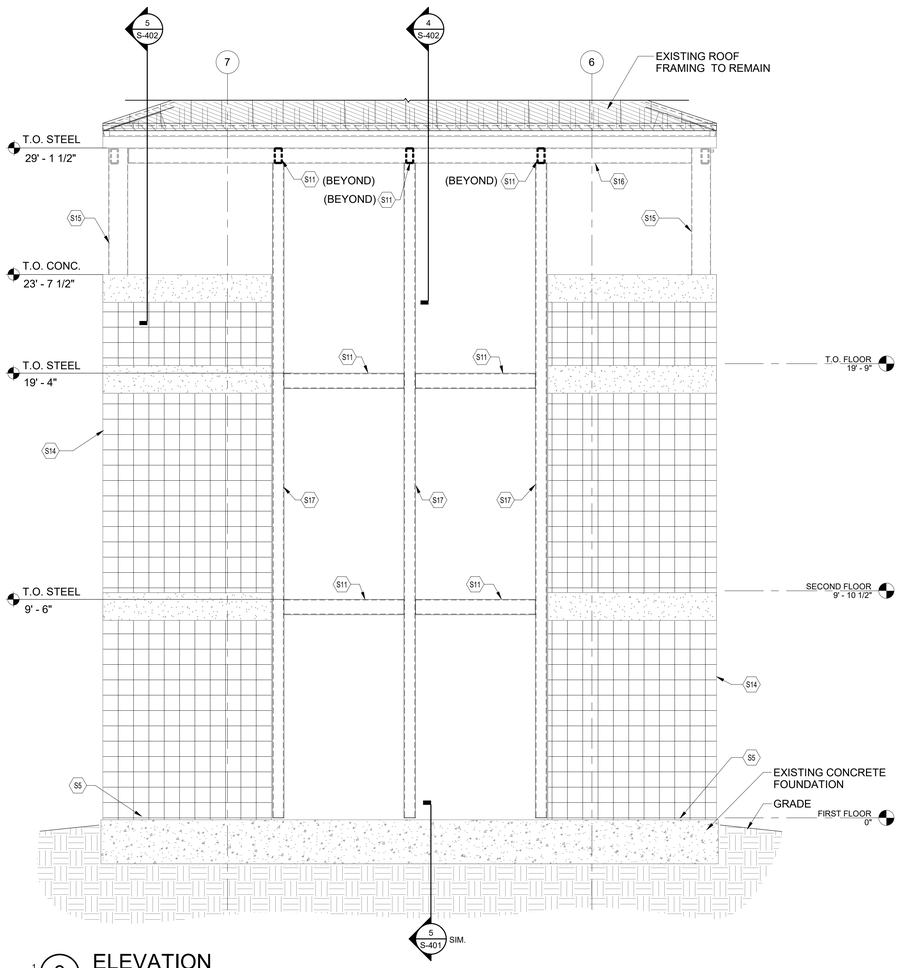
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28471 | CEMS | S-401 |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY E. SHIPMAN | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND — MARINE CORPS AIR STATION — BEAUFORT, S.C. | |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY E. SHIPMAN | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR B. MINOR | ENLARGED STAIR PLAN AND SECTIONS | |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F — CODE IDENT. NO. — N A V F A C DRAWING NO. | |
| | PUBLIC WORKS OFFICER | HQMC PROJ. NO.: — CONSTR. CONTR. NO.:N40085-18-D-0616 | |
| | | SCALE: As indicated — MACC. — T.O. # — SHEET 17 OF 53 | |



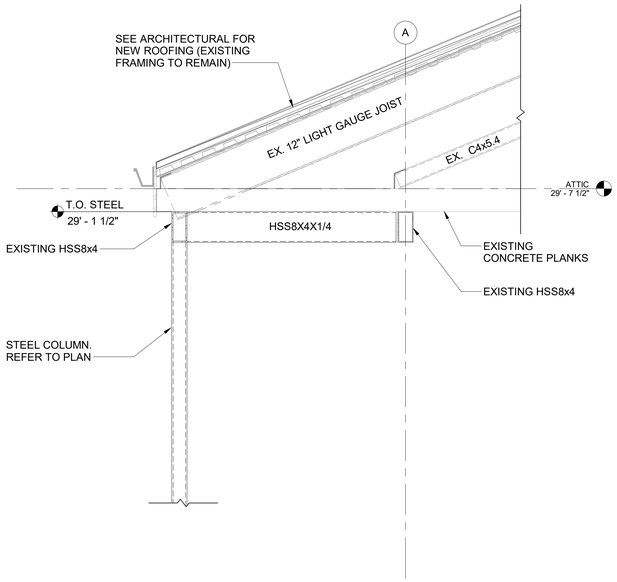

3 S-101 **1** **FOUNDATION PLAN**
SCALE: 3/8" = 1'-0"



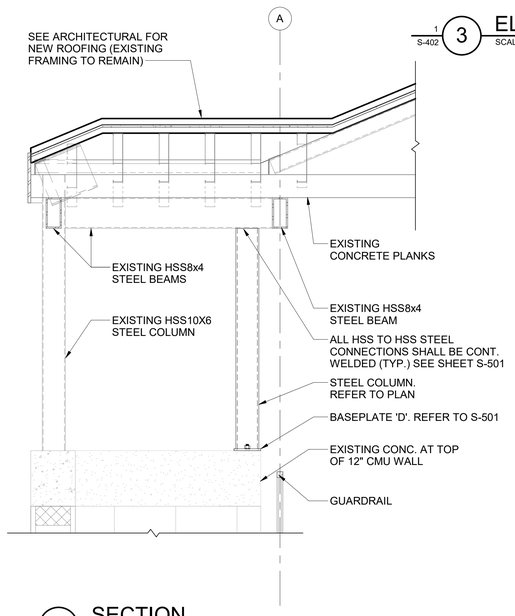

1 S-102 **2** **CEILING FRAMING PLAN**
SCALE: 3/8" = 1'-0"




1 S-402 **3** **ELEVATION**
SCALE: 3/8" = 1'-0"




2 S-402 **4** **SECTION**
SCALE: 3/4" = 1'-0"




2 S-402 **5** **SECTION**
SCALE: 3/4" = 1'-0"

PLAN NORTH

12" 0 1' 2'
**SCALE: 3/4"=1'-0"**

12" 0 1' 2' 4'
**SCALE: 3/8"=1'-0"**

**GRAPHIC SCALES**

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. ALL EXISTING CONDITIONS PER ORIGINAL DOCUMENTS AND NON-DESTRUCTIVE OBSERVATION.
2. COORDINATE DEMOLITION WITH ARCHITECTURAL DRAWINGS.
3. DO NOT CUT ANY EXISTING STRUCTURAL MEMBERS UNLESS DIRECTLY AUTHORIZED BY ENGINEER OF RECORD.
4. ALL HSS TO HSS STEEL CONNECTIONS SHALL BE CONT. WELDED U.N.O. (TYP.)
5. SEE SHEET S-101 FOR FOUNDATION PLAN AND SHEET S-102 FOR CEILING FRAMING PLAN.

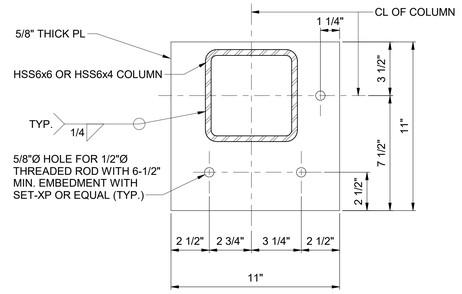
## SHEET KEYNOTES

S5 4"+/- x 4"+/- OPENINGS IN EXISTING CMU WALLS FOR DRAINAGE. INSTALL THROUGH EXISTING UNGROUTED CELLS (SEE S-401 AND S-402).
S11 STEEL BEAM HSS8x4x1/4.
S14 EXISTING 12" CMU WALL.
S15 EXISTING STEEL COLUMN HSS10x6x1/4.
S16 EXISTING HSS STEEL BEAM.
S17 STEEL COLUMN HSS6x4x3/8.
S19 BASEPLATE TYPE 'A1' OR 'A2'. REFER TO TYPICAL DETAIL ON S-501.
S20 BASEPLATE TYPE 'B'. REFER TO TYPICAL DETAIL ON S-501.
S24 BASEPLATE TYPE 'D'. REFER TO TYPICAL DETAIL ON S-501.

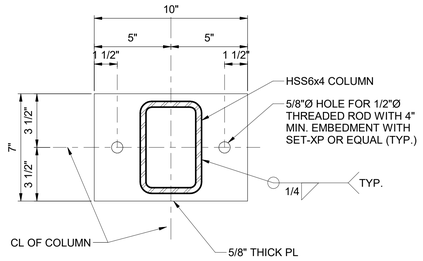
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

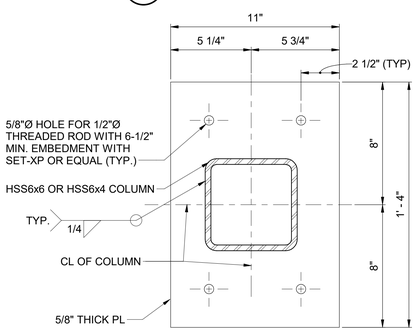
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28472 | CEMS | S-402 |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY E. SHIPMAN | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND — MARINE CORPS AIR STATION BEAUFORT, S.C. | |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY E. SHIPMAN | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR B. MINOR | ENLARGED STAIR PLAN AND SECTIONS | |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: As indicated; MACC.; T.O. #; SHEET 18 OF 53 | |



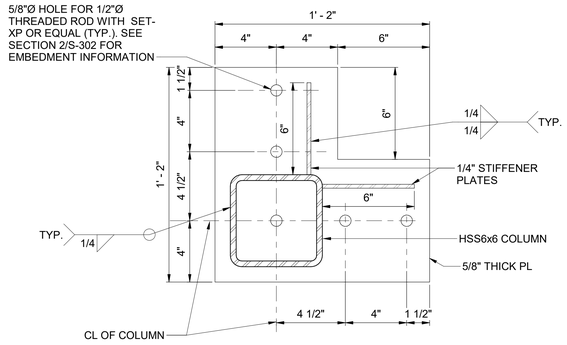

## 1 BASEPLATE TYPE 'A1'
SCALE: NOT TO SCALE



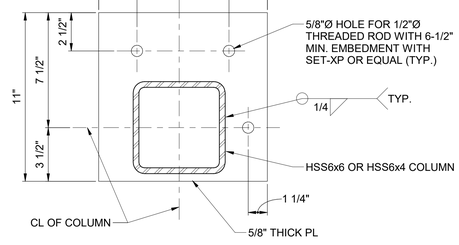

## 2 BASEPLATE TYPE 'D'
SCALE: NOT TO SCALE



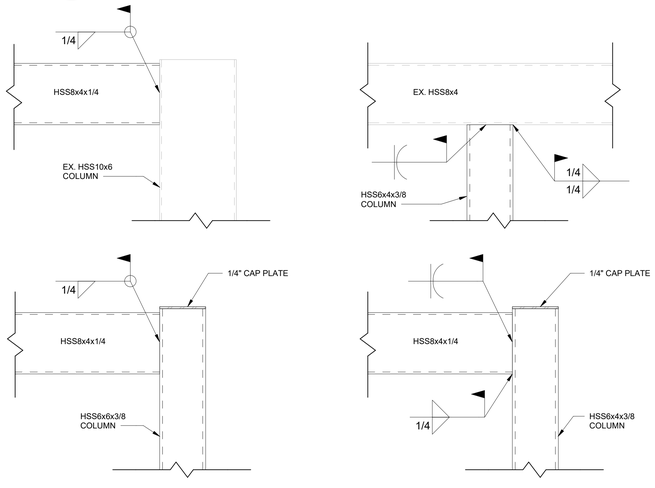

## 3 BASEPLATE TYPE 'B'
SCALE: NOT TO SCALE



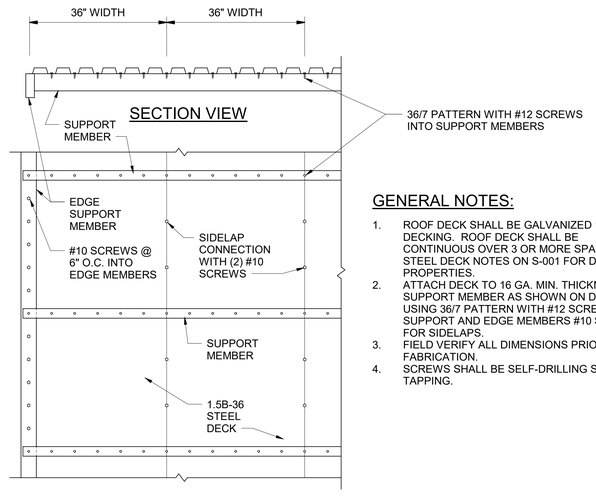

## 4 BASEPLATE TYPE 'C'
SCALE: NOT TO SCALE




## 5 BASEPLATE TYPE 'A2'
SCALE: NOT TO SCALE




## 6 BEAM TO COLUMN CONNECTIONS
SCALE: NOT TO SCALE




## 7 METAL DECK FASTENING DETAIL
SCALE: NOT TO SCALE

## GENERAL NOTES:

1. ROOF DECK SHALL BE GALVANIZED 1.5B-36 DECKING. ROOF DECK SHALL BE CONTINUOUS OVER 3 OR MORE SPANS. SEE STEEL DECK NOTES ON S-001 FOR DECK PROPERTIES.
2. ATTACH DECK TO 16 GA. MIN. THICKNESS SUPPORT MEMBER AS SHOWN ON DETAIL USING 36/7 PATTERN WITH #12 SCREWS IN SUPPORT AND EDGE MEMBERS #10 SCREWS FOR SIDELAPS.
3. FIELD VERIFY ALL DIMENSIONS PRIOR TO FABRICATION.
4. SCREWS SHALL BE SELF-DRILLING SELF-TAPPING.

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

04-OCT-2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28473 | CEMS www.cems-se.com \| PM: Dave Tenaski \| Project: 19152T \| 843.972.3637 | S-501 |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY E. SHIPMAN | DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND MARINE CORPS AIR STATION BEAUFORT, S.C. | |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY E. SHIPMAN | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR B. MINOR | TYPICAL DETAILS | |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F \| CODE IDENT. NO. | N A V F A C DRAWING NO. |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: As indicated \| MACC. \| T.O. # | SHEET 19 OF 53 |

# ARCHITECTURAL ABBREVIATIONS:

| A | |
|---|---|
| AFC | ABOVE FINISHED COUNTER |
| AFF | ABOVE FINISHED FLOOR |
| AFG | ABOVE FINISHED GRADE |
| ASC | ABOVE SUSPENDED CEILING |
| ACC | ACCESSIBLE |
| ACT | ACOUSTICAL CEILING TILE |
| AWT | ACOUSTICAL WALL TREATMENT |
| ADMIN | ADMINISTRATION |
| A/C | AIR CONDITION |
| ALT | ALTERNATE |
| ALUM | ALUMINUM |
| ADA | AMERICANS WITH DISABILITIES ACT |
| APPD | APPROVED |
| APPROX | APPROXIMATE |
| ARCH | ARCHITECT |
| ASI | ARCHITECT'S SUPPLEMENTAL INSTRUCTION |
| A/E | ARCHITECT/ENGINEER |
| AW | ARCHITECTURAL WOODWORK |
| AWI | ARCHITECTURAL WOODWORKING INSTITUTE |
| AR | AS REQUIRED |
| AHJ | AUTHORITY HAVING JURISDICTION |
| ADC | AUTOMATIC DOOR CLOSER |
| ADS | AUTOMATIC DOOR SEAL |
| AVG | AVERAGE |

| B | |
|---|---|
| BKG | BACKING |
| BM | BEAM |
| BLW CLG | BELOW CEILING |
| BFF | BELOW FINISH FLOOR |
| BITUM | BITUMINOUS |
| BLKT | BLANKET |
| BD | BOARD |
| BF | BOTH FACES |
| BS | BOTH SIDES |
| BOT | BOTTOM |
| BOS | BOTTOM OF STEEL |
| BHMA | BUILDERS HARDWARE MANUFACTURERS ASSOCIATION |
| BLDG | BUILDING |
| BU | BUILT-UP |
| BUR | BUILT-UP ROOFING |
| BLKHD | BULKHEAD |
| BN | BULLNOSE |

| C | |
|---|---|
| CAB | CABINET |
| CPT | CARPET |
| C&P | CARPET AND PAD |
| CLG | CEILING |
| CTR | CENTER |
| CL | CENTER LINE |
| CM | CENTER MATCHED |
| C TO C | CENTER TO CENTER |
| CB | CERAMIC BASE |
| CT | CERAMIC TILE |
| CTB | CERAMIC TILE BASE |
| CTF | CERAMIC TILE FLOOR |
| CH BD | CHALKBOARD |
| C | CHANNEL |
| CO | CLEANOUT |
| CLR | CLEAR |
| COL | COLUMN |
| CONC | CONCRETE |
| CMU | CONCRETE MASONRY UNIT |
| CONF | CONFERENCE |
| CD | CONSTRUCTION DOCUMENTS |
| CM | CONSTRUCTION MANAGEMENT |
| CONT | CONTINUE |
| CONTR | CONTRACTOR |
| CFE | CONTRACTOR FURNISHED EQUIPMENT |
| CF/CI | CONTRACTOR FURNISHED/CONTRACTOR INSTALLED |
| CF/OI | CONTRACTOR FURNISHED/OWNER INSTALLED |
| CJ | CONTROL JOINT |
| COORD | COORDINATE |
| CG | CORNER GUARD |
| CORR | CORRIDOR |
| CNTR | COUNTER |

| D | |
|---|---|
| DEG | DEGREE |
| DEMO | DEMOLITION |
| DEPT | DEPARTMENT |
| DESCR | DESCRIPTION |
| D-B | DESIGN-BUILD |
| DET | DETAIL |
| DIAG | DIAGONAL |
| DIA | DIAMETER |
| DIFF | DIFFERENCE |
| DIM | DIMENSION |
| DW | DISHWASHER |
| DISP | DISPENSER |
| DR CL | DOOR CLOSER |
| DR FR | DOOR FRAME |
| DRH | DOOR HOLDER |
| DRLV | DOOR LOUVER |
| DR OPNG | DOOR OPENING |
| DRST | DOOR STOP |
| DH | DOUBLE HUNG (DOOR, WINDOW) |
| DHW | DOUBLE HUNG WINDOWS |
| DS | DOWNSPOUT |
| DWG | DRAWING |
| DF | DRINKING FOUNTAIN |
| DT DR | DUTCH DOOR |

| E | |
|---|---|
| EA | EACH |
| EE | EACH END |
| EF | EACH FACE |
| E | EAST |
| ELAST | ELASTOMERIC |
| ELEC | ELECTRIC |
| EHD | ELECTRIC HAND DRYER |
| EWC | ELECTRIC WATER COOLER |
| ELEV | ELEVATOR |
| ENGR | ENGINEER |
| ENTR | ENTRANCE |
| EQ | EQUAL |
| EQL SP | EQUALLY SPACED |
| EQUIP | EQUIPMENT |
| EQUIV | EQUIVALENT |
| ESCAL | ESCALATOR |
| ETC | ET CETERA |
| EVAC | EVACUATE |
| EXH | EXHAUST |
| EXH FN | EXHAUST FAN |
| EXH HD | EXHAUST HOOD |
| EXHV | EXHAUST VENT |
| EXST | EXISTING |
| ERD | EXISTING ROOF DRAIN |
| ETR | EXISTING TO REMAIN |
| EM | EXPANDED METAL |
| EPS | EXPANDED POLYSTYRENE BOARD (INSULATION) |
| EXP | EXPANSION |
| EJ | EXPANSION JOINT |
| EXT | EXTERIOR |
| EXT GR | EXTERIOR GRADE |
| EGSB | EXTERIOR GYPSUM SHEATHING BOARD |
| EIFS | EXTERIOR INSULATION AND FINISH SYSTEM |
| EWS | EYE WASH STATION |

| F | |
|---|---|
| FAB | FABRIC |
| FC BRK | FACE BRICK |
| FOC | FACE OF CONCRETE |
| FOF | FACE OF FINISH |
| FOM | FACE OF MASONRY |
| FOS | FACE OF STUD |
| FOW | FACE OF WALL |
| FTD | FACIAL TISSUE DISPENSER |
| FCTY | FACTORY |
| FAS | FASCIA |
| FAS BD | FASCIA BOARD |
| FED | FEDERAL |
| FT | FEET |
| F | FEMALE |
| FGL | FIBERGLASS |
| FRP | FIBERGLASS REINFORCED PLASTIC |
| FC | FILE CABINET |
| FIN | FINISH |
| FIN BS | FINISH BOTH SIDES |
| FF | FINISH FACE |
| FIN FLR | FINISH FLOOR |
| FIN GR | FINISH GRADE |
| FIN WD | FINISH WOOD |
| FO | FINISHED OPENING |
| FA | FIRE ALARM |
| FDR | FIRE DOOR |
| FE | FIRE EXTINGUISHER |
| FEC | FIRE EXTINGUISHER CABINET |
| FHC | FIRE HOSE CABINET |
| FH | FIRE HYDRANT |
| FRA | FIRE RATED ASSEMBLY |
| FR | FIRE RESISTANT |
| FIXT | FIXTURE |
| FLASH | FLASHING |
| FD | FLOOR DRAIN |
| FLG | FLOORING |
| FLOUR | FLOURESCENT |
| FLMT | FLUSH MOUNT |
| FLDG | FOLDING |
| FTG | FOOTING |
| FDTN | FOUNDATION |
| FFA | FROM FLOOR ABOVE |
| FFB | FROM FLOOR BELOW |
| FURN | FURNITURE |
| FF&E | FURNITURE, FIXTURE, AND EQUIPMENT |

| G | |
|---|---|
| GA | GAGE |
| GALV | GALVANIZED |
| GL BLK | GLASS BLOCK |
| GWT | GLAZED WALL TILE |
| GLZ | GLAZING |
| GOVT | GOVERNMENT |
| GF/CI | GOVERNMENT FURNISHED/CONTRACTOR INSTALLED |
| GF/GI | GOVERNMENT FURNISHED/GOVERNMENT INSTALLED |
| GB | GRAB BAR |
| GRAN | GRANITE |
| GL | GRIDLINE |
| GR FL | GROUND FLOOR |
| GP | GROUP |
| GT | GROUT |
| GDR | GUARD RAIL |
| GUT | GUTTER |
| GYP | GYPSUM |
| GYP BD | GYPSUM BOARD |
| GYP PLAS | GYPSUM PLASTER |
| GPC | GYPSUM PLASTER CEILING |
| GSB | GYPSUM SHEATHING BOARD |

| H | |
|---|---|
| HD | HAND DRYER |
| HS | HAND SINK |
| HC | HANDICAP |
| HCP | HANDICAPPED |
| HNDRL | HANDRAIL |
| HDW | HARDWARE |
| H | HATCH (ROOF) |
| HVAC | HEATING, VENTILATING, AND AIR CONDITIONING |
| HT | HEIGHT |
| H | HIGH |
| HO | HOLD OPEN |
| HC | HOLLOW CORE |
| HCWD | HOLLOW CORE WOOD DOOR |
| HM | HOLLOW METAL |
| HMD | HOLLOW METAL DOOR |
| HMDF | HOLLOW METAL DOOR AND FRAME |
| HMF | HOLLOW METAL FRAME |
| HORIZ | HORIZONTAL |
| HB | HOSE BIBB |

| I | |
|---|---|
| IFS | INISDE FACE OF STUD |
| IF | INISIDE FACE |
| ID | INSIDE DIAMETER |
| ILTG | INSULATED LAMINATED TEMPERED GLASS |
| IMP | INSULATED METAL PANEL |
| INSUL | INSULATION |
| INS | INSURANCE |
| INT | INTERIOR |

| J | |
|---|---|
| JAN | JANITOR |
| JAN CLO | JANITOR CLOSET |
| JS | JANITOR'S SINK |

| K | |
|---|---|
| KPL | KICKPLATE |
| KOP | KNOCK OUT PANEL |
| KD | KNOCKED DOWN |
| KO | KNOCKOUT |

| L | |
|---|---|
| L | ANGLE |
| LAB | LABORATORY |
| LAM | LAMINATE |
| LAM GL | LAMINATED GLASS |
| LDG | LANDING |
| LAT | LATITUDE |
| LAU | LAUNDRY |
| LC | LAUNDRY CHUTE |
| LAV | LAVATORY |
| LH | LEFT HAND |
| LHR | LEFT HAND REVERSE |
| LT | LIGHT |
| LIN | LINEAR |
| LF | LINEAR FEET (FOOT) |
| LD BRG | LOAD-BEARING |
| LKR | LOCKER |
| LVR | LOUVER |
| LVDR | LOUVER DOOR |
| LBR | LUMBER |
| LB | POUND |

| M | |
|---|---|
| MGT | MANAGEMENT |
| MFR | MANUFACTURER |
| MRB | MARBLE BASE |
| MRT | MARBLE FHRESHOLD |
| MRF | MARBLE FLOOR |
| MKR | MARKER |
| MO | MASONRY OPENING |
| ME | MATCH EXISTING |
| MATL | MATERIAL |
| MAX | MAXIMUM |
| MEL | MELAMINE |
| MEMB | MEMBRANE |
| MWP | MEMBRANE WATERPROOFING |
| MTL | METAL |
| MD | METAL DECK |
| MTLD | METAL DOOR |
| MTLF | METAL FLASHING |
| MTLP | METAL PARTITION |
| MTLR | METAL ROOF |
| MT | METAL THRESHOLD |
| MEZZ | MEZZANINE |
| MW | MICROWAVE |
| MICRO | MICROWAVE OVEN |
| MID | MIDDLE |
| MLWK | MILLWORK |
| MIN | MINIMUM |
| MISC | MISCELLANEOUS |
| MSB | MOB SERVICE BASIN |
| MR | MOISTURE RESISTANT |
| MOPR | MOP RACK |
| MS | MOP SINK |
| MT | MOUNT |
| MTD | MOUNTED |
| MTG | MOUNTING |
| MULL | MULLION |
| MULT | MULTIPLE |

| N | |
|---|---|
| NFPA | NATIONAL FIRE PROTECTION ASSOCIATION |
| NR | NOISE REDUCTION |
| NRC | NOISE REDUCTION COEFFICIENT |
| NOM | NOMINAL |
| N | NORTH |
| NA | NOT APPLICABLE |
| NIC | NOT IN CONTRACT |
| NTS | NOT TO SCALE |

| O | |
|---|---|
| OC | ON CENTER |
| OPP | OPPOSITE |
| OD | OUTSIDE DIAMETER |
| OF | OUTSIDE FACE |
| OFS | OUTSIDE FACE OF STUDS |
| OA | OVERALL |
| OVFL | OVERFLOW |
| OFD | OVERFLOW DRAIN |
| ORD | OVERFLOW ROOF DRAIN |
| OH | OVERHANG |
| OH DR | OVERHEAD (COILING) DOOR |
| OF/CI | OWNER FURNISHED/CONTRACTOR INSTALLED |
| OF/OI | OWNER FURNISHED/OWNER INSTALLED |

| P | |
|---|---|
| PT | PAINT |
| PR | PAIR |
| PTD | PAPER TOWEL DISPENSER |
| PTDR | PAPER TOWEL DISPENSER AND RECEPTACLE |
| PTR | PAPER TOWEL RECEPTACLE |
| PAR | PARAPET |
| PRKG | PARKING |
| PTN | PARTITION |
| PAT | PATTERN |
| PH | PHASE |
| PHOTO | PHOTOGRAPH |
| PLAM | PLASTIC LAMINATE |
| PORC | PORCELAIN |
| PSF | POUNDS PER SQUARE FOOT |
| PSI | POUNDS PER SQUARE INCH |
| PCC | PRECAST CONCRETE |
| PREFAB | PREFABRICATE |
| PREFIN | PREFINISH |
| PRELIM | PRELIMINARY |
| PEJ | PREMOLDED EXPANSION JOINT |
| PREV | PREVIOUS |
| PL | PROPERTY LINE |

| Q | |
|---|---|
| QTY | QUANTITY |
| QT | QUARRY TILE |
| QTB | QUARRY TILE BASE |
| QTF | QUARRY TILE FLOOR |

| R | |
|---|---|
| R | RADIUS |
| RLG | RAILING |
| REC | RECESSED |
| RCP | REFLECTED CEILING PLAN |
| REF | REFRIGERATOR |
| REINF | REINFORCE |
| RC | REINFORCED CONCRETE |
| REBAR | REINFORCING STEEL BARS |
| REQD | REQUIRED |
| RESIL | RESILIENT |
| RF | RESILIENT FLOORING |
| REST | RESTROOM |
| REV | REVISION |
| RH | RIGHT HAND |
| RHR | RIGHT HAND REVERSE |
| RD | ROOF DRAIN |
| RV | ROOF VENT |
| RM | ROOM |
| RO | ROUGH OPENING |
| RB | RUBBER BASE |
| RTF | RUBBER TILE FLOOR |

| S | |
|---|---|
| SND | SANITARY NAPKIN DISPENSER |
| SCHED | SCHEDULE |
| SLNT | SEALANT |
| SS | SERVICE SINK |
| SM | SHEET METAL |
| SIM | SIMILAR |
| SA | SINGLE ACTING (DOOR) |
| SH | SINGLE HUNG (WINDOW) |
| SD | SOAP DISPENSER |
| SC | SOLID CORE |
| SCWD | SOLID CORE WOOD DOOR |
| STC | SOUND TRANSMISSION CLASS |
| S | SOUTH |
| SPEC | SPECIFICATION |
| SQ | SQUARE |
| SST | STAINLESS STEEL |
| STD | STANDARD |
| SAT | SUSPENDED ACOUSTICAL TILE |
| SATC | SUSPENDED ACOUSTICAL TILE CEILING |
| SYM | SYMBOL |
| SYS | SYSTEM |

| T | |
|---|---|
| TOC | TABLE OF CONTENT |
| TK BD | TACKBOARD |
| TEL | TELEPHONE |
| TV | TELEVISION |
| TMPD GL | TEMPERED GLASS |
| TEMP | TEMPORARY |
| TER | TERRAZZO |
| THK | THICKNESS |
| TPD | TOILET PAPER DISPENSER |
| TPH | TOILET PAPER HOLDER |
| TO | TOP OF |
| TOB | TOP OF BEAM |
| TOF | TOP OF FOOTING |
| TO FDN | TOP OF FOUNDATION |
| TOJ | TOP OF JOIST |
| TOM | TOP OF MASONRY |
| TOS | TOP OF STEEL |
| TOW | TOP OF WALL |
| TB | TOWEL BAR |
| TR | TOWEL RACK |
| TSH | TOWEL SHELF |
| TRANS | TRANSOM |
| TRANS | TRANSPARENT |
| TD | TRENCH DRAIN |
| TYP | TYPICAL |

| U | |
|---|---|
| UCD | UNDERCUT DOOR |
| UL | UNDERWRITERS LABORATORIES |
| UNFIN | UNFINISH |
| UNO | UNLESS NOTED OTHERWISE |
| UR | URINAL |

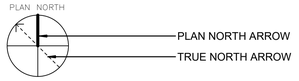
| V | |
|---|---|
| VAN | VANITY |
| VR | VAPOR RETARDER |
| VAR | VARIES |
| VTR | VENT THROUGH ROOF |
| VIF | VERIFY IN FIELD |
| VERT | VERTICAL |
| VEST | VESTIBULE |
| VB | VINYL BASE |
| VCT | VINYL COMPOSITION TILE |

| W | |
|---|---|
| WSCT | WAINSCOT |
| WH | WALL HUNG |
| WC | WATER CLOSET |
| WR | WEATHER RESISTANCE |
| WSL | WEATHER SEAL |
| WS | WEATHERSTRIP |
| WM | WIRE MESH |
| W/ | WITH |
| W/O | WITHOUT |
| WD | WOOD DOOR |

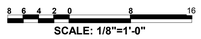
| Y | |
|---|---|
| YD | YARD |

# GRAPHIC SYMBOLS LEGEND:

## NORTH ARROW

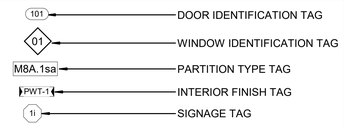



## DRAWING SCALE INDICATOR

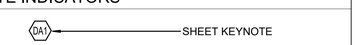



GRAPHIC SCALE

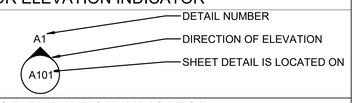
## DOOR, WINDOW AND WALL INDICATORS



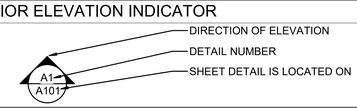

## KEYNOTE INDICATORS



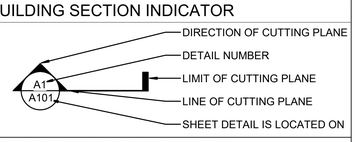

## INTERIOR ELEVATION INDICATOR



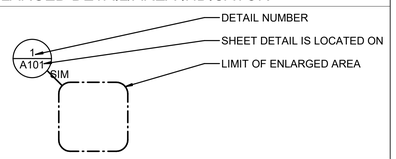

## EXTERIOR ELEVATION INDICATOR



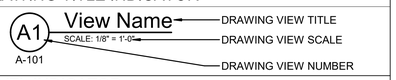

## WALL/BUILDING SECTION INDICATOR




## ENLARGED DETAIL/AREA INDICATOR

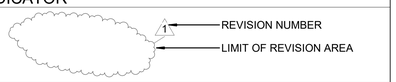



## DRAWING TITLE INDICATOR




## LEVEL INDICATOR

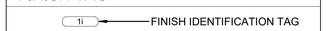
## REVISION CLOUD AND NUMBER INDICATOR




## ROOM TAG

## FURNITURE TAG

## FINISH TAG




| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |

# GENERAL NOTES

1. ABBREVIATIONS, GRAPHIC SYMBOLS LEGEND, LINEWORK LEGEND, FIRE RATED LEGEND, AND MATERIALS ON THIS SHEET APPLY TO ALL ARCHITECTURAL AND INTERIOR SHEETS.

# MATERIALS LEGEND:

| MATERIAL | NEW | | EXISTING TO REMAIN | | DEMOLITION |
|---|---|---|---|---|---|
| | IN SECTION | IN PLAN AND ELEVATION | IN SECTION | IN PLAN AND ELEVATION | IN PLAN AND ELEVATION |
| EARTH | | | | | |
| CONCRETE / GROUT | | | | | |
| FACE BRICK | | | | | |
| MASONRY | | | | | |
| WOOD | | | | | |
| PLYWOOD | | | | | |
| STEEL | | | | | |
| RIGID INSULATION | | | | | |
| BATT INSULATION | | | | | |
| SAND / E.I.F.S. / PLASTER / STUCCO / GYPSUM BOARD | | | | | |
| GRAVEL / STONE FILL | | | | | |
| ACT CEILING | | | | | |
| METAL PANEL CEILING | | | | | |

# LINEWORK LEGEND:

**AS INDICATED ON NEW WORK DRAWINGS:**

- NEW CONSTRUCTION (BLACK SOLID, DASHED, AND CENTERLINE LINEWORK
- EXISTING CONSTRUCTION TO REMAIN (GRAY SHADED SOLID, DASHED, AND CENTERLINE LINEWORK)

**AS INDICATED ON DEMOLITION DRAWINGS:**

- EXISTING CONSTRUCTION TO REMAIN (GRAY SHADED SOLID, DASHED AND CENTERLINE LINEWORK)
- DEMOLITION WORK (BLACK DASHED LINEWORK)

# FIRE RATED LEGEND:

- 30 MINUTE FIRE RATED PATTER AND LINE APPLIED TO ALL APLICABLE WALLS
- 1 HOUR FIRE RATED PATTER AND LINE APPLIED TO ALL APLICABLE WALLS
- 2 HOUR FIRE RATED PATTER AND LINE APPLIED TO ALL APLICABLE WALLS

WISCONSIN
DEAN C. VON TERSCH
E-17303
CHARLESTON
SC
ARCHITECT
04-OCT-2023

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28474 | CEMS | A-001 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | NAVAL FACILITIES ENGINEERING COMMAND |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | ABBREVIATIONS, SYMBOLS AND LEGENDS | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE: F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| PUBLIC WORKS OFFICER | | HQMC PROJ. NO.:; CONSTR. CONTR. NO.: N40085-18-D-0616 | |
| | | SCALE: As indicated; MACC.; T.O. #; SHEET 20 OF 53 | |

## GENERAL NOTES:

1. THE DRAWINGS INDICATE THE GENERAL EXTENT OF THE WORK. THE DRAWINGS ARE NOT INTENDED TO INDICATE OR DESCRIBE ALL WORK REQUIRED FOR THE FULL PERFORMANCE AND COMPLETION OF THE REQUIREMENTS OF THE CONTRACT DOCUMENTS. THE DRAWINGS INDICATE THE GENERAL INTENT OF THE WORK REQUIRED.
2. THE ENUMERATION OF PARTICULAR ITEMS OF WORK IN ONE PORTION OF THE CONTRACT DOCUMENTS SHALL NOT BE CONSTRUED TO EXCLUDE OTHER ITEMS NECESSARY OR IMPLIED.
3. THE CONTRACTOR IS RESPONSIBLE FOR THE COORDINATION OF ALL PARTS OF THE WORK SO THAT NO WORK SHALL BE LEFT IN AN UNFINISHED OR INCOMPLETE CONDITION.
4. ALL WORK SHALL CONFORM TO APPLICABLE INDUSTRY AND MANUFACTURER'S PUBLISHED STANDARDS FOR QUALITY OF MATERIALS AND WORKMANSHIP. ANY CONFLICTING REQUIREMENTS OF THE SOURCES LISTED ABOVE SHALL BE BROUGHT TO THE CONTRACTING OFFICER'S ATTENTION PRIOR TO PROCEEDING WITH THE WORK.
5. THE CONTRACTOR SHALL PROTECT EXISTING, IN-PLACE, AND NEW WORK.
6. ALL WORK NOTED "N.I.C." OR "G.F.G.I." IS MEANT TO EXCLUDE THIS SCOPE OF WORK FROM THE CONSTRUCTION CONTRACT.
7. THE CONTRACTOR SHALL PAY FOR AND COORDINATE THE COLLECTION, SEPARATION, RECYCLE, REMOVAL AND LEGAL DISPOSAL OF MATERIALS AND RUBBISH.
8. THE CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND SHALL VERIFY ALL NEW AND EXISTING CONDITIONS SHOWN. THE CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER IN WRITING OF ANY DIFFERING CONDITIONS BEFORE COMMENCEMENT OF WORK. COMMENCEMENT OF WORK SHALL CONSTITUTE ACCEPTANCE OF ALL NEW AND EXISTING CONDITIONS.
9. DO NOT SCALE DRAWINGS. DIMENSIONS GOVERN. LARGE SCALE DETAILS GOVERN OVER SMALL SCALE DETAILS. THE CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER IN WRITING OF ANY DIFFERING CONDITIONS BEFORE COMMENCEMENT OF WORK.
10. UNLESS NOTED OTHERWISE, ALL NEW GYPSUM BOARD SURFACES ARE TO RECEIVE ONE PRIMER COAT AND TWO COATS OF PAINT.
11. ALL DIMENSIONS INDICATED ARE NOT TO VARY BY MORE THAN 1/8".
12. NFPA 241, STANDARD FOR SAFEGUARDING CONSTRUCTION, ALTERATION, AND DEMOLITION OPERATIONS SHALL BE APPLIED.
13. WALL AND CEILING ASSEMBLIES THAT ARE IDENTIFIED WITH A FIRE RESISTIVE RATING SHALL BE CONSTRUCTED AS DETAILED OR REFERENCED BY UL OR OTHER REFERENCE.
14. DIMENSIONS SHOWN TO STRUCTURAL BEAMS AND COLUMNS ARE TO THE CENTERLINE OF THE MEMBER UNLESS SPECIFICALLY NOTED OTHERWISE.
15. PROVIDE EXPANSION AND CONTROL JOINTS IN ALL WORK AS PER PRODUCT MANUFACTURER'S STANDARDS.
16. THE CONTRACTOR SHALL PERFORM ALL WORK IN ACCORDANCE WITH ALL APPLICABLE CODES, ORDINANCES AND REGULATORY AGENCIES AND SHALL OBTAIN NECESSARY BUILDING AND FIRE PERMITS FROM ALL AUTHORITIES HAVING JURISDICTION.
17. ALL DISSIMILAR METALS SHALL BE ISOLATED FROM EACH OTHER TO AVOID GALVANIC CORROSION.
18. NOTES ON ANY ONE SHEET ARE TO BE APPLIED TO RELATED DRAWINGS AND DETAILS ON OTHER SHEETS.
19. ALL FINISH AND FIRE RATING REQUIREMENTS INDICATED SHALL APPLY TO THE ENTIRE LENGTH AND HEIGHT OF THE WALL.
20. WHEN NON-DIMENSIONED PARTITIONS APPEAR IN CONJUNCTION WITH DOOR OPENINGS, THE DOOR WIDTH AND THE DOOR FRAME DETAILS DETERMINE THE LOCATION OF ADJACENT WALLS AND FRAMES.
21. DETAILS INDICATED FOR CONSTRUCTION ARE TO BE USED FOR ALL LOCATIONS OF SIMILAR CONSTRUCTION. WHERE SPECIFIC DIMENSIONS, DETAILS OR DESIGN INTENT CANNOT BE DETERMINED, CONSULT THE CONTRACTING OFFICER BEFORE PROCEEDING WITH THE WORK.
22. THE CONTRACTOR SHALL COORDINATE ALL PENETRATIONS WITH ALL TRADES.
23. PROVIDE ACCESS PANELS AS REQUIRED BY APPLICABLE CODES AND AS REQUIRED FOR MECHANICAL EQUIPMENT AND PLUMBING WORK. ALL ACCESS PANEL LOCATIONS SHALL BE REVIEWED WITH THE CONTRACTING OFFICER PRIOR TO PROCEEDING.
24. ALL PIPE DUCTS AND BUS DUCTS THAT PENETRATE ASSEMBLIES SHALL BE INSTALLED IN A MANNER THAT WILL PRESERVE THE MOISTURE RESISTANCE, FIRE RATING, AESTHETICS AND STRUCTURAL INTEGRITY, AND AESTHETICS OF THE ASSEMBLY.
25. CONTRACTOR SHALL PLAN THEIR WORK TO PROVIDE ADEQUATE PROTECTION FOR PERSONS AND PROPERTY AT ALL TIMES AND EXECUTE THE WORK IN SUCH A MANNER TO AVOID ANY HAZARD TO PERSONS AND PROPERTY AS NECESSARY.
26. CONTRACTOR SHALL CONTINUOUSLY COORDINATE PHASING AND CONSTRUCTION WITH THE CONTRACTING OFFICER.
27. ALL PENETRATIONS, GAPS IN CONSTRUCTION MATERIALS, MOVEMENT JOINTS, AND SIMILAR CONSTRUCTION CONDITIONS, IN COMMUNICATIONS ROOMS, MTR ROOMS, TR ROOMS, CATV ROOMS, ELECTRICAL ROOMS, MECHANICAL ROOMS, ELEVATOR EQUIPMENT ROOMS, AND ROOMS OF SIMILAR PURPOSE, ARE TO BE SEALED AIR TIGHT. SEALANTS IN FIRE AND SMOKE RATED WALLS ARE TO BE FIRE AND SMOKE RATED. SEALANTS IN ACOUSTICALLY RATED WALLS ARE TO BE ACOUSTICAL SEALANTS. EXPANDING FOAM SEALANTS ARE NOT PERMITTED.
28. ALL WINDOW TREATMENT, BLINDS, AND DRAPERIES ARE TO BE STORED AND REINSTALLED AFTER COMPLETION OF INTERIOR PAINTING.

## GENERAL DEMOLITION NOTES:

1. THE SCOPE OF DEMOLITION WORK INCLUDES, BUT IS NOT LIMITED TO, THE DEMOLITION WORK SHOWN ON THE DRAWINGS. CONTRACTOR SHALL PERFORM ALL DEMOLITION AS REQUIRED TO COMPLETE THE WORK, INCLUDING DEMOLITION THAT IS IMPLIED BUT NOT SPECIFICALLY CALLED OUT ON THE DRAWINGS.
2. DEMOLITION SHOWN ON DRAWINGS MAY NOT REPRESENT THE FULL EXTENT OF REQUIRED DEMOLITION OR DEMOLITION PERFORMED BY OTHER TRADES. CONTRACTOR SHALL REVIEW ALL OTHER TRADES FOR FULL EXTENT OF DEMOLITION REQUIREMENTS PERTAINING TO EACH TRADE. ITEMS INCLUDED, BUT NOT LIMITED TO, ARE: PLUMBING FIXTURES, MECHANICAL EQUIPMENT, LIGHT FIXTURES, ETC.
3. EXISTING CONDITIONS AND DIMENSIONS ARE SHOWN PER ORIGINAL DESIGN DRAWINGS AND NON DESTRUCTIVE SITE INVESTIGATIONS. CONTRACTOR SHALL VERIFY ALL EXISTING CONDITIONS PRIOR TO START OF DEMOLITION/CONSTRUCTION AND NOTIFY GOVERNMENT PROMPTLY IF ANY DISCREPANCIES EXIST.
4. REMOVE EXISTING MATERIALS COMPLETE DOWN TO ORIGINAL SUBSTRATES. CLEAN AND PREP SUBSTRATES AS REQUIRED FOR NEW CONSTRUCTION.
5. ADDITIONAL DEMOLITION MAY BE REQUIRED BEYOND THAT INDICATED ON THE DEMOLITION DRAWINGS TO ACHIEVE THE FINAL NEW WORK INDICATED ON THE NEW WORK DRAWINGS. IT SHALL BE THE CONTRACTORS RESPONSIBILITY TO DETERMINE THE FINAL EXTENT OF DEMOLITION REQUIRED TO ACHIEVE THE NEW WORK SCOPE AS INDICATED ON THE NEW WORK DRAWINGS.

## DOOR AND HARDWARE FUNCTIONING REQUIREMENTS:

1. ALL EXTERIOR BEQ DOOR SLABS ARE TO BE REPLACED. ALL EXTERIOR BEQ DOOR FRAMES ARE EXISTING TO REMAIN.
2. ALL INTERIOR SLEEPING ROOM DOORS ARE EXISTING TO REMAIN AND ARE TO BE READJUSTED TO ENSURE PROPER OPENING AND CLOSING.
3. CLOSERS, WEATHERSTRIPPING, AND THRESHOLDS ON ALL EXTERIOR BEQ DOORS ARE TO BE REPLACED WITH BALANCE OF HARDWARE EXISTING TO REMAIN.
4. ALL EXISTING HARDWARE ON ALL INTERIOR AND EXTERIOR DOORS TO BE ADJUSTED, REPAIRED, AND OR REPLACED TO ENSURE PROPER DOOR FUNCTION.
5. BATTERIES IN ALL EXISTING ELECTRONIC LOCKING HARDWARE TO BE REPLACED IMMEDIATELY PRIOR TO REINSTALLATION OF ELECTRONIC DOOR HARDWARE.
6. EXISTING MORTAR BEDS AT ALL EXTERIOR BEQ DOOR THRESHOLDS IS TO BE REMOVED FOR INSTALLATION OF NEW THRESHOLDS BEING SET IN A NEW MORTAR BED.
7. ALL LOCKING AND LATCHING HARDWARE AT ALL BEQ EXTERIOR DOORS TO BE REMOVED, TAGGED, STORED, AND REAPPLIED ON TO NEW DOOR SLABS AT THE SAME LOCATION.
8. ALL ROOM SIGNAGE AND EXISTING ROOM TAGS AFFIXED TO ALL BEQ EXTERIOR DOORS TO BE REMOVED, TAGGED, STORED, AND REAPPLIED ON TO NEW DOOR SLABS AT THE SAME LOCATION AND TO HAVE NEW DOOR TAGS INSTALLED BY CONTRACTOR.

## MISCELLANEOUS EQUIPMENT:

1. CEILING FANS: PRODUCT REQUIREMENTS: BLADE SWEEP: 52"; NUMBER OF BLADES: 4; BLADE PITCH: 14 DEGREES; MOTOR SIZE: 172X17 MM; AMPS AT HIGH SPEED: 0.62; WATTS AT HIGH SPEED: 74.91; HIGH SPEED: 5494 CFM; LOW SPEED: 2012 CFM; BULB CCT: 3000; LED BULB WATTAGE: 14; LED CRI: 90; LIGHT KIT: INTEGRAL; LED LIGHT LUMENS: 1195.20; EXPOSURE RATING: INDOOR; WARRANTY: LIMITED LIFETIME. BASIS OF DESIGN: CRAFTMADE MODEL LAVAL LAV52ESP4LK-LED. SEE ELECTRICAL FOR ADDITIONAL REQUIREMENTS.
2. REFRIGERATORS: PRODUCT REQUIREMENTS: NUMBER OF DOORS: 2; DOORS FINISH: STAINLESS STEEL; TYPE: COMPACT; DEFROST TYPE: AUTOMATIC; THRU-DOOR DISPENSER: NO; ICE MAKER: NO; HEIGHT: 55.7"; WIDTH: 21.7"; OVERALL CAPACITY: 7.4 CUBIC FEET; FREEZER CAPACITY: 1.4 CUBIC FEET; REFRIGERATOR CAPACITY: 6 CUBIC FEET; SHELVES: GLASS; CAN RACK: YES; INTERIOR LIGHT: YES; REVERSIBLE DOORS: YES; VEGETABLE STORAGE: YES; ENERGY STAR CERTIFIED: YES. BASIS OF DESIGN: MAGIC CHEF MODEL MCDR740STE.

## BUILDING PRESSURE WASHING GENERAL NOTES:

1. ALL EXISTING EXPOSED TO WEATHER EXTERIOR SURFACES ARE TO BE PRESSURE WASHED.
2. ALL PRESSURE WASHING SHALL BE CONDUCTED PRIOR TO ANY EXTERIOR FINISH FIELD PAINTING.
3. DO NOT ERODE MORTAR JOINTS.
4. CONSTRUCT A 10' WIDE X 8' HIGH MOCK UP FOR REVIEW AND APPROVAL. APPROVED MOCK UP SHALL BE THE STANDARD FOR EVALUATING THE REMAINING WORK.

## METAL ROOF PANEL AND ASSOCIATED TRIM GENERAL NOTES:

1. THE ROOF PANELS SHALL BE STRUCTURAL STANDING SEAM PREFINISHED ALUMINUM WITH A MINIMUM 2 1/2" HIGH ROLL FORMED RIB WITH 180 DEGREE DOUBLE FOLD MECHANICAL CRIMPING SEAMS WITH CONCEALED CLIPS AND FASTENERS. ROOF PANEL THICKNESS NOT LESS THAN THE GREATER OF 0.032 INCH THICKNESS OR THAT DETERMINED FROM CALCULATIONS BASED ON PANEL PROFILE, LOCAL WIND LOADS, AND PANEL ATTACHMENTS. WIND LOAD PRESSURES SHALL BE IN ACCORDANCE WITH ASCE-7, MINIMUM DESIGN LOADS FOR BUILDINGS AND OTHER STRUCTURES AND AS DETERMINED BASED ON BUILDING LOCATIONS AND CONFIGURATION. ALUMINUM SHEET SHALL CONFORM TO ASTM B209, ALLOY 3004 AND AA-ASD1. ROOF PANELS TO BE SMOOTH NON-EMBOSSED.
2. ROOF PANEL AND FLASHING DETAILS ON THE DRAWINGS ARE GENERAL IN NATURE TO AVOID PROPRIETARY CONFLICTS. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO PROVIDE SPECIFIC DETAILS UNIQUE TO THE SELECTED ROOF PANEL MANUFACTURER THAT MEET THAT ROOF MANUFACTURER'S REQUIREMENTS WHILE ALSO MEETING THE OVERALL INTENT AS INDICATED ON THE DRAWINGS. THE DRAWINGS INDICATE THE OVERALL DESIGN INTENT AND DELIBERATELY AVOID INDICATING SPECIFIC INFORMATION OF ANY ONE ROOF PANEL MANUFACTURER TO ENCOURAGE ROOF PANEL MANUFACTURERS TO PROVIDE THEIR FINAL DESIGN THAT MEETS THE OVERALL INTENT INDICATED. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO PROVIDE THE ADDITIONAL REQUIRED DETAILS AND DESIGN INFORMATION, AS WELL AS ALL THE ROOFING COMPONENTS THAT SUPPORT THEIR DETAILS AND DESIGN INFORMATION, THAT COMPLETES THE OVERALL DESIGN INTENT AS INDICATED ON THE DRAWINGS.
3. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY DURING THE BIDDING PROCESS TO SECURE ANY ADDITIONAL INFORMATION THEY DEEM NECESSARY TO COMPLETE THE DESIGN INTENT AS INDICATED ON THE DRAWINGS. THE CONTRACTOR WILL NOT BE AWARDED ANY ADDITIONAL COMPENSATION AFTER BIDDING IN THEIR PURSUIT TO COMPLETE THE DESIGN OF THE ROOF SYSTEM.
4. ALL FLASHING AND TRIM SHALL BE PREFINISHED ALUMINUM. ROOF PANELS ARE TO RECEIVE A FACTORY-APPLIED POLYVINYLIDENE FLUORIDE OF KYNAR 500/HYLAR 5000 FINISH CONSISTING OF A BAKED-ON TOP-COAT WITH A MANUFACTURER'S RECOMMENDED PRIME COAT. PROVIDE COMPATIBLE FASTENERS TO AVOID GALVANIC ACTION AND CORROSION OF DISSIMILAR MATERIALS. COLORS OF PREFINISHED STANDING SEAM ROOF PANELS AND ALL EXPOSED TO VIEW ROOFING RELATED COMPONENTS ARE TO BE PER THE BEAUFORT, SC MARINE CORPS AIR STATION (MCAS) COLOR STANDARDS AND APPROVED BY MCAS PUBLIC WORKS (PW) DEPARTMENT. EXPOSED FASTENERS FOR TRIM AND ALL ASSOCIATED ACCESSORIES SHALL BE FINISHED. NO ADDITIONAL COMPENSATION WILL BE AWARDED TO THE CONTRACTOR FOR CUSTOM COLORS IN PURSUIT OF MATCHING THE MCAS COLOR STANDARD. FIELD PAINTING OF COMPONENTS INDICATED TO BE PREFINISHED IS NOT ALLOWED. WHERE COLORS ARE NOT SPECIFICALLY INDICATED, THE CONTRACTOR SHALL MAKE AVAILABLE FOR FINAL SELECTION COLORS FROM THE MANUFACTURERS FULL LINE OF STANDARD AND CUSTOM COLORS.
5. PREFINISHED ALUMINUM GUTTER AND DOWNSPOUT MINIMUM SIZES ARE INDICATED ON THE DRAWINGS. FINAL GUTTER AND DOWNSPOUT PROFILES AND ROUTING ARE TO BE DETERMINED BY THE ROOF PANEL MANUFACTURER AND PROVIDE NOT LESS THAN THE WATER FLOW AS PROVIDED BY THE GUTTER AND DOWNSPOUT SIZES INDICATED. LOCATIONS OF THE DOWNSPOUTS TO BE AS INDICATED ON THE DRAWINGS.
6. REQUESTED DEVIATIONS BY THE CONTRACTOR FROM THE INFORMATION PROVIDED ON THE DRAWINGS MUST BE APPROVED BY THE CONTRACTING OFFICER PRIOR TO THE CONTRACTOR SUBMITTING THEIR BID FOR THE WORK. NO ADDITIONAL COMPENSATION WILL BE AWARDED TO THE CONTRACTOR FOR ANY DESIGN CHANGES IF THOSE REQUESTED CHANGES WERE NOT APPROVED DURING THE BIDDING PROCESS.
7. ALL FLASHING, TRIM, ACCESSORIES, AND MISCELLANEOUS ROOFING COMPONENTS SHALL BE APPROVED BY THE ROOF PANEL MANUFACTURER AND INCORPORATED INTO THE DESIGN BY THE ROOF PANEL MANUFACTURER. THE ROOF SYSTEM, FLASHING, TRIM, GUTTER, DOWNSPOUT, AND SO FORTH SHALL BE COMPRISED OF COMPONENTS DESIGNED AND PROVIDED TO THE GREATEST EXTENT POSSIBLE BY THE ROOF PANEL MANUFACTURER.
8. ROOF PENETRATIONS ARE NOT PERMITTED EXCEPT FOR THOSE INDICATED ON THE DRAWINGS.
9. THE ROOF PANEL AND ALL ASSOCIATED ROOFING ACCESSORIES SUCH AS FLASHING, TRIM, GUTTER, DOWNSPOUTS, AND SO FOR SHALL BE WARRANTED FOR NOT LESS THAN 20 YEARS. CONTRACTOR TO SUBMIT MANUFACTURER'S STANDARD WARRANTY TO CONTRACTING OFFICER.
10. STRUCTURAL STANDING SEAM METAL ROOF MAY BE SHOWN ABBREVIATED AS "SSSMR".
11. THE EXISTING WOOD BLOCKING AT THE EAVES OF ALL BUILDINGS IS TO REMAIN AS IS UNLESS DAMAGED DUE TO WOOD ROT OR DAMAGED DURING DEMOLITION. IF DAMAGED, THE CONTRACTOR SHALL REPLACE THE WOOD WITH BENT GALVANIZED SHEET METAL OF THICKNESS NOT LESS THAN 18 GAUGE. BENT METAL TO BE FORMED AS REQUIRED TO PERMIT FASTENING OF NEW AND EXISTING TO REMAIN BUILDING COMPONENTS AT THE ROOF EAVES. NO NEW WOOD BLOCKING IS TO BE INSTALLED.

## ELEVATED WALKWAY UNDERLAYMENT AND URETHANE FINISH GENERAL NOTES:

1. AT NUMEROUS LOCATIONS AT BOTH LEVELS OF THE ELEVATED WALKWAYS WATER PONDS UP AGAINST THE EXTERIOR WALLS AND WATER FLOWS TOWARDS THE STAIR LANDINGS AND TOWARDS THE WALKWAY EDGE AT EACH STAIR.
2. AFTER THE CONTRACTOR HAS REMOVED ALL THE EXISTING FLOOR FINISH, THE CONTRACTOR SHALL FIELD VERIFY ALL LOCATIONS OF THE ELEVATED WALKWAYS WHERE THERE IS NOT POSITIVE DRAINAGE AWAY FROM THE EXTERIOR WALLS TOWARDS THE OUTER EDGE OF THE WALKWAYS AND WHERE WATER IS FLOWING ONTO THE STAIR LANDINGS AND WHERE WATER IS FLOWING OVER THE LANDING EDGE ONTO THE STAIRS BELOW.
3. PRIOR TO INSTALLATION OF URETHANE COATING, A TROWEL GRADE UNDERLAYMENT SHALL BE PLACED TO ENSURE POSITIVE DRAINAGE AWAY FROM THE BUILDING EXTERIOR WALLS AT THE ELEVATED WALKWAYS. PRIOR TO INSTALLATION OF URETHANE COATING, A TROWEL GRADE UNDERLAYMENT SHALL ALSO BE PLACED ON THE ELEVATED WALKWAYS TO ENSURE POSITIVE DRAINAGE IS PROVIDED AWAY FROM EACH OF THE STAIR LANDINGS AND AWAY FROM EACH OF THE WALKWAY EDGES AT THE STAIRS TO PREVENT WATER FROM DRAINING ONTO THE STAIRS BELOW. TO ADDRESS THE POSITIVE DRAINAGE REQUIREMENTS, THE CONTRACTOR SHALL INCLUDE IN THEIR WORK EFFORT TO PROVIDE AND INSTALL UNDERLAYMENT AT NOT LESS THAN 25% OF THE ENTIRE ELEVATED WALKWAY SURFACE.
4. AT LOCATIONS WHERE PROVIDING UNDERLAYMENT ALONG THE BUILDING WALL WILL RAISE THE FINISH FLOOR ELEVATION AND RESULT IN A TRIPPING CONDITION AT THE DOORWAY, THE CONTRACTOR SHALL TAPER THE UNDERLAYMENT NOT GREATER THAN A 1% SLOPE ON EACH SIDE OF THE DOORWAY WHILE STILL MAINTAINING POSITIVE SLOPE FROM THE DOORWAY TO THE EDGE OF THE WALKWAY.
5. ALONG THE EXISTING CMU SPLIT FACE WALL AT THE ELEVATED WALKWAYS, PROVIDE A HORIZONTAL WORKING JOINT BETWEEN 1/4" WIDE AND 1/2" WIDE AT THE NEW UNDERLAYMENT AND URETHANE FLOOR FINISH. DEPTH OF JOINT AND BACKERROD PLACEMENT SHALL BE PER SEALANT MANUFACTURER'S WRITTEN RECOMMENDATIONS.
6. AT LOCATIONS WHERE THE ABOVE REQUIREMENTS CANNOT BE MET, THE CONTRACTOR, AFTER REMOVAL OF ALL EXISTING WALKWAY FLOOR FINISHES AND PRIOR TO STARTING OF ANY UNDERLAYMENT PLACEMENT, SHALL NOTIFY THE CONTRACTING OFFICER OF ALL SUCH LOCATIONS SO DIRECTION OF AN ALTERNATE SOLUTION CAN BE PROVIDED.
7. UNDERLAYMENT WORK IS NOT REQUIRED AT THE CORRIDOR BETWEEN THE WINGS.
8. THE NEW URETHANE COATING SHALL BE INSTALLED OVER THE NEW TROWEL GRADE UNDERLAYMENT AND PREPARED CONCRETE TOPPING AT ALL ELEVATED WALKWAYS, STAIR LANDINGS AT THE ELEVATED WALKWAYS, CONNECTOR WALKWAYS BETWEEN THE WINGS, AND AT THE FRONT EDGE OF ALL ELEVATED WALKWAYS. THE SAME URETHANE COATING SHALL BE INSTALLED AT ALL LOCATIONS MENTIONED HEREIN. COLORS SHALL BE FROM URETHANE MANUFACTURER'S FULL LINE OF STANDARD AND CUSTOM COLORS. FINAL COLOR SELECTION SHALL BE MADE BY THE GOVERNMENT.
9. THE URETHANE COATING MATERIAL SHALL BE COMPATIBLE WITH ALL VARIOUS MATERIALS THE URETHANE COATING WILL BE APPLIED OVER.
10. SEE STRUCTURAL DRAWINGS FOR REQUIREMENTS REGARDING THE STRUCTURAL PATCHING AT THE ELEVATED SLAB EDGE WHERE THE EXISTING BENT PLATE WAS REMOVED.
11. PROVIDE AND INSTALL A SKIM COAT AS REQUIRED OVER THE STRUCTURAL PATCH AT THE SLAB EDGE TO ENSURE THE VERTICAL FACE OF THE SLAB EDGE STRUCTURAL PATCH IS UNIFORM IN ALL DIRECTIONS. UNIFORMITY SHALL BE PROVIDED TO A FLATNESS NOT EXCEEDING 1/8" IN 4' IN ANY DIRECTION. THE SKIM COAT PRODUCT SHALL BE COMPATIBLE WITH BOTH THE STRUCTURAL PATCH MATERIAL AND THE URETHANE FINISH COAT MATERIAL.

## SHOWER TILE REPLACEMENT GENERAL NOTES:

1. DEMOLISH WALL TILE AND BACKER BOARD WITHIN SHOWER AREA STARTING AT SHOWER FLOOR AND EXTENDING TO A HEIGHT OF 4 FULL WALL TILES. DEMOLITION OF WALL TILE AND BACKER BOARD AT THIS HEIGHT TO EXTEND TO TILED WALLS OUTSIDE SHOWER.
2. DEMOLISH WALL TILE AND BACKER BOARD WITHIN SHOWER AREA A HEIGHT OF 3 FULL WALL TILES CENTERED ON THE SHOWER MIXING VALVE. DEMOLITION OF WALL TILE AND BACKER BOARD AT THIS HEIGHT TO EXTEND TO TILED WALLS OUTSIDE SHOWER.
3. AT DEMOLISHED WALL TILE LOCATIONS PROVIDE NEW CEMENTITIOUS BACKER BOARD OF EQUAL THICKNESS TO ORIGINAL BACKER MATERIAL WITH A FLUID APPLIED WATER BARRIER MEMBRANE.
4. PROVIDE AND INSTALL WALL TILE AS INDICATED.

## FIBERGLASS BLANKET ROOF AND KNEE WALL ATTIC INSULATION REWORK GENERAL NOTES:

1. THE INSULATION IN THE BEQ BUILDING HAS BEEN DISPLACED IN NUMEROUS LOCATIONS.
2. REWORK OF THE INSULATION IS REQUIRED AT THE NEW ROOF AREAS OVER THE EXISTING END STAIRS.
3. TO ADDRESS THE TWO ABOVE CONDITIONS, THE CONTRACTOR SHALL INCLUDE IN THEIR WORK EFFORT TO REWORK AND REPOSITION NOT LESS THAN 50% OF ALL OF THE EXISTING FIBERGLASS BLANKET ROOF AND KNEE WALL INSULATION.
4. NUMEROUS LOCATIONS OF EXISTING INSULATION HAS BEEN DAMAGED AND NEEDS TO BE REPLACED. THE CONTRACTOR SHALL INCLUDE IN THEIR WORK EFFORT TO REPLACE NOT LESS THAN 5% OF ALL OF THE EXISTING FIBERGLASS BLANKET ROOF AND KNEE WALL INSULATION.

**REVISIONS**

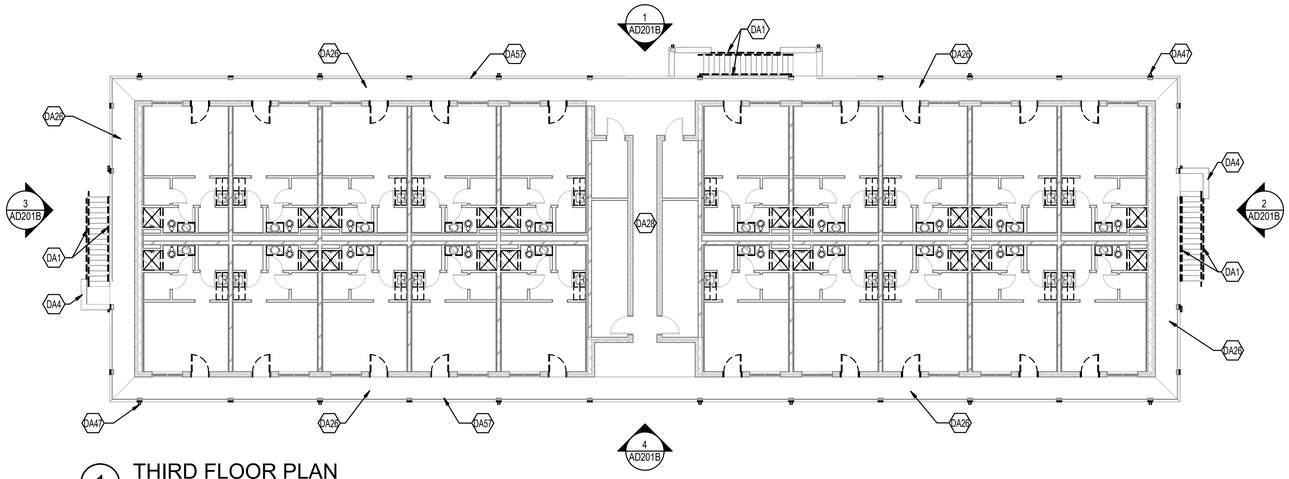
| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## PAINTING AND REFINISHING GENERAL NOTES:

1. SEE SPECIFICATIONS FOR ADDITIONAL PAINTING AND EXISTING SURFACE PREPARATION REQUIREMENTS. ALL EXISTING PAINTED METAL SURFACES ARE TO BE CHEMICALLY DEGLOSSED AND CHEMICALLY CLEANED OF ALL DELETERIOUS MATERIALS PRIOR TO REPAINTING.
2. ALL NEW EXTERIOR HOLLOW METAL DOOR SLABS ARE TO BE FIELD PAINTED AND ALL EXISTING TO REMAIN INTERIOR HOLLOW METAL DOOR SLABS AND ALL EXISTING TO REMAIN HOLLOW METAL FRAMES, INCLUDING HOLLOW METAL SIDELITE FRAMES, HOLLOW METAL TRANSOM FRAMES, AND METAL TRANSOM INFILL PANELS, ARE TO BE REPAINTED. ALL HINGE STOPS, MIRRORS, GARMENT RACKS, AND SO FORTH THAT ARE MOUNTED TO DOORS ARE TO BE REMOVED, STORED, AND REINSTALLED AFTER REPAINTING OF DOORS. ALL STICKERS ARE TO BE REMOVED AND REPLACED WITH NEW STICKERS AFTER REPAINTING OF DOORS. DOOR STRIKES AND LOCKING AND LATCHING HARDWARE IS EXISTING TO REMAIN AND SHALL BE MASKED AND PROTECTED DURING REPAINTING UNLESS NOTED OTHERWISE. EXISTING DOOR HINGES ARE TO REMAIN AND PREVIOUSLY PAINTED DOOR HINGES ARE TO BE REPAINTED.
3. ALL EXPOSED TO VIEW STRUCTURAL FRAMING (TRUSSES, COLUMNS, BEAMS, ETC.) AND UNDERSIDE OF THE STEEL ROOF DECKING OF THE BREEZEWAYS IS TO BE REPAINTED. ALL NEW AND EXISTING EXPOSED TO VIEW STRUCTURAL FRAMING (COLUMNS, BEAMS, STAIR STRINGERS, INVERTED T SUPPORTS, ETC.) AT THE BEQ BUILDING IS TO BE REPAINTED.
4. THE UNDERSIDE OF THE PLYWOOD ROOF SHEATHING, WOOD TRUSS FRAMING, STEEL COLUMNS, AND STEEL BEAMS AT BOTH PAVILIONS IS TO BE REPAINTED.
5. AT ALL EXISTING GYPSUM BOARD FINISHED WALL AND FINISHED CEILING LOCATIONS, PATCH, REPAIR, AND PREP ALL SURFACES TO A LEVEL 4 SURFACE FOR REPAINTING.
6. UNLESS NOTED OTHERWISE TO BE REPLACED WITH NEW, AT ALL SLEEPING ROOM SUITES, ALL WALL AND CEILING MOUNTED ITEMS INCLUDING, BUT NOT LIMITED TO, CURTAINS, MIRRORS, BLINDS, AND ASSOCIATED WINDOW TREATMENT HARDWARE, ELECTRICAL DEVICES, ELECTRICAL COVER PLATES, LOUVER TRIMS, WALL AND CEILING MOUNTED LIGHT FIXTURES, FIRE ALARM AND RELATED DEVICES, OCCUPANCY SENSORS, ALL RESTROOM ACCESSORIES AND FIXTURES, ALL WIRE SHELVING AND SUPPORT BRACKETS, AND SO FORTH, ARE TO BE REMOVED, STORED, AND REINSTALLED AFTER ROOM REFINISHING. AT FIXTURES AND ACCESSORIES THAT HAVE CONCEALED MOUNTING BRACKETS, THE CONCEALED MOUNTING BRACKETS CAN BE LEFT IN PLACE FOR REPAINTING. RESTROOM ACCESSORIES WITHIN TILED AREAS, CORNER GUARDS, EXPOSED ELECTRICAL RACEWAYS, WIREMOLD AND CONDUIT SERVING ELECTRICAL EQUIPMENT, FIRE ALARMS, PULL STATIONS, AND RELATED FIRE DEVICES IS EXISTING TO REMAIN UNLESS NOTED OTHERWISE.
7. IN ALL SLEEPING ROOMS, ALL EXISTING CEILING ACCESS PANELS AND PANEL FRAMES ARE TO BE CLEANED AND PREPARED FREE OF RUST AND DELETERIOUS CONDITIONS FOR REPAINTING. REPAINT ALL ACCESS PANELS AND PANEL FRAMES CEILING FINISH COLOR.
8. THE UNDERSIDE OF THE EXISTING HOLLOW CORE CONCRETE PLANKS ARE NOT TO BE PAINTED.

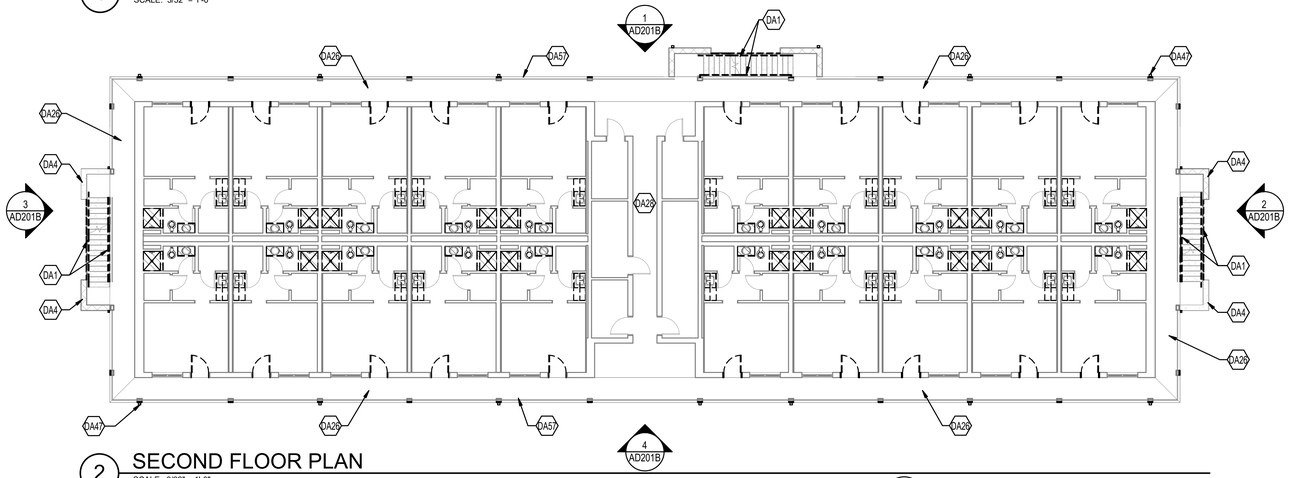
## ALUMINUM RAIN SCREEN LOUVER GENERAL NOTES:

1. LOUVERS SHALL BE CONSTRUCTED OF ALUMINUM EXTRUSIONS MEETING ASTM B211, ALLOY 6063-T5, 6063-T6 OR 6061-T6 AND ALUMINUM FORMED PROFILES MEETING ASTM B3209, ALLOY 1100, 3003 OR 5005. LOUVER HEAD, JAMBS, SILL, AND BLADES SHALL NOT EXCEED 6" IN TOTAL DEPTH. THICKNESS OF EXTRUSIONS AND FORMED PROFILES SHALL BE NOT LESS THAN THE GREATER OF THAT INDICATED IN THE SPECIFICATIONS AND THAT DETERMINED FROM CALCULATIONS BASED ON LOCAL WIND LOADS AND LOUVER ATTACHMENTS. WIND LOAD PRESSURES SHALL BE IN ACCORDANCE WITH ASCE-7, MINIMUM DESIGN LOADS FOR BUILDINGS AND OTHER STRUCTURES AND AS DETERMINED BASED ON BUILDING LOCATIONS AND CONFIGURATION. LOUVERS SHALL HAVE BEEN TESTED IN ACCORDANCE WITH DADE COUNTY PROTOCOLS TAS-201, TAS-202 AND TAS-203; AND SHALL BE DADE COUNTY APPROVED FOR OPEN STRUCTURE BUILDING ENVELOPE PROTECTION (INCLUDING MISSILE) FOR ALLOWABLE DESIGN WIND LOADING UP TO 170 PSF
2. LOUVER DETAILS ON THE DRAWINGS ARE GENERAL IN NATURE TO AVOID PROPRIETARY CONFLICTS. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO PROVIDE SPECIFIC DETAILS UNIQUE TO THE SELECTED LOUVER MANUFACTURER THAT MEET THAT LOUVER MANUFACTURER'S REQUIREMENTS WHILE ALSO MEETING THE OVERALL INTENT AS INDICATED ON THE DRAWINGS. THE DRAWINGS INDICATE THE OVERALL DESIGN INTENT AND DELIBERATELY AVOID INDICATING SPECIFIC INFORMATION OF ANY ONE LOUVER MANUFACTURER TO ENCOURAGE LOUVER MANUFACTURERS TO PROVIDE THEIR FINAL DESIGN THAT MEETS THE OVERALL INTENT INDICATED. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO PROVIDE THE ADDITIONAL REQUIRED DETAILS AND DESIGN INFORMATION, AS WELL AS ALL THE LOUVER MOUNTING COMPONENTS THAT SUPPORT THEIR DETAILS AND DESIGN INFORMATION, THAT COMPLETES THE OVERALL DESIGN INTENT AS INDICATED ON THE DRAWINGS.
3. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY DURING THE BIDDING PROCESS TO SECURE ANY ADDITIONAL INFORMATION THEY DEEM NECESSARY TO COMPLETE THE DESIGN INTENT AS INDICATED ON THE DRAWINGS. THE CONTRACTOR WILL NOT BE AWARDED ANY ADDITIONAL COMPENSATION AFTER BIDDING IN THEIR PURSUIT TO COMPLETE THE DESIGN OF THE LOUVER SYSTEM.
4. ALL LOUVER COMPONENTS INCLUDING ANY ATTACHMENT BRACKETS, SHALL BE PREFINISHED. PROVIDE COMPATIBLE FASTENERS TO AVOID GALVANIC ACTION AND CORROSION OF DISSIMILAR MATERIALS. COLORS OF PREFINISHED LOUVER COMPONENTS ARE TO MATCH THE LOUVER COLOR. EXPOSED FASTENERS FOR TRIM AND ALL ASSOCIATED LOUVER ACCESSORIES SHALL BE FINISHED MATCHING THE LOUVER COLOR. FIELD PAINTING OF COMPONENTS INDICATED TO BE PREFINISHED IS NOT ALLOWED. FINAL COLOR SELECTION TO BE BY GOVERNMENT FROM LOUVER MANUFACTURER'S FULL LINE OF STANDARD AND CUSTOM NON-METALIC COLORS. THE CONTRACTOR WILL NOT BE AWARDED ANY ADDITIONAL COMPENSATION S



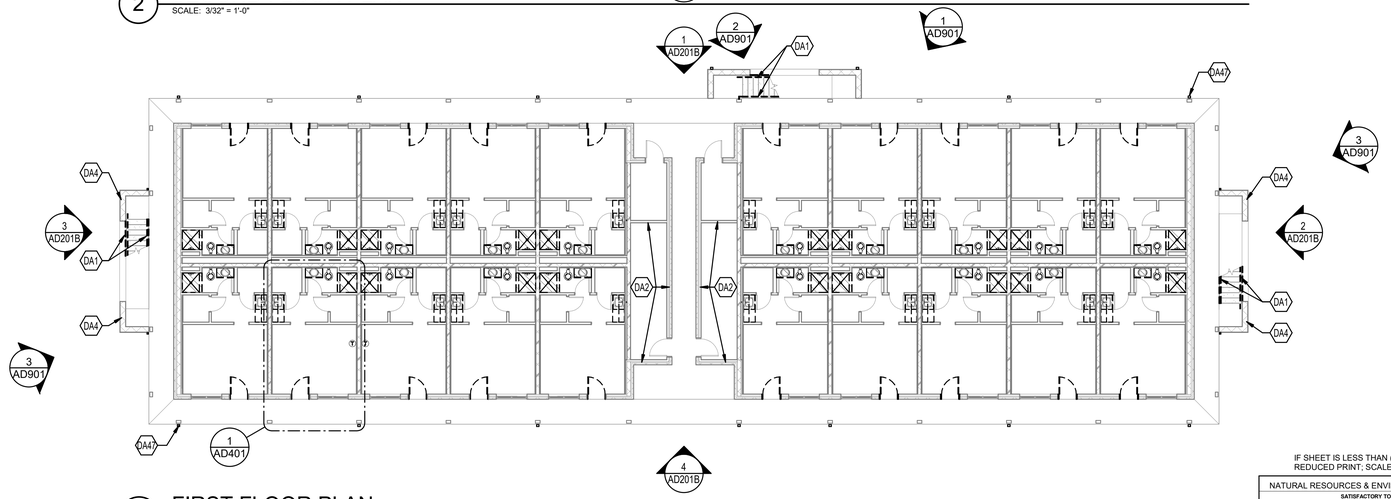

1 THIRD FLOOR PLAN

SCALE: 3/32" = 1'-0"




2 SECOND FLOOR PLAN

SCALE: 3/32" = 1'-0"




3 FIRST FLOOR PLAN

SCALE: 3/32" = 1'-0"

PLAN NORTH

24"0 4' 8' 16'

**SCALE: 3/32"=1'-0"**

**GRAPHIC SCALE**

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. CLEAN AND PREPARE ALL VISIBLE, TO REMAIN, STRUCTURAL STEEL FOR NEW PAINT. INCLUDING BUT NOT LIMITED TO COLUMNS, BEAMS, CONNECTIONS AND OTHER EXPOSED STEEL FRAMING ELEMENTS.
2. DEMOLISH WATER DAMAGED, AND DEMOLISH, OR TREAT, IDENTIFIED MOLD, IN GYPSUM BOARD TO EXTENT REQUIRED IN ALL AREAS FOUND DURING CONSTRUCTION. ASSUME 16 SQ FT OF GYP BD REQUIRED FOR EACH UNIT RESTROOM. PROVIDE UNIT COST FOR ADDITIONAL SQ FT NEEDED.
3. IN ALL SLEEPING ROOM SUITES: DEMOLISH ALL FLOOR FINISHES AND WALL BASE.
4. REMOVE EXISTING FLOOR COATING AT ALL ELEVATED PERIMETER WALKWAYS AND ALL ELEVATED WALKWAYS BETWEEN THE WINGS AND CLEAN AND PREPARE EXISTING CONCRETE FLOOR FOR NEW UNDERLAYMENT AND URETHANE FLOOR FINISH. SEE UNDERLAYMENT GENERAL NOTES FOR ADDITIONAL REQUIREMENTS.
5. DEMOLISH ALL EXISTING GUTTERS COMPLETE.
6. DEMOLISH ALL EXISTING DOWNSPOUTS COMPLETE.
7. DEMOLISH ALL EXISTING GUARDRAILS AND HANDRAILS AT EXISTING STAIRS. EXISTING STEEL SUPPORT PLATES WELDED ONTO STRINGER FLANGES ARE TO REMAIN AND THE CONTRACTOR IS RESPONSIBLE FOR COORDINATING ATTACHMENT OF THE NEW RAILINGS TO THESE EXISTING SUPPORT PLATES. EXISTING STAIR STRINGERS ARE TO REMAIN. WITH THE EXCEPTION OF THE ELEVATED WALKWAY GUARDRAILS ADJACENT TO THE BREEZEWAYS, THE EXISTING GUARDRAILS AT THE ELEVATED PERIMETER WALKWAYS ARE TO REMAIN. THE GUARDRAILS AT THE ELEVATED WALKWAYS ADJACENT TO THE BREEZEWAYS ARE TO BE REMOVED, STORED, AND REINSTALLED AFTER THE WORK IS COMPLETED ON THE WALKWAY SLAB EDGE.
8. PATCH, REPAIR, AND PREP ALL GYPSUM BOARD WALLS TO A LEVEL 4 SURFACE FOR REPAINTING.

## SHEET KEYNOTES

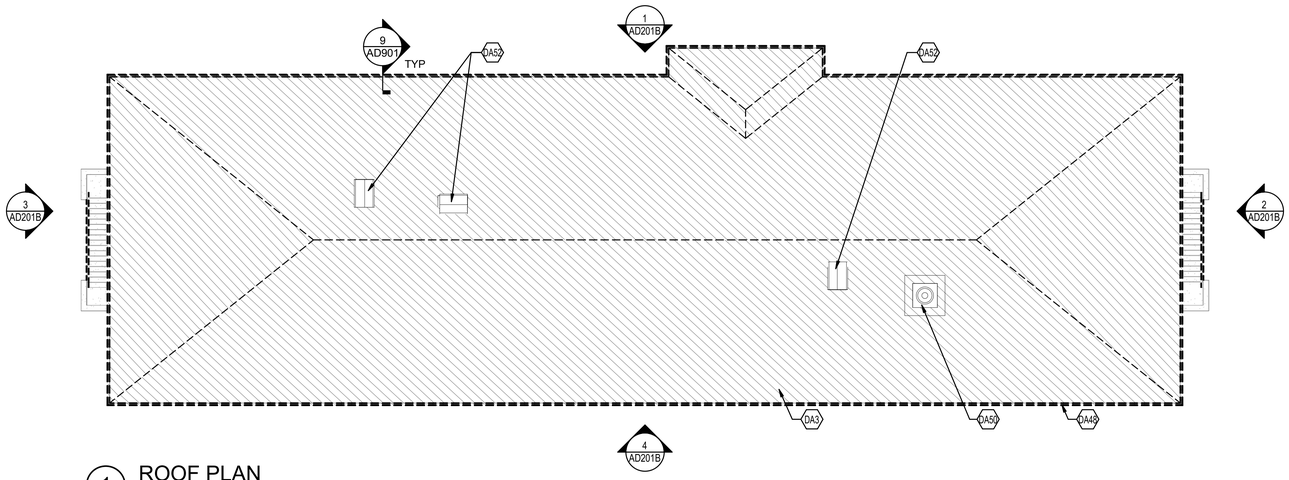
| | |
|---|---|
| DA1 | EXISTING GUARDRAIL AND HANDRAIL - DEMOLISH. STAIR AND STRINGER - ETR. |
| DA2 | WATER DAMAGED GYP BD IN MECHANICAL ROOM - DEMOLISH TO EXTENT REQUIRED. APPROXIMATELY 4'-0" HIGH X 36'-8" LENGTH PER ROOM AND TREAT BALANCE OF ROOM FOR MOLD AND PREPARE FOR NEW GYP BD. |
| DA4 | EXISTING MASONRY SPLITFACE CMU WALL. PREPARE TO RECEIVE NEW RAIN SCREEN SYSTEM. |
| DA26 | EXISTING ELEVATED PERIMETER WALKWAY. SEE GENERAL NOTES FOR WORK. |
| DA28 | EXISITNG ELEVATED WALKWAY BETWEEN WINGS. SEE GENERAL NOTES FOR WORK. |
| DA47 | EXISTING DOWNSPOUT - DEMOLISH, TYP |
| DA57 | ALL EXISTING GUARDRAILS AT EDGE OF ELEVATED WALKWAYS TO REMAIN. |

WISCONSIN
DEAN C. VON TERSCH
CA-7903
CHARLESTON SC
ARCHITECT
04 OCT 2023

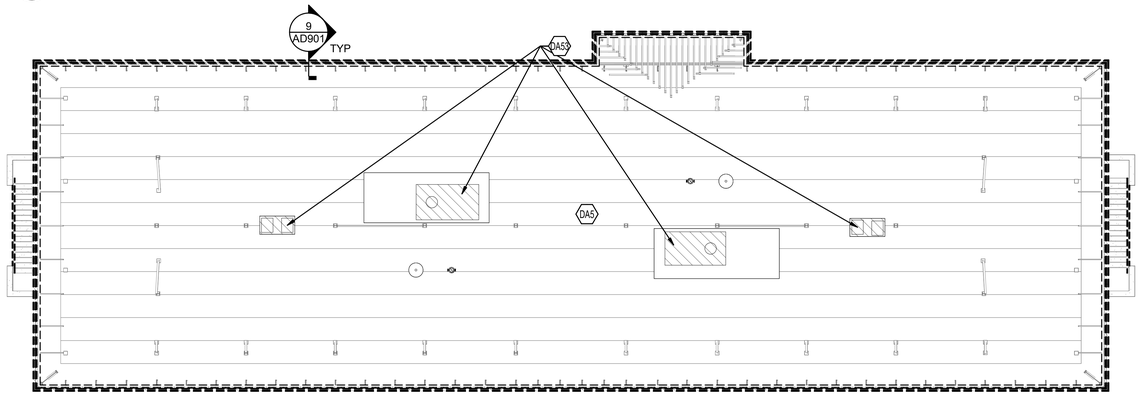
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

| | | |
|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28476 | CEMS<br>AD101B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>**MARINE CORPS AIR STATION**<br>BEAUFORT, S.C. |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | DEMOLITION FIRST, SECOND AND THIRD FLOOR PLANS - BLDG 1245G |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. |
| | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: As indicated; MACC.; T.O. #; SHEET 22 OF 53 |




1 ROOF PLAN
SCALE: 3/32" = 1'-0"




2 ATTIC PLAN
SCALE: 3/32" = 1'-0"

PLAN NORTH

24" 0 4' 8' 16'

SCALE: 1/8"=1'-0"

GRAPHIC SCALE

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. CLEAN AND PREPARE ALL VISIBLE, TO REMAIN, STRUCTURAL STEEL FOR NEW PAINT. INCLUDING BUT NOT LIMITED TO COLUMNS, BEAMS, CONNECTIONS AND OTHER EXPOSED STEEL FRAMING ELEMENTS.
2. DEMOLISH WATER DAMAGED, AND DEMOLISH, OR TREAT, IDENTIFIED MOLD, IN GYPSUM BOARD TO EXTENT REQUIRED IN ALL AREAS FOUND DURING CONSTRUCTION. ASSUME 16 SQ FT OF GYP BD REQUIRED FOR EACH UNIT RESTROOM. PROVIDE UNIT COST FOR ADDITIONAL SQ FT NEEDED.
3. IN ALL SLEEPING ROOM SUITES: DEMOLISH ALL FLOOR FINISHES AND WALL BASE.
4. REMOVE EXISTING FLOOR COATING AT ALL ELEVATED PERIMETER WALKWAYS AND ALL ELEVATED WALKWAYS BETWEEN THE WINGS AND CLEAN AND PREPARE EXISTING CONCRETE FLOOR FOR NEW UNDERLAYMENT AND URETHANE FLOOR FINISH. SEE UNDERLAYMENT GENERAL NOTES FOR ADDITIONAL REQUIREMENTS.
5. DEMOLISH ALL EXISTING GUTTERS COMPLETE.
6. DEMOLISH ALL EXISTING DOWNSPOUTS COMPLETE.
7. DEMOLISH ALL EXISTING GUARDRAILS AND HANDRAILS AT EXISTING STAIRS. EXISTING STEEL SUPPORT PLATES WELDED ONTO STRINGER FLANGES ARE TO REMAIN AND THE CONTRACTOR IS RESPONSIBLE FOR COORDINATING ATTACHMENT OF THE NEW RAILINGS TO THESE EXISTING SUPPORT PLATES. EXISTING STAIR STRINGERS ARE TO REMAIN. WITH THE EXCEPTION OF THE ELEVATED WALKWAY GUARDRAILS ADJACENT TO THE BREEZEWAYS, THE EXISTING GUARDRAILS AT THE ELEVATED PERIMETER WALKWAYS ARE TO REMAIN. THE GUARDRAILS AT THE ELEVATED WALKWAYS ADJACENT TO THE BREEZEWAYS ARE TO BE REMOVED, STORED, AND REINSTALLED AFTER THE WORK IS COMPLETED ON THE WALKWAY SLAB EDGE.
8. PATCH, REPAIR, AND PREP ALL GYPSUM BOARD WALLS TO A LEVEL 4 SURFACE FOR REPAINTING.

## SHEET KEYNOTES

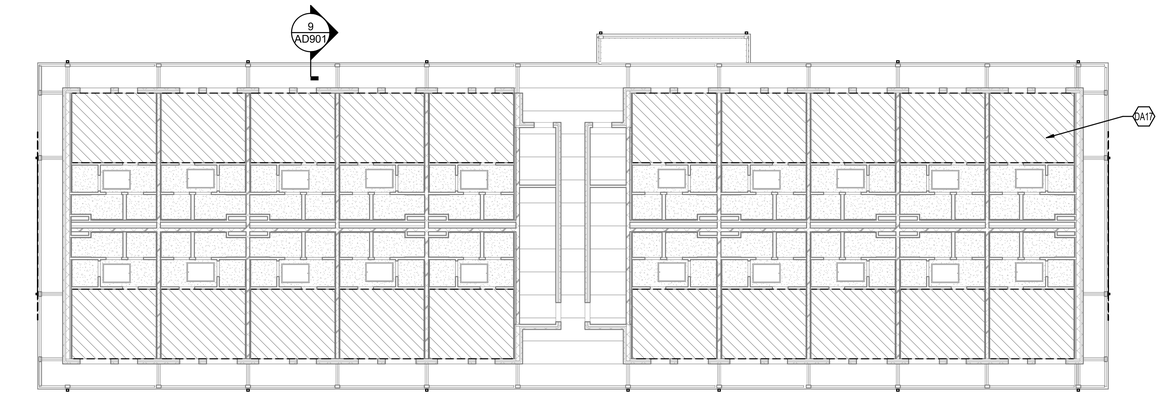
| | |
|---|---|
| DA3 | EXISTING ASPHALT SHINGLE ROOFING, UNDERLAYMENT, AND EXISTING PLYWOOD SHEATHING - DEMOLISH DOWN TO ETR METAL DECKING, TYP. |
| DA5 | NO ARCHITECTURAL DEMOLITION WORK IN ATTIC. PLAN SHOWN FOR COORDINATION PURPOSES ONLY. |
| DA48 | EXISTING GUTTERS - DEMOLISH, TYP |
| DA50 | ROOF PENETRATION ETR, TYP. PROTECT DURING DEMOLITION AND NEW WORK. REPAIR AS REQUIRED. |
| DA52 | ROOF PENETRATION ETR, TYP. SEE MECHANICAL. |
| DA53 | DEMOLISH MECHANICAL EQUIPMENT. SEE MECHANICAL. |

WISCONSIN
DEAN C. VON TERSCH
A-7903
CHARLESTON SC
ARCHITECT
04-OCT-2023

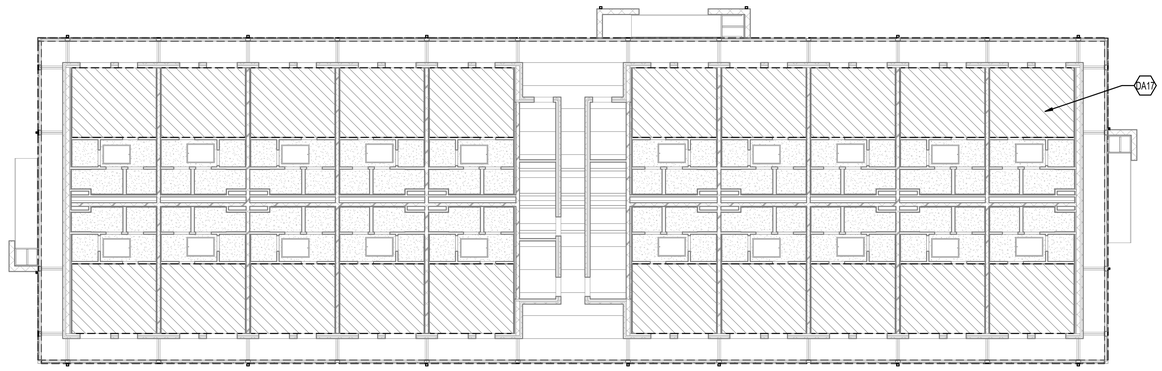
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

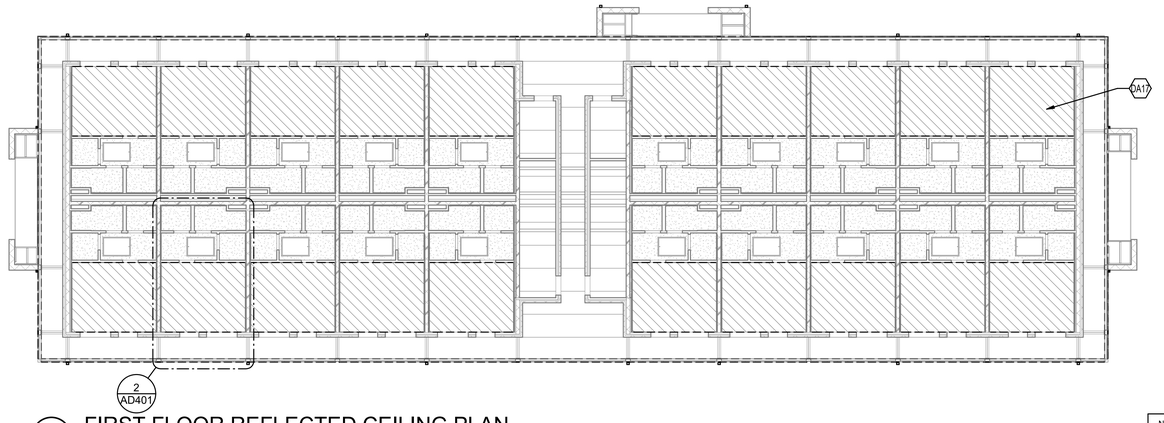
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28477 | CEMS | AD102B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY<br>NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G<br>(BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | DEMOLITION ATTIC AND ROOF PLANS - BLDG 1245G | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F / CODE IDENT. NO. / N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: As indicated / MACC. / T.O. # | SHEET 23 OF 53 |




1 THIRD FLOOR REFLECTED CEILING PLAN
SCALE: 3/32" = 1'-0"




2 SECOND FLOOR REFLECTED CEILING PLAN
SCALE: 3/32" = 1'-0"




3 FIRST FLOOR REFLECTED CEILING PLAN
SCALE: 3/32" = 1'-0"

PLAN NORTH

24"0 4' 8' 16'

SCALE: 3/32"=1'-0"

GRAPHIC SCALE

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. TYPICAL IN ALL LIVING/SLEEPING ROOMS IN ALL SLEEPING UNITS: STRIP TEXTURED CEILINGS ON UNDERSIDE OF EXISTING PRECAST CONCRETE AND PREPARE SURFACE FOR REPAINTING; PATCH, REPAIR, AND PREP ALL GYPSUM BOARD CEILINGS TO A LEVEL 4 SURFACE FOR REPAINTING.
2. REMOVE AND REPLACE TYPICAL IN ALL ROOMS: ALL SPRINKLER HEAD ESCUTCHEONS AND ALL SPRINKLER HEADS WITH NEW QUICK RESPONSE HEADS OF SAME TYPE AND TEMPERATURE.

## SHEET KEYNOTES

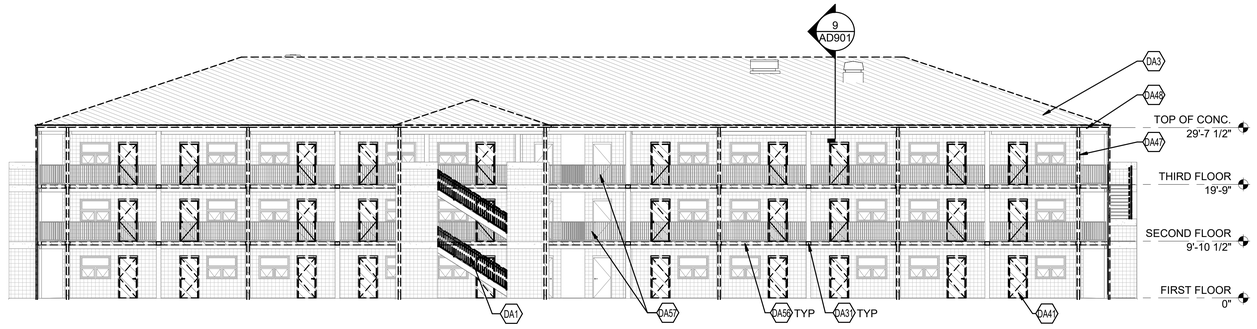
DA17 EXISTING TEXTURE COATED CEILING - REMOVE TEXTURE COATING TO STRUCTURAL HOLLOW CORE PRECAST SLAB SURFACE, PREPARE PRECAST SURFACE FOR NEW PAINT.

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

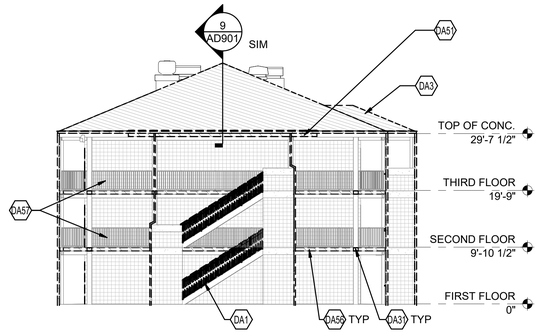
WISCONSIN
DEAN C. VON TERSCH
A-7903
CHARLESTON SC
ARCHITECT
04-OCT-2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28478 | CEMS | AD103B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G<br>(BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | DEMOLITION FIRST, SECOND AND THIRD FLOOR REFLECTED CEILING PLANS - BLDG 1245G | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: As indicated; MACC.; T.O. #; SHEET 24 OF 53 | |



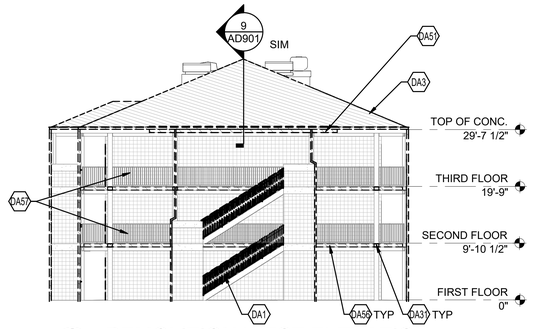

1 AD101B — 1 DEMOLITION NORTH ELEVATION

SCALE: 3/32" = 1'-0"



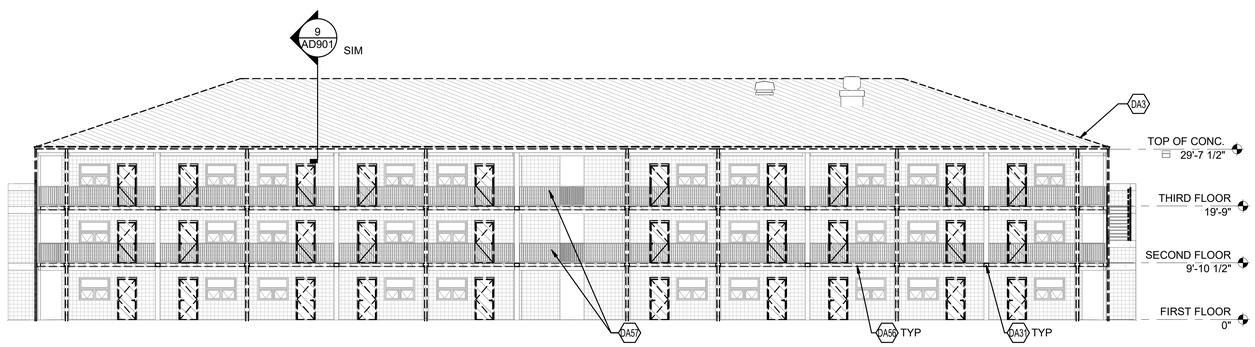

1 AD101B — 2 DEMOLITION EAST ELEVATION

SCALE: 3/32" = 1'-0"




1 AD101B — 3 DEMOLITION WEST ELEVATION

SCALE: 3/32" = 1'-0"




1 AD101B — 4 DEMOLITION SOUTH ELEVATION

SCALE: 3/32" = 1'-0"

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## SHEET KEYNOTES

| | |
|---|---|
| DA1 | EXISTING GUARDRAIL AND HANDRAIL - DEMOLISH. STAIR AND STRINGER - ETR. |
| DA3 | EXISTING ASPHALT SHINGLE ROOFING, UNDERLAYMENT, AND EXISTING PLYWOOD SHEATHING - DEMOLISH DOWN TO ETR METAL DECKING, TYP. |
| DA31 | AT ELEVATED WALKWAY SLAB LOCATIONS, PORTION OF EXISTING WALKWAY IN FRONT OF STEEL COLUMNS TO BE DEMOLISHED, TYPICAL ALL COLUMN LOCATIONS. SEE STRUCTURAL FOR ADDITIONAL REQUIREMENTS. |
| DA41 | EXISTING DOOR PANEL - DEMOLISH, FRAME - ETR, EXISTING CLOSER - DEMOLISH, THRESHOLD AND WEATHER SEALS - DEMOLISH. ALL OTHER DOOR HARDWARE TO BE RETAINED FOR REINSTALLATION. |
| DA47 | EXISTING DOWNSPOUT - DEMOLISH, TYP |
| DA48 | EXISTING GUTTERS - DEMOLISH, TYP |
| DA51 | AT NEW ROOFS OVER END STAIRS, EXISTING SHEET METAL FASCIA TO BE DEMOLISHED. PREPARE THESE AREAS FOR NEW SOFFIT AT NEW ROOF AREAS. |
| DA56 | EXISTING BENT STEEL PLATE AT 2ND AND 3RD FLOOR ELEVATED WALKWAY - DEMOLISH, SEE STRUCTURAL. |
| DA57 | ALL EXISTING GUARDRAILS AT EDGE OF ELEVATED WALKWAYS TO REMAIN. |

24"0 4' 8' 16'

**SCALE: 3/32"=1'-0"**

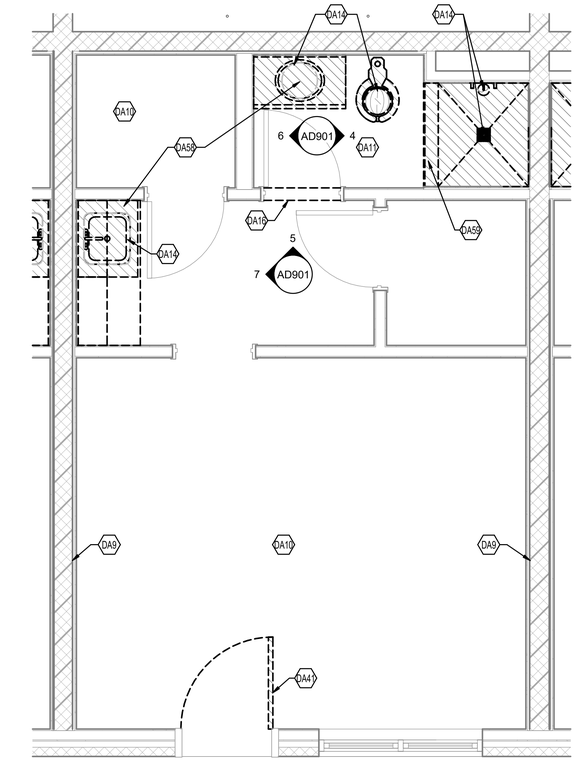
**GRAPHIC SCALE**

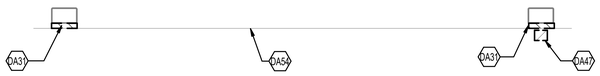
WISCONSIN
DEAN C. VON TERSCH
E-7103
CHARLESTON
SC
ARCHITECT
04-OCT-2023

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

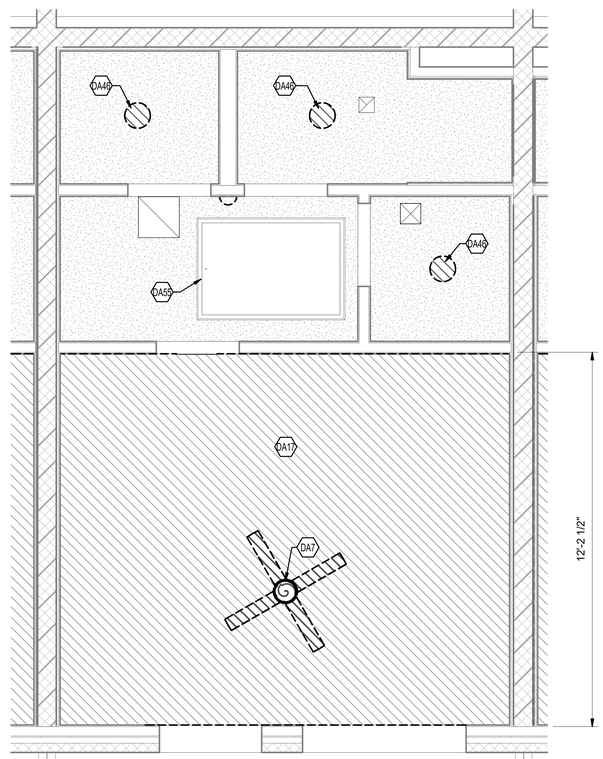
ISSUE DATE: 08 SEPT 2023

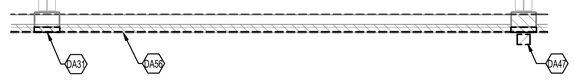
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28479 | CEMS | AD201B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>**MARINE CORPS AIR STATION**<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | DEMOLITION EXTERIOR ELEVATIONS - BLDG 1245G | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: 3/32" = 1'-0" \| MACC. \| T.O. # | SHEET 25 OF 53 |




1 ENLARGED DEMOLITION SLEEPING ROOM PLAN
SCALE: 1/2" = 1'-0"




2 ENLARGED DEMOLITION SLEEPING ROOM CEILING PLAN
SCALE: 1/2" = 1'-0"

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. PROTECT WINDOWS, MIRRORS, ETC. DURING DEMOLITION AND NEW WORK. REPLACE ANY BROKEN ITEM IN KIND SHOULD DAMAGE OCCUR DURING CONSTRUCTION.
2. WORK INDICATED ON THIS SHEET IS TYPICAL FOR ALL SLEEPING ROOMS.
3. TYPICAL IN ALL LIVING/SLEEPING ROOMS IN ALL SLEEPING UNITS: STRIP TEXTURED CEILINGS ON UNDERSIDE OF EXISTING PRECAST CONCRETE AND PREPARE SURFACE FOR REPAINTING; PATCH, REPAIR, AND PREP ALL GYPSUM BOARD CEILINGS TO A LEVEL 4 SURFACE FOR REPAINTING.
4. REMOVE AND REPLACE TYPICAL IN ALL ROOMS: ALL SPRINKLER HEAD ESCUTCHEONS AND ALL SPRINKLER HEADS WITH NEW QUICK RESPONSE HEADS OF SAME TYPE AND TEMPERATURE.

## SHEET KEYNOTES

| | |
|---|---|
| DA7 | EXISTING CEILING FAN - DEMOLISH. |
| DA9 | EXISTING TACKBOARD STRIP - REMOVE AND STORE FOR REINSTALLATION AFTER ROOM REFINISHING. |
| DA10 | EXISTING VCT FLOOR FINISH AND RUBBER BASE - DEMOLISH, CLEAN AND PREPARE EXISTING CONCRETE SLAB AND WALL FOR NEW FINISHES. |
| DA11 | EXISTING TILE FLOOR AND TILE BASE - DEMOLISH, CLEAN AND PREPARE EXISTING CONCRETE SLAB AND WALL FOR NEW FINISHES. |
| DA14 | EXISTING PLUMBING FIXTURE - DEMOLISH |
| DA16 | EXISTING MARBLE THRESHOLD - DEMOLISH |
| DA17 | EXISTING TEXTURE COATED CEILING - REMOVE TEXTURE COATING TO STRUCTURAL HOLLOW CORE PRECAST SLAB SURFACE, PREPARE PRECAST SURFACE FOR NEW PAINT. |
| DA31 | AT ELEVATED WALKWAY SLAB LOCATIONS, PORTION OF EXISTING WALKWAY IN FRONT OF STEEL COLUMNS TO BE DEMOLISHED, TYPICAL ALL COLUMN LOCATIONS. SEE STRUCTURAL FOR ADDITIONAL REQUIREMENTS. |
| DA41 | EXISTING DOOR PANEL - DEMOLISH, FRAME - ETR, EXISTING CLOSER - DEMOLISH, THRESHOLD AND WEATHER SEALS - DEMOLISH. ALL OTHER DOOR HARDWARE TO BE RETAINED FOR REINSTALLATION. |
| DA46 | EXISTING LIGHT FIXTURE - DEMOLISH |
| DA47 | EXISTING DOWNSPOUT - DEMOLISH, TYP |
| DA54 | AT SLAB ON GRADE LOCATIONS, SLAB ON GRADE IS ETR. SEE STRUCTURAL. |
| DA55 | ACCESS PANEL - ETR |
| DA56 | EXISTING BENT STEEL PLATE AT 2ND AND 3RD FLOOR ELEVATED WALKWAY - DEMOLISH, SEE STRUCTURAL. |
| DA58 | DEMOLISH EXISTING CASEWORK AND COUNTERTOP AND PREP AREA FOR NEW CASEWORK AND COUNTERTOP. |
| DA59 | DEMOLISH EXISTING SHOWER CURB AND PREP AREA FOR NEW SHOWER CURB. |

6" 0 1' 2' 4'

**SCALE: 1/2"=1'-0"**

**GRAPHIC SCALE**

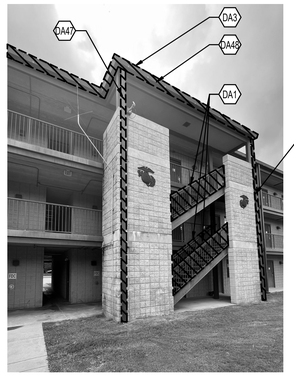
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

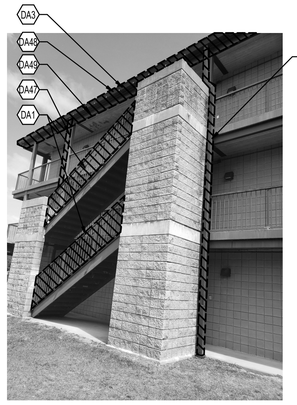
WISCONSIN — DEAN C. VON TERSCH — E-17503 — CHARLESTON SC — ARCHITECT — 04-OCT-2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28480 | CEMS | AD401 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>**MARINE CORPS AIR STATION**<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | TYPICAL DEMOLITION ENLARGED SLEEPING ROOM PLANS | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F — CODE IDENT. NO. — N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: As indicated — MACC. — T.O. # | SHEET 26 OF 53 |

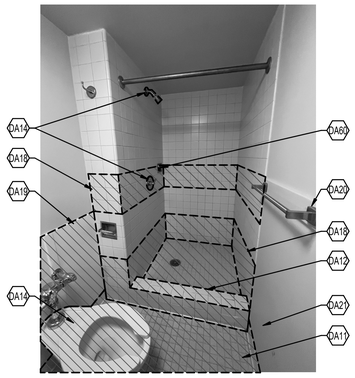
1 DEMO PHOTO 1
SCALE: NOT TO SCALE



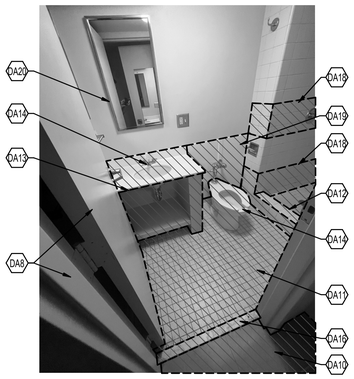

2 DEMO PHOTO 2
SCALE: NOT TO SCALE



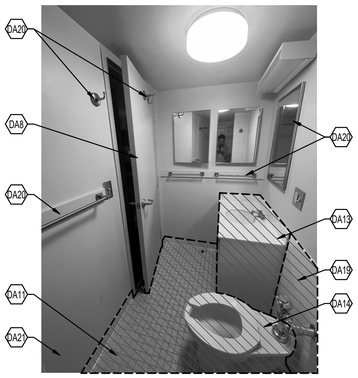

3 DEMO PHOTO 3
SCALE: NOT TO SCALE



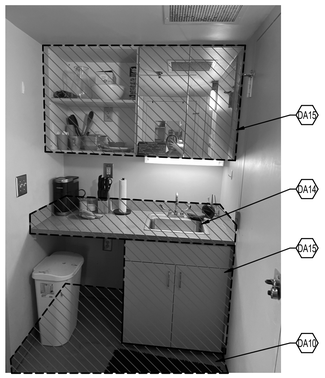

4 DEMO PHOTO 4
SCALE: NOT TO SCALE




5 DEMO PHOTO 5
SCALE: NOT TO SCALE

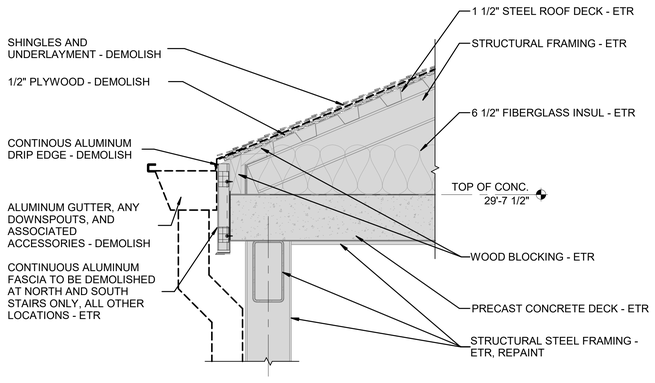



6 DEMO PHOTO 6
SCALE: NOT TO SCALE




7 DEMO PHOTO 7
SCALE: NOT TO SCALE

8 DEMO PHOTO 8
SCALE: NOT TO SCALE




9 TYPICAL DEMOLITION EAVE DETAIL
SCALE: 1 1/2" = 1'-0"

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. EXISTING PHOTOS INDICATE GENERAL SCOPE OF WORK. EXISTING CONDITIONS MAY VARY. CONTRACTOR TO FIELD VERIFY EXISTING CONDITIONS PRIOR TO COMMENCING WORK.

## SHEET KEYNOTES

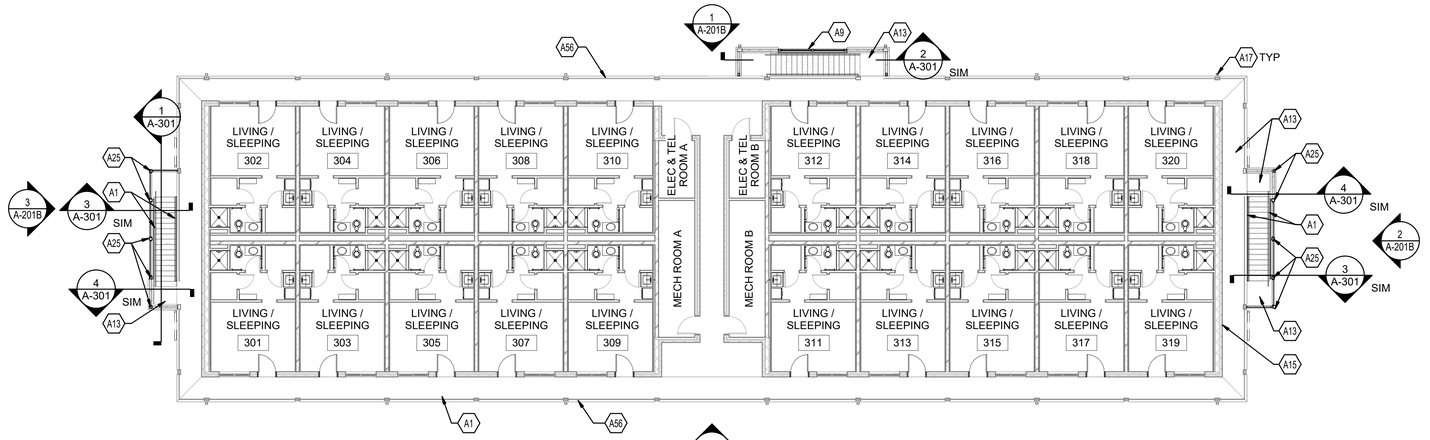
| | |
|---|---|
| DA1 | EXISTING GUARDRAIL AND HANDRAIL - DEMOLISH. STAIR AND STRINGER - ETR. |
| DA2 | WATER DAMAGED GYP BD IN MECHANICAL ROOM - DEMOLISH TO EXTENT REQUIRED. APPROXIMATELY 4'-0" HIGH X 36'-8" LENGTH PER ROOM AND TREAT BALANCE OF ROOM FOR MOLD AND PREPARE FOR NEW GYP BD. |
| DA3 | EXISTING ASPHALT SHINGLE ROOFING, UNDERLAYMENT, AND EXISTING PLYWOOD SHEATHING - DEMOLISH DOWN TO ETR METAL DECKING, TYP. |
| DA8 | EXISTING DOOR PANEL AND FRAME - CLEAN AND PREPARE FOR NEW PAINT. |
| DA10 | EXISTING VCT FLOOR FINISH AND RUBBER BASE - DEMOLISH, CLEAN AND PREPARE EXISTING CONCRETE SLAB AND WALL FOR NEW FINISHES. |
| DA11 | EXISTING TILE FLOOR AND TILE BASE - DEMOLISH, CLEAN AND PREPARE EXISTING CONCRETE SLAB AND WALL FOR NEW FINISHES. |
| DA12 | EXISTING TILE SHOWER FLOOR FINISH, GROUT BASE, AND CURB - DEMOLISH, CLEAN AND PREPARE CONCRETE SLAB FOR NEW TILE FINISHES, BASE, AND CURB. |
| DA13 | EXISTING BASE CABINET, COUNTERTOP, AND BACKSPLASH - DEMOLISH |
| DA14 | EXISTING PLUMBING FIXTURE - DEMOLISH |
| DA15 | EXISTING WALL MOUNTED UPPER CABINETS, BASE CABINETS, COUNTERTOP, END SPLASHES, AND BACKSPLASH - DEMOLISH |
| DA16 | EXISTING MARBLE THRESHOLD - DEMOLISH |
| DA18 | PORTION OF EXISTING WALL TILE INDICATED - DEMOLISH. PREPARE SURFACE FOR NEW WALL TILE. IF WATER DAMAGE DISCOVERED, REMEDIATE AND REPLACE TO EXTENT REQUIRED. |
| DA19 | EXISTING SOLID SURFACE AND GYPSUM BACKING - DEMOLISH. PREPARE STUD FRAMING FOR NEW CEMENTITIOUS BACKER. |
| DA20 | TOWEL BARS, TOILET ACCESSORIES, AND MIRRORS - REMOVE DOWN TO MOUNTING BRACKET PRIOR TO PAINTING, RETAIN FOR REINSTALLATION IN NEW WORK PHASE. |
| DA21 | CONTRACTOR TO REPLACE NOT LESS THAN 16 SF OF WATER DAMAGED GYP BD AT CERAMIC WALL BASE LOCATIONS IN EACH BATHROOM. CONTRACTOR TO FIELD VERIFY QUANTITY AND LOCATION IN EACH BATHROOM. |
| DA47 | EXISTING DOWNSPOUT - DEMOLISH, TYP |
| DA48 | EXISTING GUTTERS - DEMOLISH, TYP |
| DA49 | EXISTING METAL FASCIA - DEMOLISH AT NORTH AND SOUTH STAIRS ONLY. |
| DA60 | RECESSED SHOWER SOAP DISH - REMOVE, STORE, AND REINSTALL AS REQUIRED TO ACCOMODATE WALL TILE REPLACEMENT. |

WISCONSIN
DEAN C. VON TERSCH
#A-7903
CHARLESTON SC
ARCHITECT
04-OCT-2023

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

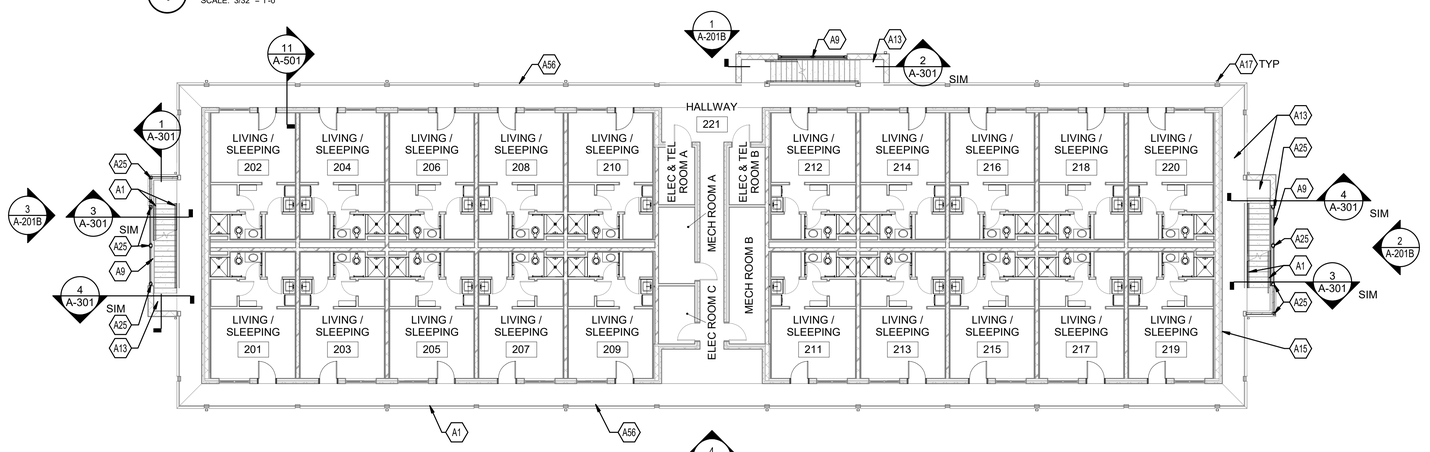
ISSUE DATE: 08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28481 | CEMS | AD901 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | NAVAL FACILITIES ENGINEERING COMMAND |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | DEMOLITION PHOTOS AND DETAIL | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-18-D-0616 |
| PUBLIC WORKS OFFICER | | SCALE: As indicated; MACC.; T.O. #; SHEET 27 OF 53 | |



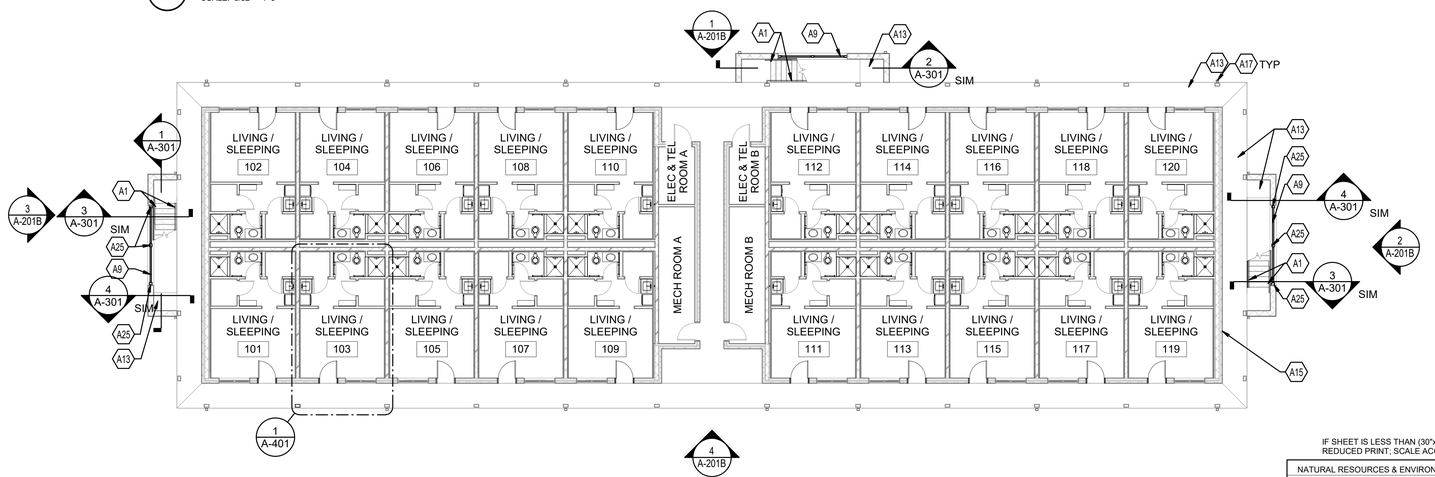

## 1 THIRD FLOOR PLAN

SCALE: 3/32" = 1'-0"




## 2 SECOND FLOOR PLAN

SCALE: 3/32" = 1'-0"




## 3 FIRST FLOOR PLAN

SCALE: 3/32" = 1'-0"

PLAN NORTH

24"0 4' 8' 16'

**SCALE: 3/32"=1'-0"**

**GRAPHIC SCALE**

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. COORDINATE WITH DEMOLITION AND CONSTRUCTION PLANS AND NOTATIONS ON ALL SHEETS.
2. SEE ENLARGED PLANS AS FOR SPECIFIC CONSTRUCTION SCOPE OF WORK TO BE COMPLETED IN ALL SLEEPING ROOM SUITES ON ALL FLOORS.
3. PAINT ALL EXTERIOR STEEL PER LEGEND OF COLORS AND FINISHES ON INTERIORS SHEETS. INCLUDING BUT NOT LIMITED TO COLUMNS, BEAMS, CONNECTIONS AND OTHER EXPOSED STEEL FRAMING ELEMENTS.
4. INSTALL TROWEL GRADE CONCRETE TOPPING MIX AND FINISH TO MATCH EXISTING TEXTURE WHERE REQUIRED TO CREATE POSITIVE SLOPE AWAY FROM LANDINGS, EXTERIOR WALLS AND STEEL COLUMNS.
5. NEW URETHANE COATING ON ALL EXPOSED ELEVATED WALKWAY FLOORS, LANDINGS AND FACE OF ELEVATED WALKWAYS WHERE BENT PLATE WAS REMOVED.
6. IN ALL SLEEPING ROOM SUITES, REPLACE THE FOLLOWING: SHOWER MIXING VALVE, SHOWER HEAD, TOILETS, VANITY CABINET, VANITY FAUCET, VANITY SINK BOWL AND TOP, KITCHENETTE SINK, KITCHENETTE FAUCET, KITCHENETTE BASE, WALL CABINETS, AND COUNTERTOP. SEE PLUMBING DRAWINGS FOR ADDITIONAL REQUIREMENTS.
7. ALL SPRINKLER PIPING MAINS AND BRANCH LINES ARE TO REMAIN.

## SHEET KEYNOTES

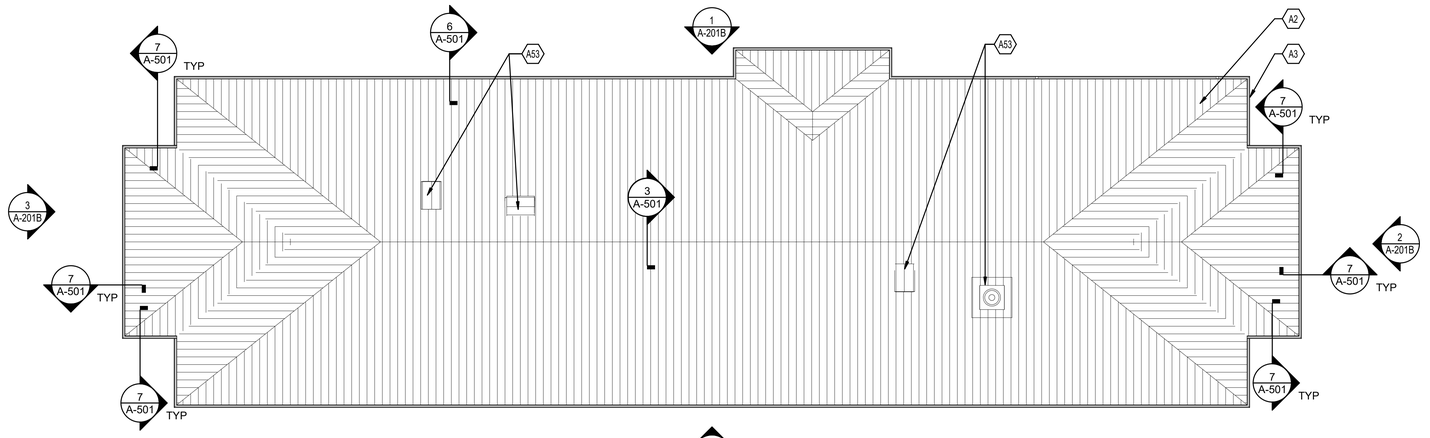
- A1 PREFINISHED ALUMINUM GUARDRAIL AND HANDRAIL SYSTEM
- A9 PREFINISHED ALUMINUM LOUVER RAIN SCREEN SYSTEM.
- A13 APPLY NEW UNDERLAYMENT TO CREATE POSITIVE FLOW AWAY FROM EXTERIOR WALLS, LANDINGS, AND STRUCTURAL COLUMNS. SEE ELEVATED WALKWAY UNDERLAYMENT GENERAL NOTES.
- A15 CONTINUOUS SEALANT ALONG BASE OF EXISTING EXTERIOR WALL AND CONCRETE AROUND ENTIRE BUILDING. SEE TYPICAL CONTINUANT SEALANT RECESS DETAIL.
- A17 NEW 4"x5" PREFINISHED ALUMINUM DOWNSPOUT TIED INTO NEW CAST IRON BOOT, WITH CLEANOUT, AND CONNECTED TO REWORKED UNDERGROUND STORM SEWER SYSTEM. NEW DOWNSPOUTS AND BOOTS TO BE IN THE SAME LOCATION AS EXISTING DOWNSPOUTS. NEW BOOTS TO BE FACTORY PRIME PAINTED AND FIELD PAINTED TO MATCH DOWNSPOUT COLOR. SEE CIVIL FOR STORM SEWER PIPING REQUIREMENTS.
- A25 STRUCTURAL COLUMN
- A56 ALL EXISTING GUARDRAILS AT EDGE OF ELEVATED WALKWAYS TO REMAIN.

WISCONSIN DEAN C. VON TERSCH A-7903 CHARLESTON SC ARCHITECT 04-OCT-2023

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

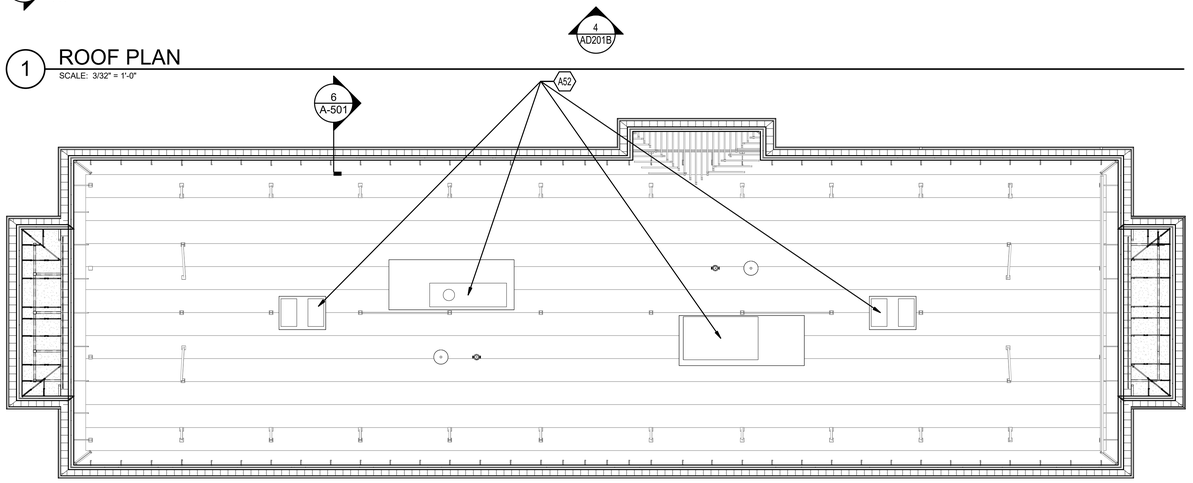
ISSUE DATE: 08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28482 | CEMS | A-101B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | FIRST, SECOND AND THIRD FLOOR PLANS - BLDG 1245G | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: As indicated; MACC.; T.O. #; SHEET 28 OF 53 | |




1 ROOF PLAN

SCALE: 3/32" = 1'-0"




2 ATTIC PLAN

SCALE: 3/32" = 1'-0"

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |

## GENERAL NOTES

1. COORDINATE WITH DEMOLITION AND CONSTRUCTION PLANS AND NOTATIONS ON ALL SHEETS.
2. SEE ENLARGED CONSTRUCTION PLANS AS DENOTED FOR SPECIFIC CONSTRUCTION SCOPE OF WORK AT INDICATED AREAS.
3. CLEAN AND PREPARE ALL EXISTING TO REMAIN STEEL FOR NEW PAINT. INCLUDING BUT NOT LIMITED TO COLUMNS, BEAMS, CONNECTIONS AND OTHER EXPOSED STEEL FRAMING ELEMENTS.

## SHEET KEYNOTES

| | |
|---|---|
| A2 | STRUCTURAL STANDING SEAM ALUMINUM ROOF |
| A3 | PREFINISHED ALUMINUM GUTTER, TYP |
| A52 | MECHANICAL EQUIPMENT. SEE MECHANICAL. |
| A53 | EXISTING ROOF PENETRATION, MODIFY EXISTING ROOF CURB PER NEW STANDING SEAM METAL ROOF REQUIREMENTS. |

PLAN NORTH

24"0 4' 8' 16'

SCALE: 3/32"=1'-0"

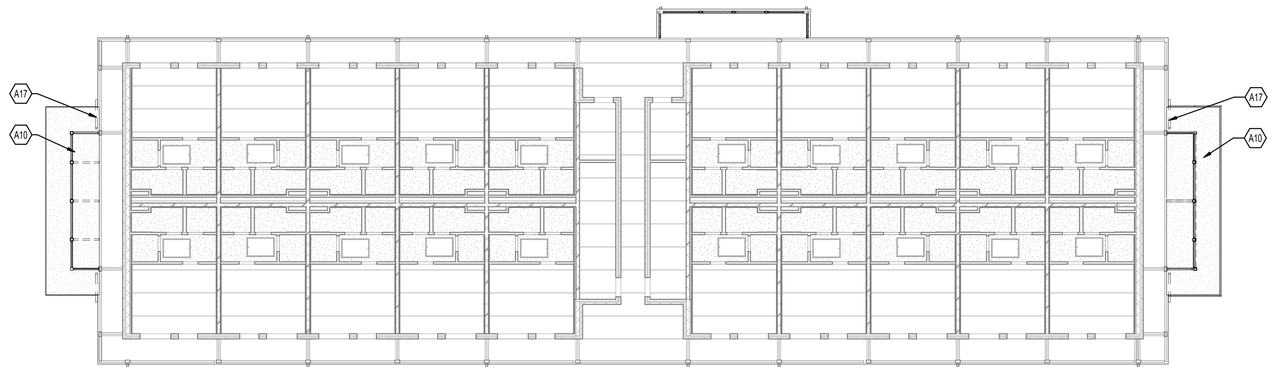
GRAPHIC SCALE

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

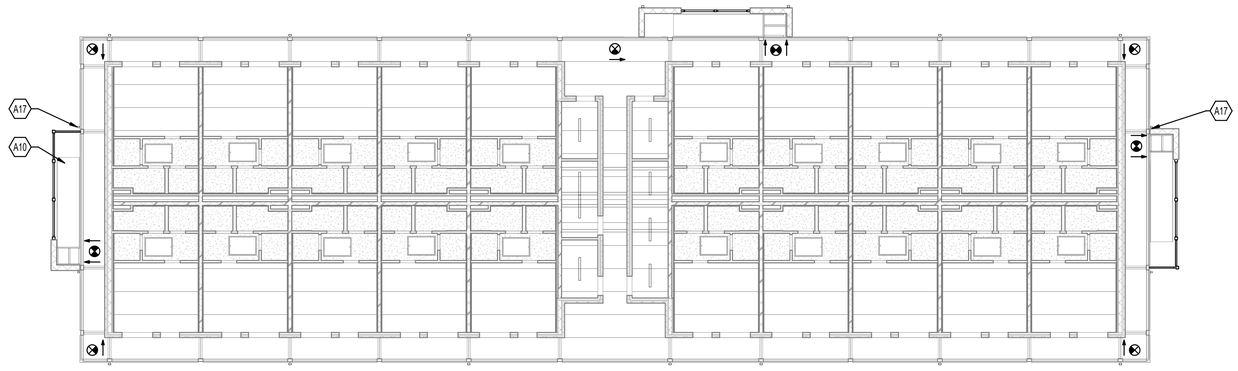
ISSUE DATE: 08 SEPT 2023

WISCONSIN
DEAN C. VON TERSCH
A-7903
CHARLESTON SC
ARCHITECT
04-OCT-2023

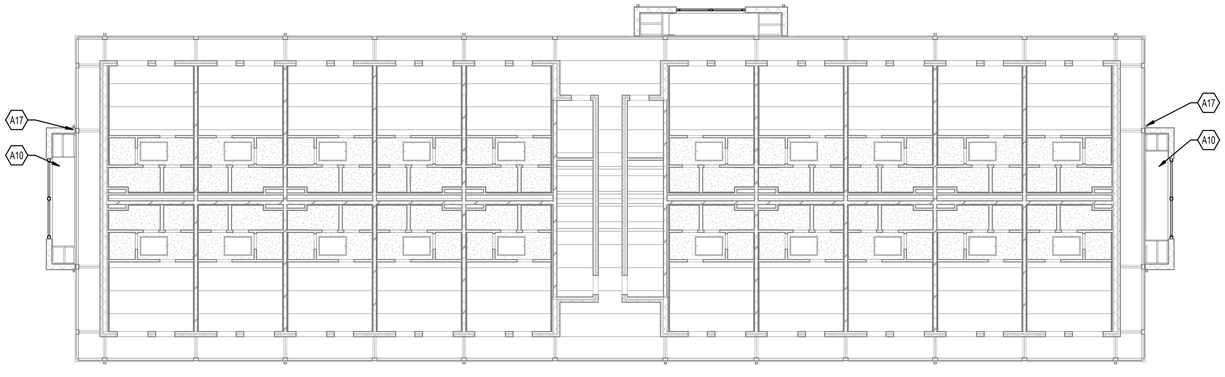
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28483 | CEMS | A-102B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY<br>NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G<br>(BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | ATTIC AND ROOF PLANS - BLDG 1245G | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F / CODE IDENT. NO. / N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: 3/32" = 1'-0" / MACC. / T.O. # | SHEET 29 OF 53 |




1 THIRD FLOOR REFLECTED CEILING PLAN
SCALE: 3/32" = 1'-0"




2 SECOND FLOOR REFLECTED CEILING PLAN
SCALE: 3/32" = 1'-0"




3 FIRST FLOOR REFLECTED CEILING PLAN
SCALE: 3/32" = 1'-0"

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. COORDINATE WITH DEMOLITION AND CONSTRUCTION PLANS AND NOTATIONS ON ALL SHEETS.
2. SEE ENLARGED PLANS AS FOR SPECIFIC CONSTRUCTION SCOPE OF WORK TO BE COMPLETED IN ALL SLEEPING ROOM SUITES ON ALL FLOORS.
3. PAINT ALL EXTERIOR STEEL PER LEGEND OF COLORS AND FINISHES ON INTERIORS SHEETS. INCLUDING BUT NOT LIMITED TO COLUMNS, BEAMS, CONNECTIONS AND OTHER EXPOSED STEEL FRAMING ELEMENTS.
4. REMOVE AND REPLACE TYPICAL IN ALL ROOMS: ALL SPRINKLER HEAD ESCUTCHEONS AND ALL SPRINKLER HEADS WITH NEW QUICK RESPONSE HEADS OF SAME TYPE AND TEMPERATURE.
5. PROVIDE 6 SPARE ESCUTCHEONS AND 6 SPARE SPRINKLER HEADS OF EACH TYPE AND PLACE IN EXISTING CABINET NEAR RISER.

## SHEET KEYNOTES

A10 5/8" WR PAINTED GYP BD SOFFIT. MATCH TEXTURE AND FINISH ON EXISTING PRECAST CONCRETE CENTER STAIR SOFFIT.

A17 NEW 4"x5" PREFINISHED ALUMINUM DOWNSPOUT TIED INTO NEW CAST IRON BOOT, WITH CLEANOUT, AND CONNECTED TO REWORKED UNDERGROUND STORM SEWER SYSTEM. NEW DOWNSPOUTS AND BOOTS TO BE IN THE SAME LOCATION AS EXISTING DOWNSPOUTS. NEW BOOTS TO BE FACTORY PRIME PAINTED AND FIELD PAINTED TO MATCH DOWNSPOUT COLOR. SEE CIVIL FOR STORM SEWER PIPING REQUIREMENTS.

PLAN NORTH

24"0 4' 8' 16'

SCALE: 3/32"=1'-0"

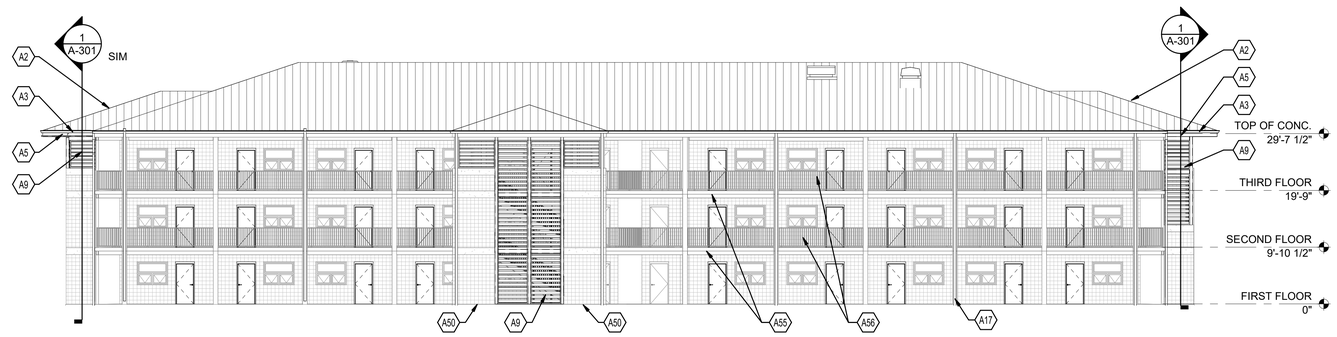
GRAPHIC SCALE

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

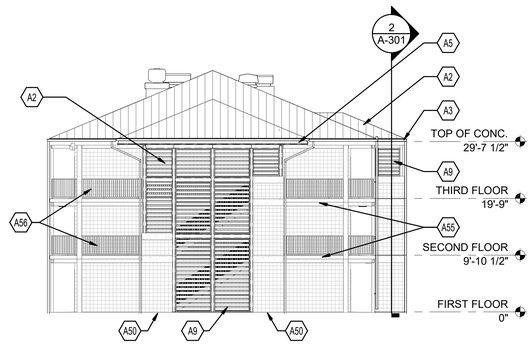
WISCONSIN DEAN C. VON TERSCH A-7903 CHARLESTON SC ARCHITECT 04-OCT-2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28484 | CEMS | A-103B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | FIRST, SECOND AND THIRD FLOOR REFLECTED CEILING PLANS - BLDG 1245G | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: As indicated; MACC.; T.O. # | SHEET 30 OF 53 |



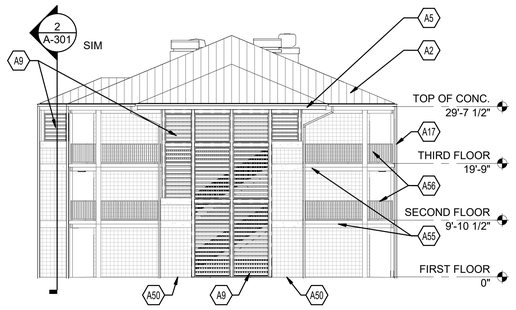

1 A-101B **1** NORTH ELEVATION

SCALE: 3/32" = 1'-0"



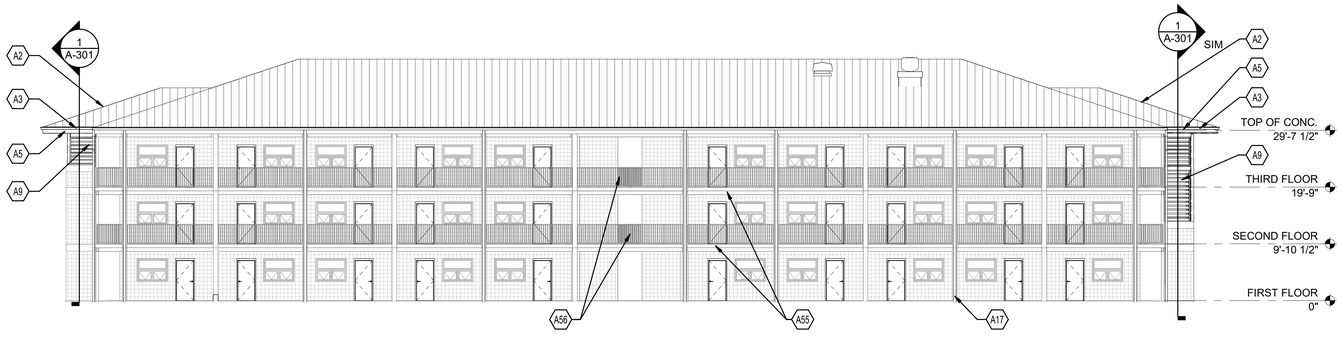

1 A-101B **2** EAST ELEVATION

SCALE: 3/32" = 1'-0"




1 A-101B **3** WEST ELEVATION

SCALE: 3/32" = 1'-0"




1 A-101B **4** SOUTH ELEVATION

SCALE: 3/32" = 1'-0"

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. PRESSURE WASH AND CLEAN ALL EXTERIOR SURFACES AT THE COMPLETION OF CONSTRUCTION.
2. ALL EXTERIOR STRUCTURAL STEEL TO BE PAINTED.

## SHEET KEYNOTES

A2 STRUCTURAL STANDING SEAM ALUMINUM ROOF

A3 PREFINISHED ALUMINUM GUTTER, TYP

A5 ALUMINUM FASCIA. - ETR

A9 PREFINISHED ALUMINUM LOUVER RAIN SCREEN SYSTEM.

A17 NEW 4"x5" PREFINISHED ALUMINUM DOWNSPOUT TIED INTO NEW CAST IRON BOOT, WITH CLEANOUT, AND CONNECTED TO REWORKED UNDERGROUND STORM SEWER SYSTEM. NEW DOWNSPOUTS AND BOOTS TO BE IN THE SAME LOCATION AS EXISTING DOWNSPOUTS. NEW BOOTS TO BE FACTORY PRIME PAINTED AND FIELD PAINTED TO MATCH DOWNSPOUT COLOR. SEE CIVIL FOR STORM SEWER PIPING REQUIREMENTS.

A50 CONSTRUCT DRAINAGE SCUPPER. SAW CUT WITH A CONCRETE CUTTING CHAIN SAW 4" HIGH X 4" WIDE FULL DEPTH OPENING AT HALFWAY POINT OF WALL. BOTTOM OF SAW CUT TO MATCH TOP OF CONCRETE WALK ON BACKSIDE OF CMU WALL. INFILL WITH MORTAR CMU OPENINGS OPEN CORES AT ALL FOUR SIDES OF OPENING TO ENSURE WATER DRAINAGE FROM BACKSIDE OF WALL TO FRONT SIDE OF WALL. SEE SAWCUT CMU OPENING DETAIL.

A55 STRUCTURAL CONCRETE PATCH, SKIM COAT, AND URETHANE FINISH AT SLAB EDGE - TYPICAL ELEVATED SLAB EDGES.

A56 ALL EXISTING GUARDRAILS AT EDGE OF ELEVATED WALKWAYS TO REMAIN.

**24"0 4' 8' 16'**

**SCALE: 3/32"=1'-0"**

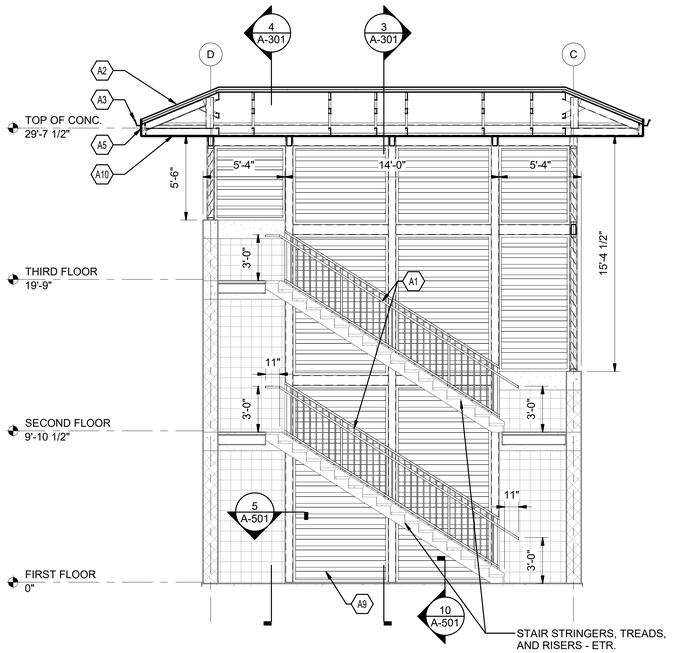
**GRAPHIC SCALE**

DEAN C. VON TERSCH A-7903 CHARLESTON SC WISCONSIN ARCHITECT 04-OCT-2023

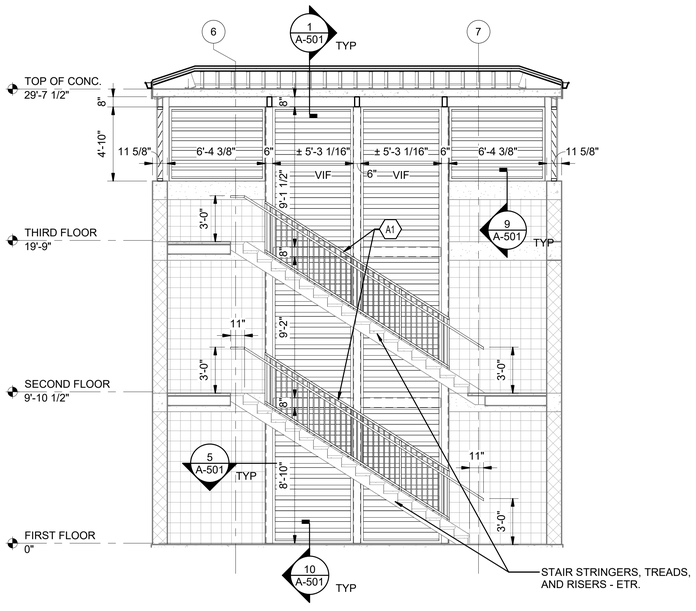
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

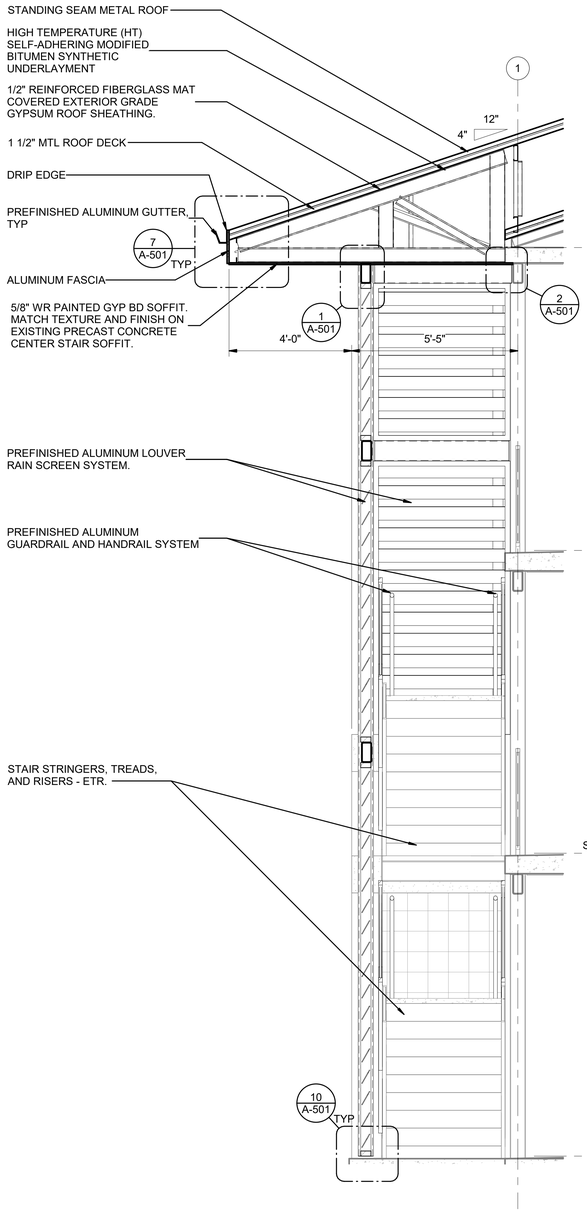
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28485 | CEMS | A-201B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M)<br>EXTERIOR ELEVATIONS - BLDG 1245G | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: As indicated; MACC.; T.O. #; SHEET 31 OF 53 | |



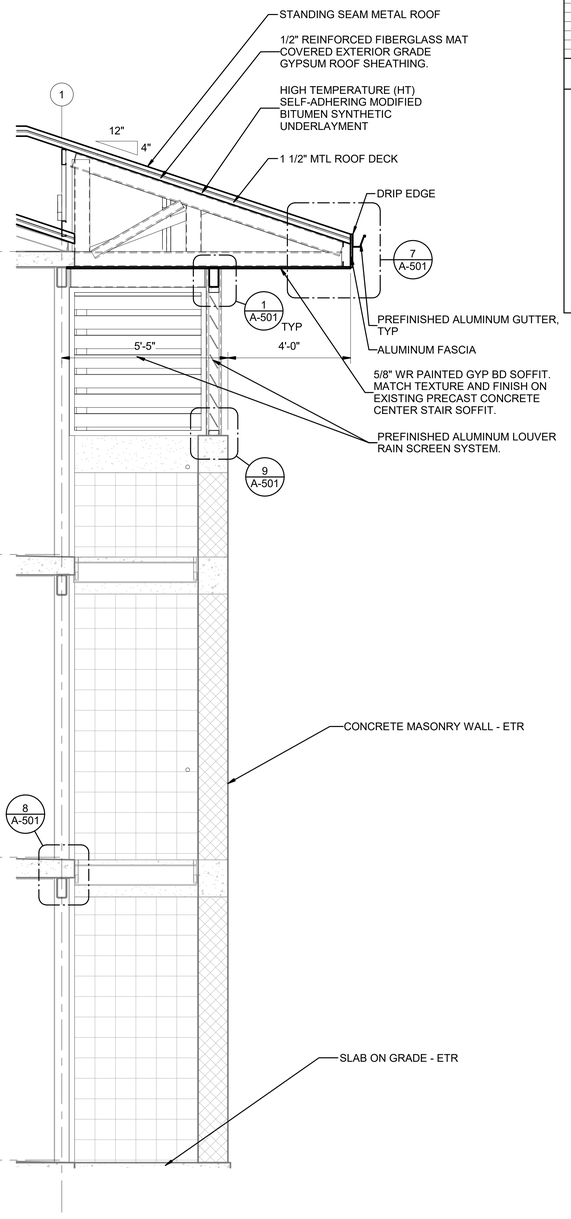

1 SECTION
SCALE: 1/4" = 1'-0"




2 SECTION
SCALE: 1/4" = 1'-0"




3 SECTION
SCALE: 1/2" = 1'-0"




4 SECTION
SCALE: 1/2" = 1'-0"

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

## SHEET KEYNOTES

| | |
|---|---|
| A1 | PREFINISHED ALUMINUM GUARDRAIL AND HANDRAIL SYSTEM |
| A2 | STRUCTURAL STANDING SEAM ALUMINUM ROOF |
| A3 | PREFINISHED ALUMINUM GUTTER, TYP |
| A5 | ALUMINUM FASCIA. - ETR |
| A9 | PREFINISHED ALUMINUM LOUVER RAIN SCREEN SYSTEM. |
| A10 | 5/8" WR PAINTED GYP BD SOFFIT. MATCH TEXTURE AND FINISH ON EXISTING PRECAST CONCRETE CENTER STAIR SOFFIT. |
| A44 | HIGH TEMPERATURE (HT) SELF-ADHERING MODIFIED BITUMEN SYNTHETIC UNDERLAYMENT |
| A49 | 1/2" REINFORCED FIBERGLASS MAT COVERED EXTERIOR GRADE GYPSUM ROOF SHEATHING. |

6" 0 1' 2' 4'
SCALE: 1/2"=1'-0"

12" 0 2' 4' 8'
SCALE: 1/4"=1'-0"

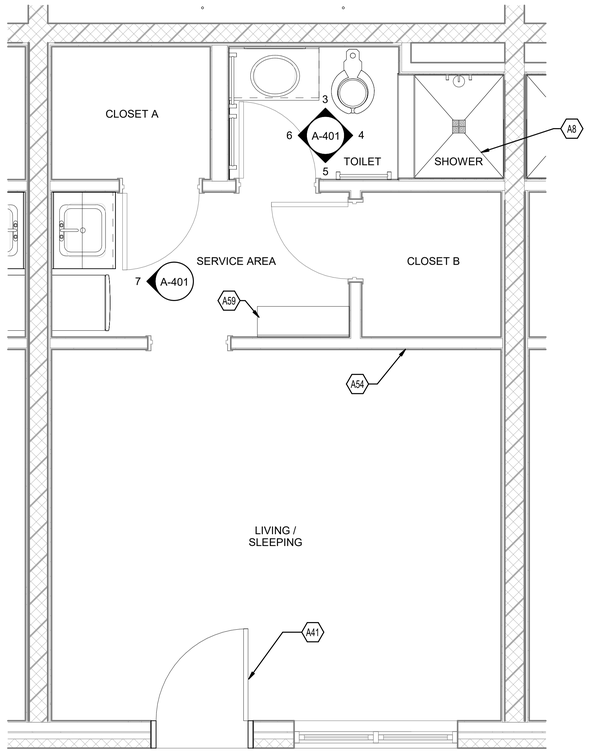
GRAPHIC SCALES

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

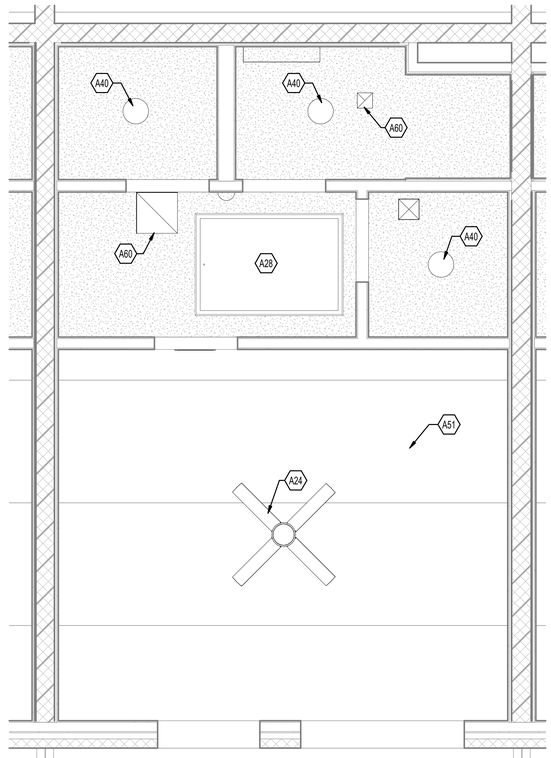
DEAN C. VON TERSCH
A-7903
CHARLESTON SC
WISCONSIN
ARCHITECT
04-OCT-2023

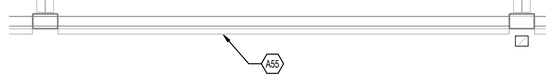
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28486 | CEMS | A-301 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY<br>NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M)<br>SECTIONS | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: As indicated; MACC.; T.O. #; SHEET 32 OF 53 | |



## 1 ENLARGED SLEEPING ROOM PLAN TYP

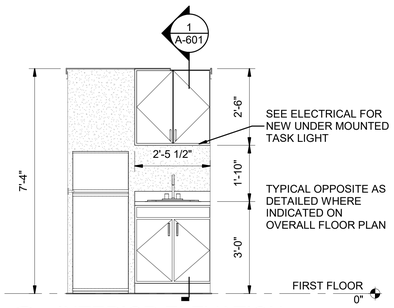
SCALE: 1/2" = 1'-0"



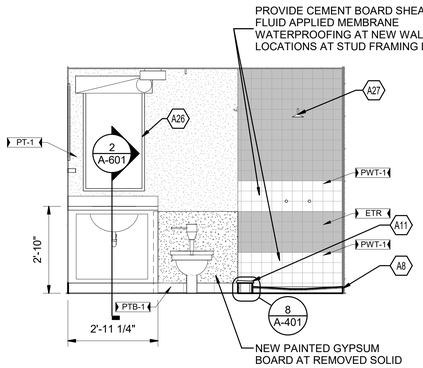

## 2 ENLARGED SLEEPING ROOM CEILING PLAN TYP

SCALE: 1/2" = 1'-0"



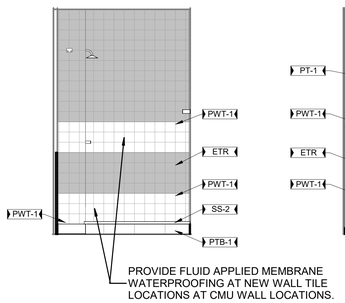

## 1/A-401 7 INTERIOR ELEVATION

SCALE: 1/2" = 1'-0"



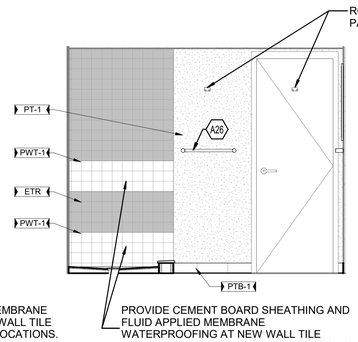

## 1/A-401 3 INTERIOR ELEVATION

SCALE: 1/2" = 1'-0"



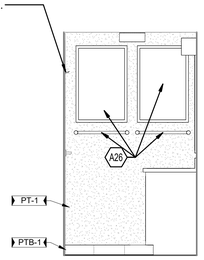

## 1/A-401 4 INTERIOR ELEVATION

SCALE: 1/2" = 1'-0"



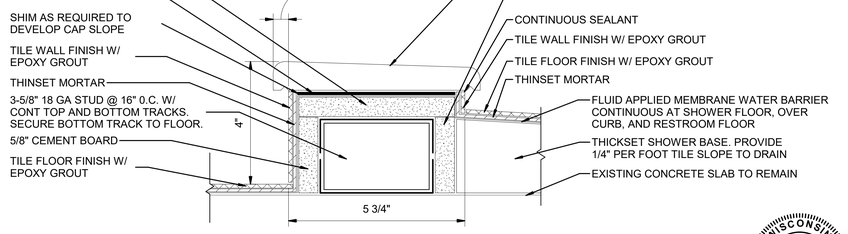

## 1/A-401 5 INTERIOR ELEVATION

SCALE: 1/2" = 1'-0"




## 1/A-401 6 INTERIOR ELEVATION

SCALE: 1/2" = 1'-0"




## 3/A-401 8 SHOWER CURB DETAIL

SCALE: NOT TO SCALE

3" 0 6" 1'

**SCALE: 3" = 1'-0"**

6" 0 1' 2' 4'

**SCALE: 1/2"=1'-0"**

**GRAPHIC SCALE**

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

# GENERAL NOTES

1. TYPICAL IN ALL SLEEPING ROOMS: CONTRACTOR TO TEMPORARILY REMOVE ALL WINDOW TREATMENTS, TAC STRIPS, TOILET ACCESSORIES, ETC. BEFORE PAINTING AND REINSTALL ALL AFTER PAINTING IS COMPLETE.
2. WORK INDICATED ON THIS SHEET IS TYPICAL FOR ALL SLEEPING ROOM SUITES.
3. ALL WINDOW TREATMENT, BLINDS, AND DRAPERIES ARE TO BE STORED AND REINSTALLED AFTER COMPLETION OF INTERIOR PAINTING.
4. UNLESS NOTED OTHERWISE TO BE REPLACED WITH NEW, AT ALL SLEEPING ROOM SUITES, ALL WALL AND CEILING MOUNTED ITEMS INCLUDING, BUT NOT LIMITED TO, CURTAINS, BLINDS, AND ASSOCIATED WINDOW TREATMENT HARDWARE, ELECTRICAL DEVICES, ELECTRICAL COVER PLATES, LOUVER TRIMS, WALL AND CEILING MOUNTED LIGHT FIXTURES, FIRE ALARM AND RELATED DEVICES, OCCUPANCY SENSORS, ALL RESTROOM ACCESSORIES AND FIXTURES, ALL WIRE SHELVING AND SUPPORT BRACKETS, AND SO FORTH, ARE TO BE REMOVED, STORED, AND REINSTALLED AFTER ROOM REFINISHING. AT FIXTURES AND ACCESSORIES THAT HAVE CONCEALED MOUNTING BRACKETS, THE CONCEALED MOUNTING BRACKETS CAN BE LEFT IN PLACE FOR REPAINTING. RESTROOM ACCESSORIES WITHIN TILED AREAS, CORNER GUARDS, EXPOSED ELECTRICAL RACEWAYS, WIREMOLD AND CONDUIT SERVING ELECTRICAL EQUIPMENT, FIRE ALARMS, PULL STATIONS, AND RELATED FIRE DEVICES IS EXISTING TO REMAIN UNLESS NOTED OTHERWISE.

# SHEET KEYNOTES

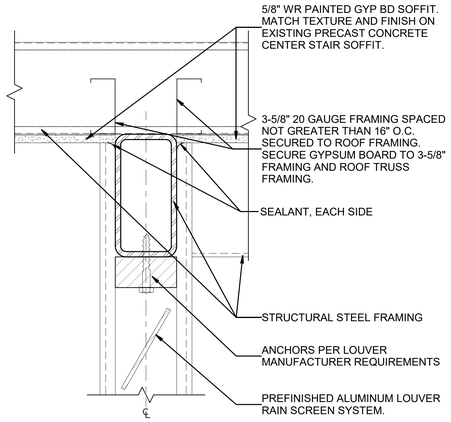
| | |
|---|---|
| A8 | NEW TILE FLOOR OVER FLUID APPLIED WATERPROOFING MEMBRANE ON NEW SLOPED GROUT BED. SLOPE 1/4" PER FOOT TO FLOOR DRAIN, COORDINATE WITH PLUMBING. |
| A11 | 3 5/8" METAL STUD CURB W/ TILE ON CEMENTITIOUS BACKER BOARD EACH SIDE AND SOLID SURFACE CAP. |
| A24 | CEILING FAN |
| A26 | REINSTALL MIRRORS AND TOWEL BAR |
| A27 | PLUMBING FIXTURE |
| A28 | EXISTING ACCESS PANEL - PAINT, PT-5 |
| A40 | SURFACE MOUNTED LIGHTING. |
| A41 | EXTERIOR HOLLOW METAL SLABS FOR DOORS |
| A51 | PAINT UNDERSIDE OF CONCRETE HOLLOW CORE PLANK CEILING. |
| A54 | TYPICAL ALL SLEEPING ROOMS: 12"x12" KEYED ACCESS PANEL FOR HVAC CONTROLS; CONTRACTOR TO FIELD VERIFY REQUIRED LOCATION; ACCESS PANEL TO BE PAINTED SURROUNDING WALL PAINT COLOR. |
| A55 | STRUCTURAL CONCRETE PATCH, SKIM COAT, AND URETHANE FINISH AT SLAB EDGE - TYPICAL ELEVATED SLAB EDGES. |
| A59 | 1'-0"x3'-0" COUNTERTOP RUN. |
| A60 | ALL NEW EXPOSED TO VIEW MECHANICAL DIFFUSERS, GRILLS, AND SIMILAR MECHANICAL EQUIPMENT TO BE MANUFACTURER'S STANDARD PREFINISHED WHITE. |

WISCONSIN
DEAN C. VON TERSCH
CA-7103
CHARLESTON SC
ARCHITECT
04-OCT-2023

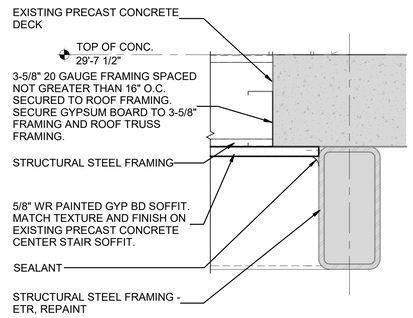
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

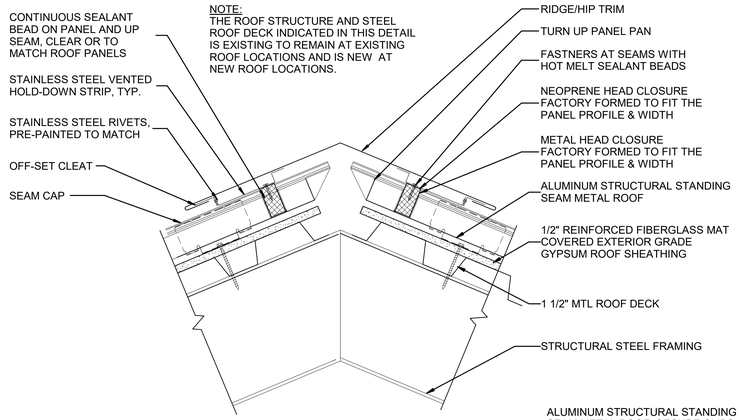
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28487 | CEMS | A-401 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY<br>NAVAL FACILITIES ENGINEERING COMMAND<br>**MARINE CORPS AIR STATION**<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G<br>(BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | ENLARGED SLEEPING ROOM PLANS | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F / CODE IDENT. NO. / N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: / CONSTR. CONTR. NO.: N40085-18-D-0616 | |
| | PUBLIC WORKS OFFICER | SCALE: As indicated / MACC. / T.O. # / SHEET 33 OF 53 | |




2 A-301 **1** LOUVER HEAD DETAIL
SCALE: 3" = 1'-0"

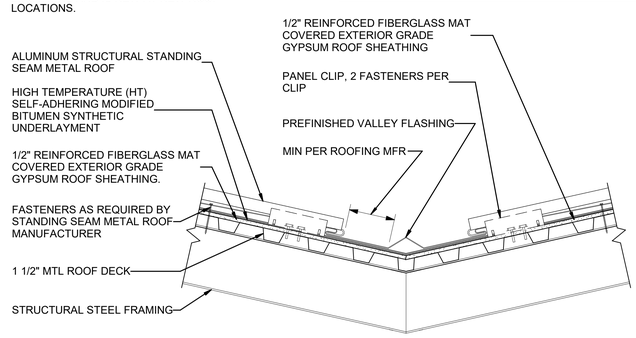



3 A-301 **2** SOFFIT BEAM DETAIL
SCALE: 3" = 1'-0"

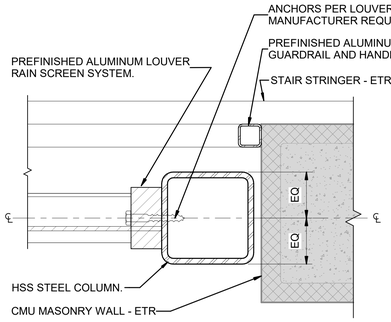



3 **TYP HIP/RIDGE DETAIL**
SCALE: 3" = 1'-0"

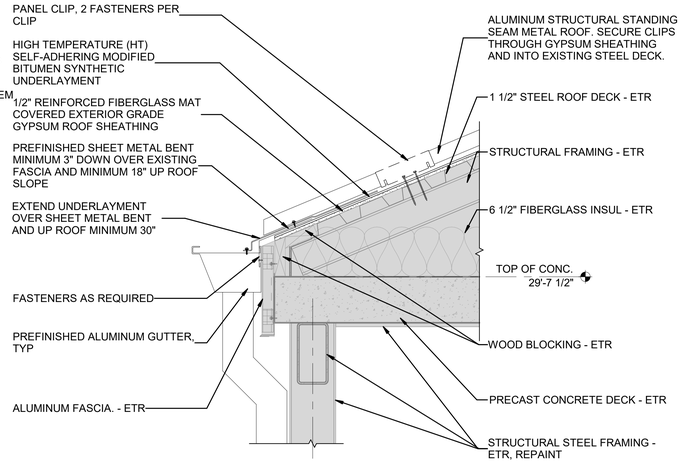
NOTE:
THE ROOF STRUCTURE AND STEEL ROOF DECK INDICATED IN THIS DETAIL IS EXISTING TO REMAIN AT EXISTING ROOF LOCATIONS AND IS NEW AT NEW ROOF LOCATIONS.



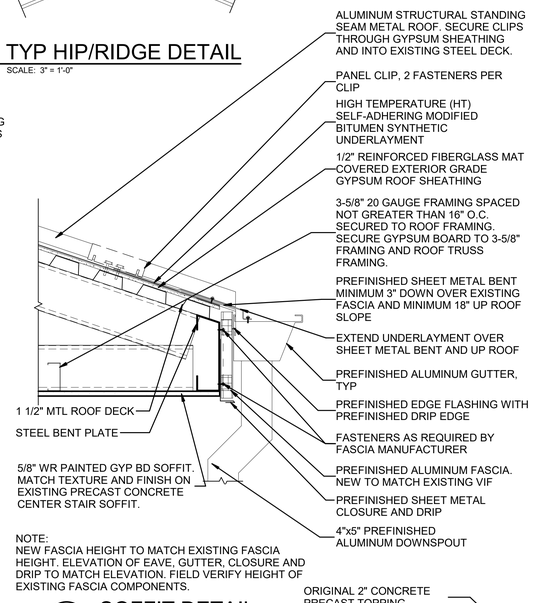

4 **TYP VALLEY DETAIL**
SCALE: 1 1/2" = 1'-0"




1 A-301 **5** LOUVER JAMB DETAIL
SCALE: 3" = 1'-0"

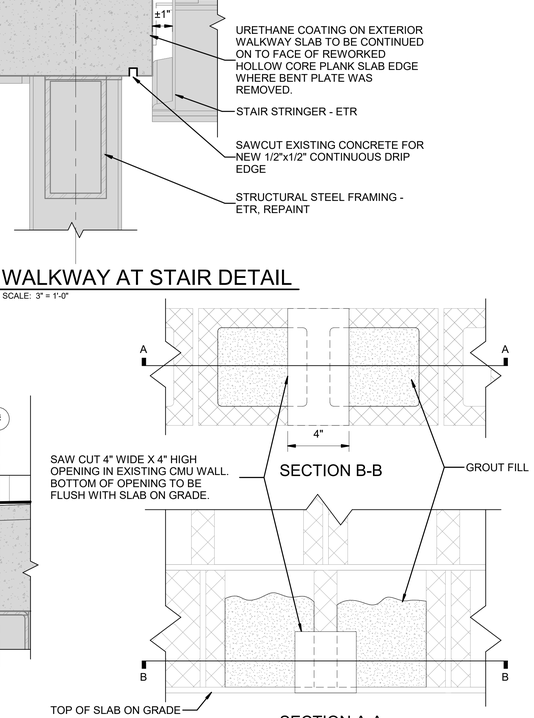



6 **EAVE DETAIL**
SCALE: 1 1/2" = 1'-0"

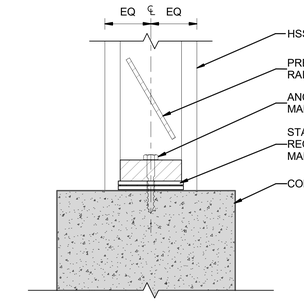



NOTE:
NEW FASCIA HEIGHT TO MATCH EXISTING FASCIA HEIGHT. ELEVATION OF EAVE, GUTTER, CLOSURE AND DRIP TO MATCH ELEVATION. FIELD VERIFY HEIGHT OF EXISTING FASCIA COMPONENTS.

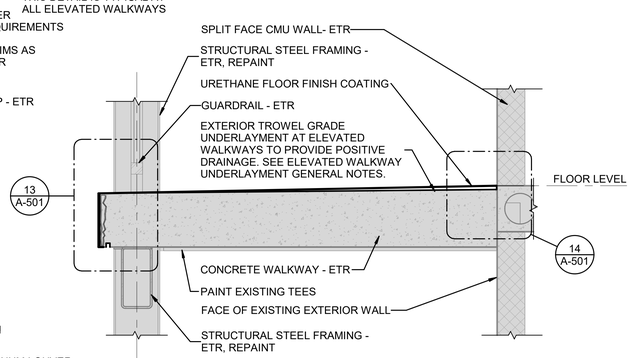
7 **SOFFIT DETAIL**
SCALE: 1 1/2" = 1'-0"



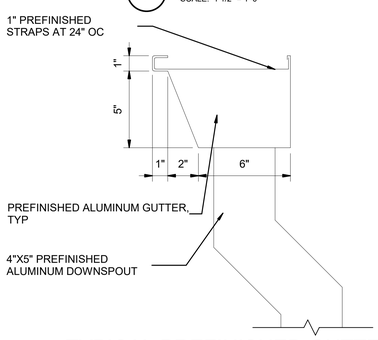

4 A-301 **8** WALKWAY AT STAIR DETAIL
SCALE: 3" = 1'-0"



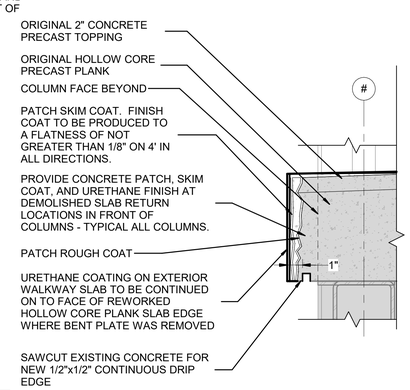

2 A-301 **9** LOUVER WALL SILL DETAIL
SCALE: 3" = 1'-0"




11 **TYPICAL WALKWAY DETAIL**
SCALE: 1 1/2" = 1'-0"



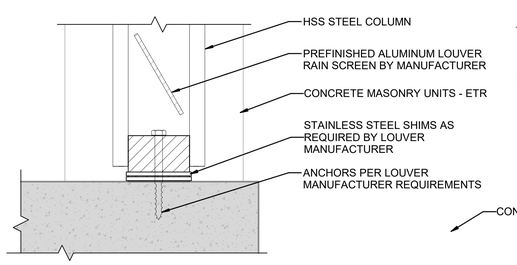

12 **TYPICAL PREFINISHED GUTTER AND DOWNSPOUT DETAIL**
SCALE: 3" = 1'-0"



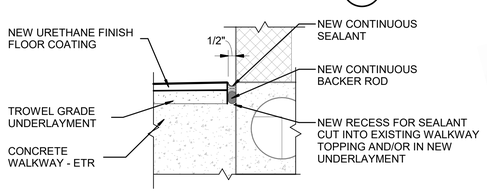

11 A-501 **13** ENLARGED SLAB EDGE DETAIL
SCALE: 3" = 1'-0"




**15** SAW CUT CMU OPENING DETAIL
SCALE: 3" = 1'-0"




1 A-301 **10** LOUVER FLOOR SILL DETAIL
SCALE: 3" = 1'-0"




11 A-501 **14** TYPICAL CONTINUOUS SEALANT RECESS DETAIL
SCALE: 3" = 1'-0"

WISCONSIN
DEAN C. VON TERSCH
E-17503
CHARLESTON, SC
ARCHITECT
04-OCT-2023

12" 0 6" 1'
**SCALE: 1 1/2"=1'-0"**

3" 0 6" 1'
**SCALE: 3" = 1'-0"**

**GRAPHIC SCALES**

**REVISIONS**

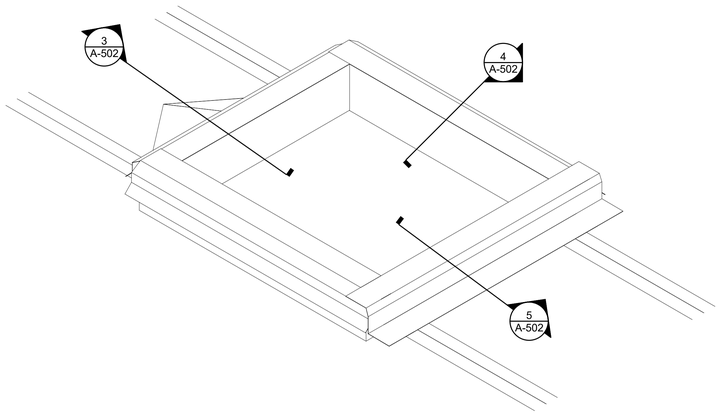
| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

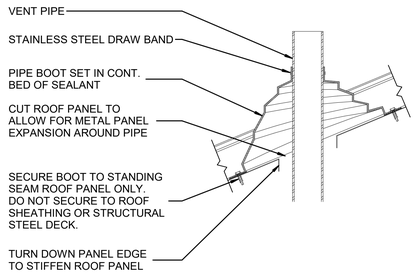
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

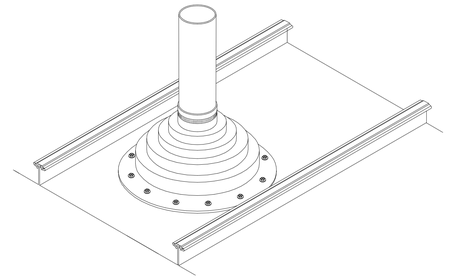
| | | |
|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28488 | CEMS — A-501 |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY M. JOHANNES | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR D. VON TERSCH | DETAILS |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. |
| PUBLIC WORKS OFFICER | | HQMC PROJ. NO.:; CONSTR. CONTR. NO.:N40085-18-D-0616 |
| | | SCALE: As indicated; MACC.; T.O. #; SHEET 34 OF 53 |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |



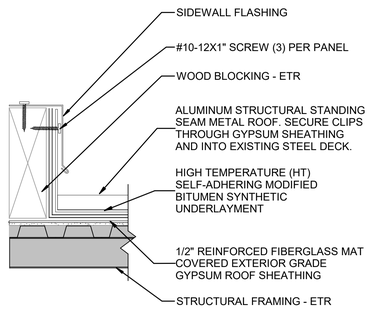

**SECTION VIEW**



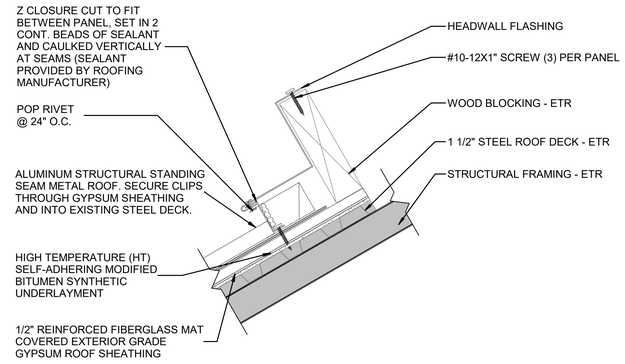
**ISOMETRIC VIEW**

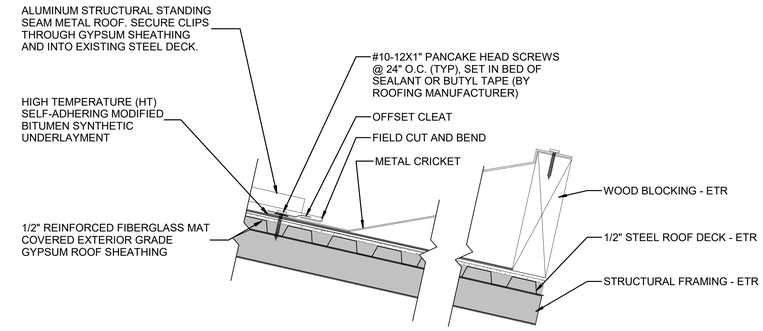
NOTE: CONTRACTOR TO FIELD VERIFY QUANTITY AND LOCATION OF EXISTING THROUGH ROOF PIPE PENETRATIONS AND PROVIDE NEW THROUGH ROOF PIPE PENETRATION FLASHING AS DETAILED HEREIN AT EACH EXISTING PIPE PENETRATION. CONTRACTOR TO MAKE MINOR ADJUSTMENTS IN LOCATION OF EXISTING THROUGH ROOF PIPE PENETRATION TO AVOID PENETRATION FALLING AT LOCATION OF STANDING SEAM IN NEW STANDING SEAM METAL ROOF. THROUGH ROOF PIPE PENETRATION TO FOLLOW ALL REQUIREMENTS OF STANDING SEAM METAL ROOF MANUFACTURER.

## 1 CURB DETAIL
SCALE: NOT TO SCALE




## 2 VENT THROUGH ROOF DETAIL
SCALE: 3" = 1'-0"




## 1/A-502 3 CURB DETAIL
SCALE: 1 1/2" = 1'-0"

## 1/A-502 4 CURB DETAIL
SCALE: 1 1/2" = 1'-0"

## 1/A-502 5 CURB DETAIL
SCALE: 1 1/2" = 1'-0"

12" 0 6" 1'

**SCALE: 1 1/2"=1'-0"**

**GRAPHIC SCALE**

DEAN C. VON TERSCH LA-7303 CHARLESTON SC WISCONSIN ARCHITECT 04-OCT-2023

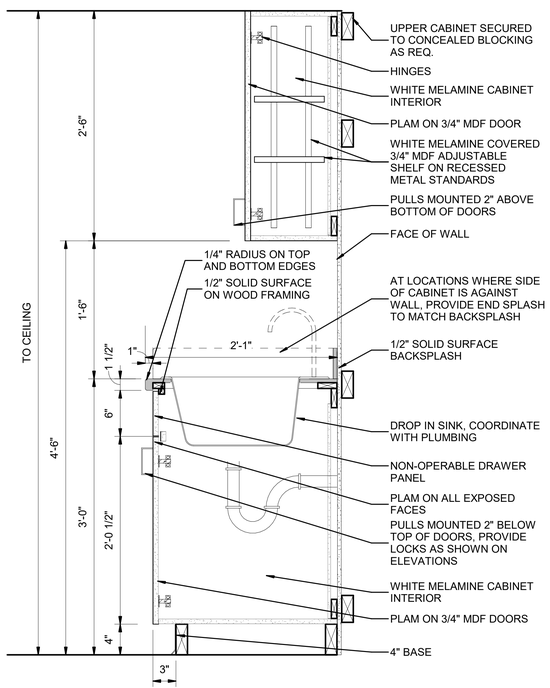
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28489 | CEMS | A-502 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>M. JOHANNES | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>**MARINE CORPS AIR STATION**<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M)<br>DETAILS | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: As indicated; MACC.; T.O. #; SHEET 35 OF 53 | |

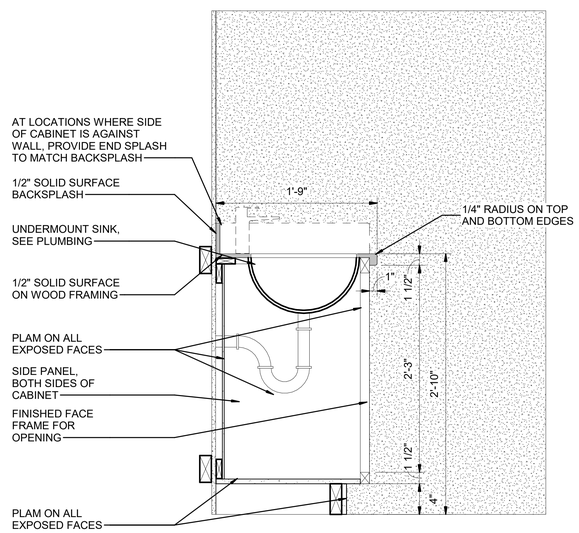
# FINISH SCHEDULE

| ROOM NAME | WALL FINISH | BASE FINISH | FLOOR MAT. | CEILING MAT. | NOTES & REMARKS |
|---|---|---|---|---|---|
| FIRST FLOOR | | | | | |
| CLOSET A | PT-1 | RB-1 | LVT-1 | PT-5 | |
| CLOSET B | PT-1 | RB-1 | LVT-1 | PT-5 | |
| ELEC & TEL ROOM A | PT-1 | ETR | ETR | PT-5 | |
| ELEC & TEL ROOM B | PT-1 | ETR | ETR | PT-5 | |
| HALLWAY | ETR | ETR | SC-1 | ETR | |
| LIVING / SLEEPING | PT-1 | RB-1 | LVT-1 | PT-5 | |
| MECH ROOM A | PT-1 | ETR | ETR | PT-5 | |
| MECH ROOM B | PT-1 | ETR | ETR | PT-5 | |
| SERVICE AREA | PT-1 | RB-1 | LVT-1 | PT-5 | |
| SHOWER | PWT-1 | PTB-1 | PFT-1 | PT-5 | |
| TOILET | PT-2, PWT-1 | PTB-1 | PFT-2 / PFT-3 | PT-5 | PFT-3 IS A 4" WIDE BORDER AROUND PERIMETER OF ROOM |
| SECOND FLOOR | | | | | |
| CLOSET A | PT-1 | RB-1 | LVT-1 | PT-5 | |
| CLOSET B | PT-1 | RB-1 | LVT-1 | PT-5 | |
| ELEC & TEL ROOM A | PT-1 | ETR | ETR | PT-5 | |
| ELEC & TEL ROOM B | PT-1 | ETR | ETR | PT-5 | |
| ELEC ROOM C | PT-1 | ETR | ETR | PT-5 | |
| HALLWAY | ETR | ETR | SC-1 | ETR | |
| LIVING / SLEEPING | PT-1 | RB-1 | LVT-1 | PT-5 | |
| MECH ROOM A | PT-1 | ETR | ETR | PT-5 | |
| MECH ROOM B | PT-1 | ETR | ETR | PT-5 | |
| SERVICE AREA | PT-1 | RB-1 | LVT-1 | PT-5 | |
| SHOWER | PWT-1 | PTB-1 | PFT-1 | PT-5 | |
| TOILET | PT-2, PWT-1 | PTB-1 | PFT-2 / PFT-3 | PT-5 | PFT-3 IS A 4" WIDE BORDER AROUND PERIMETER OF ROOM |
| THIRD FLOOR | | | | | |
| CLOSET A | PT-1 | RB-1 | LVT-1 | PT-5 | |
| CLOSET B | PT-1 | RB-1 | LVT-1 | PT-5 | |
| ELEC & TEL ROOM A | ETR | ETR | ETR | PT-5 | |
| ELEC & TEL ROOM B | ETR | ETR | ETR | PT-5 | |
| HALLWAY | ETR | ETR | SC-1 | ETR | |
| LIVING / SLEEPING | PT-1 | RB-1 | LVT-1 | PT-5 | |
| MECH ROOM A | ETR | ETR | ETR | PT-5 | |
| MECH ROOM B | ETR | ETR | ETR | PT-5 | |
| SERVICE AREA | PT-1 | RB-1 | LVT-1 | PT-5 | |
| SHOWER | PWT-1 | PTB-1 | PFT-1 | PT-5 | |
| TOILET | PT-2, PWT-1 | PTB-1 | PFT-2 / PFT-3 | PT-5 | PFT-3 IS A 4" WIDE BORDER AROUND PERIMETER OF ROOM |




7 / A-401 — 1 — UPPER / LOWER - DROP-IN SINK

SCALE: 1 1/2" = 1'-0"




2 — VANITY - DROP-IN SINK

SCALE: 1 1/2" = 1'-0"

12" 0 6" 1'

**SCALE: 1 1/2"=1'-0"**

**GRAPHIC SCALE**

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. SEE INTERIORS SHEET FOR LEGEND OF COLORS AND FINISHES.
2. SEE INTERIORS SHEET FOR GENERAL FINISH NOTES.

WISCONSIN — DEAN C. VON TERSCH — A-7903 — CHARLESTON SC — ARCHITECT — 04 OCT 2023

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28490 | CEMS | A-601 |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY M. JOHANNES | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND — MARINE CORPS AIR STATION — BEAUFORT, S.C. | |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR D. VON TERSCH | FINISH SCHEDULE AND CABINET DETAILS | |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F — CODE IDENT. NO. — N A V F A C DRAWING NO. | |
| | PUBLIC WORKS OFFICER | HOMC PROJ. NO.: — CONSTR. CONTR. NO.: N40085-18-D-0616 | |
| | | SCALE: As indicated — MACC. — T.O. # — SHEET 36 OF 53 | |

# INTERIOR ABBREVIATIONS

| ABBREV. | DESCRIPTION |
|---|---|
| ACT | ACOUSTICAL CEILING TILE AND GRID SYSTEM |
| AFS | RAISED ACCESS FLOORING SYSTEM |
| AWP | ACOUSTICAL WALL PANEL |
| BRK | BRICK |
| CFT | CERAMIC FLOOR TILE |
| CMU | CONCRETE MASONRY UNIT |
| CONC | EXPOSED CONCRETE |
| CPT | CARPET |
| CR | CHAIRRAIL |
| CRW | CROWN MOLDING |
| CTB | CERAMIC TILE BASE |
| CWT | CERAMIC WALL TILE |
| EF | ENTRANCE FLOORING |
| EPX | EPOXY FLOORING |
| EXPS | EXPOSED STRUCTURE |
| FRP | FIBERGLASS REINFORCED PANELS |
| GCMU | GLAZED MASONRY UNIT WITH INTEGRAL COVE BASE |
| GT | GROUT |
| GWB | GYPSUM WALLBOARD |
| GWT | GLASS WALL TILE |
| LVT | LUXURY VINYL TILE |
| M | MORTAR |
| MTL | METAL |
| OP | OPERABLE PARTITION |
| PFT | PORCELAIN FLOOR TILE |
| PLAM | PLASTIC LAMINATE |
| PT | PAINT |
| PTB | PORCELIAN TILE BASE |
| PW | PUBLIC WORKS |
| PWT | PORCELAIN WALL TILE |
| QT | QUARRY TILE |
| RB | RESILENT BASE |
| RES | RESINOUS FLOORING |
| RF | ROOF |
| RFT | RESILIENT FLOOR TILE |
| RSTR | RUBBER STAIR TREAD |
| SC | SEALED CONCRETE |
| SCS | SECURITY CEILING SYSTEM |
| SDT | STATIC DISSIPATIVE VINYL TILE |
| SF | SEATING FABRIC |
| SRT | SOLID RUBBER TILE |
| SS | SOLID SURFACE |
| ST | TRANSPARENT STAIN |
| STCO | STUCCO |
| SVF | SHEET VINYL FLOORING |
| SVT | SOLID VINYL TILE |
| TP | TRIM PAINT |
| TRZ | TERRAZZO |
| VCT | VINYL COMPOSITION TILE |
| VWC | VINYL WALLCOVERING |
| WB | WINDOW BLIND |
| WC | WALLCOVERING |
| WDB | WOOD BASE |
| WG | WALL GUARDS |
| WM | WALK-OFF MAT |
| WSCT | WAINSCOT |
| WST | WALL STONE |
| WV | WOOD VENEER WALLS |

## GENERAL FINISH NOTES:

1. CONTRACTOR SHALL READ ALL NOTES BEFORE BEGINNING WORK.
2. FURNISH AND INSTALL MATERIALS IN COMPLIANCE WITH MANUFACTURER'S PRINTED SPECIFICATIONS AND INSTALLATION INSTRUCTIONS.
3. ANY DISCONTINUED ITEMS SHALL BE BROUGHT TO THE ATTENTION OF THE ARCHITECT IMMEDIATELY. CONTRACTOR TO NOTIFY GOVERNMENT OF ANY ITEMS REQUIRING FURTHER CLARIFICATION OF FINISH SELECTION.
4. CONTRACTOR SHALL FIELD VERIFY ALL DIMENSIONS PRIOR TO ORDERING OF MATERIALS. DO NOT SCALE ARCHITECTURAL DRAWINGS. REFER TO DIMENSIONED PLANS
5. CONTRACTOR SHALL PROVIDE TO CONTRACTING OFFICER ALL MANUFACTURER'S RECOMMENDED MAINTENANCE INSTRUCTIONS ON ALL FINISH MATERIALS.
6. INTERIOR CONSTRUCTION REQUIRING FINISH / COLOR SELECTION NOT NOTED WITHIN THE SCHEDULE SHALL BE SUBMITTED TO THE CONTRACTING OFFICER IN WRITING, WITH SAMPLES OR FINISH / COLORS AVAILABLE, CLEARLY IDENTIFYING ANY THAT MIGHT BE AT A PREMIUM ABOVE THE PRODUCT BID.
7. ALL PRODUCT SPECIFICATIONS ARE GIVEN TO DEFINE DESIGN, COLOR AND QUALITY. SUBSTITUTIONS REQUIRE THE PRIOR APPROVAL OF THE ARCHITECT. PROPOSED SUBSTITUTIONS SHALL ADHERE TO THE INTENT OF THE QUALITY, DESIGN, COLOR AND PATTERN.
8. ELECTRICAL PANELS SHALL BE PAINTED TO MATCH THE ADJACENT WALL FINISHES UNLESS PRE-FINISHED.
9. ALL FINISHES ARE DESIGNATED USING "PLAN NORTH".
10. THE CONTRACTOR SHALL INCLUDE IN THEIR EFFORT TO REPLACE IN LIKE AND KIND NOT LESS THAN 20% OF ALL THE WINDOW TREATMENTS IN ALL BUILDINGS. THE CONTRACTOR SHALL CONFIRM WITH THE GOVERNMENT WHICH WINDOW TREATMENTS WILL BE REPLACED.

## FLOOR FINISH NOTES:

1. CONTRACTOR TO FOLLOW MANUFACTURER'S PRINTED INSTALLATION INSTRUCTIONS ON ALL FLOORING PRODUCTS AND USE THE ADHESIVES, EQUIPMENT / TOOLS REQUIRED / RECOMMENDED BY THE MANUFACTURER'S INSTALLATION INSTRUCTIONS.
2. CONTRACTOR TO REMOVE EXISTING FLOORING MATERIALS AS REQUIRED AND PREPARE SUB FLOOR PER MANUFACTURER'S INSTRUCTIONS.
3. CONTRACTOR TO FOLLOW MANUFACTURER'S PRINTED INSTRUCTIONS ON PROTECTION OF FLOORING MATERIAL AFTER INSTALLATION. PROTECT FLOORING AGAINST MARS, INDENTATIONS AND OTHER DAMAGE FROM CONSTRUCTION OPERATIONS, PLACEMENT OF EQUIPMENT AND FIXTURES DURING THE REMAINDER OF THE CONSTRUCTION PERIOD. USE PROTECTION METHODS INDICATED OR RECOMMENDED IN WRITING BY THE FLOOR COVERING MANUFACTURER.
4. CONTRACTOR TO PROVIDE AND INSTALL APPROPRIATE TRANSITION STRIPS WHERE DIFFERING MATERIALS ABUT IF NOT NOTED OTHERWISE. PROVIDE ARCHITECT WITH COLOR SAMPLES TO SELECT FROM.
5. FLOOR FINISH CHANGES SHALL OCCUR UNDER THE DOOR CENTERLINE (OR CENTER OF THE WALL OPENING IF NO DOOR) BETWEEN ROOMS.
6. CONTRACTOR SHALL FOLLOW MANUFACTURER'S RECOMMENDATION ON WIDTHS OF GROUT LINES. WHEN USING STAGGERED BRICK JOINT PATTERN ON PLANK FLOORING, THE OVERLAP SHALL NOT EXCEED 33%.

## WALL FINISH NOTES:

1. MISCELLANEOUS GRILLS AND ACCESSORIES SHALL BE PAINTED TO MATCH ADJACENT WALL FINISHES UNLESS THEY ARE PREFINISHED. CONTRACTOR TO VERIFY PAINT COLOR WITH CONTRACTING OFFICER.

## DOORS AND DOOR FRAMES:

1. ALL EXTERIOR HOLLOW METAL DOOR SLABS AND ALL EXTERIOR DOOR FRAMES SHALL BE PAINTED TP-2. THE INTERIOR SIDE OF ALL EXTERIOR HOLLOW METAL DOOR SLABS AND ALL INTERIOR DOORS AND METAL FRAMES SHALL BE PAINTED TP-1. SEE LEGEND OF COLORS AND FINISHES. COLOR CHANGE TO TAKE PLACE AT INTERIOR INSIDE CORNER OF THE STOP.
2. FINISH CHANGES BETWEEN AREAS SHALL OCCUR UNDER THE DOOR OR AT THE CENTERLINE OF DOOR FRAME.
3. WHERE NEW GYP.BD. HAS BEEN INSTALLED ON ADJACENT WALLS RETURN GYP. BD. TO METAL FRAME.
4. VERIFY WITH THE BASE PUBLIC WORKS DEPARTMENT (PW) FOR THE APPROVAL OF FINAL COLOR SELECTIONS.

## CEILING NOTES:

1. CEILING ACCESS PANELS, HVAC REGISTERS AND GRILLS SHALL BE PAINTED IN ALL AREAS AFFECTED.
2. ALL EXISTING AND NEW PAINTED GYPSUM BOARD CEILING TO RECEIVE NEW CEILING PAINT PT-5, FLAT.
3. EXPOSED DECK SHALL BE PAINTED PT-5.

## CORNER GUARDS NOTES:

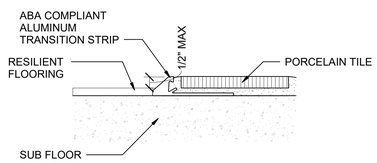
1. CORNER GUARDS ARE TO REMAIN IN PLACE, MASK OFF TO PROTECT WHILE PAINTING UP TO THE EDGES.

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

# LEGEND OF COLORS AND FINISHES

| MARK | MATERIAL | MANUFACTURER | STYLE | COLOR | SIZE | COMMENTS |
|---|---|---|---|---|---|---|
| BASE | | | | | | |
| PTB-1 | PORCELAIN TILE BASE | CROSSVILLE | COLOR BLOX 2.0 | CBX07 MUD PIE | 4" X 12" | --- |
| RB-1 | RUBBER BASE | ROPPE | 1/8" PINNACLE | 147 LIGHT BROWN | 4" HEIGHT | 120'-0" COILS |
| CEILING | | | | | | |
| ACG-1 | ACOUSTICAL CEILING TILE GRID | ARMSTRONG | | PRELUDE 15/16, WHITE | --- | --- |
| ACT-1 | ACOUSTICAL CEILING TILE | ARMSTRONG | 1732 | FINE FISSURE TEGULAR, WHITE | 2'-0" X 2'-0" | --- |
| PT-5 | CEILING PAINT | SHERWIN WILLIAMS | SW7007 | CEILING BRIGHT WHITE | --- | INTERIOR CEILINGS |
| FLOORING | | | | | | |
| ETR | EXISTING TO REMAIN | | | | | |
| LVT-1 | LUXURY VINYL TILE | ARMSTRONG | DIAMOND TECHNOLOGY VERONA | NA350 ROCK DUST | 18" X 18" | --- |
| PFT-1 | PORCELAIN SHOWER FLOOR TILE | DALTILE | KEYSTONES | URBAN PUTTY D201 | 2" X 2" | SHOWER FLOOR TILE |
| PFT-2 | PORCELAIN FLOOR TILE | CROSSVILLE | COLOR BLOX 2.0 | CBX01 SANDBOX | 12" X 12" | RESTROOM FLOOR FIELD COLOR |
| PFT-3 | PORCELAIN FLOOR TILE | CROSSVILLE | COLOR BLOX 2.0 | CBX07 MUD PIE | 4" X 12" | RESTROOM FLOOR BORDER |
| SC-1 | URETHANE | STONHARD OR EQUAL | URETHANE COATING | TO MATCH BASE STANDARD TAUPE | | POWER WASH WALKWAYS AND DRY PRIOR TO APPLICATION AND USE TWO COATS. COORDINATE WITH PW ON FINAL FINISH SELECTION |
| MISC. | | | | | | |
| GT-1 | WALL TILE GROUT TOLIET ROOM FLOORS | LATICRETE | | 23 ANTIQUE WHITE | --- | RESTROOM FLOOR TILE PFT-2 |
| GT-2 | WALL TILE GROUT TILE BASE AND BORDER | LATICRETE | | 30 SAND BEIGE | --- | RESTROOM BASE AND BORDER PFT-3 |
| GT-3 | WALL TILE GROUT SHOWER WALLS | LATICRETE | | 03 SILK | --- | SHOWER WALL TILE PWT-1 AND SHOWER FLOOR PFT-1 |
| PLAM-1 | PLASTIC LAMINATE | WILSONART | D439-60 | WALLABY | --- | LAUNDRY ROOM COUNTERTOPS KITCHEN CASEWORK |
| SS-1 | SOLID SURFACE | WILSONART | | MYSTIQUE 9200CS | --- | SERVICE AREA COUNTERTOPS AND TOILET VANITY TOPS |
| SS-2 | SOLID SURFACE | WILSONART | | ANTIQUE WHITE 1572SL | --- | SHOWER CURB TOPS |
| TP-1 | TRIM PAINT | SHERWIN WILLIAMS | GLOSS | BUCKRAM BINDING SW0036 | --- | INTERIOR SIDE OF EXTERIOR HOLLOW METAL ENTRY DOORS AND ALL INTERIOR HOLLOW METAL DOORS AND FRAMES AT SLEEPING ROOM SUITES. COORDINATE WITH BASE PUBLIC WORKS ON FINAL FINISH SELECTION. |
| TP-2 | TRIM PAINT | SHERWIN WILLIAMS | | | --- | EXTERIOR HOLLOW METAL DOORS AND EXTERIOR FRAMES TO MATCH EXISTING BASE STANDARD DARK TAN/ BRONZE. COORDINATE WITH PUBLIC WORKS |
| WALLS | | | | | | |
| PT-1 | WALL PAINT | SHERWIN WILLIAMS | SATIN | CREME SW7556 | --- | FIELD PAINT FOR WALLS |
| PT-2 | WALL PAINT | SHERWIN WILLIAMS | B73-360 PRO INDUSTRIAL WATERBASED CATALYZED EPOXY EGGSHELL | CREME SW7556 | --- | FIELD PAINT FOR RESTROOM WALLS |
| PT-3 | WALL PAINT | SHERWIN WILLIAMS | SATIN | LABRODORITE SW7619 | --- | ACCENT PAINT IN LAUNDRY BUILDING |
| PT-4 | WALL PAINT | SHERWIN WILLIAMS | SATIN | ROOKWOOD BLUE GREEN SW2811 | --- | ACCENT PAINT IN LAUNDRY BUILDING |
| PWT-1 | PORCELAIN WALL TILE | DALTILE | CLASSIC | MATTE ALMOND X735 | 4" X 4" | BOTTOM 4 ROWS OF RESTROOM SHOWER WALLS |



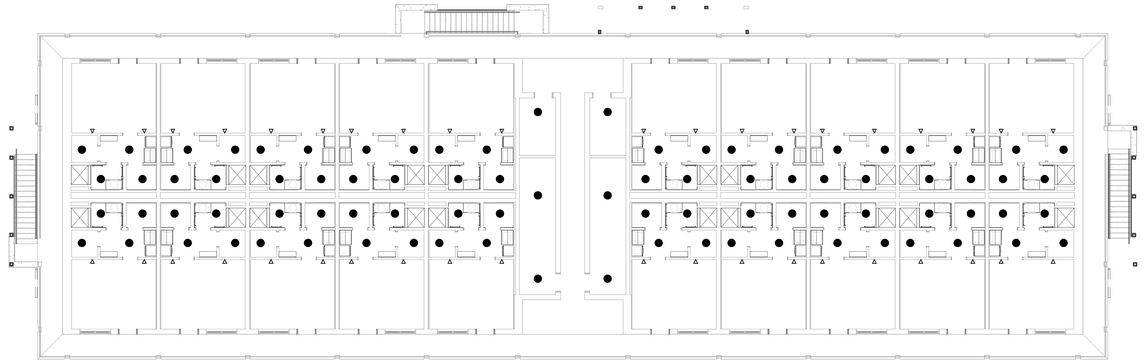

1 TRANSITION DETAIL
SCALE: NOT TO SCALE

WISCONSIN
DEAN C. VON TERSCH
(A-7303)
CHARLESTON
SC
ARCHITECT
04-OCT-2023

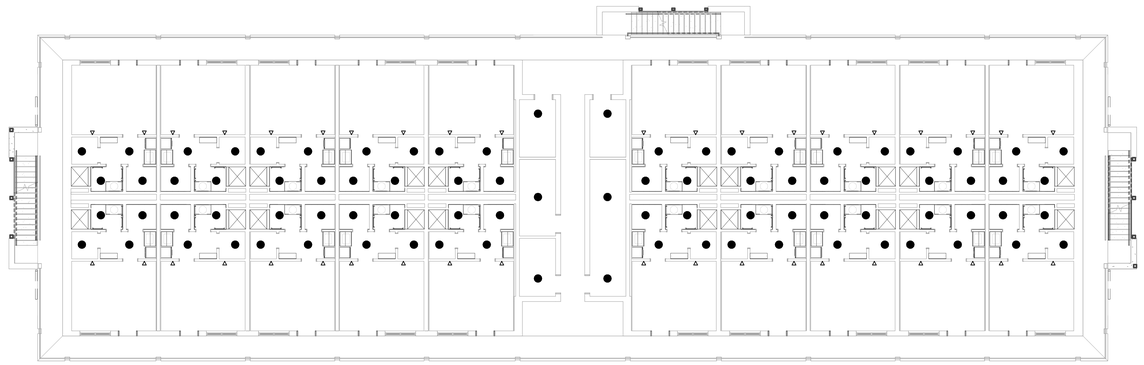
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

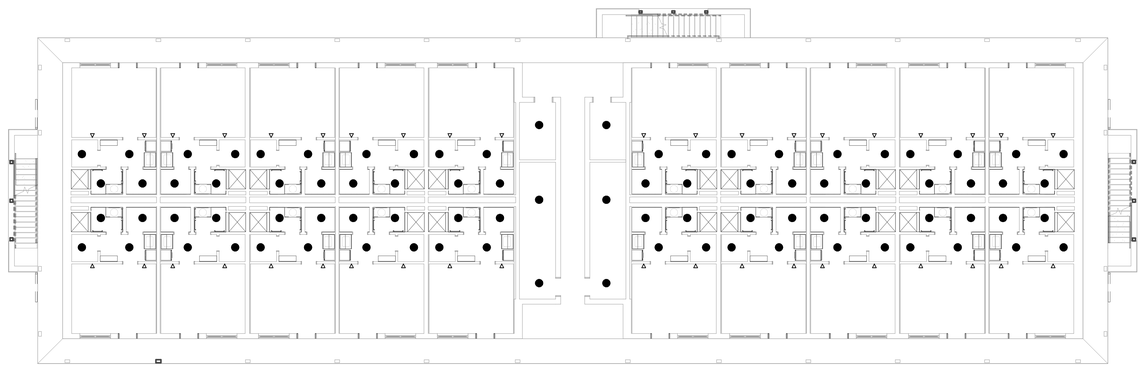
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28491 | CEMS | I-001 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>K. CALHOUN | DEPARTMENT OF THE NAVY<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | NAVAL FACILITIES ENGINEERING COMMAND |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>D. VON TERSCH | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>D. VON TERSCH | INTERIORS GENERAL NOTES AND ABBREVIATIONS | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F / CODE IDENT. NO. / N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| PUBLIC WORKS OFFICER | | SCALE: As indicated / MACC. / T.O. # | SHEET 37 OF 53 |



1 THIRD FLOOR
SCALE: 3/32" = 1'-0"



2 SECOND FLOOR
SCALE: 3/32" = 1'-0"



3 FIRST FLOOR
SCALE: 3/32" = 1'-0"

PLAN NORTH

24"0 4' 8' 16'

SCALE: 3/32"=1'-0"

GRAPHIC SCALE

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. AUTOMATIC SPRINKLER SYSTEMS HAVE BEEN SPECIFIED IN ACCORDANCE WITH THE CURRENT REQUIREMENTS OF NFPA 13 (2022 EDITION) .
2. CONTRACTOR MUST FIELD VERIFY ALL SPRINKLER HEADS AND LOCATIONS.
3. REMOVE AND REPLACE TYPICAL IN ALL ROOMS: ALL SPRINKLER HEADS ESCUTCHEONS AND ALL SPRINKLER HEADS WITH NEW QUICK RESPONSE HEADS OF SAME TYPE AND TEMPERATURE.

## FIRE SUPPRESSION LEGEND:

● PENDENT SPRINKLER HEAD

▽ SIDEWALL SPRINKLER HEAD

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE: 08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28492 | CEMS | F-101B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>C. STOCKMAN | DEPARTMENT OF THE NAVY<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | NAVAL FACILITIES ENGINEERING COMMAND |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>J. TORNER | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>M. LAZO | FIRST, SECOND AND THIRD FLOOR SUPPRESSION PLAN -BLDG 1245G | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F / CODE IDENT. NO. / N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: As indicated / MACC. / T.O. # | SHEET 38 OF 53 |

## LEGEND:

- TO BE REMOVED/DEMOLISHED
- EXISTING DUCTWORK / MECHANICAL (TO REMAIN)
- NEW DUCTWORK / MECHANICAL
- SUPPLY DIFFUSER
- RETURN GRILLE
- EXHAUST GRILLE
- SUPPLY DUCT UP OR DOWN
- RETURN DUCT UP OR DOWN
- EXHAUST DUCT UP OR DOWN
- CHILLED WATER RETURN, CHWR
- (E) CHILLED WATER RETURN, CHWR
- CHILLED WATER SUPPLY, CHWS
- (E) CHILLED WATER SUPPLY, CHWS
- CONDENSER WATER SUPPLY, CWS
- (E) CONDENSER WATER SUPPLY, CWS
- CONDENSER WATER RETURN, CWR
- (E) CONDENSER WATER RETURN, CWR
- SPLITTER VANE ELBOW
- FLEXIBLE DUCT - MAX 5' LONG. SEE AIR DISTRIBUTION SCHEDULE FOR SIZES
- SINGLE THICKNESS TURNING VANES ELBOW, TURNING VANES
- T — THERMOSTAT (T); HUMIDISTAT (H); CARBON DIOXIDE SENSOR (C)
- DUCT TRANSITION
- PIPE TURNING DOWN
- PIPE TURNING UP
- POINT OF CONNECTION BETWEEN NEW AND EXISTING
- POINT OF DISCONNECTION BETWEEN EXISTING TO REMAIN AND EXISTING TO BE REMOVED
- ? — KEYNOTE
- Ø — ROUND DUCTWORK
- ESS — HVAC EMERGENCY SHUTOFF SWITCH

## ABBREVIATIONS

| | |
|---|---|
| (E) | EXISTING |
| AC | AIR CONDITIONING UNIT |
| AFMS | AIRFLOW MEASUREMENT STATION |
| CHWR | CHILLED WATER RETURN |
| CHWS | CHILLED WATER SUPPLY |
| CU | CONDENSING UNIT |
| CWR | CONDENSER WATER RETURN |
| CWS | CONDENSER WATER SUPPLY |
| DDC | DIRECT DIGITAL CONTROL |
| DOAU | DEDICATED OUTDOOR AIR UNIT |
| EA | EXHAUST AIR |
| EF | EXHAUST FAN |
| ER | ENERGY RECOVERY VENTILATOR |
| F.C. | FLEXIBLE CONNECTION |
| HCC | DUAL HEATING/COOLING COIL |
| HP | WATER TO WATER HEAT PUMP |
| OA | OUTDOOR AIR |
| OAT | OUTDOOR AIR TEMPERATURE |
| OF | OUTSIDE AIR FAN |
| P | PUMP |
| RA | RETURN AIR |
| RHC | REHEAT COIL |
| SA | SUPPLY AIR |
| SF | SUPPLY FAN |
| TYP. | TYPICAL |
| W/ | WITH |
| WSHP | WATER SOURCE HEAT PUMP |

## GENERAL NOTES:

1. DRAWINGS ARE DIAGRAMMATIC IN NATURE AND ARE NOT TO BE SCALED FOR DIMENSIONS, UNLESS DIMENSIONED. ALL EXISTING PIPING AND MECHANICAL EQUIPMENT LOCATIONS AND SIZING BASED ON AS-BUILT DRAWINGS AND FIELD VERIFICATION. CONTRACTOR SHALL BECOME TOTALLY FAMILIARIZED WITH AND VERIFY ALL EXISTING CONDITIONS AND DIMENSIONS THAT AFFECT THIS CONTRACT AND SHALL MAKE ADJUSTMENTS AS REQUIRED BASED ON EXISTING CONDITIONS.
2. ALL WORK SHOWN ON THESE DRAWINGS SHALL BE NEW UNLESS NOTED OTHERWISE AS EXISTING. CONTRACTOR SHALL PROVIDE ALL MATERIALS AND LABOR FOR A COMPLETE AND OPERABLE SYSTEM. ALL WORK TO BE PERFORMED SHALL BE PROVIDED BY THE CONTRACTOR UNLESS OTHERWISE NOTED.
3. WHEN WORKING IN AND AROUND THE EXISTING BUILDING, EXTREME CARE SHALL BE EXERCISED WITH REGARD TO PROTECTION OF THE EXISTING STRUCTURE AND MECHANICAL AND ELECTRICAL SERVICES WHICH WILL REMAIN. REPAIR, REPLACE, OR RESTORE TO THE SATISFACTION OF THE ENGINEER ALL EXISTING WORK DAMAGED IN THE PERFORMANCE OF DEMOLITION AND/OR NEW WORK.
4. COORDINATE LOCATION OF MECHANICAL WORK WITH OTHER TRADES TO AVOID CONFLICTS AND INTERFERENCES. COORDINATE ALL ROOF PENETRATION SIZES AND LOCATIONS WITH APPROVED EQUIPMENT SHOP DRAWINGS. INSTALL ALL MECHANICAL EQUIPMENT SUCH THAT MANUFACTURER'S MAINTENANCE AREA IS CLEAR.
5. DUCT SIZES INDICATED ARE NET FREE INSIDE DIMENSIONS. TAKEOFFS FOR CEILING DIFFUSERS ARE TO BE THE SAME SIZE AS THE DIFFUSER NECK SIZE UNLESS OTHERWISE INDICATED ON PLANS. DUCTWORK SHALL BE A MINIMUM OF 26 GAUGE. ALL BRANCH DUCT TAKEOFFS ARE TO BE PROVIDED WITH BALANCING DAMPERS. PROVIDE ACCESS DOORS IN DUCTWORK TO SERVICE DAMPERS AND DEVICES WITHIN DUCTS.
6. REFER TO SPECIFICATIONS FOR HAZARDOUS MATERIAL REMOVAL PRIOR TO CONSTRUCTION.
7. LOCATE ALL ABOVE CEILING EQUIPMENT WHICH MUST BE SERVICED, OPERATED, OR MAINTAINED, IN FULLY ACCESSIBLE POSITIONS. WHERE OVERHEAD EQUIPMENT CANNOT BE LOCATED ABOVE SPACES WITH EITHER NO CEILINGS OR REMOVABLE CEILING TILES, CONTRACTOR SHALL PROVIDE ACCESS DOORS AS REQUIRED (COLOR SHALL MATCH CEILING). WITHIN THE SAME ROOM, EQUIPMENT SHALL BE GROUPED TOGETHER AS MUCH AS POSSIBLE TO MINIMIZE THE NUMBER OF PANELS.
8. SIZE REFRIGERANT PIPING IN ACCORDANCE WITH MANUFACTURER'S WRITTEN INSTRUCTIONS BASED ON THE PROJECT SPECIFIC LAYOUT AND LENGTH OF REFRIGERANT PIPING. REFER TO DRAWINGS FOR ADDITIONAL INFORMATION.
9. PROVIDE VFDS, STARTERS, AND DISCONNECT SWITCHES FOR ALL MECHANICAL EQUIPMENT AS SHOWN AND AS SPECIFIED. REFER TO SPECIFICATIONS FOR REQUIRED QUALITY, CONTROL FUNCTIONS, AND ADDITIONAL FEATURES. ALL VFD'S FOR MECHANICAL EQUIPMENT SHALL BE PROVIDED BY THE MECHANICAL CONTRACTOR AND INSTALLED BY THE ELECTRICAL CONTRACTOR.
10. FIELD CHECK ALL ROUGH AND/OR FINISH DIMENSIONS FOR ACCURATE FITTING OF EQUIPMENT, CABINETS, COUNTERS, FIXTURES AND ACCESSORIES BEFORE FABRICATION. PROVIDE AND INSTALL ALL NECESSARY FILLERS, SCRIBE STRIPS, PANELS, BASES, OR TRIM TO COMPLETE AND FINISH INSTALLATIONS.

## DESIGN CONDITIONS

### OUTDOOR DESIGN CONDITIONS

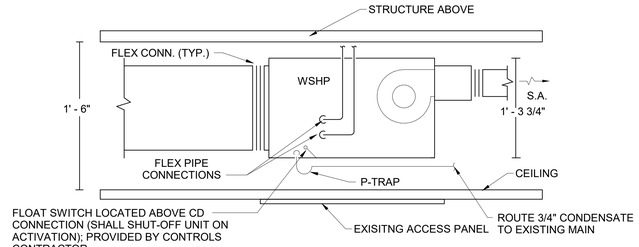
| | |
|---|---|
| SEASON | DB/WB (°F) |
| SUMMER | 92.7/77.8 |
| WINTER | 28 |

### INDOOR DESIGN CONDITIONS

| SPACE TYPE | SEASON | OCCUPIED | UNOCCUPIED |
|---|---|---|---|
| ALL | COOLING | 75°F | 84°F |
| | HEATING | 70°F | 55°F |

### DESIGN VENTILATION RATES

| SPACE TYPE | CFM/PPL | CFM/FT² |
|---|---|---|
| SLEEPING ROOM | 5 | 0.06 |
| CORRIDOR | 0 | 0.06 |
| MULTI-PURPOSE ASSEMBLY | 5 | 0.06 |




NOTES:
1. UNIT SHALL BE SUSPENDED FROM THE STRUCTURE ABOVE.
2. UNIT SHALL BE TIGHT TO STRUCTURE.
3. PROVIDE WITH EXPANSION ANCHORS OR CONCRETE SCREWS, THREADED RODS, AND VIBRATION ISOLATION AT STRUTURE W/ NEOPRENE PADS.

## 1 LOW-PROFILE WATER SOURCE HEAT PUMP DETAIL

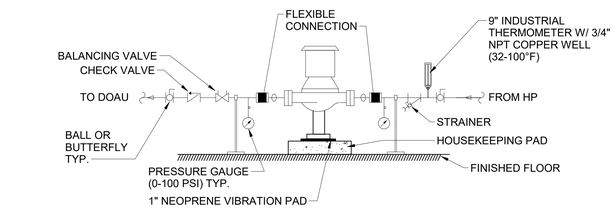
SCALE: NOT TO SCALE




NOTES:
1. SINGLE THICKNESS TYPE TURNING VANES RADIUS AND SPACING SHALL BE PROVIDED PER THIS DETAIL.

## 2 SINGLE THICKNESS VANE ELBOW DETAIL

SCALE: NOT TO SCALE



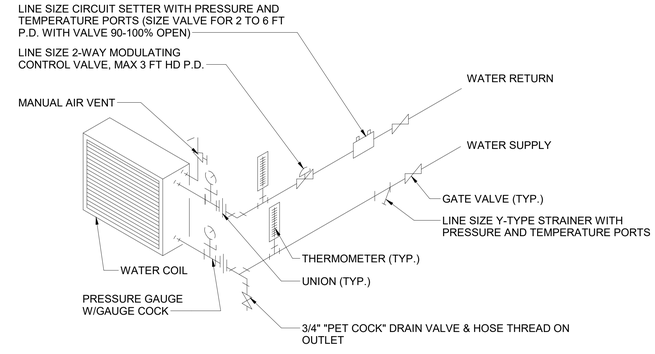

## 3 INLINE PUMP DETAIL

SCALE: NOT TO SCALE




## 4 TYP. FLEXIBLE CONNECTION

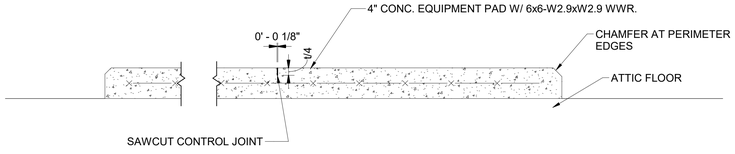
SCALE: NOT TO SCALE




NOTES:
1. COIL SHALL BE PIPED COUNTERFLOW.
2. COIL SHALL BE FURNISHED WTIH MANUAL AIR VENT PIPED TO ABOVE PIPE MAIN.
3. SET CIRCUIT SETTER DURING INSTALLATION FOR MIN 2FT. P.D.
4. PROVIDE WITH FLEXIBLE STAINLESS STEEL BRAIDED HOSES.

## 5 TYPICAL 2-WAY COIL PIPING DETAIL

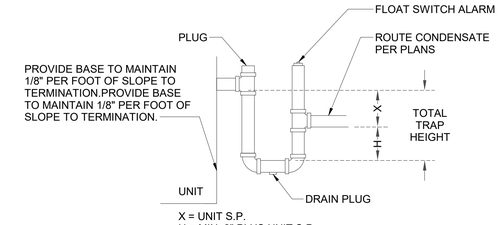
SCALE: NOT TO SCALE




NOTES:
1. INSTALL CONTROL JOINTS SUCH THAT NO DIMENSION BETWEEN JOINTS EXCEEDS 16' OR 3t MEASURED IN FEET, WHICHEVER IS LESSER (i.e. FOR A 4" THICK SLAB, MAX. CONTROL JOINT SPACING = 3*4 = 12').
2. INSTALL CONTROL JOINTS SUCH THAT THE RECTANGULAR SECTIONS CREATED BETWEEN THE JOINTS DO NOT EXCEED A 1:1.25 MAXIMUM RATIO (i.e. 10'x12.5' SECTION).
3. 4" SLAB SHOWN IS FOR TYPICAL CONSTRUCTION. ADJUST CONTROL JOINT DEPTH AND SPACING BASED ON ACTUAL SLAB THICKNESS (t) PROVIDED.
4. PROVIDE 6" MIN. BETWEEN EQUIPMENT AND EDGE OF SLAB.
5. CONCRETE MUST BE CURED FOR A MINIMUM 21 DAYS PRIOR TO ANY ADHESIVE ANCHOR INSTALLATION.
6. SEE SHEET S-001 FOR ADDITIONAL REQUIREMENTS.

## 6 EQUIPMENT PAD DETAIL

SCALE: NOT TO SCALE




X = UNIT S.P.
H = MIN. 2" PLUS UNIT S.P.
TOTAL TRAP HEIGHT = X + H

## 7 CONDENSATE DRAIN TRAP DETAIL

SCALE: NOT TO SCALE

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

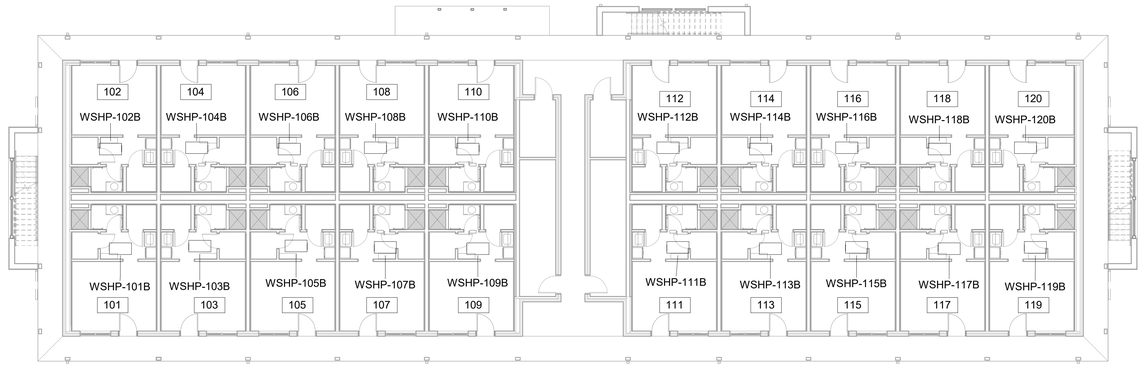
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS SATISFACTORY TO DATE | P.W. DWG. NO. 28493 | CEMS www.cemsite.com | M-001 |
| SAFETY DEPARTMENT SATISFACTORY TO DATE | DRAWN BY D. ILDERTON | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND — MARINE CORPS AIR STATION BEAUFORT, S.C. | |
| FIRE DEPARTMENT SATISFACTORY TO DATE | CHECKED BY P. MAHAFFEY | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY SATISFACTORY TO DATE | SUPERVISOR P. MAHAFFEY | LEGEND, ABBREVIATIONS, AND GENERAL NOTES | |
| PUBLIC WORKS SUPERVISOR SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: As indicated; MACC.; T.O. #; SHEET 39 OF 53 | |

10-04-2023

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |

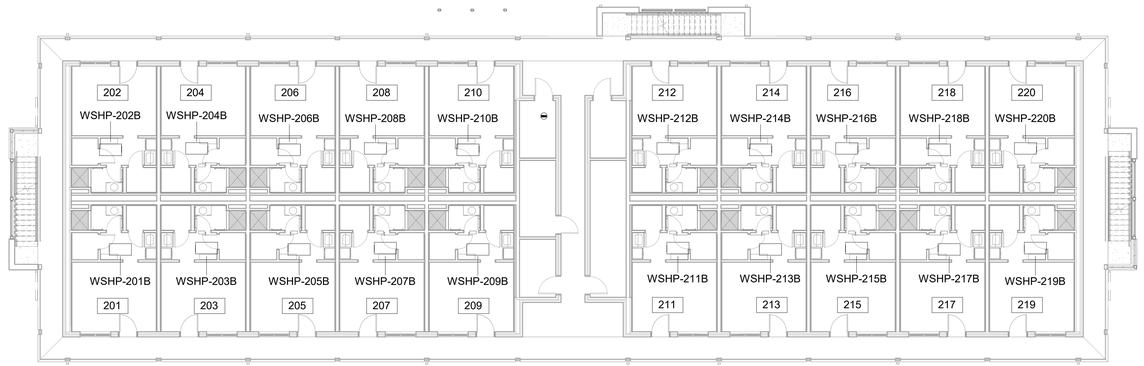
## GENERAL NOTES

1. REFER TO SHEET MP402 FOR PLUMBING WORK REQUIRED IN EACH SLEEPING ROOM.
2. ALL EXISTING REUSED DUCTWORK SHALL BE CLEANED AS PER SPECIFICATIONS.
3. REFER TO SHEETS M-401 AND M-501 FOR ENLARGED VIEWS OF SLEEPING ROOMS AND ATTIC.




## 1 FIRST FLOOR PLAN

SCALE: 3/32" = 1'-0"




## 2 SECOND FLOOR PLAN

SCALE: 3/32" = 1'-0"

## 3 THIRD FLOOR PLAN

SCALE: 3/32" = 1'-0"

PLAN NORTH

24"0 4' 8' 16'

**SCALE: 3/32"=1'-0"**

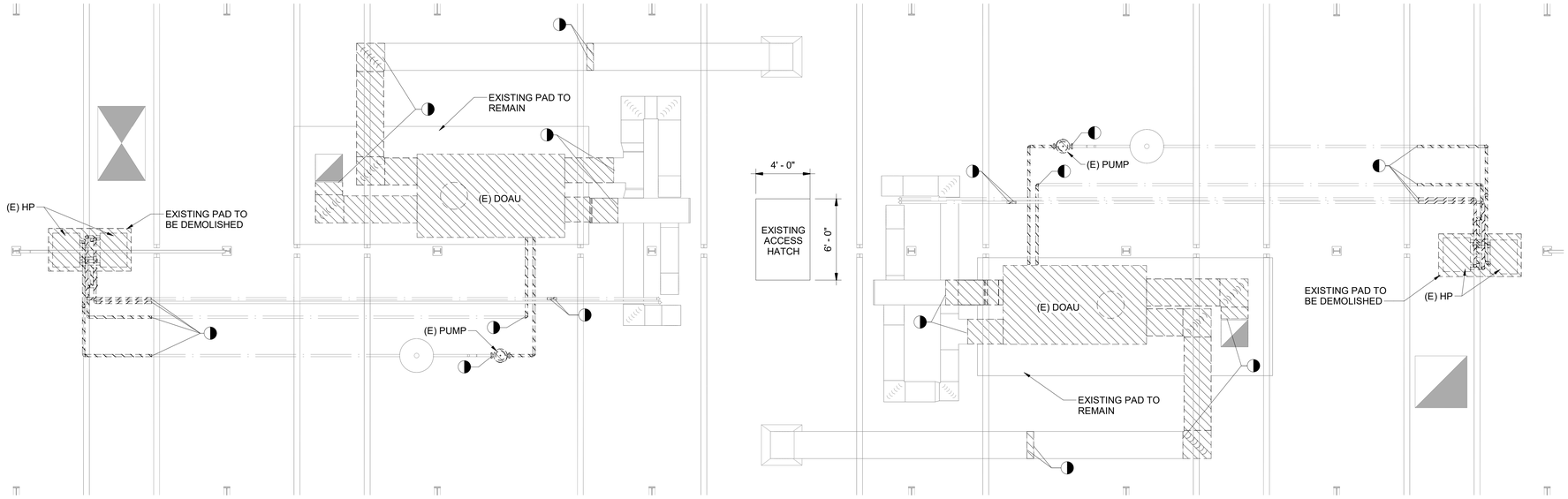
**GRAPHIC SCALE**

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

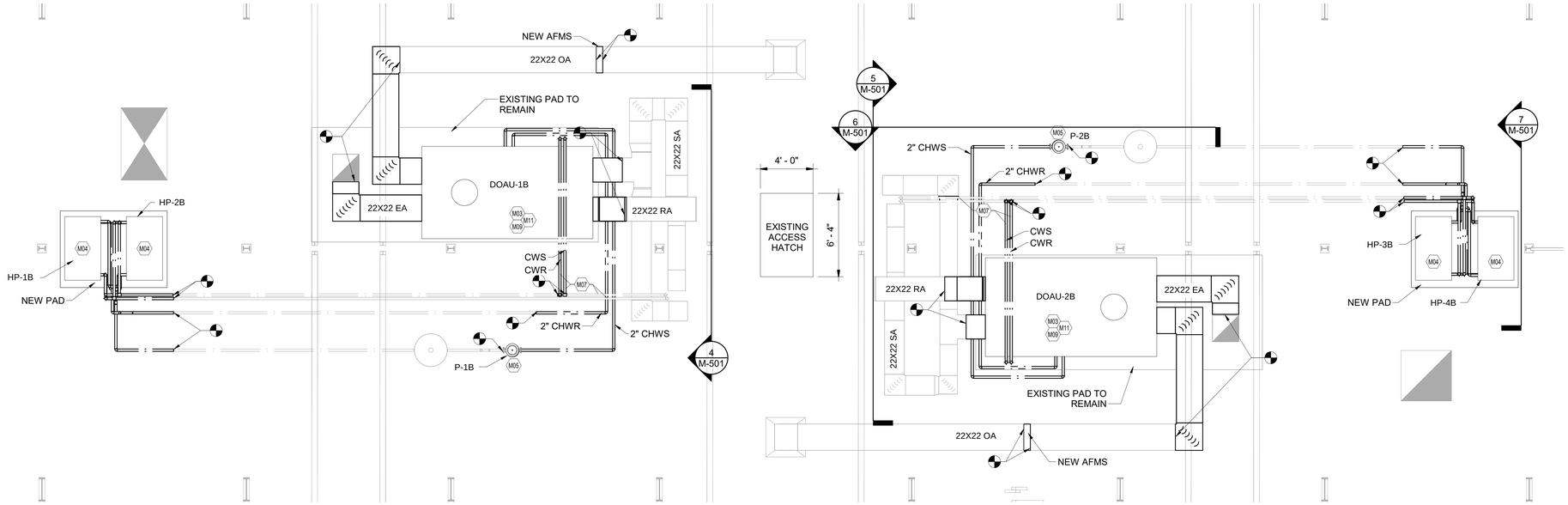
10-04-2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28494 | CEMS<br>www.cems-se.com \ PW Dave Tehoon \ Project \ 2019 \ 1943-875-3637 | M-101B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>D. ILDERTON | DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND<br>**MARINE CORPS AIR STATION**<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>P. MAHAFFEY | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>P. MAHAFFEY | FIRST, SECOND, AND THIRD FLOOR PLANS - BLDG 1245G | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F \| CODE IDENT. NO. \| N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: \| CONSTR. CONTR. NO.: N40085-18-D-0616 | |
| | PUBLIC WORKS OFFICER | SCALE: 3/32" = 1'-0" \| MACC. \| T.O. # \| SHEET 40 OF 53 | |




## 1 ENLARGED DEMO ATTIC PLAN

SCALE: 1/4" = 1'-0"




## 2 ENLARGED ATTIC PLAN

SCALE: 1/4" = 1'-0"

PLAN NORTH

12" 0 2' 4' 8'

**SCALE: 1/4"=1'-0"**

**GRAPHIC SCALE**

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

# GENERAL NOTES

1. RECONNECT ALL NEW EQUIPMENT TO EXISTING DUCTWORK AND HYDRONIC SYSTEMS.
2. ALL EXISTING REUSED DUCTWORK SHALL BE CLEANED AS PER SPECIFICATIONS.

# SHEET KEYNOTES

M03 TRANSITION AND DEMOLISH OR ADD DUCTWORK AND PIPING AS NECESSARY FOR INSTALLATION OF NEW EQUIPMENT, MATCH NEW PIPE AND DUCT SIZE TO EXISTING PIPE AND DUCT SIZE. PROVIDE FLEXIBLE CONNECTIONS FOR DUCTWORK AND PIPING AT POINT OF RECONNECTION.

M04 TRANSITION AND DEMOLISH OR ADD PIPING AS NECESSARY FOR INSTALLATION OF NEW EQUIPMENT, MATCH SIZE OF NEW PIPE TO EXISTING PIPE SIZE. PROVIDE FLEXIBLE CONNECTIONS FOR PIPING AT POINT OF RECONNECTION. CONTRACTOR TO COORDINATE INSTALLATION OF NEW WWHP'S WITH EXISTING STRUCTURAL FRAMING AND BRACING.

M05 TRANSITION AND DEMOLISH OR ADD PIPING AS NECESSARY FOR INSTALLATION OF NEW EQUIPMENT, MATCH NEW PIPE AND DUCT SIZE TO EXISTING PIPE AND DUCT SIZE. PROVIDE FLEXIBLE CONNECTIONS FOR PIPING AT POINT OF RECONNECTION.

M07 SIZE OF NEW CONDENSER WATER PIPING SERVING DOAU REHEAT COIL SHALL MATCH SIZE OF EXISTING CONDENSER WATER PIPING.

M09 CONTRACTOR TO VERIFY NEW UNIT SIZE, WHICH WILL REQUIRE FIELD INSTALLATION OF MODULAR SECTIONS THROUGH THE EXISTING OPENING ABOVE THE HALLWAY (SEE PLAN FOR OPENING SIZE).

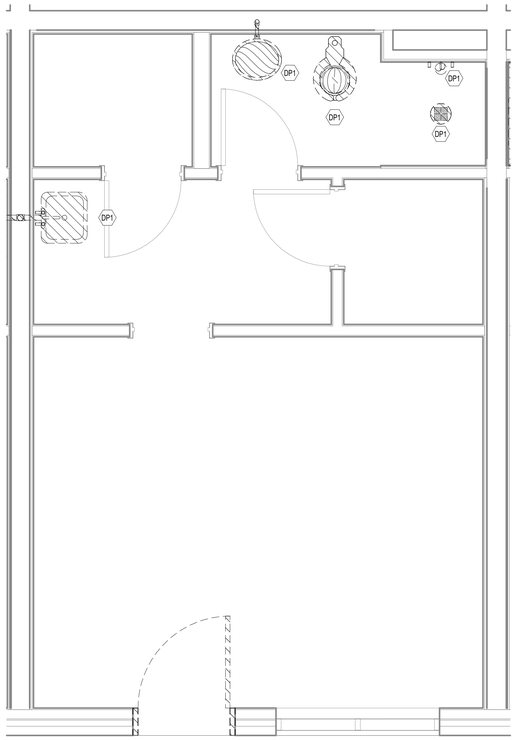
M11 COORDINATE LOCATION OF EXISTING STRUCTURE WITH ACCESS HATCHES TO ENSURE EQUIPMENT MAINTENANCE ACCESS IS MAINTAINED.

STATE OF SOUTH CAROLINA No. 23532 10-04-2023

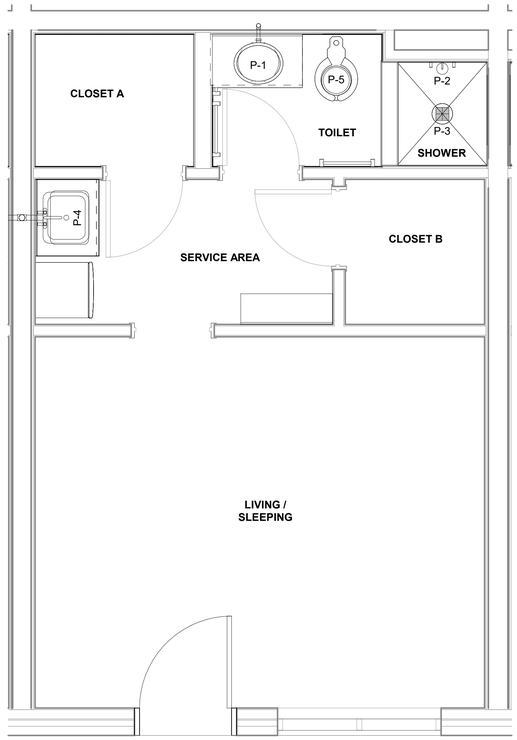
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28495 | CEMS<br>www.cems-eng.com 1766 Sam Rittenberg Blvd. Charleston, SC 29407 (843) 871-9007 | M-401 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>D. ILDERTON | DEPARTMENT OF THE NAVY<br>**MARINE CORPS AIR STATION**<br>BEAUFORT, S.C. | NAVAL FACILITIES ENGINEERING COMMAND |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>P. MAHAFFEY | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M)<br>ENLARGED PLANS | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>P. MAHAFFEY | | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE<br>PUBLIC WORKS OFFICER | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO.<br>HQMC PROJ. NO.:; CONSTR. CONTR. NO.: N40085-18-D-0616<br>SCALE: As indicated; MACC.; T.O. #; SHEET 41 OF 53 | |




1 TYPICAL DEMOLITION SLEEPING ROOM PLAN
SCALE: 1/2" = 1'-0"




2 TYPICAL SLEEPING ROOM PLAN
SCALE: 1/2" = 1'-0"

## PLUMBING FIXTURE SCHEDULE

| MARK | FIXTURE | ROUGH-IN SIZE | | | | REMARKS |
|---|---|---|---|---|---|---|
| | | DCW | DHW | SS | V | |
| P-1 | LAVATORY | 1/2" | 1/2" | 1 1/4" | 1 1/4" | SEE 22.00.00 |
| P-2 | SHOWER MIXING VALVE AND HEAD | 1/2" | 1/2" | 1 1/4" | 1 1/2" | SEE 22.00.00 |
| P-3 | SHOWER DRAIN | | | 3" | | SEE 22.00.00 |
| P-4 | SINK | 1/2" | 1/2" | 1 1/2" | 1 1/2" | SEE 22.00.00 |
| P-5 | WATER CLOSET | 1" | | 4" | 2" | SEE 22.00.00 |

NOTES:
1. NEW ANGLE STOP ISOLATION VALVES, STAINLESS STEEL BRAIDED HOSES, WAX RINGS, AND TRAPS. SEE SPECIFICATIONS FOR COMPLETE REQUIREMENTS.

## ABBREVIATIONS

| | |
|---|---|
| (E) | EXISTING |
| P-# | PLUMBING FIXTURE DESIGNATION |
| TYP. | TYPICAL |

## LEGEND:

- TO BE REMOVED/DEMOLISHED
- FLOOR DRAIN
- PIPE TURNING UP
- PIPE TURNING DOWN
- SHEET KEYNOTE

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. CONTRACTOR SHALL INSPECT AND THOROUGHLY CLEAN EXISTING UNDERFLOOR SANITARY LINE PRIOR TO INSTALLING NEW PLUMBING SERVICES.
2. SHOULD THE CONTRACTOR DISCOVER ANY ISSUES WITH REUSING THE EXISTING UNDERFLOOR SANITARY LINE, CONTRACTOR SHALL REPORT ISSUE TO THE CONTRACTING OFFICER FOR RESOLUTION PRIOR TO CONTINUATION OF WORK.
3. RECONNECT ALL NEW PLUMBING FIXTURES TO EXISTING DOMESTIC PIPING, SANITARY AND WASTE VENT.

## SHEET KEYNOTES

DP1 DEMOLISH AND REMOVE EXISTING PLUMBING FIXTURE. ASSOCIATED SUPPLY, SANITARY SEWER, AND VENT PIPING TO REMAIN FOR REUSE.

6" 0 1' 2' 4'

SCALE: 1/2"=1'-0"

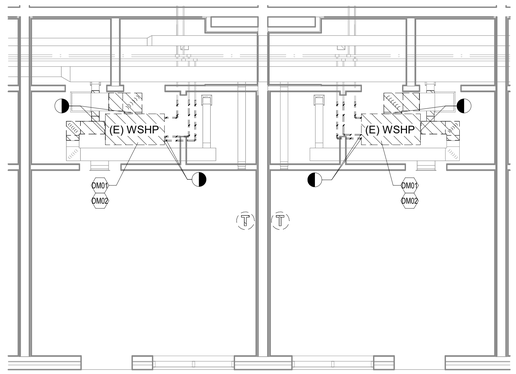
GRAPHIC SCALE

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

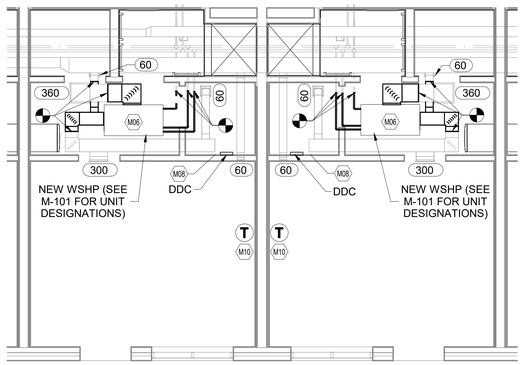
ISSUE DATE:08 SEPT 2023

10-04-2023

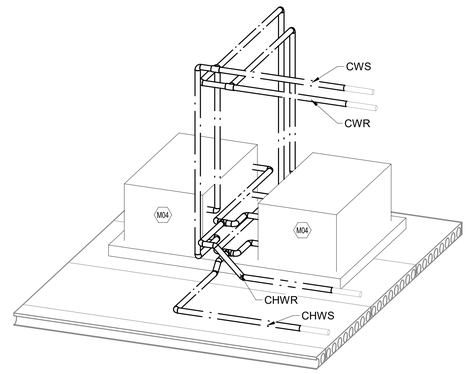
| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28496 | CEMS | MP402 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>J. GREENE | DEPARTMENT OF THE NAVY<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | NAVAL FACILITIES ENGINEERING COMMAND |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>P. MAHAFFEY | REPAIR PHASE 4 BEQ BUILDING 1245G<br>(BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>P. MAHAFFEY | TYPICAL SLEEPING ROOM PLAN,<br>ABBREVIATIONS, LEGEND AND SCHEDULE | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE<br>PUBLIC WORKS OFFICER | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO.<br>HQMC PROJ. NO.:; CONSTR. CONTR. NO.:N40085-18-D-0616<br>SCALE: As indicated; MACC.; T.O. #; SHEET 42 OF 53 | |



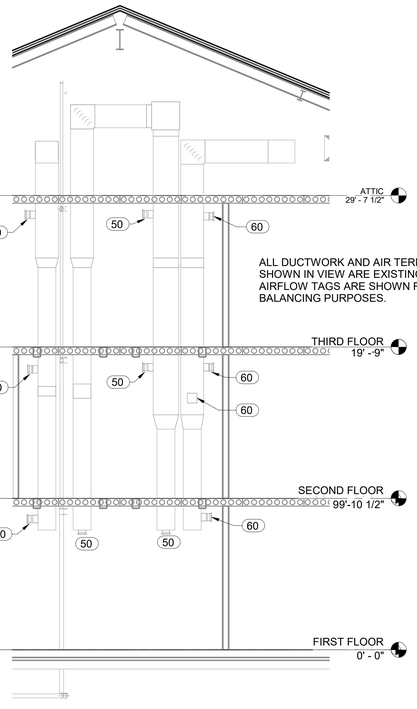

1 ENLARGED DEMO SLEEPING ROOM PLAN
SCALE: 1/4" = 1'-0"



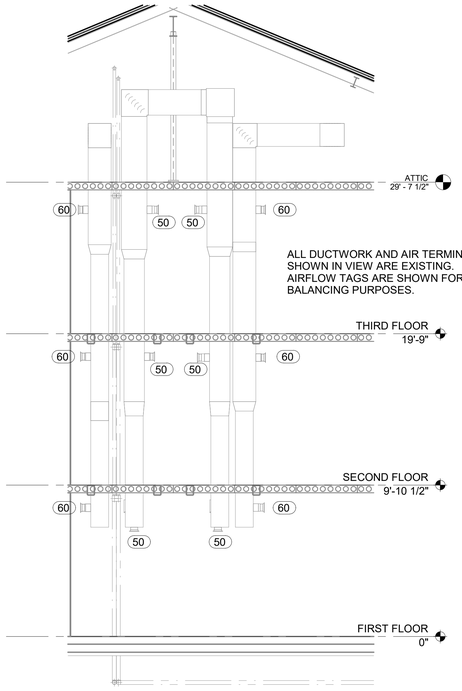

2 ENLARGED SLEEPING ROOM PLAN
SCALE: 1/4" = 1'-0"



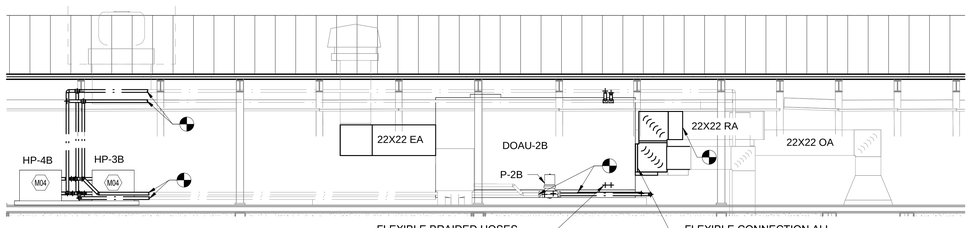

3 HP ISOMETRIC VIEW
SCALE: NOT TO SCALE



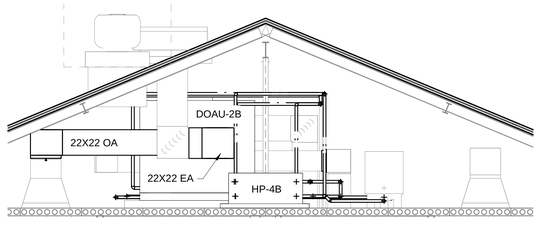

4 DOAU NORTH RISER
SCALE: 1/4" = 1'-0"




5 DOAU SOUTH RISER
SCALE: 1/4" = 1'-0"




2 / M-401 — 6 SECTION (TYP. DOAU)
SCALE: 1/4" = 1'-0"




2 / M-401 — 7 SECTION (TYP. DOAU)
SCALE: 1/4" = 1'-0"

PLAN NORTH

12" 0 2' 4' 8'

SCALE: 1/4"=1'-0"

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. RECONNECT ALL NEW EQUIPMENT TO EXISTING DUCTWORK AND HYDRONIC SYSTEMS.
2. ALL EXISTING REUSED DUCTWORK SHALL BE CLEANED AS PER SPECIFICATIONS.

## SHEET KEYNOTES

| | |
|---|---|
| DM01 | DISCONNECT EXISTING WATER-SOURCE HEAT PUMP FROM EXISTING DUCTWORK AND PIPING, DEMOLISH DUCTWORK AND PIPING AS NEEDED FOR REMOVAL OF UNIT. |
| DM02 | DEMOLISH AND REMOVE EXISING WATER-SOURCE HEAT PUMP, INCLUDING EXISTING CONTROLS LOCATED IN CEILING ABOVE CLOSET ALONG WITH EXISING WALL MOUNTED THERMOSTAT. |
| M04 | TRANSITION AND DEMOLISH OR ADD PIPING AS NECESSARY FOR INSTALLATION OF NEW EQUIPMENT. MATCH SIZE OF NEW PIPE TO EXISTING PIPE SIZE. PROVIDE FLEXIBLE CONNECTIONS FOR PIPING AT POINT OF RECONNECTION. CONTRACTOR TO COORDINATE INSTALLATION OF NEW WWHP'S WITH EXISTING STRUCTURAL FRAMING AND BRACING. |
| M06 | TRANSITION AND DEMOLISH OR ADD DUCTWORK AND PIPING AS NECESSARY FOR INSTALLATION OF NEW EQUIPMENT THROUGH EXISTING ACCESS HATCH, MATCH SIZE OF NEW DUCTWORK AND PIPING TO EXISTING DUCTWORK AND PIPING DIMENSIONS. PROVIDE FLEXIBLE CONNECTIONS FOR DUCTWORK AND PIPING AT POINT OF RECONNECTION. CONTRACTOR TO VERIFY LEFT OR RIGHT-HAND RETURN CONFIGURATION FOR EACH ROOM WSHP. |
| M08 | NEW CONTROLS FOR WATER SOURCE HEAT PUMP SHALL BE LOCATED WITHIN THE SOFFIT, ADJACENT TO EXISTING GRILLE. A 12 X 12 ACCESS HATCH SHALL BE PROVIDED ON THE SIDE OF THE SOFFIT. |
| M10 | EXISTING THERMOSTAT SHALL BE DEMOLISHED AND REPLACED WITH NEW THERMOSTAT. |

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28497 | CEMS | M-501 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>D. ILDERTON | DEPARTMENT OF THE NAVY<br>NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>P. MAHAFFEY | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>P. MAHAFFEY | SECTIONS AND ENLARGED PLANS | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: 1/4" = 1'-0" | MACC. T.O. # SHEET 43 OF 53 |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |

## BEQ B WATER-SOURCE HEAT PUMP SCHEDULE

| TYPICAL ASSOCIATED SPACE (SPACE NUMBER) | SA CFM | MIN ESP | COOLING | | | | | HEATING | MOTOR HP | EER | FLA | MCA | V/Ø/HZ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | CAPACITY (SENS./TOTAL MBH) | EAT (DB/WB) (°F) | LAT (°F) | EWT/LWT (°F) | FLOW (GPM) | CAPACITY (MBH) | | | | | |
| WSHP-101-120, 201-220, 301-320 | 360 | 0.25 in-wg | 8.7 | 75/62.5 | 55 °F | 85/95 | 2.25 | 11.2 | 1/3 | 15.5 | 6 A | 6.8 | 208/1/60 |

NOTES:
1. PROVIDE SINGLE POINT POWER CONNECTION.
2. PROVIDE UNIT WITH 2" MERV 13 FILTER.
3. PROVIDE LOW-PROFILE UNIT SELECTION.
4. PROVIDE WITH HOT GAS REHEAT AND ECM MOTOR.
5. ALL CONTROLS BY CONTROLS CONTRACTOR (HARRIS CONTROLS).
6. SEE M-700 SERIES DRAWINGS FOR CONTROL REQUIREMENTS.
7. SEE SHEET M-101 FOR UNIT DESIGNATIONS. TOTAL NUMBER OF WSHP: 60.
8. PROVIDE WITH BALL VALVE CONTROL HOSE KITS, STAINLESS STEEL BRAIDED HOSES, BRASS BALL VALVES, Y-BALL STRAINER WITH BLOW DOWN VALVE AND HOSE CONNECTOR.

## BEQ B DOAU HYDRONIC COIL SCHEDULE

| | | Air | | | | | | | | | | | Hydronic | | | | | | | | Physical | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Heating | | | | Cooling | | | Entering | | Leaving | | Heating | | | | Cooling | | | | | | | | |
| ID | Location | Airflow | Load | Entering DB Temp | Leaving DB Temp | Airflow | Load | Sensible Load | DB Temp | WB Temp | DB Temp | WB Temp | Flow | Entering Temp | Leaving Temp | Head Loss | Flow | Entering Temp | Leaving Temp | Head Loss | Working Fluid | Number of Coils | FLA | MCA | MOP | V/Ø/HZ |
| RHC-DOAU-1B, HCC-DOAU-1B | DOAU-1B | 2220 cfm | 59940 Btu/h, 51000 Btu/h | 50°F, 59.3°F | 75°F, 80°F | 2220 cfm | 123000 Btu/h | 73000 Btu/h | 79.8°F | 68.3°F | 50°F | 50°F | 12 gpm, 9 gpm | 90°F, 112°F | 80°F, 100°F | 10 ft | 25 gpm | 42°F | 52°F | 15 ft | Water | 2 | 27 | 34 | 35 | 208/3/60 |
| RHC-DOAU-2B, HCC-DOAU-2B | DOAU-2B | 2160 cfm | 58320 Btu/h, 49000 Btu/h | 50°F, 59.4°F | 75°F, 80°F | 2160 cfm | 119000 Btu/h | 71000 Btu/h | 79.8°F | 68.3°F | 50°F | 50°F | 11.70 gpm, 8 gpm | 90°F, 112°F | 80°F, 100°F | 10 ft | 24 gpm | 42°F | 52°F | 15 ft | Water | 2 | 27 | 34 | 35 | 208/3/60 |

## BEQ B DOAU ERV SCHEDULE

| | | | Return/Exhaust | | | | | Outside/Supply | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Heating | | Cooling | | | Heating | | | Cooling | | | | |
| | | | | | | Entering | | | | | | Entering | | Leaving | |
| ID | Location | Type | Airflow | Entering DB Temp | Airflow | DB Temp | WB Temp | Airflow | Entering DB Temp | Leaving DB Temp | Airflow | DB Temp | WB Temp | DB Temp | WB Temp |
| ER-DOAU-1B | DOAU-1B | Wheel, Counter | 1850 cfm | 70°F | 1850 cfm | 75°F | 62.5°F | 2220 cfm | 28°F | 59.3°F | 2220 cfm | 92.7°F | 77.8°F | 79.8°F | 68.3°F |
| ER-DOAU-2B | DOAU-2B | Wheel, Counter | 1800 cfm | 70°F | 1800 cfm | 75°F | 62.5°F | 2160 cfm | 28°F | 59.3°F | 2160 cfm | 92.7°F | 77.8°F | 79.8°F | 68.3°F |

## BEQ B DOAU EXHAUST FAN SCHEDULE

| | | | | Air | | |
|---|---|---|---|---|---|---|
| ID | Location | Qty | Type | Maximum Airflow | Motor HP | External Static Pressure |
| EF-DOAU-1B | DOAU-1B | 1 | PLUG, VFD | 1850 cfm | 3 HP | 3 in. H2O |
| EF-DOAU-2B | DOAU-2B | 1 | PLUG, VFD | 1800 cfm | 3 HP | 3 in. H2O |

## BEQ B DOAU OUTSIDE AIR FAN SCHEDULE

| | | | | Air | | |
|---|---|---|---|---|---|---|
| ID | Location | Qty | Type | Maximum Airflow | Motor HP | External Static Pressure |
| OF-DOAU-1B | DOAU-1B | 1 | PLUG, VFD | 2220 cfm | 3 HP | 4 in. H2O |
| OF-DOAU-2B | DOAU-2B | 1 | PLUG, VFD | 2160 cfm | 3 HP | 4 in. H2O |

NOTES:
1. PROVIDE SINGLE POINT POWER CONNECTION.
2. PROVIDE UNIT WITH 2" MERV-8 FILTERS IN EXHAUST AND SUPPLY AIRSTREAMS.
3. SEE M-700 SERIES DRAWINGS FOR CONTROL REQUIREMENTS.
4. ALL CONTROLS BY CONTROLS CONTRACTOR (HARRIS CONTROLS).
5. PROVIDE IN MODULAR SECTIONS CAPABLE OF FITTING THROUGH EXISTING ATTIC ACCESS OPENING (4'X6').
6. PROVIDE ROOM FOR CONTRACTOR TO INSTALL SENSOR BETWEEN COILS.
7. PROVIDE 6 INCH EQUIPMENT RAIL AND ROUTE CONDENSATE DRAIN LINE TO NEAREST DRAIN.

## BEQ B WATER TO WATER SOURCE HEAT PUMP SCHEDULE

| | | | | | Hydronic (Transfer) Load | | | | | | | | | | | Hydronic (Source) Load | | | | | | | | | | | Electrical | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Heating | | | | | Cooling | | | | | | Heating | | | | | Cooling | | | | | | | | |
| ID | Location | Type | Refrigerant | Nominal Tonnage | Load | Flow | Entering Temp | Leaving Temp | Head Loss | Load | Flow | Entering Temp | Leaving Temp | Head Loss | Working Fluid | Load | Flow | Entering Temp | Leaving Temp | Head Loss | Load | Flow | Entering Temp | Leaving Temp | Head Loss | Working Fluid | MCA | FLA | V/Ø/HZ |
| HP-1B | ATTIC | Scroll | R-410A | 10 | 25000 Btu/h | 5 gpm | 100°F | 112°F | 15 ft | 62500 Btu/h | 12.5 gpm | 52°F | 42°F | 15 ft | Water | 50000 Btu/h | 10 gpm | 70°F | 60°F | 15 ft | 100000 Btu/h | 20 gpm | 80°F | 90°F | 15 ft | Water | 60 | 52.6 | 208/1/60 |
| HP-2B | ATTIC | Scroll | R-410A | 10 | 25000 Btu/h | 5 gpm | 100°F | 112°F | 15 ft | 62500 Btu/h | 12.5 gpm | 52°F | 42°F | 15 ft | Water | 50000 Btu/h | 10 gpm | 70°F | 60°F | 15 ft | 100000 Btu/h | 20 gpm | 80°F | 90°F | 15 ft | Water | 60 | 52.6 | 208/1/60 |
| HP-3B | ATTIC | Scroll | R-410A | 10 | 25000 Btu/h | 5 gpm | 100°F | 112°F | 15 ft | 60000 Btu/h | 12 gpm | 52°F | 42°F | 15 ft | Water | 50000 Btu/h | 10 gpm | 70°F | 60°F | 15 ft | 100000 Btu/h | 20 gpm | 80°F | 90°F | 15 ft | Water | 60 | 52.6 | 208/1/60 |
| HP-4B | ATTIC | Scroll | R-410A | 10 | 25000 Btu/h | 5 gpm | 100°F | 112°F | 15 ft | 60000 Btu/h | 12 gpm | 52°F | 42°F | 15 ft | Water | 50000 Btu/h | 10 gpm | 70°F | 60°F | 15 ft | 100000 Btu/h | 20 gpm | 80°F | 90°F | 15 ft | Water | 60 | 52.6 | 208/1/60 |

NOTES:
1. PROVIDE SINGLE POINT POWER CONNECTION.
2. PROVIDE 2-STAGE COMPRESSOR. 10 TON NOMINAL COMPRESSOR WITH STANDARD 5 YEAR WARRANTY.
3. SEE M-700 SERIES DRAWINGS FOR CONTROL REQUIREMENTS.
4. ALL CONTROLS BY CONTROLS CONTRACTOR (HARRIS CONTROLS).

## BEQ B DOAU HYDRONIC PUMPS SCHEDULE

| | | | Fluid | | | Electrical | | |
|---|---|---|---|---|---|---|---|---|
| ID | Qty | Type | Flow | Working Fluid | Head Loss | Motor Size | Motor Speed | V/Ø/HZ |
| P-1B | 1 | Base-mounted, Close-Coupled, In-Line | 25 gpm | Water | 20 ft | 0.75 hp | 931 rpm | 208/3/60 |
| P-2B | 1 | Base-mounted, Close-Coupled, In-Line | 24 gpm | Water | 20 ft | 0.75 hp | 931 rpm | 208/3/60 |

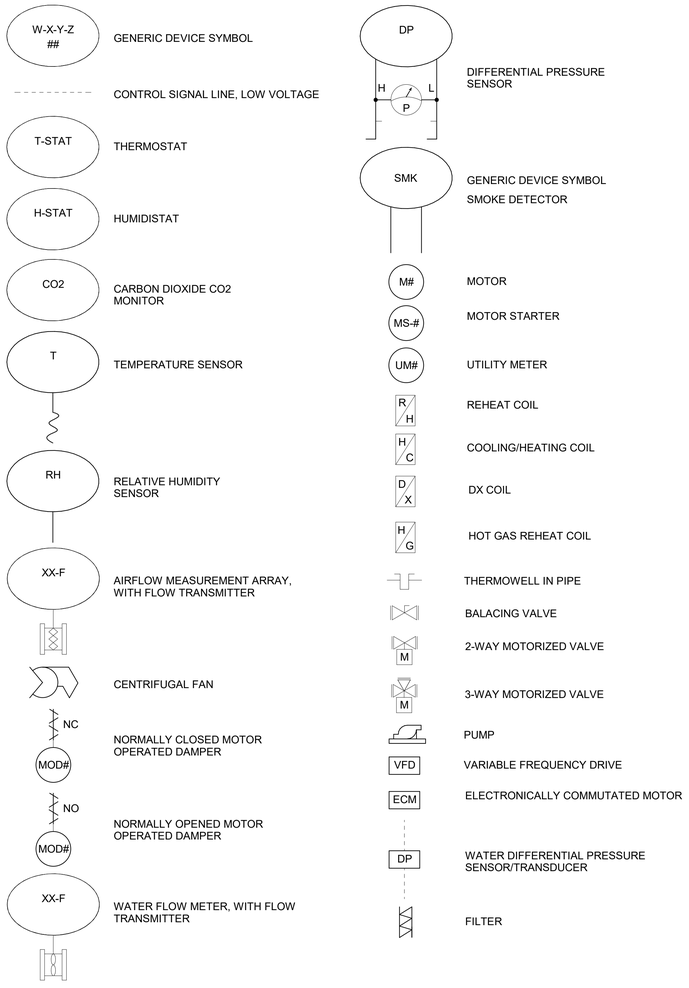
NOTES:
1. PROVIDE SINGLE POINT POWER CONNECTION.
2. MOTOR SHALL BE PROVIDED WITH ECM MOTOR.
3. SEE INLINE PUMP DETAIL SHEET M-001.

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

| | | |
|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28498 | CEMS<br>M-601B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>D. ILDERTON | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>P. MAHAFFEY | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M)<br>SCHEDULES - BLDG 1245G |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>P. MAHAFFEY | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. |
| | | HQMC PROJ. NO.:; CONSTR. CONTR. NO.:N40085-18-D-0616 |
| PUBLIC WORKS OFFICER | SCALE: | MACC.; T.O. #; SHEET 44 OF 53 |

## LEGEND:





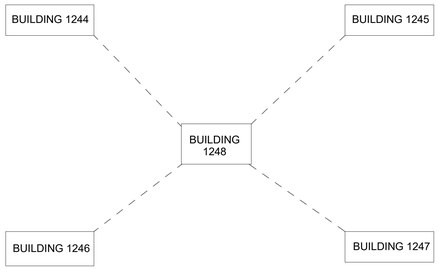


## ABBREVIATIONS

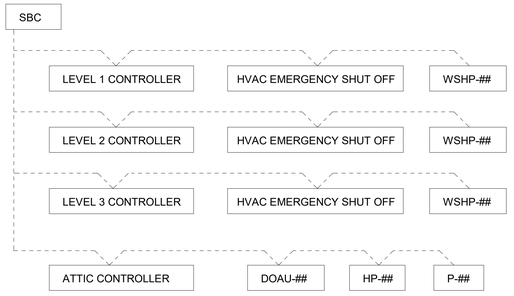
| | |
|---|---|
| 2P | TWO-POSITION (CONTROL SIGNAL) |
| ADJ. | ADJUSTABLE |
| AFMS | AIRFLOW MEASUREMENT STATION |
| ALM | ALARM |
| ATFP | ANTITERRORISM AND FORCE PROTECTION |
| BP | BYPASS |
| C | COMMAND (MODULATING CONTROL SIGNAL) |
| CHWR | CHILLED WATER RETURN |
| CHWS | CHILLED WATER SUPPLY |
| CP | CONTROL POINT |
| CV | CONTROL VALVE |
| D | DAMPER |
| DA | DISCHARGE AIR |
| DDC | DIRECT DIGITAL CONTROL(LER) |
| DOAU | DEDICATED OUTDOOR AIR UNIT |
| DP | DIFFERENTIAL PRESSURE |
| EA | EXHAUST AIR |
| ECM | ELECTRONICALLY COMMUTATED MOTOR |
| EF | EXHAUST FAN |
| ENA | ENABLE |
| ESW | END SWITCH |
| EW | ENTHALPY WHEEL |
| EWT | ENTERING WATER TEMPERATURE |
| FBK | FEEDBACK |
| H-STAT | HUMIDISTAT |
| HP | WATER TO WATER HEAT PUMP |

## ABBREVIATIONS

| | |
|---|---|
| LWT | LEAVING WATER TEMPERATURE |
| MA | MIXED AIR |
| MOV | MOTOR OPERATED VALVE |
| MS | MOTOR STARTER |
| N/A | NOT APPLICABLE |
| NC | NORMALLY CLOSED |
| NO | NORMALLY OPEN |
| OA | OUTDOOR AIR |
| P | PUMP |
| POS | POSITION |
| RA | RETURN AIR |
| RF | RETURN FAN |
| RH | RELATIVE HUMIDITY |
| RHT | REHEAT |
| S | STATUS |
| SA | SUPPLY AIR |
| SBC | SYSTEM BUILDING CONTROLLER |
| SF | SUPPLY FAN |
| SMK | SMOKE |
| SP | SETPOINT |
| SPD | SPEED |
| SS | START/STOP COMMAND |
| STS | STATUS |
| SYS | SYSTEM |
| T | TEMPERATURE |
| T-STAT | THERMOSTAT |
| WSHP | WATER SOURCE HEAT PUMP |




1 NETWORK DIAGRAM
SCALE: NOT TO SCALE




2 BUILDING CONTROL DIAGRAM
SCALE: NOT TO SCALE

**REVISIONS**

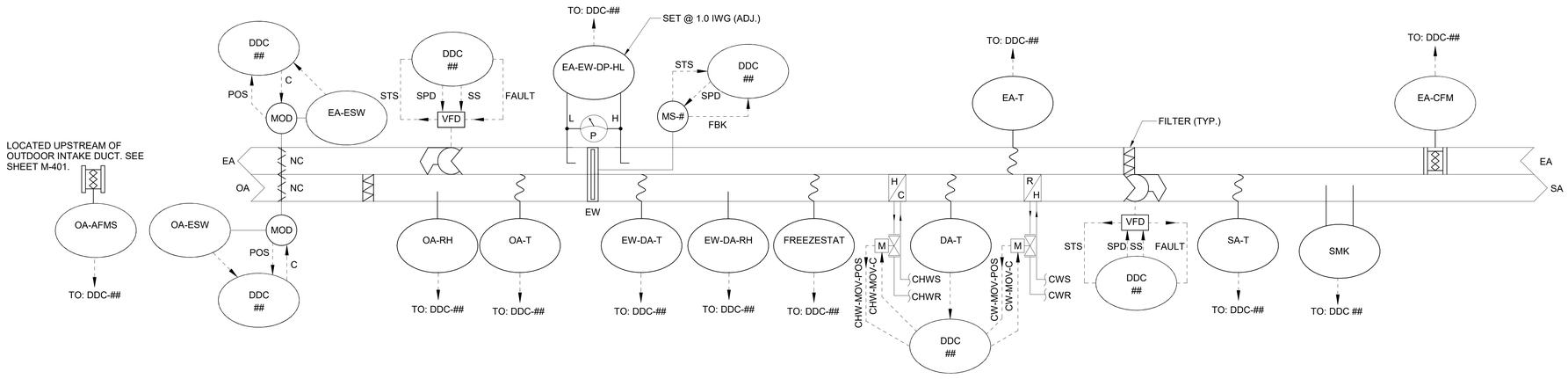
| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28499 | CEMS | M-701 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>D. ILDERTON | DEPARTMENT OF THE NAVY<br>NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>P. MAHAFFEY | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>P. MAHAFFEY | CONTROLS LEGEND AND ABBREVIATIONS | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F / CODE IDENT. NO. / N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: 12" = 1'-0" / MACC. / T.O. # | SHEET 45 OF 53 |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |




1 DOAS CONTROLS DIAGRAM
SCALE: NOT TO SCALE

DEDICATED OUTDOOR AIR SYSTEM (DOAS-1) SEQUENCE OF OPERATION (SEE DOAS-1 CONTROL DIAGRAM):

GENERAL: THIS UNIT IS A 100% OUTDOOR AIR WHICH IS TO SUPPLY NEUTRAL, DEHUMIDIFIED SUPPLY AIR DIRECTLY INTO THE OCCUPIED SPACES. THE UNIT IS A CONSTANT VOLUME DOAU WITH ENERGY RECOVERY ENTHALPY WHEEL, FILTRATION, SUPPLY/EXHAUST FANS, REHEAT COIL, AND COOLING/HEATING COIL. THE DOAU WILL OPERATE 24/7. ALL CONTROLS TO BE PROVIDED BY HARRIS INTEGRATED CONTROLS.

ENTHALPY WHEEL OPERATION: ENTHALPY WHEEL SHALL BE IN CONSTANT OPERATION, EXCEPT DURING ECONOMIZER MODE OR WHEN OPERATION OF ENTHALPY WHEEL WILL CAUSE THE COOLING/HEATING COIL TO WORK HARDER.

SUPPLY AIR TEMPERATURE CONTROL: UNIT SHALL MAINTAIN A SUPPLY AIR TEMPERATURE BASED ON AMBIENT OUTDOOR AIR TEMPERATURES.

WHEN THE AMBIENT OUTDOOR AIR TEMPERATURE IS BELOW 55°F (ADJ.) THE DDC SYSTEM SHALL MODULATE THE HEATING COIL VALVE TO MAINTAIN THE DISCHARGE AIR TEMPERATURE (DA-T) SETPOINT OF 70°F (ADJ.).

WHEN THE AMBIENT OUTDOOR AIR TEMPERATURE RISES ABOVE 70°F (ADJ.), THE DDC SYSTEM SHALL MODULATE THE COOLING/HEATING COIL VALVE TO MAINTAIN THE DISCHARGE AIR TEMPERATURE (DA-T) SETPOINT OF 50°F (ADJ.) AND MODULATE THE REHEAT COIL VALVE TO MAINTAIN THE SUPPLY AIR TEMPERATURE (SA-T) SETPOINT OF 70°F (ADJ.).

VALVE INPUT AND POSITION FEEDBACK SHALL BE MONITORED AND GRAPHICALLY DISPLAYED IN THE DDC SYSTEM. MODULATING CONTROL VALVES SHALL BE SET TO SHUT IN THE EVENT THERE IS NO CALL FOR CONDITIONING.

MOTOR OPERATED DAMPER POSITION FEEDBACK SHALL BE MONITORED AND GRAPHICALLY DISPLAYED IN THE DDC SYSTEM. UPON CLOSURE OF EXHAUST AND/OR OUTDOOR AIR DAMPER, THE EXHAUST FAN AND/OR SUPPLY FAN SHALL BE DE-ACTIVATED, RESPECTIVELY.

ECONOMIZER MODE: IF THE AMBIENT OUTDOOR AIR TEMPERATURE IS BETWEEN 55°F (ADJ.) AND 70°F (ADJ.) AND THE OUTDOOR AIR RELATIVE HUMIDITY IS BELOW 55%RH (ADJ.), THE UNIT SHALL ENTER AN ECONOMIZER MODE. THE COOLING/HEATING COIL VALVE SHALL CLOSE, THE ENTHALPY WHEEL SHALL BE OFF, AND THE REHEAT COIL VALVE SHALL BE CLOSED. THE ENERGY WHEEL SHALL MODULATE A MINIMUM OF 3 MINUTES (ADJ.) PER HOUR TO PREVENT THE WHEEL FROM BECOMING IMPACTED.

DEHUMIDIFICATION MODE: WHEN THE RELATIVE HUMIDITY OF THE ENTHALPY WHEEL DISCHARGE AIR (EW-DA-RH) IS ABOVE 55% (ADJ.), THE COOLING/HEATING COIL VALVE SHALL MAINTAIN A DA-T OF 50°F (ADJ.) AND THE REHEAT VALVE TO MAINTAIN A SA-T OF 70°F (ADJ.).

SENSORS: ALL SENSORS AND DEVICES TO BE PROVIDED BY HARRIS INTEGRATED CONTROLS, AS REQUIRED BY THIS SEQUENCE OF OPERATION AND THE CONTROL DIAGRAM.

SAFETY SHUTDOWN AND ALARMS: ALL ALARMS SHALL BE SENT TO THE DDC SYSTEM IF THE ALARM CONDITIONS OCCUR.

1. EMERGENCY SHUTDOWN: IF THE HVAC EMERGENCY SHUTDOWN SIGNAL IS RECEIVED, THE SUPPLY/EXHAUST FANS SHALL BE DE-ENERGIZED, THE ENTHALPY WHEEL SHALL DE-ENERGIZE, DAMPERS SHALL CLOSE, CHILLED AND HEATING HOT WATER VALVES SHALL CLOSE, AND AN ALARM SHALL BE SENT TO THE DDC SYSTEM.
2. SMOKE DETECTION SHUTDOWN/ALARM: UPON SMOKE DETECTION HVAC UNIT SHUTDOWN SHALL BE ACCOMPLISHED VIA A HARDWIRE INTERLOCK AND A DDC COMMAND. WHEN THE SENSOR SENSES SMOKE, THE FANS SHALL BE DE-ENERGIZED AND AN ALARM SHALL BE SENT TO THE DDC SYSTEM AND BUILDING FIRE ALARM CONTROL PANEL. MANUAL RESET OF SMOKE DETECTOR IS REQUIRED. OUTSIDE AIR AND EXHAUST DAMPERS SHALL CLOSE.
3. IF THE FREEZE STAT IS TRIGGERED (BELOW 38°F), THE SUPPLY/EXHAUST FANS SHALL BE DE-ENERGIZED, DAMPERS SHALL CLOSE, CHILLED/HOT WATER VALVE SHALL BE OPENED TO 100%, AND AN ALARM SHALL BE SENT TO THE DDC SYSTEM. MANUAL RESET OF THE ALARM SHALL BE REQUIRED.
4. ENERGY RECOVERY ENTHALPY WHEEL ALARM: IF THE ENTHALPY WHEEL COMMAND IS ON AND STATUS IS OFF, AN ALARM SHALL BE SENT TO THE DDC SYSTEM.
5. DIFFERENTIAL PRESSURE HIGH LIMIT ALARM: IF ANY DIFFERENTIAL PRESSURE HIGH LIMITS ARE TRIGGERED, AN ALARM SHALL BE SENT TO THE DDC SYSTEM.
6. HIGH/LOW SUPPLY AIR TEMPERATURE ALARM: SUPPLY AIR TEMPERATURE SENSOR SHALL BE INSTALLED AFTER THE SUPPLY FANS AND IN A LOCATION TO PROVIDE AN ACCURATE SUPPLY AIR TEMPERATURE REPRESENTATIVE OF AIR DELIVERED. IF THE SUPPLY AIR TEMPERATURE IS NOT WITHIN +/- 5°F OF SA-T SETPOINT AN ALARM SHALL BE SENT TO THE DDC SYSTEM.
7. SUPPLY/EXHAUST FAN FAILURE: IF ANY OF THE FOLLOWING OCCUR, AN ALARM SHALL BE SENT TO THE DDC SYSTEM.
   A. SUPPLY/EXHAUST FANS COMMAND IS ON AND STATUS IS OFF, AN ALARM SHALL BE SENT TO THE DDC.
   B. SUPPLY/EXHAUST FAN COMMAND IS OFF AND STATUS IS ON, AN ALARM SHALL BE SENT TO THE DDC.

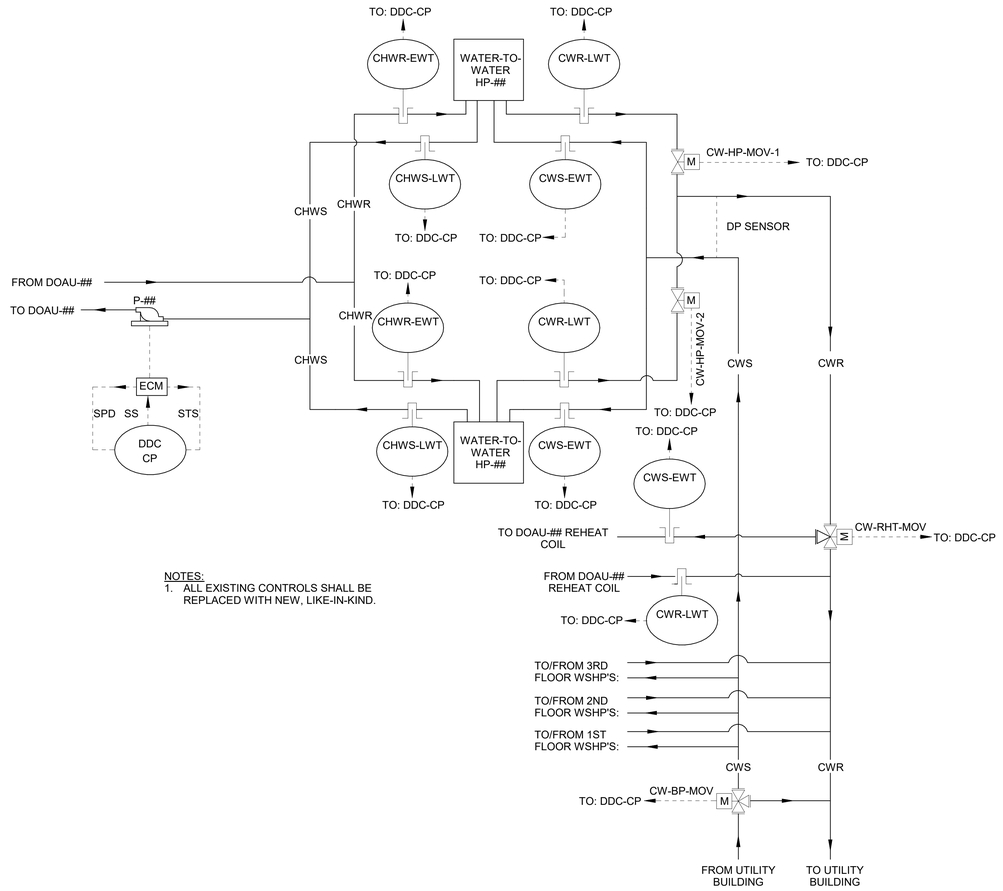
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

10-04-2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28500 | CEMS | M-702 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>D. ILDERTON | DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>P. MAHAFFEY | REPAIR PHASE 4 BEQ BUILDING 1245G<br>(BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>P. MAHAFFEY | CONTROLS | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: 12" = 1'-0" | MACC. T.O. # SHEET 46 OF 53 |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |




NOTES:
1. ALL EXISTING CONTROLS SHALL BE REPLACED WITH NEW, LIKE-IN-KIND.

HYDRONIC SYSTEM SEQUENCE OF OPERATION:

GENERAL: THE HYDRONIC EQUIPMENT CONSISTS OF THE FOLLOWING: (2) WATER-TO-WATER HEAT PUMPS, (1) INLINE HYDRONIC PUMP. AN AUTOMATIC LOGIC CONTROLLER CAPABLE OF STANDALONE OPERATION SHALL CONTROL THE SYSTEM. ALL CONTROLS SHALL BE BY HARRIS INTEGRATED CONTROLS. CONTROLS SHALL BE AUTOMATED LOGIC.

ALL OF THE WATER-TO-WATER HEAT PUMPS SHALL OPERATE IN TANDEM AND HAVE MOTORIZED 2-WAY CONTROL VALVES ON THE CONDENSER WATER LOOPS TO PROVIDE CONDENSER WATER TO THE WATER-TO-WATER HEAT PUMPS. THE VALVES SHALL BE LINKED TO THE EQUIPMENT VIA DIRECT CONNECTION TO THE SUPPLIED EQUIPMENT OR THE BAS AND COMMANDED TO OPEN/CLOSE WHEN THE EQUIPMENT CALLS FOR CONDENSER WATER. IF AT ANYTIME THE VALVE COMMAND IS NOT EQUAL TO THE VALVE STATUS WITHIN 5% VARIANCE (ADJ.), AN ALARM WILL ACTIVATE AT THE BAS. THE VALVES SHALL CLOSE WHEN THE WATER-TO-WATER HEAT PUMP SHUTS DOWN.

WATER-TO-WATER HEAT PUMP OPERATION:

COOLING MODE: THE HYDRONIC PUMP SHALL BE ACTIVATED WHEN THE DOAU IS IN COOLING OR DEHUMIDIFICATION MODE, THE SYSTEM SHALL FLOW CONSTANT WATER, AND THE HEAT PUMP COMPRESSOR AND CONDENSER WATER CONTROL VALVES (CW-HP-MOV-1 AND CW-HP-MOV-2) SHALL MODULATE TO MAINTAIN A CHILLED L.W.T. OF 42°F (ADJ.). ON A DROP IN CHILLED WATER TEMPERATURE BELOW 40°F (ADJ.), THE HEAT PUMP COMPRESSOR SHALL BE DE-ACTIVATED AND CONDENSER WATER CONTROL VALVES (CW-HP-MOV-1 AND CW-HP-MOV-2) SHALL CLOSE; THE HYDRONIC PUMP SHALL CONTINUE OPERATION.

HEATING MODE: THE HYDRONIC PUMP SHALL BE ACTIVATED WHEN THE DOAU IS IN HEATING MODE, THE SYSTEM SHALL FLOW CONSTANT WATER, THE HEAT PUMP REVERSING VALVE SHALL BE ACTIVATED, AND THE HEAT PUMP COMPRESSOR AND CONDENSER WATER CONTROL VALVES (CW-HP-MOV-1 AND CW-HP-MOV-2) SHALL MODULATE TO MAINTAIN A HEATING L.W.T. OF 112°F (ADJ.). ON A RISE IN HEATING HOT WATER TEMPERATURE ABOVE 115°F (ADJ.), THE HEAT PUMP COMPRESSOR SHALL BE DE-ACTIVATED AND CONDENSER WATER CONTROL VALVES (CW-HP-MOV-1 AND CW-HP-MOV-2) SHALL CLOSE; THE HYDRONIC PUMP SHALL CONTINUE OPERATION.

CONDENSER WATER (DOAU-## REHEAT COIL) OPERATION: DURING THE DOAU COOLING MODE OR DEHUMIDIFICATION MODE, THE CONTROL VALVE (CW-RH-MOV) SHALL MODULATE TO MAINTAIN A CONDENSER LEAVING WATER TEMPERATURE OF 90°F (ADJ.). DURING HEATING MODE THE CONTROL VALVE (CW-RH-MOV) SHALL BE CLOSED.

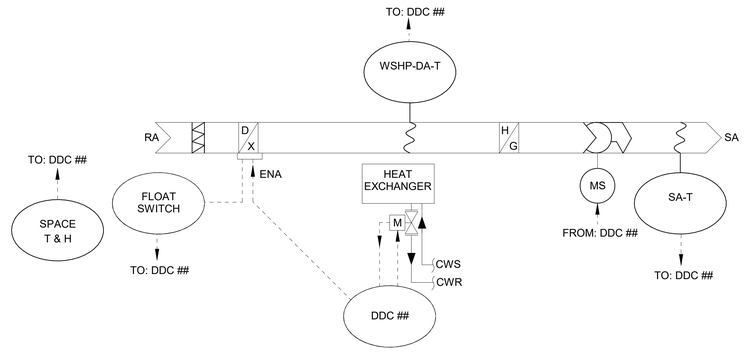
CW BYPASS CONTROL VALVE: THE CONDENSER WATER BYPASS CONTROL VALVE (CW-BP-MOV) SHALL BE CONTROLLED TO MODULATE VALVE POSITION TO MAINTAIN A MINIMUM DIFFERENTIAL PRESSURE SETPOINT OF 5 PSI (ADJ.) AT THE DIFFERENTIAL PRESSURE SENSOR. THIS SETPOINT IS TO BE PROVIDED DURING TEST AND BALANCE (TAB).

ALARMS: PUMP FAILURE, HIGH CONDENSER WATER TEMP (100°F ADJ.), OR HIGH CHILLED WATER TEMP (50°F ADJ.), VALVE POSITION.

FRONT END GRAPHICS: ALL POINTS INDICATED IN CONTROL SEQUENCE AND CONTROL DIAGRAM SHALL BE PROVIDED IN GRAHICS OF FRONT END SYSTEM.

## 1 CONDENSER WATER SYSTEM CONTROLS

SCALE: NOT TO SCALE




WATER-SOURCE HEAT PUMP SEQUENCE OF OPERATION:

GENERAL: THE WATER-SOURCE HEAT PUMP UNITS INCLUDE THE FOLLOWING: FILTER, COMPRESSOR, COIL, REVERSING VALVE AND DRAW THROUGH SUPPLY FAN. ALL CONTROLS SHALL BE BY HARRIS INTEGRATED CONTROLS. CONTROLS PROVIDED SHALL BE AUTOMATED LOGIC.

COOLING/HEATING MODE: THE FAN SHALL BE ACTIVATED ON A CALL FOR COOLING OR HEATING. THE HEAT PUMP CONTROL VALVE SHALL OPEN AND THE COMPRESSOR(S) SHALL BE STAGED TO MAINTAIN SPACE COOLING SETPOINT OF 75°F (ADJ.). ON A DROP IN SPACE TEMPERATURE BELOW 68°F, THE UNIT'S HEAT MODE SHALL BE ACTIVATED AND THE UNIT SHALL RUN TO MAINTAIN SPACE HEAT SETPOINT OF 70°F (ADJ.). WHEN THERE IS NO CALL FOR COOLING OR HEATING, THE CONTROL VALVE SHALL CLOSE AND THE FAN SHALL BE DEACTIVATED.

DEHUMIDIFICATION MODE: ON AN INCREASE IN SPACE HUMIDITY ABOVE 60% RH (ADJ.), THE FAN SHALL ENERGIZE AND THE COMPRESSOR SHALL OPERATE IN COOLING MODE. THE HGR VALVE SHALL MODULATE TO MAINTAIN A SA-T OF 70°F (ADJ.). UPON RH SETPOINT BEING SATISFIED, THE HGR VALVE SHALL CLOSE AND UNIT SHALL RETURN TO NORMAL OPERATION.

SAFETY SHUTDOWN AND ALARMS: ALL ALARMS SHALL BE SENT TO THE DDC SYSTEM IF THE ALARM CONDITIONS OCCUR.

FLOAT SWITCH: IF THE FLOAT SWITCH IS TRIGGERED THE UNIT SHALL SHUT DOWN AND AN ALARM SHALL BE TRIGGERED AT THE DDC.

FRONT END GRAPHICS: ALL POINTS INDICATED IN CONTROL SEQUENCE AND CONTROL DIAGRAM SHALL BE PROVIDED IN GRAHICS OF FRONT END SYSTEM.

## 2 WATER-SOURCE HEAT PUMP CONTROLS

SCALE: NOT TO SCALE

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28501 | CEMS | M-703 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>D. ILDERTON | DEPARTMENT OF THE NAVY<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | NAVAL FACILITIES ENGINEERING COMMAND |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>P. MAHAFFEY | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M)<br>CONTROLS | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>P. MAHAFFEY | | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-18-D-0616 |
| | PUBLIC WORKS OFFICER | SCALE: 12" = 1'-0" MACC. T.O. # | SHEET 47 OF 53 |

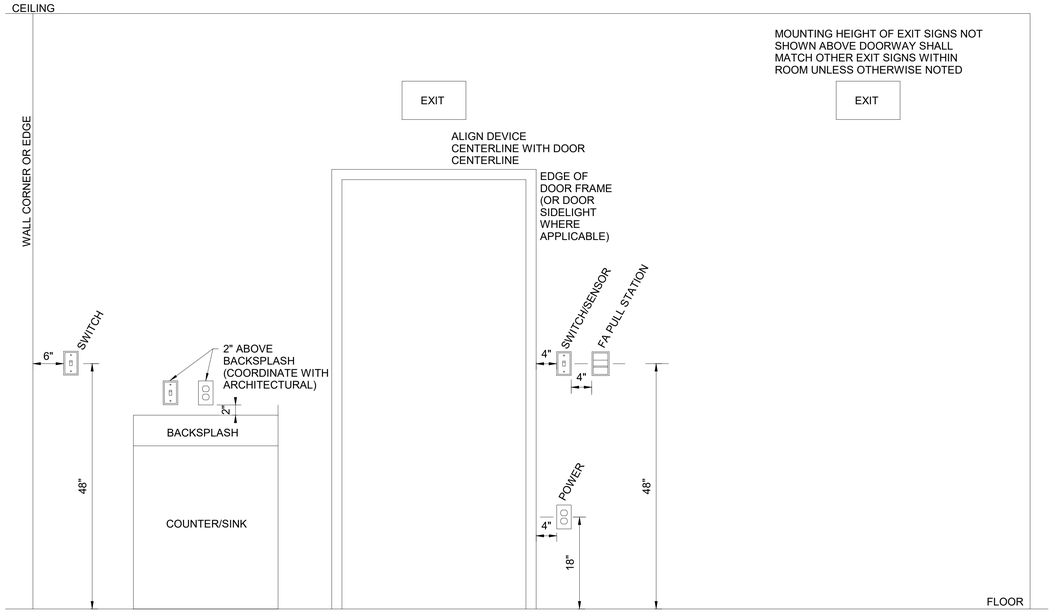
| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |

## LEGEND:

- RECEPTACLE, 20A, 120VOLT ('G' - INDICATES GROUND FAULT RECEPTACLE; 'W' - INDICATES A WEATHER PROOF WHILE IN USE RECEPTACLE)
- RECEPTACLE, 20A, 120VOLT MOUNTED AT 48" AFF. ('G' - INDICATES A GROUND FAULT RECEPTACLE)
- QUADRUPLEX 20A, 120VOLT RECEPTACLE, 18"AFF
- SWITCH, 20A ('3' - DENOTES THREEWAY, 'OS' - DENOTES OCCUPANCY SENSOR; 'm' - DENOTES MOTOR RATED SWITCH)
- 2'x4' LED LIGHT FIXTURE. 'E' DENOTES FIXTURE TO BE FED FROM EXISTING CENTRAL LIGHTING INVERTER UPS4. SEE FIXTURE SCHEDULE ON SHEET E-103A.
- 4' LED INDUSTRIAL STRIP LIGHT FIXTURE. 'E' DENOTES FIXTURE TO BE FED FROM EXISTING CENTRAL LIGHTING INVERTER UPS4. SEE FIXTURE SCHEDULE ON SHEET E-103A.
- 4-BLADE,52-IN LED CEILING FAN BASIS OF DESIGN: CRAFTMADE MODEL #LAV52ESP4LK-LED
- WALL MOUNTED LED FIXTURE. SEE FIXTURE SCHEDULE ON SHEET E-501.
- PANELBOARD SURFACE MOUNTED
- PANELBOARD RECESS MOUNTED
- EXIT SIGN WITH 90 MINUTES BATTERY BACKUP MINIMUM. SEE FIXTURE SCHEDULE ON SHEET E-501.
- SURFACE MOUNT ROUND LED FIXTURE. SEE FIXTURE SCHEDULE ON SHEET E-501.
- JUNCTION BOX
- EXISTING CAN LIGHT
- SURFACE MOUNT LED CANOPY LIGHT. 'E' DENOTES INTEGRAL BATTERY BACKUP WITH 90 MINUTES OPERATION MINIMUM. SEE FIXTURE SCHEDULE ON SHEET E-501.
- DISCONNECT SWITCH, NON-FUSED
- DISCONNECT SWITCH, FUSED
- CLASS 1 LIGHTNING PROTECTION SYSTEM BARE ALUMINUM CONDUCTOR, #1 AWG
- CLASS 1 ALUMINUM AIR TERMINAL 12"H MINIMUM, 1/2" DIAMETER MINIMUM
- DOWNLEAD TO GROUND, 1" PVC CONDUIT WITH CONDUCTORS TO COPPER-CLAD GROUND ROD
- SHEET KEYNOTE

## GENERAL ELECTRICAL NOTES (APPLIES TO ALL DRAWINGS):

1. ALL ELECTRICAL WORK IS TO BE DONE IN ACCORDANCE WITH THE CURRENT EDITION OF NFPA 70: NATIONAL ELECTRICAL CODE.
2. MATERIALS ARE TO BE NEW AND CONFORM TO THE STANDARDS OF THE UNDERWRITER'S LABORATORIES, INC. (U.L.) AND THE NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA).
3. CONTRACTOR SHALL CONSIDER VOLTAGE DROP WHEN DETERMINING THE EXACT ROUTING OF BRANCH CIRCUIT AND FEEDER WIRING. ADJUST WIRE AND CONDUIT SIZE AS NECESSARY TO PREVENT A VOLTAGE DROP EXCEEDING 3% AT THE FURTHEST DEVICE LOCATION.
4. ALL PENETRATIONS THROUGH FIRE RATED WALLS, FLOORS AND SMOKE PARTITIONS SHALL BE SEALED WITH AN APPROVED FIRE PROOFING SEALANT OR U.L. LISTED PENETRATION DEVICE.
5. ELECTRICAL DRAWING IS GENERALLY DIAGRAMMATIC. COORDINATE WORK WITH ALL TRADES PRIOR TO START OF CONSTRUCTION. BRANCH CIRCUIT CONDUIT ROUTING SHALL BE COORDINATED IN THE FIELD BY THE CONTRACTOR TO MEET SPECIFICATIONS, CODE REQUIREMENTS, AND TO PROVIDE A NEAT, WORKMAN LIKE, FULLY OPERATIONAL SYSTEM.
6. PROVIDE A TYPED SCHEDULE IN PANELBOARDS CLEARLY DESCRIBING THE LOCATION AND TYPE OF LOAD BEING SERVED FOR ALL CIRCUITS.
7. CONTRACTOR IS TO BE AWARE OF ALL RULES, REGULATIONS, AND CODES REQUIRED FOR SUCCESSFUL COMPLETION OF CONSTRUCTION.
8. COORDINATE WITH THE MECHANICAL PLANS FOR THE EXACT LOCATION OF ALL MECHANICAL EQUIPMENT BEFORE ROUGHING IN. ALL MECHANICAL EQUIPMENT SHALL BE ROUGHED IN AND WIRED ACCORDING TO THE SHOP DRAWINGS.
9. ALL SWITCHBOARDS, SWITCHGEAR, AND PANELBOARDS SUPPLIED BY A FEEDER ARE TO BE MARKED AS TO THE DEVICE OR EQUIPMENT WHERE THE POWER SUPPLY ORIGINATES. IDENTIFICATION MUST BE PERMANENTLY AFFIXED AND BE OF SUFFICIENT DURABILITY TO WITHSTAND THE ENVIRONMENT INVOLVED.



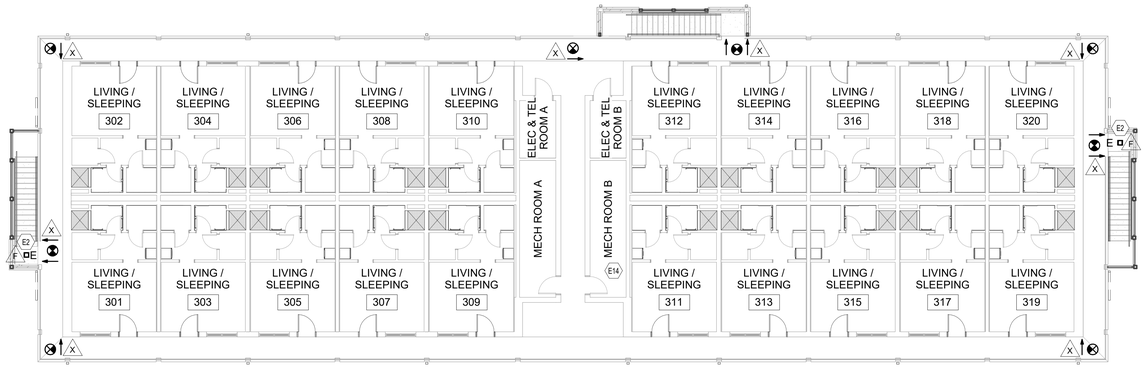

1 TYPICAL MOUNTING HEIGHTS

SCALE: NOT TO SCALE

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

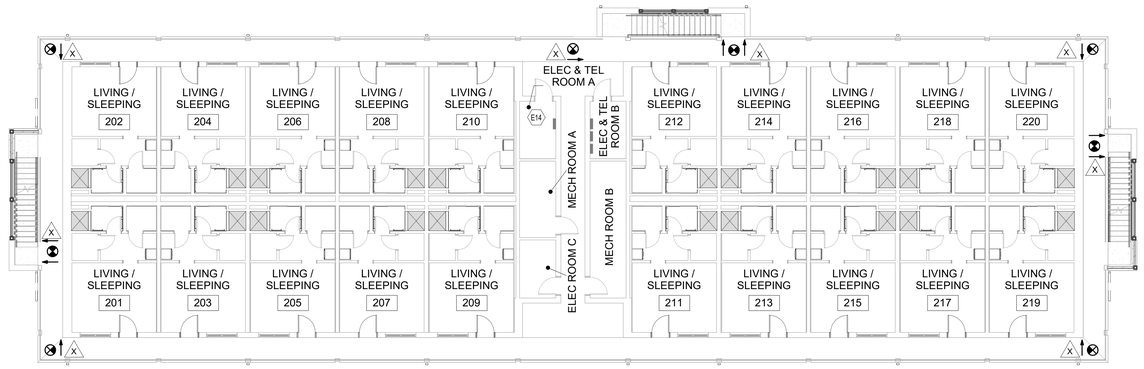
ISSUE DATE:08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28502 | CEMS | E-001 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>C. FISK | DEPARTMENT OF THE NAVY<br>NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>M. ACHORD | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>J. BONGIORNO | GENERAL NOTES, LEGENDS AND MOUNTING HEIGHTS | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.:; CONSTR. CONTR. NO.:N40085-17-B-0019 | |
| | PUBLIC WORKS OFFICER | SCALE: As indicated; MACC.; T.O. #; SHEET 48 OF 53 | |



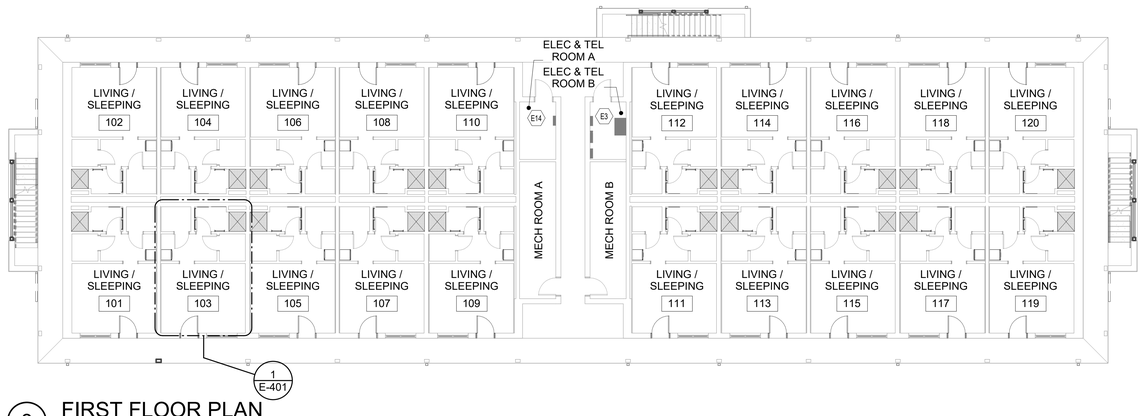

## 1 THIRD FLOOR PLAN

SCALE: 3/32" = 1'-0"




## 2 SECOND FLOOR PLAN

SCALE: 3/32" = 1'-0"




## 3 FIRST FLOOR PLAN

SCALE: 3/32" = 1'-0"

PLAN NORTH

24"0 4' 8' 16'

**SCALE: 3/32"=1'-0"**

**GRAPHIC SCALE**

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

# GENERAL NOTES

1. SEE SHEET E-001 FOR GENERAL NOTES AND LEGEND.
2. LIGHT FIXTURES INDICATED ARE PROVIDED AS A REPLACEMENT TO EXISTING FIXTURES AT SAME LOCATION. CONTRACTOR IS TO REMOVE EXISTING LIGHT FIXTURE AND REPLACE WITH NEW.
3. RE-USE EXISTING CIRCUITS AND CONTROLS PRESERVED DURING DEMOLITION TO SERVE NEW LIGHTING FIXTURES THROUGHOUT.

# SHEET KEYNOTES

E2 EXTEND EXISTING LOCAL LIGHTING CIRCUIT TO NEW FIXTURE.

E3 PROVIDE SPD AT SERVICE ENTRANCE WITH MINIMUM SURGE CURRENT RATING OF 80,000A FOR L-L MODE MINIMUM AND 40,000A FOR OTHER MODES (L-N,L-G,N-G). SPD TO HAVE MAXIMUM L-N, AND N-G VOLTAGE PROTECTION RATING OF 600V, MAXIMUM L-G PROTECTION RATING OF 700V, AND MAXIMUM L-L VOLTAGE PROTECTION RATING OF 1,200V.

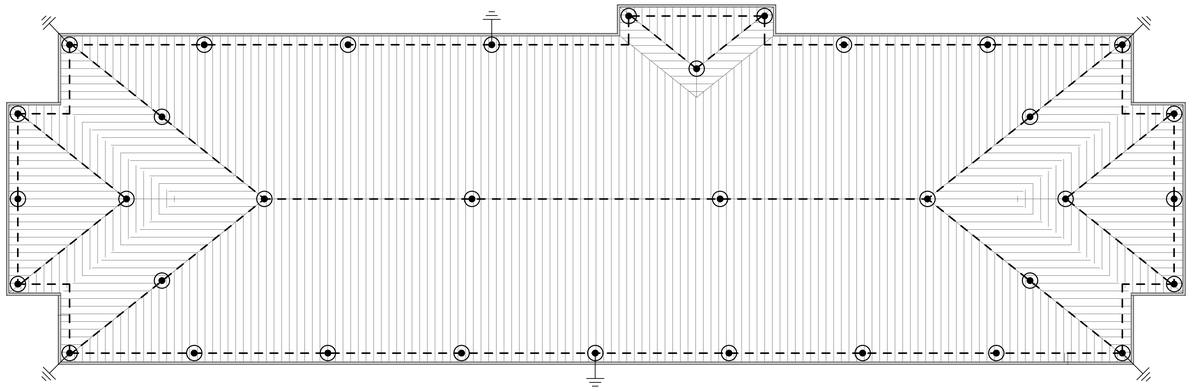
E14 DISCONNECT EXISTING 120V CIRCUIT FROM WSHP CONTROLLER TO BE REMOVED. CONNECT EXISTING, PRESERVED CIRCUIT TO NEW WSHP CONTROLLER TO BE INSTALLED BY DIVISION 23.

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

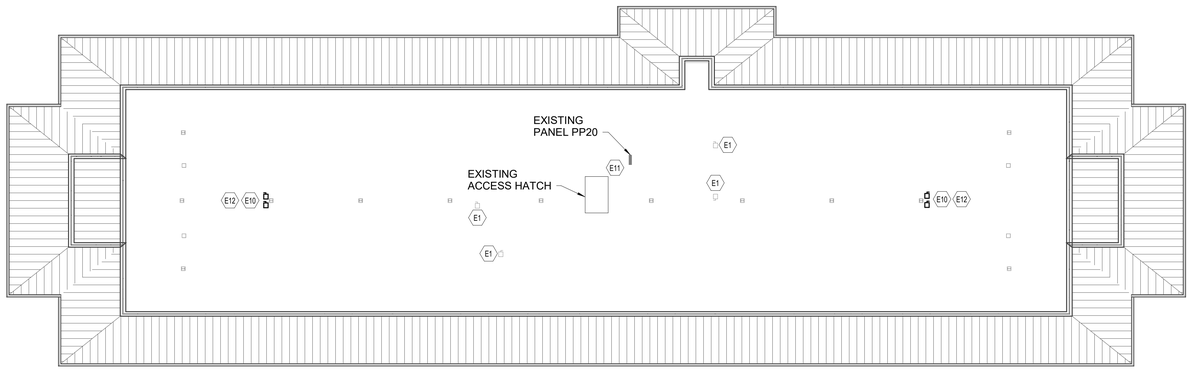
ISSUE DATE:08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28503 | CEMS | E-101B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>C. FISK | DEPARTMENT OF THE NAVY / NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>M. ACHORD | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>J. BONGIORNO | FIRST, SECOND AND THIRD FLOOR PLANS - BLDG 1245G | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F / CODE IDENT. NO. / N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.:N40085-17-B-0019 |
| | PUBLIC WORKS OFFICER | SCALE: 3/32" = 1'-0" / MACC. / T.O. # | SHEET 49 OF 53 |

10/09/2023



1 ROOF PLAN
SCALE: 3/32" = 1'-0"




2 ATTIC FLOOR PLAN
SCALE: 3/32" = 1'-0"

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. SEE SHEET E-001 FOR GENERAL NOTES AND LEGEND.
2. PROVIDE LIGHTNING PROTECTION SYSTEM TO MEET THE REQUIREMENTS OF NFPA 780.
3. SYSTEM SHALL BE PROVIDED WITH UL MASTER LABEL.
4. ALL ROOF PENETRATIONS SHALL BE MADE WATERTIGHT.
5. BOND LPS TO VENTS, METAL APPETENCIES, FANS AND OTHER METAL COMPONENTS AS REQUIRED BY NFPA 780.
6. ALL GROUNDING MEDIA IN OR ON STRUCTURE INCLUDING LIGHTNING PROTECTION, ELECTRICAL SERVICE AND TELEPHONE SHALL BE INTERCONNECTED TO PROVIDE A COMMON GROUND POTENTIAL.
7. INSTALLATION TO BE IN ACCORDANCE WITH U.L. REQUIREMENTS AND TO RECEIVE U.L. MASTER LABEL 'C'.
8. BOND ALL BODIES OF CONDUCTANCE, AND BODIES OF INDUCTANCE LOCATED WITHIN 6' OF MAIN CONDUCTOR.
9. ALL CONDUCTOR BENDS TO BE A MINIMUM OF 90° WITH A MINIMUM 8" RADIUS.
10. CONDUCTOR TO BE FASTENED A MAXIMUM OF 3'-0" ON CENTER.
11. CONDUCTORS TO BE CLASS 1 ALUMINUM, 24 STRAND MINIMUM.
12. AIR TERMINAL SPACING SHALL CONFORM TO NFPA 780 AND UL96A REQUIREMENTS.
13. MATERIAL COMPATIBILITY SHALL CONFORM TO NFPA 780 AND UL96A REQUIREMENTS.
14. MATERIAL CLASS SHALL CONFORM TO NFPA 780 AND UL96A REQUIREMENTS.
15. MAINTAIN SEPARATION OF DISSIMILAR MATERIAL, I.E. ALUMINUM CONDUCTOR VS. STEEL ROOF.

## SHEET KEYNOTES

E1 EXISTING DISCONNECT AND CIRCUIT TO REMAIN. CONTRACTOR IS TO DISCONNECT EXISTING FLEXIBLE CONNECTION FROM EXISTING EQUIPMENT TO BE REMOVED AND CONNECT TO NEW EQUIPMENT.

E10 REMOVE EXISTING CIRCUITS FEEDING EXISTING WATER SOURCE HEAT PUMPS. REMOVE MOTOR STARTERS AND CONDUIT AND CONDUCTORS BACK TO PANEL. REMOVE CIRCUIT BREAKERS.

E11 PROVIDE NEW 80A, 250V, 2 POLE CIRCUIT BREAKERS IN EXISTING PANEL FOR NEW WATER SOURCE HEAT PUMPS.

E12 PROVIDE NEW MOTOR STARTER FOR EACH NEW HEAT PUMP WITH 2#4, #8G, 1"C BACK TO EXISTING PANEL.

PLAN NORTH

24"0 4' 8' 16'

**SCALE: 3/32"=1'-0"**

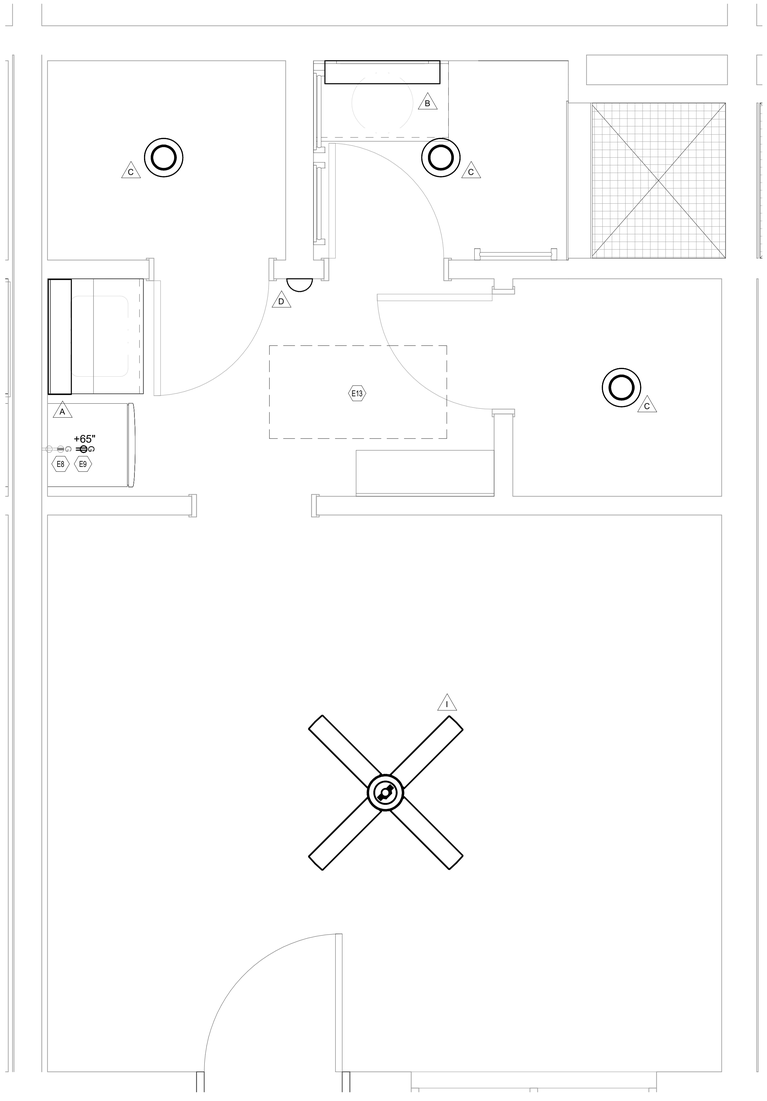
**GRAPHIC SCALE**

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28504 | CEMS | E-102B |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>C. FISK | DEPARTMENT OF THE NAVY<br>NAVAL FACILITIES ENGINEERING COMMAND<br>**MARINE CORPS AIR STATION**<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>M. ACHORD | REPAIR PHASE 4 BEQ BUILDING 1245G<br>(BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>J. BONGIORNO | ATTIC AND ROOF PLANS - BLDG 1245G | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE F / CODE IDENT. NO. / N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.: | CONSTR. CONTR. NO.: N40085-17-B-0019 |
| | PUBLIC WORKS OFFICER | SCALE: 3/32" = 1'-0" / MACC. / T.O. # / SHEET 50 OF 53 | |

REGISTERED PROFESSIONAL ENGINEER No. 28176 10/04/2023




1 ENLARGED PLAN
SCALE: 3/4" = 1'-0"

**REVISIONS**

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## GENERAL NOTES

1. SEE SHEET E-001 FOR GENERAL NOTES AND LEGEND.
2. LIGHT FIXTURES INDICATED ARE PROVIDED AS A REPLACEMENT TO EXISTING FIXTURES AT SAME LOCATION. CONTRACTOR IS TO REMOVE EXISTING LIGHT FIXTURE AND REPLACE WITH NEW.
3. RE-USE EXISTING CIRCUITS AND CONTROLS PRESERVED DURING DEMOLITION TO SERVE NEW LIGHTING FIXTURES THROUGHOUT.

## SHEET KEYNOTES

E8 REMOVE EXISTING GFI OUTLET AND ASSOCIATED CONDUCTORS INSTALLED AT COUNTER HEIGHT. PROVIDE BLANK PLATE AT EXISTING BOX.

E9 REPLACE EXISTING OUTLET INSTALLED 65" AFF WITH GFI OUTLET.

E13 DISCONNECT EXISTING FEEDER FROM WATER SOURCE HEAT PUMP AND PRESERVE CIRCUIT FOR RE-USE. CONNECT NEW HEAT PUMP TO PRESERVED CIRCUIT.

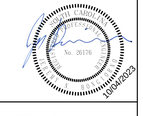
12" 0 1' 2'

SCALE: 3/4"=1'-0"

GRAPHIC SCALE

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023




| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28505 | CEMS | E-401 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>C. FISK | DEPARTMENT OF THE NAVY<br>NAVAL FACILITIES ENGINEERING COMMAND<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>M. ACHORD | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M)<br>ENLARGED PLAN | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>J. BONGIORNO | | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE<br>PUBLIC WORKS OFFICER | SIZE F / CODE IDENT. NO. / N A V F A C DRAWING NO.<br>HQMC PROJ. NO.: / CONSTR. CONTR. NO.: N40085-17-B-0019<br>SCALE: 3/4" = 1'-0" / MACC. / T.O. # / SHEET 51 OF 53 | |

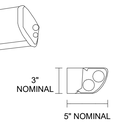
| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |

LUMINAIRE REQUIREMENTS:

1. HOUSING - DIE-FORMED, COLD-ROLLED STEEL, WITH REINFORCEMENT RIBS FOR RIGIDITY; END CAPS SAME MATERIAL AS HOUSING, SECURED WITH TABS, SCREWS OR RIVETS. WHITE ENAMEL FINISH APPLIED AFTER FABRICATION. OPTIONAL LENGTHS OF 2FT OR 4FT.
2. LENS - ACRYLIC PRISMATIC OR WHITE OPAL TO PROVIDE EVEN ILLUMINATION AND TO ELIMINATE PIXILATION FROM LED LIGHT SOURCE.
3. LIGHT SOURCE - SOLID STATE LEDS WITH MINIMUM 50K HOURS RATED LIFE AT L70, 3500K CCT UON, MINIMUM 80 CRI, MAXIMUM 4-STEP MCADAM ELLIPSE BINNING TOLERANCE FOR COLOR CONSISTENCY, AND MINIMUM EFFICACY OF 90 LUMENS/WATT. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
4. DRIVER - REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, < 20% TOTAL HARMONIC DISTORTION. ON-OFF CONTROL, STEP-DIMMABLE OR FULLY DIMMABLE AS INDICATED.
5. CERTIFICATION - UL 1598, DAMP LOCATION, DLC QUALIFIED, AND ROHS COMPLIANT. COMPLIES WITH LM79, LM80 AND TM21 TESTING STANDARDS. UL 924 WHEN EQUIPPED WITH EMERGENCY BATTERY BACK-UP.
6. MOUNTING - SURFACE.
7. OPTIONS - INTEGRAL MOTION SENSOR, EMERGENCY BACK-UP.
8. THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

## WALL MOUNTED LED

| REVISED: | APRIL 2016 | LIGHTING PLATE: | NL-9 |
|---|---|---|---|




LUMINAIRE REQUIREMENTS:

1. HOUSING - DIE-FORMED, COLD-ROLLED STEEL. ENDCAPS WHEN USED SHALL BE MADE FROM SAME MATERIAL AS HOUSING EXCEPT WHEN OPTIONAL FINISH SUCH AS CHROME IS SPECIFIED.
2. FINISH - MULTI-STAGE PHOSPHATE BONDING TREATMENT FINISHED WITH HIGH REFLECTANCE (MINIMUM 85%), BAKED WHITE ENAMEL FINISH.
3. DIFFUSER - 100% ACRYLIC, CLEAR PRISMATIC, OR WHITE OPAL ACRYLIC. DECORATIVE TREATMENTS ON DIFFUSER ENDS OPTIONAL.
4. FIXTURE TYPES -

   TYPE A - 2' LENGTH
5. LIGHT SOURCE - LED BOARD OR ARRAY WITH MINIMUM 50K HOURS RATED LIFE AT L70, 3500K COLOR TEMPERATURE UON, MAXIMUM 4-STEP MCADAM ELLIPSE BINNING TOLERANCE FOR COLOR CONSISTENCY, MINIMUM EFFICACY OF 90 LUMENS/WATT, WITH A MINIMUM CRI OF 80. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
6. DRIVER - REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, < 20% TOTAL HARMONIC DISTORTION. STEP-DIMMABLE OR FULLY DIMMABLE AS INDICATED IN LUMINAIRE SCHEDULE.
7. CERTIFICATION - UL 1598, DAMP LOCATION, DLC QUALIFIED, AND ROHS COMPLIANT, COMPLIES WITH LM79, LM80 AND TM21 TESTING STANDARDS. UL 924 WHEN EQUIPPED WITH EMERGENCY BATTERY BACK-UP. IC RATED WHEN INDICATED.
8. THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

## LED VANITY LIGHT

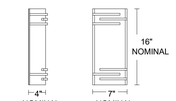
| REVISED: | AUGUST 2004 | LIGHTING PLATE: | NL-24 |
|---|---|---|---|

LUMINAIRE REQUIREMENTS:

1. HOUSING - DIE-FORMED ALUMINUM WITH KEYHOLE SLOTS FOR DIRECT MOUNTING TO 4" OCTAGONAL OUTLET BOX.
2. FINISH - MULTI-STAGE PHOSPHATE BONDING TREATMENT FINISHED WITH BAKED WHITE ENAMEL FINISH.
3. LENS - FORMED WHITE TRANSLUCENT OPAL ACRYLIC DIFFUSER.
4. LIGHT SOURCE - LED BOARD OR ARRAY WITH MINIMUM 50K HOURS RATED LIFE AT L70, 3500K COLOR TEMPERATURE UON, MAXIMUM 4-STEP MCADAM ELLIPSE BINNING TOLERANCE FOR COLOR CONSISTENCY, MINIMUM EFFICACY OF 90 LUMENS/WATT, WITH A MINIMUM CRI OF 80. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
5. DRIVER - REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, < 20% TOTAL HARMONIC DISTORTION. STEP-DIMMABLE OR FULLY DIMMABLE AS INDICATED IN LUMINAIRE SCHEDULE.
6. CERTIFICATION - UL 1598, DAMP LOCATION, DLC QUALIFIED, AND ROHS COMPLIANT. COMPLIES WITH LM79, LM80 AND TM21 TESTING STANDARDS. UL 924 WHEN EQUIPPED WITH EMERGENCY BATTERY BACK-UP. IC RATED WHEN INDICATED.
7. THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

## OPAL ACRYLIC CEILING-MOUNTED LED

| REVISED: | AUGUST 2004 | LIGHTING PLATE: | NL-26 |
|---|---|---|---|



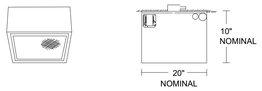

LUMINAIRE REQUIREMENTS:

1. HOUSING/BACKPLATE - DIE-FORMED, COLD-ROLLED STEEL, WITH HIGHLY-REFLECTIVE WHITE ENAMEL FINISH.
2. TRIM - OPTIONS INCLUDE BRUSHED SOLID ALUMINUM, BRASS OR CHROME-PLATED BRASS, SATIN NICKEL, AND OTHERS. FINISH OPTIONS VARY BY MANUFACTURER.
3. DIFFUSER - TRANSLUCENT WHITE ACRYLIC OR WHITE OPAL GLASS.
4. LIGHT SOURCE - LED BOARD OR ARRAY WITH MINIMUM 50K HOURS RATED LIFE AT L70, 3500K COLOR TEMPERATURE UON, MAXIMUM 4-STEP MCADAM ELLIPSE BINNING TOLERANCE FOR COLOR CONSISTENCY, MINIMUM EFFICACY OF 90 LUMENS/WATT, WITH A MINIMUM CRI OF 80. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
5. DRIVER - REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, < 20% TOTAL HARMONIC DISTORTION. STEP-DIMMABLE OR FULLY DIMMABLE AS INDICATED IN LUMINAIRE SCHEDULE.
6. CERTIFICATION - UL 1598, DAMP LOCATION, DLC QUALIFIED, AND ROHS COMPLIANT. COMPLIES WITH LM79, LM80 AND TM21 TESTING STANDARDS. UL 924 WHEN EQUIPPED WITH EMERGENCY BATTERY BACK-UP. IC RATED WHEN INDICATED.
7. MOUNTING - FIXTURE SHALL MOUNT DIRECTLY TO 4" OCTAGONAL OUTLET BOX WITH FOUR SEPARATE MOUNTING SCREWS.
8. OPTIONS/OTHER - OPTIONS INCLUDE REMOTE EMERGENCY BATTERY BACK-UP AND VARIOUS OTHER FINISHES.

   NOTE: FIXTURE DEPICTED IS ONE OF MANY TYPES AVAILABLE. PROVIDE ADDITIONAL DETAILS AND SKETCHES WHERE NECESSARY TO DEPICT EXACT SCONCE TO BE UTILIZED.

## LED WALL SCONCE

| REVISED: | AUGUST 2004 | LIGHTING PLATE: | NL-46 |
|---|---|---|---|

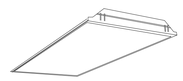



LUMINAIRE REQUIREMENTS:

1. HOUSING - TEMPERED, RIGID RUST-PROOF ALUMINUM WITH NO VISIBLE HOUSING HARDWARE. FINISHED IN NATURAL ALUMINUM, DARK BRONZE, OR BLACK ENAMEL.
2. REFLECTOR - CLEAR SPECULAR ALUMINUM.
3. LENS - UV-ABSORBING, TEMPERED PRISMATIC FLAT OR REGRESSED GLASS OR DROP POLYCARBONATE PRISMATIC LENS. PROVIDE FULLY GASKETED, ALUMINUM DOOR SECURED WITH CAM LATCHES.
4. LIGHT SOURCE - LED BOARD OR ARRAY WITH MINIMUM 50K HOURS RATED LIFE AT L70, 4000K COLOR TEMPERATURE UON, MAXIMUM 4-STEP MCADAM ELLIPSE BINNING TOLERANCE FOR COLOR CONSISTENCY, MINIMUM EFFICACY OF 90 LUMENS/WATT, WITH A MINIMUM CRI OF 80. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
5. DRIVER - REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, < 20% TOTAL HARMONIC DISTORTION. STEP-DIMMABLE OR FULLY DIMMABLE AS INDICATED IN LUMINAIRE SCHEDULE.
6. CERTIFICATION - UL LISTED AND LABELED FOR WET LOCATIONS.
7. THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

## LED CANOPY

| REVISED: | DECEMBER 2022 | LIGHTING PLATE: | NL-54 |
|---|---|---|---|



LUMINAIRE REQUIREMENTS:

1. HOUSING - DIE-FORMED, COLD ROLLED STEEL, PAINTED AFTER FABRICATION. EXTRUDED ALUMINUM DOOR WITH SPRING-LOADED CAM LATCHES. HIGH REFLECTANCE, WHITE POLYESTER POWDER-COATED FINISH. OPTIONAL SIZES OF 1FT x 4FT, 2FT x 2FT, AND 2FT x 4FT AVAILABLE.
2. LIGHT SOURCE - LED BOARD OR ARRAY WITH MINIMUM 50K HOURS RATED LIFE AT L70, 3500K COLOR TEMPERATURE UON, MAXIMUM 4-STEP MCADAM ELLIPSE BINNING TOLERANCE FOR COLOR CONSISTENCY, MINIMUM EFFICACY OF 90 LUMENS/WATT, WITH A MINIMUM CRI OF 80. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
3. DRIVER - REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, < 20% TOTAL HARMONIC DISTORTION. STEP-DIMMABLE OR FULLY DIMMABLE AS INDICATED IN LUMINAIRE SCHEDULE.
4. CERTIFICATION - UL 1598, DAMP LOCATION, DLC QUALIFIED, AND ROHS COMPLIANT. COMPLIES WITH LM79, LM80 AND TM21 TESTING STANDARDS. UL 924 WHEN EQUIPPED WITH EMERGENCY BATTERY BACK-UP. IC RATED WHEN INDICATED.
5. MOUNTING - RECESSED IN SUSPENDED ACOUSTICAL TILE OR HARD CEILING.
6. THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

## PRISMATIC LENS LED TROFFER

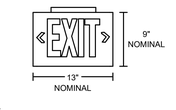
| REVISED: | APRIL 2016 | LIGHTING PLATE: | NL-3 |
|---|---|---|---|

LUMINAIRE REQUIREMENTS:

1. HOUSING - DIE-FORMED, COLD-ROLLED STEEL, WITH REINFORCEMENT RIBS FOR RIGIDITY AND SPECULAR ALUMINUM OR HIGHLY-REFLECTIVE PAINTED STEEL REFLECTORS. OPTIONAL LENGTHS OF 4FT OR 8FT.
2. FINISH - MULTI-STAGE PHOSPHATE BONDING TREATMENT FINISHED WITH HIGH-REFLECTANCE, WHITE POLYESTER POWDER COAT, PAINTED AFTER FABRICATION.
3. LIGHT SOURCE - SOLID STATE LEDS WITH MINIMUM 50K HOURS RATED LIFE AT L70, 3500K CCT UON, MINIMUM 80 CRI, MAXIMUM 4-STEP MCADAM ELLIPSE BINNING TOLERANCE FOR COLOR CONSISTENCY, AND MINIMUM EFFICACY OF 100 LUMENS/WATT. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
4. DRIVER - REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, < 20% TOTAL HARMONIC DISTORTION. ON-OFF CONTROL, STEP-DIMMABLE OR FULLY DIMMABLE AS INDICATED.
5. CERTIFICATION - UL 1598, DAMP LOCATION, DLC QUALIFIED, AND ROHS COMPLIANT. COMPLIES WITH LM79, LM80 AND TM21 TESTING STANDARDS. UL 924 WHEN EQUIPPED WITH EMERGENCY BATTERY BACK-UP.
6. MOUNTING - SURFACE ON CEILING OR SUSPENDED.
7. OPTIONS - WIRE GUARD, CHAIN, STEM OR SWIVEL STEM HANGERS.
8. THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

## LED INDUSTRIAL STRIP

| REVISED: | APRIL 2016 | LIGHTING PLATE: | NL-7 |
|---|---|---|---|




LUMINAIRE REQUIREMENTS:

1. HOUSING - DIE-CAST ALUMINUM OR HIGH-IMPACT, UV-STABILIZED, INJECTION-MOLDED THERMOPLASTIC. SINGLE OR DOUBLE-FACED AS INDICATED.
2. FINISH (ON CAST ALUMINUM HOUSING ONLY) - TEXTURED POWDER COAT FINISH OPTIONS INCLUDE WHITE, WHITE WITH BRUSHED ALUMINUM FACE, BLACK, OR BLACK WITH BRUSHED ALUMINUM FACE.
3. LETTERS/CHEVRONS - MINIMUM 6" HIGH WITH 3/4" STROKE. RED OR GREEN LETTERS AS INDICATED. PROVIDE CHEVRONS AS INDICATED EITHER LEFT, RIGHT OR BOTH DIRECTIONS AS INDICATED. CHEVRONS PUNCHED OUT THROUGH HOUSING AS REQUIRED.
4. EMERGENCY PACK - SOLID-STATE, CONSTANT-CURRENT TYPE BATTERY CHARGER WITH MAINTENANCE-FREE, NICKEL-CADMIUM BATTERY, AC-ON INDICATOR LAMP AND TEST SWITCH.
5. MOUNTING - UNIVERSAL MOUNTING KIT FOR CEILING, WALL OR END-OF-FIXTURE MOUNTING.
6. ILLUMINATION - PROVIDED BY RED, GREEN OR WHITE HIGH-OUTPUT LEDS INSIDE OF FIXTURE HOUSING. PROVIDE POLYSTYRENE DIFFUSER IN COLOR INDICATED WITH FREQUENCY-MATCHED SILKSCREEN COATING FOR MAXIMUM LED LIGHT OUTPUT.
7. CERTIFICATION - UL LISTED AND CERTIFIED FOR DAMP LOCATIONS.

## LED EXIT SIGN

| REVISED: | AUGUST 2004 | LIGHTING PLATE: | NL-63 |
|---|---|---|---|

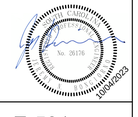
## LIGHTING FIXTURE SCHEDULE

| FIXTURE SYMBOL | STYLE NO. AND TYPE | NUMBER AND TYPE OF LAMPS | VOLTAGE | MOUNTING | NOTES |
|---|---|---|---|---|---|
| A | NL-9 | LED - 900 LUM | 120 | UNDERCABINET | |
| B | NL-24 | LED - 3000 LUM | 120 | WALL MTD AT EXISTING BOX | |
| C | NL-26 | LED - 1000 LUM | 120 | SURFACE | |
| D | NL-46 | LED - 1300 LUM | 120 | WALL MTD AT EXISTING BOX | |
| E | NL-54 | LED - 2800 LUM | 120 | SURFACE | "E" DENOTES FIXTURE TO BE PROVIDED WITH 90 MINUTES MINIMUM INTEGRAL BATTERY BACKUP |
| EG | NL-3 | LED - 5400 LUM | 120 | RECESSED 9'-0" AFF | "E" DENOTES FIXTURE TO BE FED FROM EXISTING CENTRAL LIGHTING INVERTER UPS4 |
| EG | NL-3 | LED - 5400 LUM | 120 | RECESSED 9'-0" AFF | "E" DENOTES FIXTURE TO BE FED FROM EXISTING CENTRAL LIGHTING INVERTER UPS4. PROVIDE FIXTURE SUITABLE FOR HARD CEILING MOUNTING. |
| EL | NL-7 | LED - 2800 LUM | 120 | SURFACE | "E" DENOTES FIXTURE TO BE FED FROM EXISTING CENTRAL LIGHTING INVERTER UPS4 |
| I | N/A | TWO - A19 BASE LED BULBS 10W | 120 | SURFACE | BASIS OF DESIGN: CRAFTMADE MODEL #LAVS2ESP4LK-LED IN MATTE WHITE |
| X | NL-63 | RED LED | 120 | UNIVERSAL | PROVIDE FIXTURE WITH 90 MINUTES MINIMUM INTEGRAL BATTERY BACKUP |

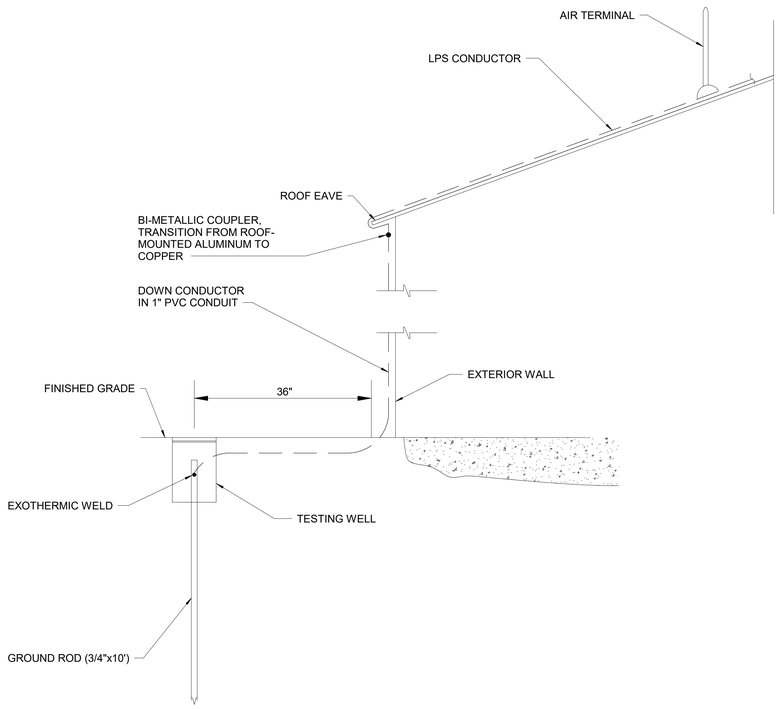
IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28506 | CEMS | E-501 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>C. FISK | DEPARTMENT OF THE NAVY<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | NAVAL FACILITIES ENGINEERING COMMAND |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>M. ACHORD | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>J. BONGIORNO | LIGHTING SCHEDULES AND DETAILS | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE | SIZE: F; CODE IDENT. NO.; N A V F A C DRAWING NO. | |
| | | HQMC PROJ. NO.:; CONSTR. CONTR. NO.: N40085-17-B-0019 | |
| | PUBLIC WORKS OFFICER | SCALE: 12" = 1'-0"; MACC.; T.O. #; SHEET 52 OF 53 | |

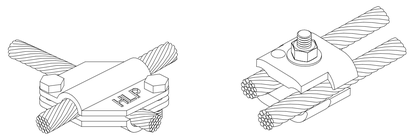


| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| | | | |
| | | | |
| | | | |
| | | | |




## 1 ROOF EXPOSED DOWN LEAD TO GROUND ROD CONNECTION

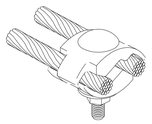
SCALE: NOT TO SCALE



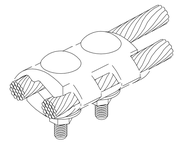
NO. 210 STYLE CROSS RUN CONNECTOR



NO. B1BC STYLE PARALLEL CONNECTOR



NO. 208 STYLE BONDING CONNECTOR

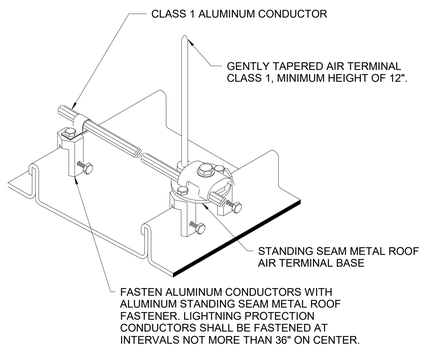


NO. 204 STYLE PARALLEL CONNECTOR

NO. 211, BI-METAL CONNECTOR

## 2 TYPICAL CABLE CONNECTIONS

SCALE: NOT TO SCALE




NOTES:
1. AIR TERMINAL SPACING SHALL CONFORM TO NFPA 780 AND UL 96A REQUIREMENTS.
2. MATERIAL COMPATIBILITY SHALL CONFORM TO NFPA 780 AND 96A REQUIREMENTS.
3. MATERIAL CLASS SHALL CONFORM TO NFPA 780 AND UL 96A REQUIREMENTS.
4. AIR TERMINALS SHALL BE UL LISTED AND CAL-OSHA APPROVED.

## 3 UBS - STANDING SEAM AIR TERMINAL

SCALE: NOT TO SCALE

IF SHEET IS LESS THAN (30"x42") IT IS A REDUCED PRINT; SCALE ACCORDINGLY

ISSUE DATE:08 SEPT 2023

| | | | |
|---|---|---|---|
| NATURAL RESOURCES & ENVIRONMENTAL AFFAIRS<br>SATISFACTORY TO DATE | P.W. DWG. NO.<br>28507 | CEMS | E-502 |
| SAFETY DEPARTMENT<br>SATISFACTORY TO DATE | DRAWN BY<br>C. FISK | DEPARTMENT OF THE NAVY<br>MARINE CORPS AIR STATION<br>BEAUFORT, S.C. | NAVAL FACILITIES ENGINEERING COMMAND |
| FIRE DEPARTMENT<br>SATISFACTORY TO DATE | CHECKED BY<br>M. ACHORD | REPAIR PHASE 4 BEQ BUILDING 1245G (BE1813M) | |
| PHYSICAL SECURITY<br>SATISFACTORY TO DATE | SUPERVISOR<br>J. BONGIORNO | DETAILS | |
| PUBLIC WORKS SUPERVISOR<br>SATISFACTORY TO DATE | APPROVED DATE<br>PUBLIC WORKS OFFICER | SIZE F; CODE IDENT. NO.; N A V F A C DRAWING NO.<br>HQMC PROJ. NO.:<br>CONSTR. CONTR. NO.: N40085-17-B-0019<br>SCALE: As indicated; MACC.; T.O. #; SHEET 53 OF 53 | |